

# UNIQUENESS OF GERVER'S SOFA

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**ABSTRACT.** The moving sofa problem asks for the largest area of a closed connected planar shape that can be moved around the right-angled corner of a hallway of unit width. Gerver found a shape of area  $2.2195\dots$  in 1992 and conjectured that it is optimal; Baek proved this in 2024. We prove that Gerver's sofa is the only optimal shape: every moving sofa of the same area is a rotated and translated copy of it. To prove optimality, Baek first bounds the area of any large moving sofa by that of a special shape with two geometric properties. Baek then bounds the area of every shape with these properties by a concave functional whose largest value is the area of Gerver's sofa. Since this chain of inequalities passes through the special shape, it says nothing about the shape of a moving sofa of maximal area. To prove uniqueness, we show that every moving sofa of maximal area has the same two properties, so that the functional bounds its area directly. Equality now forces the sofa to be Gerver's. This argument does not use the value of the maximal area, so it also gives a second proof of optimality. We prove, moreover, that the uniqueness is stable: a moving sofa whose area is  $\varepsilon$  less than the maximum lies, after a translation, within Hausdorff distance  $O(\sqrt{\varepsilon})$  of Gerver's sofa, and the exponent  $1/2$  cannot be improved. We also derive optimality, uniqueness and stability from one estimate for this functional. The proofs, together with Baek's, are formalized in Lean 4 with Mathlib and checked by Lean's kernel, using only Lean's standard axioms.

## 1. INTRODUCTION

**1.1. The moving sofa problem.** Let  $L$  be the L-shaped hallway of unit width, the union of the horizontal side  $H_L = (-\infty, 1] \times [0, 1]$  and the vertical side  $V_L = [0, 1] \times (-\infty, 1]$ . A *moving sofa* is a closed connected planar set that can be moved rigidly from the horizontal side to the vertical side without leaving  $L$  (Definition 2.2). For example, the unit square is a moving sofa of area 1. Moser [8] asked for the largest area of a moving sofa. Hammersley [5] found a sofa of area  $\pi/2 + 2/\pi = 2.2074\dots$ . Gerver [3] found one of area  $2.21953\dots$ , bounded by 18 analytic arcs and segments, and conjectured that it is optimal. Romik [10] derived Gerver's sofa  $G$  from differential equations that express the local optimality of its sides, and described its boundary

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explicitly in terms of two angles that solve transcendental equations (Figure 1, Appendix B). Kallus and Romik [6] proved by a computer-assisted method that every moving sofa has area at most 2.37. Baek [1] proved Gerver's conjecture: every moving sofa has area at most  $|G| = 2.2195\dots$ , where  $|X|$  denotes the area (Lebesgue measure) of a set  $X \subset \mathbb{R}^2$ .

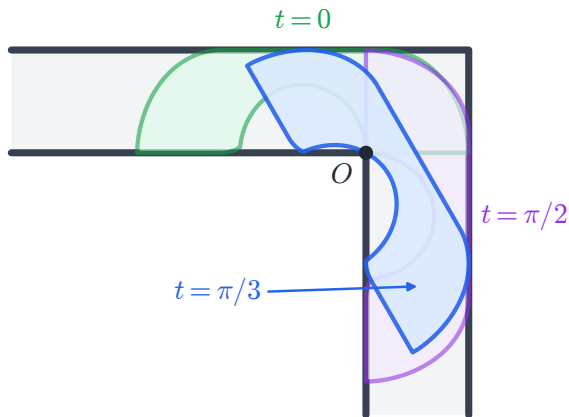


FIGURE 1. Gerver's sofa  $G$  in the hallway  $L$ , turned clockwise by the angle  $t$ : at  $t = 0$  (green), where it slides along the horizontal side, at  $t = \pi/3$  (blue), turning round the inner corner  $O$ , and at  $t = \pi/2$  (purple), where it leaves along the vertical side.

**1.2. Uniqueness.** We prove that Gerver's sofa is the only optimal sofa, up to a rotation and a translation. We write  $R_\theta$  for the counterclockwise rotation of the plane by the angle  $\theta$  about the origin.

**Theorem 1.1** (uniqueness of Gerver's sofa). *Let  $S$  be a moving sofa (Definition 2.2) with  $|S| = |G|$ . Then there are an angle  $\theta$  and a vector  $v$  such that*

$$R_\theta S + v = G.$$

Equivalently, a moving sofa has maximum area if and only if it is a rotated and translated copy of  $G$  (Corollary 9.4).

Note that Theorem 1.1 gives an equality of sets. This is stronger than what an equality of areas yields, which allows the two sets to differ by a set of measure zero, such as a point or a segment.

The uniqueness is stable (Section 10): a moving sofa whose area is  $\varepsilon$  less than  $|G|$  lies, after a translation, within Hausdorff distance  $O(\sqrt{\varepsilon})$  of  $G$  (Theorem 10.1), and the exponent  $1/2$  cannot be improved (Theorem 10.3).

**1.3. Baek's proof of optimality.** We recall the outline of Baek's proof [1], since ours follows it. Appendix A illustrates the notions and the steps of this subsection (Figures 10 to 14).

When a moving sofa is moved from the horizontal side to the vertical side, it is rotated by a total angle  $\omega$ , its *rotation angle*; in Figure 1,  $\omega = \pi/2$ . Baek's proof fixes the sofa and moves the hallway around it. For each angle  $t$  between 0 and  $\omega$ , place a copy of the hallway, rotated by  $t$ , so that its two outer walls touch the sofa; the sofa lies inside this copy. The *cap*  $K$  is the set of points that lie in the horizontal side of the hallway at the start, in its vertical side at the end, and inside the outer walls of all the copies. Figure 10 shows this for Gerver's sofa, with one of the copies in grey: the cap is the union of the blue and the orange regions. Since  $K$  touches both outer walls of every copy, placing the copies against  $K$  instead of the sofa gives the same copies. So the copies, and their inner walls, one unit inside the outer walls, depend only on  $K$  and  $\omega$ . The *niche*  $\mathcal{N}(K)$  is the part of  $K$  that, for at least one copy, lies outside the copy, in the wedge between its two inner walls (in Figure 10, the wedge below the orange dot, between the two grey lines that meet there). The set  $K \setminus \mathcal{N}(K)$  thus consists of the points of  $K$  that lie inside every copy; a set of this form is a *monotone sofa*, and its area  $\mathcal{A}_\omega(K) = |K| - |\mathcal{N}(K)|$  is the *sofa area* of  $K$ . Gerver's sofa is a monotone sofa; in Figure 10 it is the blue region, and its niche is the orange region.

Baek proves that  $G$  is optimal in four steps.

- (1) Baek shows that, after a translation, every moving sofa  $S$  with rotation angle  $\omega$  lies in the monotone sofa  $K \setminus \mathcal{N}(K)$  of the cap  $K$  of  $S$ . Baek also shows that this monotone sofa is again a moving sofa with rotation angle  $\omega$ . So the area of  $S$  is at most  $\mathcal{A}_\omega(K)$ , and the problem becomes the maximization of  $\mathcal{A}_\omega$  over caps with rotation angle  $\omega$ .
- (2) For every  $\omega$ , Baek obtains a cap with the largest sofa area as a limit of polygonal caps, and shows that these polygonal caps are balanced.
  - (a) Fix a finite set of angles strictly between 0 and  $\omega$ . Any placement of copies of the hallway at these angles, together with the start and the end, cuts out a polygonal cap and a polygonal niche (Figure 11), so the placement can be chosen freely and determines the candidate sofa. By a compactness argument, Baek shows that some placement gives the largest sofa area; its cap is a *maximum polygon cap*. Baek also shows that, as the sets of angles become dense, a limit of these caps gives a sofa with the largest area.
  - (b) Maximum polygon caps are *balanced*: each slanted side of the cap lies on an outer wall of one of the copies, and it is as long as the side of the niche on the inner wall of the same copy parallel to it, since otherwise moving these two walls slightly outward or inward would increase the sofa area (Gerver's balancing argument; Figure 12).

- (3) From the balance, Baek shows two properties of this cap, for a sofa of large enough area (which Gerver's sofa is).
- (a) One may assume, without loss of generality, that the rotation angle is  $\pi/2$ : a rotated copy of the sofa  $K \setminus \mathcal{N}(K)$  of this cap, which has the same area, turns by the full right angle (Figure 13).
  - (b) For the rotation angle  $\pi/2$ , the cap satisfies the *injectivity condition*: the inner corner of the hallway, as seen from the sofa, moves steadily to the left and never crosses its own path (the dashed curve in Figure 10).
- (4) Baek bounds the sofa area of every cap with the injectivity condition by a concave functional  $\mathcal{Q}$  of the cap. The functional  $\mathcal{Q}$  is built from areas swept by segments tangent to the boundary of the cap (Figure 14). The largest value of  $\mathcal{Q}$  is attained at *Gerver's cap*, the cap of  $G$ , where  $\mathcal{Q}$  equals  $|G|$ .

For a moving sofa  $S$  of large enough area, with rotation angle  $\omega$  and cap  $K_S$ , the four steps give

$$\begin{aligned}
 |S| &\stackrel{(1)}{\leq} \mathcal{A}_\omega(K_S) \stackrel{(2a)}{\leq} \mathcal{A}_\omega(K_\omega) \stackrel{(3a)}{\leq} \mathcal{A}_{\pi/2}(K_{\pi/2}) \\
 &\stackrel{(3b),(4)}{\leq} \mathcal{Q}(K_{\pi/2}) \stackrel{(4)}{\leq} \mathcal{Q}(K_G) = |G|, \quad (1.1)
 \end{aligned}$$

where  $K_\omega$  and  $K_{\pi/2}$  are Baek's caps of step (2) and  $K_G$  is Gerver's cap. Only the first two terms involve  $S$ .

**1.4. The strategy of the proof.** Suppose now that  $S$  has area  $|G|$ . Then every inequality in (1.1) is an equality, but this says only that  $K_S$ ,  $K_\omega$  and  $K_{\pi/2}$  have the largest sofa area, and that  $\mathcal{Q}$  is largest at  $K_{\pi/2}$ ; it says nothing about the shape of  $S$ . To use the equality case of  $\mathcal{Q}$  for  $S$  itself, we need the properties of step (3) for the cap  $K_S$ , which has the largest sofa area but need not be a limit of maximum polygon caps as in step (2a).

Baek derives step (3) from the balance of the maximum polygon caps in step (2b). We replace this balance by bounds that we prove for every cap  $K_*$  with the largest sofa area. We approximate  $K_*$  by polygon caps  $K_n$  that, in place of the sofa area of step (2a), maximize the sofa area minus a penalty for straying from  $K_*$  (Section 4). The penalty measures, at finitely many angles, how far the walls of the copies placed against  $K_n$  are from those placed against  $K_*$ . Moving one wall of  $K_n$  as in step (2b) now also changes the penalty, so  $K_n$  is only almost balanced: each side of the cap exceeds the corresponding side of the niche by at most a small error, and the weights of the penalty are chosen so that these errors add up to  $O(d_H(K_n, K_*))$ , where  $d_H$  is the Hausdorff distance (Section 5).

In the limit this gives two inequalities for  $K_*$ , the *pinned bounds*, and these are all that Baek's argument for step (3a) uses of the balance: they put a small triangle at the inner corner  $O$  into the niche, which keeps the sofa narrow enough to make the missing turn by  $\pi/2 - \omega$  inside the horizontal

side before its own movement (Section 6). This produces a second monotone sofa  $U$  of area  $|G|$  with rotation angle  $\pi/2$  that contains a rigid image of  $S$ . For the cap  $K_U$  of  $U$ , which also has the largest sofa area, the same approximation bounds how much boundary  $K_U$  has at each angle by the distances from the outer corner of the hallway to the points where  $K_U$  touches its outer walls; from these *curvature bounds*, Baek’s argument for step (3b) gives the injectivity condition (Section 7).

Now the chain runs through the cap  $K_U$  of  $U$  itself: as  $U$  has area  $|G|$ , we have  $|G| = |U| = \mathcal{A}_{\pi/2}(K_U)$ , and step (4), which applies as  $K_U$  satisfies the injectivity condition, gives  $\mathcal{A}_{\pi/2}(K_U) \leq \mathcal{Q}(K_U) \leq \mathcal{Q}(K_G) = |G|$ , where  $K_G$  is Gerver’s cap. So every inequality is an equality; in particular,  $\mathcal{Q}(K_U) = \mathcal{Q}(K_G)$ , so  $\mathcal{Q}$  is largest at  $K_U$  as well as at  $K_G$ . Since  $\mathcal{Q}$  is concave, and strictly so in the lengths of the tangent segments of Figure 14, this forces the tangent segments of  $K_U$  and of  $K_G$  to have the same lengths. Hence  $K_U$  is a horizontal translate of Gerver’s cap, and  $U$  of Gerver’s sofa (Section 8). Finally  $G$  is the closure of its interior, which turns the inclusion of a rigid image of  $S$  in  $G$  into an equality (Section 9).

**1.5. Formal verification and the use of AI.** The proofs, together with Baek’s, are formalized in Lean 4 with Mathlib [2, 13], in the repository [14]: every result of [1] used here, with the corrections of Appendix C, the proof of Theorem 1.1, a second proof of optimality (Section 8.4), the stability theorems of Section 10, and the derivation of optimality, uniqueness and stability from one estimate (Section 11). Section 12 describes the formalization, and Appendix D gives the formal statement of Theorem 1.1 and of the stability theorems, and the formal counterpart of each result of this paper. An earlier version of the formalization, without the stability theorems, is registered in the Palomar registry, which checks mechanically that it proves exactly the stated theorems, using only Lean’s standard axioms, and uses AI to check that each formal statement says the same as its informal version (Section 12). The formalization also proves that its definitions agree with those of Google DeepMind’s formal-conjectures project [4], whose statements of the moving sofa problem were written and reviewed by people independently of this work. The project lists the uniqueness of Gerver’s sofa as an open problem, and the author has opened a pull request there so that the project’s maintainers can check this solution.

Much of this work was done with AI systems. They wrote the first version of the proof, of the formalization and of this text. The author has proofread and edited the abstract and Section 1, and has not yet read the other sections. Appendix E names the systems and says what was checked, and by whom.

**1.6. Organization.** Section 2 recalls the definitions and the results of Baek’s work [1] that we use. Section 3 reduces the theorem to caps and outlines the rest; Sections 4 to 9 carry out the strategy described above, and Section 8.4, the last subsection of Section 8, gives the second proof of optimality; Section 10 proves that the uniqueness is stable and bounds the rotation angle of

a moving sofa of nearly maximal area (Theorem 10.2); Section 11 derives optimality, uniqueness and stability from one estimate of Section 10; Section 12 describes the formalization and lists open questions. The appendices give pictures of Baek's argument (Appendix A), the explicit description of Gerver's sofa (Appendix B), the corrections to Baek's work (Appendix C), the dictionary to the Lean declarations (Appendix D), and the account of the use of AI (Appendix E).

## 2. SETTING, AND WHAT WE USE FROM BAEK'S PAPER

This section fixes the notation and records the results that the proof uses and does not prove; we call them *Facts*. Most are results of Baek's paper [1], quoted in the form in which we use them and with the numbering of version 1 of the preprint (chapter.section.number). Where a printed statement or proof of [1] differs from the form used here, or is missing, Appendix C says so. The reader who knows [1] can skim this section and return to it for the notation.

**2.1. Notation and convex bodies.** The plane is  $\mathbb{R}^2$ , with points  $p = (x, y)$ , inner product  $\langle p, q \rangle$  and norm  $\|p\|$ ;  $|X|$  is the area (Lebesgue measure) of  $X \subset \mathbb{R}^2$ . For an angle  $t \in \mathbb{R}$  put

$$u_t = (\cos t, \sin t), \quad v_t = (-\sin t, \cos t) = u_{t+\pi/2},$$

and let  $R_t$  be the counterclockwise rotation by  $t$ . For  $c \in \mathbb{R}$  let  $l(t, c) = \{p : \langle p, u_t \rangle = c\}$ ,  $H_-(t, c) = \{p : \langle p, u_t \rangle \leq c\}$ ,  $H_+(t, c) = \{p : \langle p, u_t \rangle \geq c\}$ , and let  $H_\pm^\circ$  be the open half-planes. The half-plane  $H_-(t, c)$  has *normal angle*  $t$ .

A *convex body* is a nonempty compact convex subset of  $\mathbb{R}^2$ ; it may be a segment or a point. Its *support function* is  $h_K(t) = \max_{p \in K} \langle p, u_t \rangle$ , its *supporting line* is  $l_K(t) = l(t, h_K(t))$ , and  $H_K(t) = H_-(t, h_K(t))$ . Then  $K = \bigcap_t H_K(t)$ , so  $h_K$  determines  $K$ ;  $h_{K+w}(t) = h_K(t) + \langle w, u_t \rangle$  and  $h_{(1-\lambda)K+\lambda L} = (1-\lambda)h_K + \lambda h_L$  for Minkowski combinations with  $\lambda \in [0, 1]$ . The Hausdorff distance is  $d_H(K, L) = \sup_t |h_K(t) - h_L(t)|$ , and  $K_n \rightarrow K$  means  $d_H(K_n, K) \rightarrow 0$ . The *edge* of  $K$  with normal angle  $t$  is the segment  $e_K(t) = K \cap l_K(t) = [v_K^-(t), v_K^+(t)]$ , where  $v_K^\pm(t)$  is its endpoint farthest in the direction  $v_t$ ; it may be a point. For angles  $a, b$  with  $\sin(b-a) \neq 0$  the supporting lines  $l_K(a)$  and  $l_K(b)$  meet at

$$v_K(a, b) = h_K(a) u_a + \frac{h_K(b) - h_K(a) \cos(b-a)}{\sin(b-a)} v_a.$$

**Fact 2.1** (convex bodies). *Let  $K$  be a convex body.*

- (a)  $h_K$  has the one-sided derivatives  $h'_K(t^\pm) = \langle v_K^\pm(t), v_t \rangle$ ;  $v_K^\pm(s) \rightarrow v_K^\pm(t)$  as  $s \downarrow t$  and  $v_K^\pm(s) \rightarrow v_K^\pm(t)$  as  $s \uparrow t$ ;  $v_K(t, s) \rightarrow v_K^+(t)$  as  $s \downarrow t$  and  $v_K(s, t) \rightarrow v_K^-(t)$  as  $s \uparrow t$ . [1, Thm. 2.1.3]
- (b) The surface area measure  $\sigma_K$  is the Lebesgue–Stieltjes measure on  $\mathbb{R}$  of  $G_K(t) = \langle v_K^+(t), v_t \rangle + \int_0^t h_K$  (formally  $\sigma_K = h''_K + h_K$ ); it is  $2\pi$ -periodic. We write  $\sigma_K(t) = \sigma_K(\{t\})$ . Its atoms are the lengths of the edges:  $\sigma_K(t) = \|v_K^+(t) - v_K^-(t)\|$  and  $v_K^\pm(t) = v_K^-(t) + \sigma_K(t) v_t$ . Moreover

$v_K^+(d) - v_K^+(c) = \int_{(c,d]} v_s d\sigma_K(s)$  for  $c \leq d$ , and  $|K| = \frac{1}{2} \int_{[0,2\pi)} h_K d\sigma_K$ . [1, Prop. 2.1.2, Thm. 5.2.2, Thm. 7.1.3], [11, Rem. 5.1.2]

(c) Every sequence of convex bodies in a bounded set has a subsequence that converges to a convex body (Blaschke's selection theorem), and  $K \mapsto |K|$  is continuous for  $d_H$ . [11, Section 1.8, in particular Thm. 1.8.20]

The formulas for the one-sided derivatives in (a) follow from these limits and from  $\langle v_K(t, s), v_t \rangle = \langle v_K(s, t), v_t \rangle = (h_K(s) - h_K(t) \cos(s - t)) / \sin(s - t)$ . For  $a < b$ , (b) gives  $\sigma_K((a, b)) = G_K(b^-) - G_K(a)$ , and  $G_K(b^-) = \langle v_K^-(b), v_b \rangle + \int_0^b h_K$  since  $v_K^+(s) \rightarrow v_K^-(b)$  as  $s \uparrow b$ ; so

$$\sigma_K((a, b)) = \langle v_K^-(b), v_b \rangle - \langle v_K^+(a), v_a \rangle + \int_a^b h_K. \quad (2.1)$$

**2.2. Moving sofas, monotonicization, caps and niches.** The *hallway* is  $L = H_L \cup V_L$ , the union of its horizontal side  $H_L = (-\infty, 1] \times [0, 1]$  and its vertical side  $V_L = [0, 1] \times (-\infty, 1]$ ; its inner corner is the origin  $O$  and its outer corner is  $(1, 1)$ .

**Definition 2.2** (moving sofa). A *movement* of a set  $S \subset \mathbb{R}^2$  is a pair of continuous maps  $\theta : [0, 1] \rightarrow \mathbb{R}$ ,  $c : [0, 1] \rightarrow \mathbb{R}^2$  with  $\theta(0) = 0$  such that the rigid motions  $\Phi_s(p) = R_{\theta(s)}p + c(s)$  satisfy

$$\Phi_0(S) \subset H_L, \quad \Phi_s(S) \subset L \quad (0 \leq s \leq 1), \quad \Phi_1(S) \subset V_L.$$

Its *rotation angle* is  $\omega = -\theta(1)$ . A *moving sofa* is a nonempty, closed and connected set that has a movement; it is a moving sofa *with rotation angle*  $\omega$  if it has a movement with that rotation angle.

This is [1, Def. 1.1.2]: Baek moves the set along a continuous curve of orientation-preserving isometries that starts at a translation, and the angle of such a curve lifts to a continuous real function. A moving sofa  $S$  is bounded, hence compact. Indeed, the start position  $\Phi_0(S) \subset H_L$  bounds the ordinate on  $S$  and bounds the abscissa from above. If  $\theta(1) = 0$ , the end position  $\Phi_1(S) \subset V_L$  bounds the abscissa from below. Otherwise, by the intermediate value theorem,  $0 < |\theta(s)| \leq 1/2$  for some  $s \in [0, 1]$ , with  $\theta(s)$  of the sign of  $\theta(1)$ , and  $\Phi_s(S) \subset L$  bounds the abscissa from below: through the bound  $y \leq 1$  on  $L$  if  $\theta(s) < 0$ , and because  $L$  does not meet the open quarter-plane  $\{x < 0, y < 0\}$  if  $\theta(s) > 0$ . A translate of a moving sofa with rotation angle  $\omega$  is again one. Let  $\omega_0 = \sec^{-1}(11/5) = \arccos(5/11) = 1.0989 \dots$  ( $63^\circ$  approximately).

For  $\omega \in (0, \pi/2]$  let  $H = \mathbb{R} \times [0, 1]$  be the horizontal strip,  $V_\omega = \{p : 0 \leq \langle p, u_\omega \rangle \leq 1\}$ ,  $P_\omega = H \cap V_\omega$  the parallelogram (a strip if  $\omega = \pi/2$ ), and  $F_\omega = \{p : y \geq 0, \langle p, u_\omega \rangle \geq 0\}$  the *fan*. If  $\omega < \pi/2$  the two upper sides of  $P_\omega$  meet at  $o_\omega = (c_\omega, 1)$ ,  $c_\omega = \sec \omega - \tan \omega$ , and the two lower sides at  $O$ . A moving sofa  $S$  with rotation angle  $\omega$  is in *standard position* if  $h_S(\omega) = h_S(\pi/2) = 1$ .

- Fact 2.3** (rotation angle and standard position). (a) Every moving sofa  $S$  with  $|S| \geq 11/5$  is a moving sofa with some rotation angle  $\omega \in [\omega_0, \pi/2]$ . [1, Thm. 1.5.1]
- (b) If  $S$  is a moving sofa with rotation angle  $\omega \in (0, \pi/2]$ , some translate  $S + v$  is in standard position, and a moving sofa with rotation angle  $\omega$  in standard position lies in  $P_\omega$ . [1, Prop. 1.2.1, Prop. 2.3.1]

*Supporting hallways.* For a compact set  $S$  and an angle  $t$  let

$$f_{S,t}(p) = R_t p + (h_S(t) - 1)u_t + (h_S(t + \pi/2) - 1)v_t,$$

and let  $L_S(t) = f_{S,t}(L)$  be the *supporting hallway* of  $S$  at  $t$ : the translate of  $R_t L$  whose two outer walls  $a_S(t) = l_S(t)$  and  $c_S(t) = l_S(t + \pi/2)$  are supporting lines of  $S$ . Its inner corner and outer corner are

$$\mathbf{x}_S(t) = (h_S(t) - 1)u_t + (h_S(t + \pi/2) - 1)v_t,$$

$$\mathbf{y}_S(t) = h_S(t)u_t + h_S(t + \pi/2)v_t = \mathbf{x}_S(t) + u_t + v_t,$$

its inner walls lie on the lines  $b_S(t) = l(t, h_S(t) - 1)$  and  $d_S(t) = l(t + \pi/2, h_S(t + \pi/2) - 1)$  (the half-lines from  $\mathbf{x}_S(t)$  in the directions  $-v_t$  and  $-u_t$  are  $\vec{b}_S(t)$  and  $\vec{d}_S(t)$ ), and

$$L_S(t) = Q_S^+(t) \setminus Q_S^-(t), \quad Q_S^+(t) = H_S(t) \cap H_S(t + \frac{\pi}{2}),$$

$$Q_S^-(t) = H_-^\circ(t, h_S(t) - 1) \cap H_-^\circ(t + \frac{\pi}{2}, h_S(t + \frac{\pi}{2}) - 1).$$

$Q_S^-(t)$  is the open quarter-plane below the inner corner. A moving sofa in standard position lies in  $L_S(t)$  for every  $t \in [0, \omega]$ , and the *monotonization* of  $S$  is

$$\mathcal{I}(S) = P_\omega \cap \bigcap_{t \in [0, \omega]} L_S(t).$$

A *monotone sofa* with rotation angle  $\omega$  is a set  $\mathcal{I}(S')$  for a moving sofa  $S'$  in standard position with rotation angle  $\omega$ .

*Caps and niches.* Let  $J_\omega = [0, \omega] \cup [\pi/2, \omega + \pi/2]$ . A *cap* with rotation angle  $\omega \in (0, \pi/2]$  is a convex body  $K$  with

$$h_K(\omega) = h_K(\pi/2) = 1, \quad h_K(\omega + \pi) = h_K(3\pi/2) = 0,$$

that is an intersection of closed half-planes with normal angles in  $J_\omega \cup \{\omega + \pi, 3\pi/2\}$ . The caps with rotation angle  $\omega$  form the set  $\mathcal{K}_\omega^c$ . A cap lies in  $P_\omega$ , and

$$K = F_\omega \cap \bigcap_{s \in J_\omega} H_K(s). \quad (2.2)$$

The *niche* of a cap  $K$  and its *sofa area* are

$$\mathcal{N}(K) = F_\omega \cap \bigcup_{t \in (0, \omega)} Q_K^-(t), \quad \mathcal{A}_\omega(K) = |K| - |\mathcal{N}(K)|.$$

The niche depends on  $\omega$  as well as on  $K$ ; we omit  $\omega$  from the notation, since it is fixed whenever a cap is considered. For  $S$  in standard position, the *cap* of  $S$  is  $\mathcal{C}(S) = P_\omega \cap \bigcap_{t \in [0, \omega]} Q_S^+(t)$ . The reflection  $M_\omega(x, y) =$

$(-x \sin \omega + y \cos \omega, x \cos \omega + y \sin \omega)$  in the line through  $O$  at the angle  $\pi/4 + \omega/2$  (the line through  $O$  and  $o_\omega$  if  $\omega < \pi/2$ ) maps  $F_\omega$  to itself; for  $\omega = \pi/2$  it is  $(x, y) \mapsto (-x, y)$ . The *mirror image* of a cap  $K$  is  $K^m = M_\omega(K)$ . For a convex body  $K$  we write  $A_K^\pm(t) = v_K^\pm(t)$  and  $C_K^\pm(t) = v_K^\pm(t + \pi/2)$  [1, Def. 2.5.1].

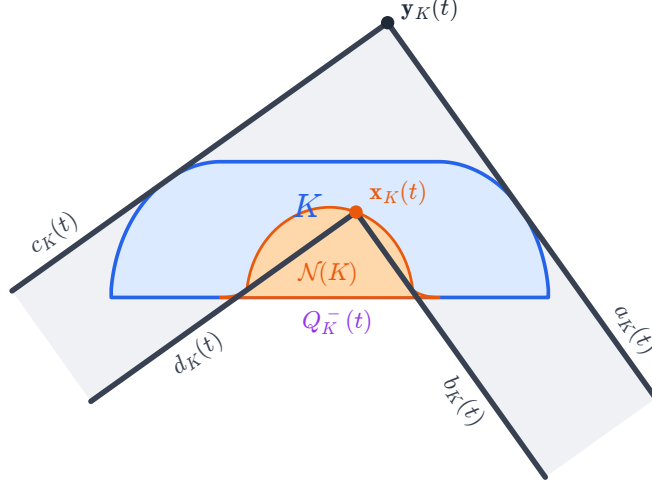


FIGURE 2. A cap, its niche and a supporting hallway: Gerver's cap  $K = K_G$  (blue), its niche  $\mathcal{N}(K)$  (orange), and the supporting hallway  $L_K(t)$  at  $t = 0.62$  (grey). The outer walls  $a_K(t)$  and  $c_K(t)$  of the hallway touch  $K$  and meet at the outer corner  $y_K(t)$ ; its inner walls  $b_K(t)$  and  $d_K(t)$  meet at the inner corner  $x_K(t)$ , on the boundary of the niche.  $Q_K^-(t)$  is the open quarter-plane below the inner corner; the niche is the union of these quarter-planes over  $t \in (0, \pi/2)$ , cut at the  $x$ -axis.

**Fact 2.4** (monotone sofas and caps). *Let  $\omega \in (0, \pi/2]$ .*

- (a) *If  $S$  is a moving sofa with rotation angle  $\omega$  in standard position, then  $\mathcal{I}(S)$  is a moving sofa with rotation angle  $\omega$  in standard position and  $S \subset \mathcal{I}(S)$ . [1, Thm. 2.3.2]*
- (b) *For such an  $S$ ,  $K = \mathcal{C}(S)$  is a cap with  $h_K = h_S$  on  $J_\omega$ , and  $\mathcal{I}(S) = K \setminus \mathcal{N}(K)$ . [1, Lemma 2.3.5, Thm. 2.4.1, Thm. 2.4.2]*
- (c) *If  $T$  is a monotone sofa with rotation angle  $\omega$  and  $K = \mathcal{C}(T)$ , then  $T = K \setminus \mathcal{N}(K)$ ,  $\mathcal{N}(K) \subset K$  and  $|T| = \mathcal{A}_\omega(K)$ . [1, Thm. 2.4.3, Thm. 2.5.9, Thm. 2.5.10]*
- (d) *If  $K$  is a cap, so is  $K^m$ , with  $h_{K^m}(t) = h_K(\omega + \pi/2 - t)$ ,  $C_{K^m}^+(t) = M_\omega(A_K^-(\omega - t))$ ,  $\mathcal{N}(K^m) = M_\omega(\mathcal{N}(K))$ ,  $\mathcal{A}_\omega(K^m) = \mathcal{A}_\omega(K)$  and  $\sigma_{K^m}(E) = \sigma_K(\omega + \pi/2 - E)$  for Borel sets  $E$ . [1, Prop. 2.5.4]*

Two more facts about caps are needed: that  $O$  and  $o_\omega$  lie in every cap if  $\omega < \pi/2$ , and where the ends of the *upper boundary*  $\bigcup_{t \in [0, \omega + \pi/2]} e_K(t)$  of a cap  $K$  [1, Def. 2.5.2] are.

**Lemma 2.5** (every cap contains  $O$  and  $o_\omega$ ). *If  $\omega < \pi/2$ , every cap  $K \in \mathcal{K}_\omega^c$  contains  $O$  and  $o_\omega$ .*

**Lemma 2.6** (the ends of the upper boundary). *For every cap  $K \in \mathcal{K}_\omega^c$ ,*

$$A_K^-(0) = (h_K(0), 0) \quad \text{and} \quad C_K^+(\omega) = h_K(\omega + \pi/2) v_\omega. \quad (2.3)$$

*Proof of Lemma 2.6.* Put  $m = h_K(\omega + \pi/2)$  and  $p = mv_\omega$ , so that  $\langle p, u_\omega \rangle = 0$  and  $\langle p, v_\omega \rangle = m$ . We show that  $p \in K$ , by (2.2). Let  $q \in K \cap l(\pi/2, 1)$ . As  $u_{\pi/2} = \sin \omega u_\omega + \cos \omega v_\omega$ ,  $\langle q, u_\omega \rangle \leq h_K(\omega) = 1$  and  $\langle q, v_\omega \rangle \leq m$ ,

$$1 = \langle q, u_{\pi/2} \rangle = \sin \omega \langle q, u_\omega \rangle + \cos \omega \langle q, v_\omega \rangle \leq \sin \omega + m \cos \omega,$$

so  $p_y = m \cos \omega \geq 1 - \sin \omega \geq 0$ , and  $p \in F_\omega$ . Let  $r \in e_K(\omega + \pi/2)$ ; then  $\langle r, v_\omega \rangle = m$ , and  $\langle r, u_\omega \rangle \geq 0$  as  $K \subset P_\omega$ . For  $s \in J_\omega$ ,  $\cos(s - \omega) \geq 0$ , so

$$\langle p, u_s \rangle = m \sin(s - \omega) = \langle r, u_s \rangle - \langle r, u_\omega \rangle \cos(s - \omega) \leq \langle r, u_s \rangle \leq h_K(s).$$

Hence  $p \in K$  by (2.2), and  $p \in e_K(\omega + \pi/2)$ . The end  $C = C_K^+(\omega)$  of this edge is its point farthest in the direction  $v_{\omega + \pi/2} = -u_\omega$ , so  $\langle C, u_\omega \rangle \leq \langle p, u_\omega \rangle = 0$ , while  $\langle C, u_\omega \rangle \geq 0$  as  $C \in K$ . With  $\langle C, v_\omega \rangle = m$ , this gives  $C = mv_\omega$ .

By Fact 2.4(d), the mirror image  $K^m$  is a cap, with  $h_{K^m}(\omega + \pi/2) = h_K(0)$  and  $C_{K^m}^+(\omega) = M_\omega(A_K^-(0))$ . The first part, applied to  $K^m$ , gives  $M_\omega(A_K^-(0)) = h_K(0)v_\omega$ . As  $M_\omega$  is a linear involution and  $M_\omega(v_\omega) = u_0$ ,  $A_K^-(0) = h_K(0)u_0 = (h_K(0), 0)$ .  $\square$

*Proof of Lemma 2.5.* Let  $\omega < \pi/2$ , so that  $\cos \omega > 0$ . By Lemma 2.6,  $A = A_K^-(0) = (h_K(0), 0)$  and  $C = C_K^+(\omega) = h_K(\omega + \pi/2)v_\omega$ ; they are ends of edges of  $K$ , so they lie in  $K \subset P_\omega$ . Hence  $h_K(0) \cos \omega = \langle A, u_\omega \rangle \geq 0$  and  $h_K(\omega + \pi/2) \cos \omega = C_y \geq 0$ , that is,  $h_K(0) \geq 0$  and  $h_K(\omega + \pi/2) \geq 0$ . For  $s \in [0, \omega]$ ,  $h_K(s) \geq \langle A, u_s \rangle = h_K(0) \cos s \geq 0$ , and for  $s \in [\pi/2, \omega + \pi/2]$ ,  $h_K(s) \geq \langle C, u_s \rangle = h_K(\omega + \pi/2) \sin(s - \omega) \geq 0$ . So  $O \in F_\omega$  satisfies  $\langle O, u_s \rangle = 0 \leq h_K(s)$  for  $s \in J_\omega$ , and  $O \in K$  by (2.2).

Let  $q \in K \cap l(\pi/2, 1)$  and  $q' \in K \cap l(\omega, 1)$ . As  $K \subset P_\omega$ ,  $q = o_\omega - \lambda u_0$  and  $q' = o_\omega - \mu v_\omega$  with  $\lambda, \mu \geq 0$ . For  $s \in [0, \omega]$ ,  $\langle o_\omega, u_s \rangle = \langle q', u_s \rangle + \mu \sin(s - \omega) \leq \langle q', u_s \rangle \leq h_K(s)$ , and for  $s \in [\pi/2, \omega + \pi/2]$ ,  $\langle o_\omega, u_s \rangle = \langle q, u_s \rangle + \lambda \cos s \leq \langle q, u_s \rangle \leq h_K(s)$ . As  $o_\omega \in F_\omega$ , (2.2) gives  $o_\omega \in K$ .  $\square$

For every rotation angle, Baek obtains a cap with the largest sofa area as a limit of maximum polygon caps ([1, Thm. 3.5.2, Thm. 3.5.4]); it is the cap  $K_\omega$  of step (2) in Section 1.3.

**Fact 2.7** (a cap with the largest sofa area). *For every  $\omega \in (0, \pi/2]$  there is a cap  $K_\omega \in \mathcal{K}_\omega^c$  with  $\mathcal{A}_\omega(K) \leq \mathcal{A}_\omega(K_\omega)$  for every  $K \in \mathcal{K}_\omega^c$ . The set  $S_\omega = K_\omega \setminus \mathcal{N}(K_\omega)$  is a monotone sofa with rotation angle  $\omega$  and cap  $\mathcal{C}(S_\omega) = K_\omega$ , and every moving sofa  $S$  with rotation angle  $\omega$  has  $|S| \leq |S_\omega|$ . [1, Thm. 3.5.5, Thm. 3.5.6]*

The next fact is the main theorem of [1], the optimality of Gerver's sofa  $G$  ([1, Thm. 1.1.1]), together with its form for caps;  $G$  is defined in Section 2.6.

**Fact 2.8** (optimality). *Every moving sofa  $S$  has  $|S| \leq |G|$ , and  $|G| > 11/5$ . For every  $\omega \in (0, \pi/2]$  and every cap  $K \in \mathcal{K}_\omega^c$ ,  $\mathcal{A}_\omega(K) \leq |G|$ .*

For the last claim, let  $K_\omega$  and  $S_\omega$  be as in Fact 2.7. Then  $\mathcal{A}_\omega(K) \leq \mathcal{A}_\omega(K_\omega)$  by Fact 2.7, and  $\mathcal{A}_\omega(K_\omega) = |S_\omega|$  by Fact 2.4(c), as  $\mathcal{C}(S_\omega) = K_\omega$ . As  $S_\omega$  is a moving sofa (Fact 2.4(a)),  $|S_\omega| \leq |G|$  by [1, Thm. 1.1.1]. The bound  $|G| > 11/5$  holds because  $|G| \geq 2.2192$  (Fact 2.17(a)). Section 8.4 gives a second proof of Fact 2.8 that does not use [1, Thm. 1.1.1], and Section 11 a variant of it.

**2.3. Polygon caps.** Let  $\omega \in (0, \pi/2]$ . An *angle set* is a finite nonempty  $\Theta \subset (0, \omega)$ , and  $\Theta^\diamond = \Theta \cup (\Theta + \pi/2) \cup \{\omega, \pi/2\} \subset (0, \pi)$  is its set of *defining normals*; the normals  $\omega$  and  $\pi/2$ , which carry the strips of  $P_\omega$ , are *pinned*, the others *floating*. A *polygon cap* with angle set  $\Theta$  is a cap that is an intersection of closed half-planes with normal angles in  $\Theta^\diamond \cup \{\omega + \pi, 3\pi/2\}$ ; they form the set  $\mathcal{K}_\Theta^c \subset \mathcal{K}_\omega^c$ . A polygon cap is the intersection of its supporting half-planes  $H_K(s)$  at these normal angles. For a cap  $K$  put

$$\begin{aligned} \mathcal{C}_\Theta(K) &= P_\omega \cap \bigcap_{t \in \Theta} Q_K^+(t), & \mathcal{N}_\Theta(K) &= F_\omega \cap \bigcup_{t \in \Theta} Q_K^-(t), \\ \mathcal{A}_\Theta(K) &= |\mathcal{C}_\Theta(K)| - |\mathcal{N}_\Theta(K)|, \end{aligned}$$

with the fan  $F_\omega$  in the polygon niche.

**Fact 2.9** (the polygon bound). *For every cap  $K$  and angle set  $\Theta$ ,  $\mathcal{C}_\Theta(K)$  is a polygon cap that contains  $K$  and has the same support values as  $K$  on  $\Theta^\diamond$ , and  $\mathcal{C}_\Theta(K) = K$  if  $K \in \mathcal{K}_\Theta^c$ ;  $\mathcal{N}_\Theta(K) = \mathcal{N}_\Theta(\mathcal{C}_\Theta(K)) \subset \mathcal{N}(K)$ . Hence  $\mathcal{A}_\omega(K) \leq \mathcal{A}_\Theta(K) = \mathcal{A}_\Theta(\mathcal{C}_\Theta(K))$ , and  $\mathcal{A}_\Theta(K) = |K| - |\mathcal{N}_\Theta(K)|$  for a polygon cap. [1, Prop. 3.2.1, Prop. 3.2.2, Thm. 3.2.3]*

**Fact 2.10** (bounded width). *For  $t \in (0, \omega)$  there is a constant  $c_{\omega,t} > 0$  such that every polygon cap  $K$  with an angle set  $\Theta \ni t$  and  $\mathcal{A}_\Theta(K) > 0$  has width  $h_K(0) + h_K(\pi) \leq c_{\omega,t}$ . [1, Lemma 3.4.2]*

*Sides of the polygon niche.* For a polygon cap  $K$  with angle set  $\Theta$ , the boundary of  $F_\omega \setminus \mathcal{N}_\Theta(K)$  consists of an open half-line, a polyline  $\mathbf{p}_K$  from  $C_K^+(\omega)$  to  $A_K^-(0)$  (see (2.3)), along which the abscissa increases and whose segments have normal angles in  $\Theta^\diamond$ , and an open half-line [1, Thm. 3.4.4]. For  $t \in \Theta^\diamond$  let  $\tau_K(t)$  be the total length of the segments of  $\mathbf{p}_K$  with normal angle  $t$ ; recall that  $\sigma_K(t)$  is the length of the edge  $e_K(t)$  of  $K$ .

**Fact 2.11** (sides of the polygon niche). *Let  $K \in \mathcal{K}_\Theta^c$ .*

- (a) *For  $t \in \Theta$ ,  $\mathcal{H}^1(\partial \mathcal{N}_\Theta(K) \cap \vec{b}_K(t)) = \tau_K(t)$  and  $\mathcal{H}^1(\partial \mathcal{N}_\Theta(K) \cap \vec{d}_K(t)) = \tau_K(t + \pi/2)$ . [1, Lemma 3.4.5 (1)]*
- (b) *For  $t \in \{\omega, \pi/2\}$ ,  $\mathcal{H}^1(\mathcal{N}_\Theta(K) \cap l(t, 0)) = \sigma_K(t + \pi) - \tau_K(t)$ . [1, Lemma 3.4.5 (2)]*

$$(c) \ C_K^+(\omega) - A_K^-(0) = \sum_{t \in \Theta^\diamond} \sigma_K(t) v_t, \text{ and}$$

$$\sum_{t \in \Theta^\diamond} \sigma_K(t) \sin t = A_K^-(0)_x - C_K^+(\omega)_x = \sum_{t \in \Theta^\diamond} \tau_K(t) \sin t.$$

[1, Proof of Lemma 3.4.6]

In (c), the vector identity is a sum along the upper boundary of  $K$ : taken in the increasing order of  $t \in \Theta^\diamond$ , the edges  $e_K(t)$  go from  $v_K^-(t)$  to  $v_K^+(t) = v_K^-(t) + \sigma_K(t)v_t$ , the first starts at  $A_K^-(0)$ , each ends where the next starts, and the last ends at  $C_K^+(\omega)$ . Its first coordinate is the first of the two equalities, as  $\langle v_t, u_0 \rangle = -\sin t$ . In the second,  $\tau_K(t) \sin t$  is the total length of the projections to the  $x$ -axis of the segments of  $\mathbf{p}_K$  with normal angle  $t$ ; since the abscissa increases along  $\mathbf{p}_K$ , the projections of all its segments cover  $[C_K^+(\omega)_x, A_K^-(0)_x]$  and meet only at their ends.

*Support values and first variations.* For  $h : \Theta^\diamond \rightarrow \mathbb{R}$  let

$$\begin{aligned} P_h &= \bigcap_{s \in \{\omega, \pi/2\}} (H_-(s, h(s)) \cap H_+(s, h(s) - 1)), \\ \mathcal{C}_\Theta(h) &= P_h \cap \bigcap_{s \in \Theta^\diamond \setminus \{\omega, \pi/2\}} H_-(s, h(s)), \\ F_h &= \bigcap_{s \in \{\omega, \pi/2\}} H_+(s, h(s) - 1), \\ \mathcal{N}_\Theta(h) &= F_h \cap \bigcup_{t \in \Theta} (H_-^\circ(t, h(t) - 1) \cap H_-^\circ(t + \frac{\pi}{2}, h(t + \frac{\pi}{2}) - 1)), \end{aligned}$$

and  $\mathcal{A}_\Theta(h) = |\mathcal{C}_\Theta(h)| - |\mathcal{N}_\Theta(h)|$  ([1, Def. 3.3.3]). For a polygon cap  $K$  and  $h = h_K|_{\Theta^\diamond}$ ,  $\mathcal{C}_\Theta(h) = K$ ,  $\mathcal{N}_\Theta(h) = \mathcal{N}_\Theta(K)$  and  $\mathcal{A}_\Theta(h) = \mathcal{A}_\Theta(K)$  ([1, Prop. 3.3.4, Prop. 3.3.5]). For  $\varepsilon \geq 0$  and  $t \in \Theta^\diamond$  let  $h_K^{t, \varepsilon}$  be  $h_K$  with the value at  $t$  replaced by  $h_K(t) + \varepsilon$ .

**Fact 2.12** (pushing one side). *Let  $K \in \mathcal{K}_\Theta^c$  and  $t \in \Theta^\diamond$ .*

(a) *There are  $\varepsilon_0 > 0$  and  $M \in \mathbb{R}$  such that for  $0 < \varepsilon \leq \varepsilon_0$*

$$|\mathcal{A}_\Theta(h_K^{t, \varepsilon}) - \mathcal{A}_\Theta(K) - (\sigma_K(t) - \tau_K(t))\varepsilon| \leq M\varepsilon^2.$$

[1, Lemma 3.4.7]

(b) *If  $\sigma_K(t) > 0$ , then for small  $\varepsilon > 0$  the set  $\mathcal{C}_\Theta(h_K^{t, \varepsilon})$  is a translate  $C + v$  of a polygon cap  $C \in \mathcal{K}_\Theta^c$ . [1, Lemma 3.4.8, Prop. 3.3.1]*

(c) *If  $\mathcal{C}_\Theta(h_K^{t, \varepsilon}) = C + v$  with  $C \in \mathcal{K}_\Theta^c$ , then  $\mathcal{A}_\Theta(h_K^{t, \varepsilon}) \leq \mathcal{A}_\Theta(C)$ . [1, Prop. 3.3.7, Thm. 3.3.6]*

**2.4. The injectivity condition and arm lengths.** The rest of this section, and Sections 7 and 8, concern caps with rotation angle  $\pi/2$ , and  $\mathcal{A} = \mathcal{A}_{\pi/2}$ . For a convex body  $K$  the *arm lengths* are

$$f_K^\pm(t) = \langle \mathbf{y}_K(t) - A_K^\pm(t), v_t \rangle, \quad g_K^\pm(t) = \langle \mathbf{y}_K(t) - C_K^\pm(t), u_t \rangle,$$

with  $\mathbf{y}_K(t) = h_K(t)u_t + h_K(t + \pi/2)v_t$  the outer corner of the supporting hallway and  $A_K^\pm, C_K^\pm$  as in Section 2.2. Thus  $\mathbf{y}_K(t) = A_K^\pm(t) + f_K^\pm(t)v_t = C_K^\pm(t) + g_K^\pm(t)u_t$  [1, Prop. 6.2.1]. The arm lengths are nonnegative, since  $A_K^\pm(t)$  and  $C_K^\pm(t)$  lie in  $K$ , so that  $\langle A_K^\pm(t), v_t \rangle \leq h_K(t + \pi/2)$  and  $\langle C_K^\pm(t), u_t \rangle \leq h_K(t)$ ; they are the distances from the outer corner to the points where  $K$  touches the two outer walls. Moreover  $g_K^+(t) - g_K^-(t) = \sigma_K(t + \pi/2) \geq 0$ , by Fact 2.1(b) and  $v_{t+\pi/2} = -u_t$ .

**Definition 2.13** (injectivity condition). A cap  $K \in \mathcal{K}_{\pi/2}^c$  satisfies the *injectivity condition* if

- (1)  $\sigma_K$  has a density on  $[0, \pi/2)$  and on  $(\pi/2, \pi]$  (the top edge, at  $\pi/2$ , may carry an atom);
- (2) the inner corner  $\mathbf{x}_K$  is continuously differentiable on  $[0, \pi/2]$ ;
- (3)  $\langle \mathbf{x}'_K(t), u_t \rangle < 0 < \langle \mathbf{x}'_K(t), v_t \rangle$  for  $0 < t < \pi/2$ .

We write  $\mathcal{K}^i$  for the set of caps in  $\mathcal{K}_{\pi/2}^c$  that satisfy the injectivity condition and have  $|K| \geq 11/5$ .

**Fact 2.14** (arms). Let  $K \in \mathcal{K}_{\pi/2}^c$ .

- (a) For every  $t$ ,  $\mathbf{x}_K$  has the right derivative  $-(f_K^+(t) - 1)u_t + (g_K^+(t) - 1)v_t$ , and the left derivative given by the same formula with  $f_K^-, g_K^-$ . [1, Thm. 6.2.3]
- (b)  $f_K^+(b) - f_K^+(a) = \int_a^b g_K^+ - \sigma_K((a, b])$  for  $0 \leq a \leq b \leq \pi/2$ . [1, Thm. 6.2.5]
- (c) For the reflection  $K^m$  in  $x \mapsto -x$ ,  $f_{K^m}^\pm(t) = g_K^\mp(\pi/2 - t)$  and  $g_{K^m}^\pm(t) = f_K^\mp(\pi/2 - t)$ . [1, Prop. 6.2.2]
- (d) If  $K$  satisfies (1), then  $A_K^+ = A_K^-$  and  $f_K^+ = f_K^-$  on  $[0, \pi/2)$ ,  $C_K^+ = C_K^-$  and  $g_K^+ = g_K^-$  on  $(0, \pi/2]$ . With  $A_K = A_K^-$ ,  $C_K = C_K^+$ ,  $f_K = f_K^-$  and  $g_K = g_K^+$  these four functions are continuous on  $[0, \pi/2]$ ;  $f_K(0) = g_K(\pi/2) = 1$ , and  $\mathbf{x}'_K(t) = -(f_K(t) - 1)u_t + (g_K(t) - 1)v_t$  for  $t \in [0, \pi/2]$ , the derivative being one-sided at 0 and  $\pi/2$ . In particular (2) holds, and (3) holds if  $f_K, g_K > 1$  on  $(0, \pi/2)$ . [1, Def. 6.4.1, Prop. 6.4.5, Prop. 6.4.6]

The end values  $f_K(0) = g_K(\pi/2) = 1$  in (d) hold for every  $K \in \mathcal{K}_{\pi/2}^c$ . The half-planes whose intersection is  $K$  have normal angles in  $[0, \pi] \cup \{3\pi/2\}$ , none of them strictly between  $-\pi/2$  and 0 modulo  $2\pi$ ; so the supporting lines  $l_K(-\pi/2)$  and  $l_K(0)$  meet at a point of  $K$ , the vertex  $v_K^-(0)$ , and this point is  $v_K(-\pi/2, 0) = (h_K(0), 0)$ , as  $h_K(-\pi/2) = h_K(3\pi/2) = 0$ . Hence  $f_K(0) = f_K^-(0) = h_K(\pi/2) - \langle v_K^-(0), v_0 \rangle = 1$ . The same holds for the cap  $K^m$  (Fact 2.4(d)), and (c) gives  $g_K(\pi/2) = g_K^+(\pi/2) = f_{K^m}^-(0) = 1$ . Let

$$k_0(x) = \max\left(|x - 1|, \frac{|x-1|+1}{2}\right), \quad m_0(x) = x - k_0(x).$$

The function  $k_0$  is 1-Lipschitz, and  $m_0$  is nondecreasing, with  $m_0(x) = \frac{3}{2}x - 1$  on  $[0, 1]$ . For a continuous  $f$  on  $[0, \pi/2]$  let  $\mathcal{F}f(x) = 1 + \int_0^x m_0(f(\pi/2 - u))du$ , and let  $f_0 = 0$ ,  $f_{j+1} = \max(f_j, \mathcal{F}f_j)$ ; each  $f_j$  is continuous.

**Fact 2.15** (convergence of arms and the iteration). (a) Let  $K_n$  be polygon caps, with any angle sets, and  $K \in \mathcal{K}_{\pi/2}^c$  a cap with  $d_H(K_n, K) \rightarrow 0$ .

Then  $\int_0^{\pi/2} |g_{K_n}^+ - g_K^+| \rightarrow 0$ . [1, Lemma 6.4.2]

(b)  $f_{11}(x) > 1$  for  $x \in (0, \pi/2]$ . [1, Lemma 6.5.5]

**2.5. Baek's upper bound.** Let  $\varphi = \varphi_G = 0.03917\dots$  be Gerver's angle (Section 2.6),  $\varphi^R = \varphi$  and  $\varphi^L = \pi/2 - \varphi$ . For  $K \in \mathcal{K}_{\pi/2}^c$  let  $H_K^b(t) = \{\langle p, u_t \rangle \geq h_K(t) - 1\}$  and  $H_K^d(t) = \{\langle p, v_t \rangle \geq h_K(t + \pi/2) - 1\}$  be the closed half-planes bounded by the inner walls  $b_K(t)$  and  $d_K(t)$ , on the side away from  $Q_K^-(t)$ , and

$$B_K = K \cap \bigcap_{t \in [\varphi^R, \pi/2]} H_K^b(t), \quad D_K = K \cap \bigcap_{t \in [0, \varphi^L]} H_K^d(t).$$

Baek's upper bound is the functional  $\mathcal{Q}$  of [1, Def. 8.2.2] on the set  $\mathcal{T}$  of triples  $(K, B, D)$  of convex bodies with  $K \in \mathcal{K}^i$ ,  $B, D \subset K$ ,  $h_K(t) + h_B(\pi + t) \leq 1$  on  $[\varphi^R, \pi/2]$  with equality at the ends, and  $h_K(\pi/2 + t) + h_D(3\pi/2 + t) \leq 1$  on  $[0, \varphi^L]$  with equality at the ends. The sets  $\mathcal{K}^i$  and  $\mathcal{T}$  are closed under Minkowski combinations  $c_\lambda((K, B, D), (K', B', D')) = ((1 - \lambda)K + \lambda K', \dots)$ ,  $\lambda \in [0, 1]$  [1, Thm. 8.1.1 (1), Prop. 8.1.2]. A functional on such a set is *convex-linear* if it commutes with  $c_\lambda$ , and *convex* or *concave* if it satisfies the corresponding inequality for  $c_\lambda$ . The *canonical triple* of  $K \in \mathcal{K}^i$  is  $\xi_K = (K, B_K, D_K)$ . For a convex body  $K$ ,  $a < b < a + \pi$ , and a continuous curve  $\mathbf{z} : [a, b] \rightarrow \mathbb{R}^2$  of bounded variation with  $\mathbf{z}(t) \in l_K(t)$ , [1, Def. 7.4.2] defines the signed combination of curve areas

$$\mathcal{M}_K(a, b; \mathbf{z}) = \mathcal{J}([v_K^+(a), \mathbf{z}(a)]) + \mathcal{J}(\mathbf{z}) + \mathcal{J}([\mathbf{z}(b), v_K^-(b)]) - \frac{1}{2} \int_{(a,b)} h_K d\sigma_K,$$

where  $\mathcal{J}(\gamma) = \frac{1}{2} \int \gamma \times d\gamma$  for an oriented curve  $\gamma$ , with  $p \times q = p_x q_y - p_y q_x$  [1, Def. 7.2.6, Def. 7.2.8], and  $[p, q]$  is the segment from  $p$  to  $q$ ; the last term is  $-\mathcal{J}$  of the arc  $\mathbf{u}_K^{a,b}$  of the boundary of  $K$  from  $v_K^+(a)$  to  $v_K^-(b)$  [1, Def. 7.3.1, Thm. 7.3.2]. Mamikon's theorem, in Baek's form [1, Thm. 7.4.1] (see [7]), evaluates it:

$$\mathcal{M}_K(a, b; \mathbf{z}) = \frac{1}{2} \int_a^b \varrho_{K, \mathbf{z}}(t)^2 dt, \quad \varrho_{K, \mathbf{z}}(t) = \langle \mathbf{z}(t) - v_K^+(t), v_t \rangle.$$

We use  $\mathcal{M}$  only through this formula, which makes  $\mathcal{M}_K(a, b; \mathbf{z})$  a convex function of  $\varrho_{K, \mathbf{z}}$ . The curves that occur are the outer corner  $\mathbf{y}_K$  and the tangent line parametrizations  $\mathbf{l}_K^T(t) = v_K(t, T)$  for  $T - \pi < t < T$  (the point where  $l_K(t)$  meets  $l_K(T)$ ) and  $\mathbf{l}_K^T(T) = v_K^-(T)$ . Let  $\mathcal{S}_K, \mathcal{R}_B, \mathcal{L}_D$  be the *Mamikon terms*

$$\begin{aligned} \mathcal{S}_K &= \mathcal{M}_K(0, \varphi^R; \mathbf{l}_K^{\pi/2}) + \mathcal{M}_K(\varphi^R, \varphi^L; \mathbf{y}_K) \\ &\quad + \mathcal{M}_K(\varphi^L, \frac{\pi}{2}; \mathbf{l}_K^{\pi-\varphi}) + \mathcal{M}_K(\frac{\pi}{2}, \pi; \mathbf{l}_K^\pi), \\ \mathcal{R}_B &= \mathcal{M}_B(\pi + \varphi^R, \frac{3\pi}{2}; \mathbf{l}_B^{3\pi/2}), \quad \mathcal{L}_D = \mathcal{M}_D(\frac{3\pi}{2}, \frac{3\pi}{2} + \varphi^L; \mathbf{l}_D^{3\pi/2+\varphi^L}). \end{aligned}$$

(In the third term  $\pi/2 + \varphi^L = \pi - \varphi$ .)

**Fact 2.16** (the upper bound). *There is a functional  $\mathcal{Q}$  on  $\mathcal{T}$  such that:*

- (a)  $\xi_K \in \mathcal{T}$  for  $K \in \mathcal{K}^i$ , and  $\mathcal{A}(K) \leq \mathcal{Q}(\xi_K)$ . [1, Thm. 8.1.8, Thm. 8.2.4]
- (b)  $\mathcal{Q}(K, B, D) = \Lambda(K) - \mathcal{S}_K - \mathcal{R}_B - \mathcal{L}_D$ , where  $\Lambda$  (Baek's  $\mathcal{P}_K + \mathcal{S}_K$ , [1, Def. 8.3.4]) is convex-linear on  $\mathcal{K}^i$ . For each of the four terms of  $\mathcal{S}_K$  (resp. for  $\mathcal{R}_B$ , for  $\mathcal{L}_D$ ) the curve  $\mathbf{z}$  above is convex-linear in  $K$  (resp. in  $B$ , in  $D$ ), so that  $\varrho(t)$  is convex-linear for every  $t$ , and each of the terms  $\mathcal{M}$ , being  $\frac{1}{2} \int \varrho^2$ , is convex. [1, Lemma 8.3.3, Lemma 8.3.4, Lemma 8.3.7, Thm. 8.3.1, Thm. 8.3.2, Thm. 7.1.2, Thm. 7.4.1, Thm. 7.4.2]
- (c)  $\mathcal{Q} \leq \mathcal{Q}(\xi_G) = |G|$  on  $\mathcal{T}$ , where  $\xi_G = \xi_{K_G}$  is the canonical triple of Gerver's cap. [1, Cor. 8.5.8, Thm. 8.4.6]

**2.6. Gerver's sofa.** Seen from a sofa that turns clockwise by  $t \in [0, \pi/2]$ , the hallway occupies  $\mathbf{x}(t) + R_t L$ ; the curve  $\mathbf{x} : [0, \pi/2] \rightarrow \mathbb{R}^2$  is the *rotation path*, and its *shape* is

$$\text{shape}(\mathbf{x}) = H_L \cap \bigcap_{t \in [0, \pi/2]} (\mathbf{x}(t) + R_t L) \cap (\mathbf{x}(\pi/2) + R_{\pi/2} V_L).$$

Gerver's sofa  $G$  is the shape of Romik's rotation path [10], which is glued from five explicit curves, on  $[0, \varphi]$ ,  $[\varphi, \theta]$ ,  $[\theta, \pi/2 - \theta]$ ,  $[\pi/2 - \theta, \pi/2 - \varphi]$  and  $[\pi/2 - \varphi, \pi/2]$ , with  $\varphi = 0.03917\dots$  and  $\theta = 0.68130\dots$  (the formulas and Romik's equations are in Appendix B). Write  $t_0, \dots, t_5 = 0, \varphi, \theta, \pi/2 - \theta, \pi/2 - \varphi, \pi/2$ ,  $\alpha = \langle \mathbf{x}', u_t \rangle$ ,  $\beta = \langle \mathbf{x}', v_t \rangle$ , and

$$\mathbf{A} = \mathbf{x} + \alpha v_t + u_t, \quad \mathbf{B} = \mathbf{x} + \alpha v_t, \quad \mathbf{C} = \mathbf{x} - \beta u_t + v_t, \quad \mathbf{D} = \mathbf{x} - \beta u_t$$

for the contact points of the sofa with the walls of the hallway.

**Fact 2.17** (Gerver's sofa). (a) *Romik's equations have exactly one solution with  $\varphi \in [0.039, 0.04]$ ,  $\theta \in [0.68, 0.69]$ , so  $G$  is well defined, and  $2.2192 \leq |G| \leq 2.2199$ .*

- (b)  $G$  is a monotone sofa with rotation angle  $\pi/2$ . Its cap  $K_G = \mathcal{C}(G)$  satisfies  $G = K_G \setminus \mathcal{N}(K_G)$  (Fact 2.4(c)) and has  $A_{K_G} = \mathbf{A}$ ,  $C_{K_G} = \mathbf{C}$  and  $\mathbf{x}_{K_G} = \mathbf{x}$  on  $[0, \pi/2]$ ; and  $K_G$  satisfies the injectivity condition; as  $|K_G| \geq \mathcal{A}(K_G) = |G| > 11/5$ ,  $K_G \in \mathcal{K}^i$ . [1, Thm. 8.4.1 (1), Thm. 6.1.2, Thm. 8.1.1 (3)]

- (c) Let  $\Gamma = \mathbf{D}([t_0, t_2]) \cup \mathbf{x}([t_1, t_4]) \cup \mathbf{B}([t_3, t_5])$ . Then  $\mathbf{D}(t_0)$  and  $\mathbf{B}(t_5)$  lie on the  $x$ -axis,  $\mathbf{D}(t_2) = \mathbf{x}(t_4)$  and  $\mathbf{B}(t_3) = \mathbf{x}(t_1)$ ; the abscissa is strictly increasing along  $\mathbf{D}$  on  $[t_0, t_2]$ , strictly decreasing along  $\mathbf{x}$  on  $[t_1, t_4]$  and strictly increasing along  $\mathbf{B}$  on  $[t_3, t_5]$ ; at every point of  $[t_0, t_1]$  and of  $[t_1, t_2]$  the curve  $\mathbf{D}$  has a derivative within that interval that is a positive multiple of  $u_t$ , and at every point of  $[t_3, t_4]$  and of  $[t_4, t_5]$  the curve  $\mathbf{B}$  has a derivative within that interval that is a negative multiple of  $v_t$  (one-sided at the ends of these intervals); and

$$\mathcal{N}(K_G) = \{q : q_y \geq 0, q_y < \gamma_y \text{ for some } \gamma \in \Gamma \text{ with } \gamma_x = q_x\}.$$

[1, Thm. 8.4.1 (2), (4)]

(d)  $\mathbf{x}(\pi/2)_y = 0$  and  $\mathbf{C}(\pi/2) = (\mathbf{x}(\pi/2)_x - 1, 0)$ .

We also use one numerical fact:

**Fact 2.18** (height of the rotation path).  $\mathbf{x}(t)_y < 1$  for every  $t \in [0, \pi/2]$ , and  $\mathbf{x}(t)_y > 0$  for  $t \in [t_1, t_4]$ .

### 3. FROM A MAXIMAL SOFA TO A MAXIMIZING CAP

Throughout this section  $S$  is a moving sofa with  $|S| = |G|$ . We reduce the problem to caps and then outline the rest of the proof.

**Definition 3.1** (maximizing cap). Let  $\omega \in (0, \pi/2]$ . A cap  $K \in \mathcal{K}_\omega^c$  is *maximizing* if  $\mathcal{A}_\omega(C) \leq \mathcal{A}_\omega(K)$  for every  $C \in \mathcal{K}_\omega^c$ .

By Fact 2.8, every cap with  $\mathcal{A}_\omega(K) = |G|$  is maximizing. A maximizing cap maximizes  $\mathcal{A}_\omega$  over all caps, including those that do not contain their niche. Baek obtains a particular maximizing cap, a limit of maximum polygon caps (Fact 2.7). A given maximizing cap need not be such a limit.

**Proposition 3.2** (the monotonization of a maximal sofa). *Let  $S'$  be a moving sofa with rotation angle  $\omega \in (0, \pi/2]$  and  $|S'| = |G|$ . Then there are  $w \in \mathbb{R}^2$  and a monotone sofa  $T$  with rotation angle  $\omega$  such that*

$$S' + w \subset T, \quad |T| = |G|,$$

*and the cap  $K = \mathcal{C}(T)$  is a maximizing cap, with  $T = K \setminus \mathcal{N}(K)$  and  $\mathcal{A}_\omega(K) = |G|$ . In particular, if  $S$  is any moving sofa with  $|S| = |G|$ , there is such a  $T$  for  $S' = S$  and some  $\omega \in [\omega_0, \pi/2]$ .*

*Proof.* By Fact 2.3(b),  $S' + w$  is in standard position for some  $w$ , and is a moving sofa with rotation angle  $\omega$ . By Fact 2.4(a),  $T = \mathcal{I}(S' + w)$  is a moving sofa in standard position containing  $S' + w$ , and it is a monotone sofa. So  $|G| = |S' + w| \leq |T| \leq |G|$  by Fact 2.8. By Fact 2.4(b),  $K = \mathcal{C}(T)$  is a cap, and Fact 2.4(c) gives  $T = K \setminus \mathcal{N}(K)$  and  $\mathcal{A}_\omega(K) = |T| = |G|$ ;  $K$  is maximizing by Fact 2.8. Finally, if  $|S| = |G|$  then  $|S| > 11/5$ , so  $S$  has a rotation angle  $\omega \in [\omega_0, \pi/2]$  by Fact 2.3(a).  $\square$

So the problem is to show that  $S$  is a rigid image of  $G$ . The proof follows  $S$  through a chain of sets, each of which contains a rigid image of  $S$  and has area  $|G|$  (Figure 3). Steps (1) and (2) are used twice: first for the maximizing cap  $K$  of Proposition 3.2, which has the rotation angle  $\omega$ , to obtain step (3), and then, after step (3), for the maximizing cap of the new monotone sofa  $U$  and for its mirror image, which have the rotation angle  $\pi/2$ , to obtain step (4).

- (1) *Selection* (Section 4). A given maximizing cap  $K_*$  with  $\mathcal{A}_\omega(K_*) > 0$  is the Hausdorff limit of polygon caps  $K_n$  that maximize the polygon sofa area minus a penalty for straying from  $K_*$ .

- (2) *Variations* (Section 5). Pushing one side of  $K_n$  shows that  $K_n$  is almost balanced: the defect  $\sigma_{K_n} - \tau_{K_n}$  at a normal is bounded by a multiple of the distance from  $K_n$  to  $K_*$  at the sampled normals, and the weights are chosen so that these bounds add up.
- (3) *The right angle* (Section 6). If  $\omega < \pi/2$ , the limit of the defect bounds gives the two *pinned bounds*  $w_{K_*}^\circ \leq \sigma_{K_*}(\pi/2)$  and  $z_{K_*}^\circ \leq \sigma_{K_*}(\omega)$  between wedge gaps and side lengths of  $K_*$ . They give  $T$  width at most one in the directions  $u_t$ ,  $t \in [\omega, \pi/2]$ , so that a rotated copy  $R_\psi T$ ,  $\psi = \pi/2 - \omega$ , turns by  $\pi/2$ . Monotonizing  $R_\psi T$  gives a monotone sofa  $U$  with rotation angle  $\pi/2$  and  $|U| = |G|$  that contains a rigid image of  $S$ .
- (4) *Injectivity* (Section 7). For the cap  $K$  of  $U$ , the defect bounds become the curvature bounds  $\sigma_K \leq k_0(g_K^+(t))dt$  on  $[0, \pi/2)$  and  $\sigma_K \leq k_0(f_K^-(t - \pi/2))dt$  on  $(\pi/2, \pi]$ , and Baek's iteration gives the injectivity condition.
- (5) *Equality* (Section 8). Then  $|G| = \mathcal{A}(K) \leq \mathcal{Q}(\xi_K) \leq \mathcal{Q}(\xi_G) = |G|$ , and equality in the concave functional  $\mathcal{Q}$  forces equality in each Mamikon term, a convex integral  $\frac{1}{2} \int \varrho^2$ . This gives differential equations for the difference of the support functions of  $K$  and of Gerver's cap, whose solutions make  $K$  a horizontal translate of Gerver's cap and  $U$  a horizontal translate of  $G$ .
- (6) *Recovery* (Section 9). A closed subset of  $G$  with the area of  $G$  is  $G$ , since  $G$  is the closure of its interior.

#### 4. APPROXIMATING A MAXIMIZING CAP BY POLYGON CAPS

In this section and in Section 5 we fix  $\omega \in (0, \pi/2]$  and a cap  $K_* \in \mathcal{K}_\omega^c$  with  $\mathcal{A}_\omega(K_*) > 0$ , and write  $h_* = h_{K_*}$ ; Proposition 4.8 assumes in addition that  $K_*$  maximizes  $\mathcal{A}_\omega$ . Section 6 applies these results, when  $\omega < \pi/2$ , to the cap of Proposition 3.2, a maximizing cap with sofa area  $|G| > 11/5$ . Section 7 applies them, with  $\omega = \pi/2$ , to the cap of the monotone sofa  $U$  of Proposition 6.9 and to the mirror image of that cap; both are maximizing, with sofa area  $|G| > 0$ .

Baek obtains his cap as a limit of maximum polygon caps, which he proves to be balanced. This procedure need not reach a given maximizing cap: if  $F_n \downarrow F$  uniformly on  $[0, 1]$ , a maximizer of  $F$  need not be a limit of maximizers of the  $F_n$  (take  $F_n(x) = x/n$  and  $F = 0$ , whose maximizers are all of  $[0, 1]$  while those of  $F_n$  are  $\{1\}$ ). We therefore subtract a penalty that vanishes at the chosen maximizer, as in the example  $F_n(x) - (x - x_*)^2$ , whose maximizers converge to  $x_*$ . The penalty is a weighted sum of squared differences of support values at dyadic normals. Section 5 bounds the resulting departure from balance.

**4.1. Dyadic angles and the penalty.** For  $k \geq 0$  let  $\Theta_k = \{i\omega/2^{k+1} : 0 < i < 2^{k+1}\}$ , an angle set with  $2^{k+1}$  intervals. Then  $\Theta_k \subset \Theta_{k+1}$ ,  $\omega/2 \in \Theta_k$ , and  $\bigcup_k \Theta_k$  is dense in  $[0, \omega]$ . Write  $\mathcal{K}_k^c = \mathcal{K}_{\Theta_k}^c$  for the polygon caps with angle set  $\Theta_k$  and  $\Theta_k^\circ$  for the defining normals.

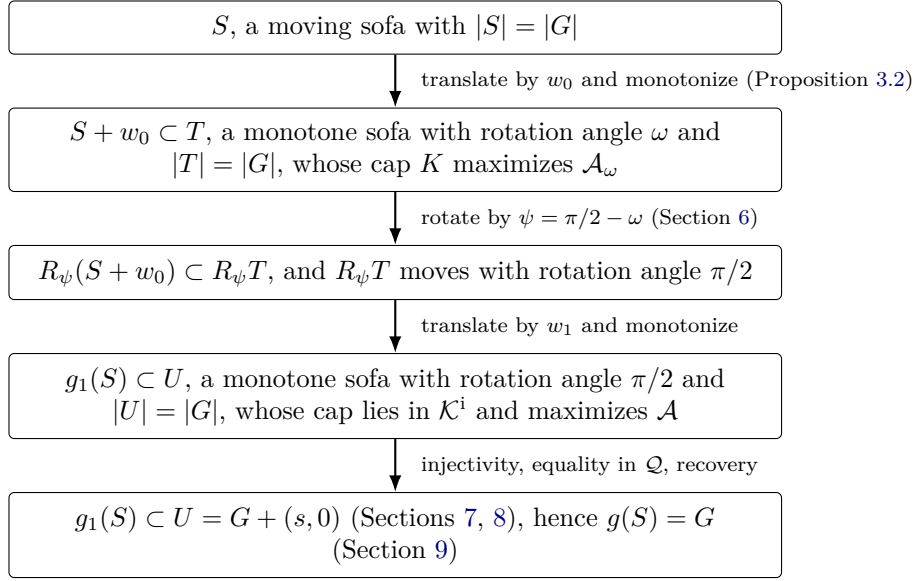


FIGURE 3. The containment chain of the proof. Every set contains the image of  $S$  under the maps so far, and all the sets have area  $|G|$ ; the first three arrows are a translation followed by a monotonization, a rotation, and a translation followed by a monotonization. Here  $g_1(p) = R_\psi(p + w_0) + w_1$  and  $g(p) = g_1(p) - (s, 0)$ .

For  $m \geq 0$  let  $\nu_m = 2^{-m-1}/(2|\Theta_m|)$ . For a convex body  $K$  and  $k \geq 0$  let

$$\Pi_k(K) = \sum_{m=0}^k \nu_m \sum_{t \in \Theta_m} \left[ (h_K(t) - h_*(t))^2 + (h_K(t + \frac{\pi}{2}) - h_*(t + \frac{\pi}{2}))^2 \right].$$

The terms are the *samples*; an angle of level  $m$  also lies in every later level, so it carries one sample from each level, from the first level that contains it up to  $k$ . For  $s \in \mathbb{R}$  let  $\nu_k(s) \geq 0$  be the total weight of the samples taken at the normal  $s$ , so that  $\nu_k(s) = 0$  unless  $s \in \Theta_k \cup (\Theta_k + \pi/2)$ . The samples of level  $m$  have total weight  $2|\Theta_m|\nu_m = 2^{-m-1}$ , hence

$$\sum_s \nu_k(s) = \sum_{m=0}^k 2^{-m-1} = 1 - 2^{-k-1} \leq 1. \quad (4.1)$$

Since every term of  $\Pi_k$  is nonnegative, each sample is at most the penalty (*persistence*): if  $m \leq k$  and  $t \in \Theta_m$ , then

$$\nu_m(h_K(s) - h_*(s))^2 \leq \Pi_k(K) \quad (s = t \text{ and } s = t + \pi/2). \quad (4.2)$$

Let  $K$  and  $K'$  be convex bodies and  $\eta \geq 0$  with  $|h_K(s) - h_{K'}(s)| \leq \eta$  at every sampled normal  $s$ . For reals  $a, b, c, r$  with  $|a - c| \leq \eta$  and  $|b - a| \leq r$ , the identity  $(b - c)^2 - (a - c)^2 = 2(a - c)(b - a) + (b - a)^2$  gives  $|(b - c)^2 - (a - c)^2| \leq$

$2\eta r + r^2$ ; we apply it to each sample, with  $a = h_K(s)$ ,  $b = h_{K'}(s)$  and  $c = h_*(s)$ . If  $|h_{K'}(s) - h_K(s)| \leq r$  at every sampled normal  $s$ , this gives

$$|\Pi_k(K') - \Pi_k(K)| \leq \left( \sum_s \nu_k(s) \right) (2\eta r + r^2). \quad (4.3)$$

If  $h_{K'}(s) = h_K(s)$  at every sampled normal  $s \neq s_0$  and  $|h_{K'}(s_0) - h_K(s_0)| \leq r$ , only the samples at  $s_0$ , of total weight  $\nu_k(s_0)$ , contribute, and

$$|\Pi_k(K') - \Pi_k(K)| \leq \nu_k(s_0) (2\eta r + r^2). \quad (4.4)$$

**Definition 4.1** (penalized maximizer). A polygon cap  $K \in \mathcal{K}_k^c$  is a *penalized maximizer at stage  $k$*  if  $\mathcal{A}_{\Theta_k}(C) - \Pi_k(C) \leq \mathcal{A}_{\Theta_k}(K) - \Pi_k(K)$  for every  $C \in \mathcal{K}_k^c$ .

**Lemma 4.2** (recovery polygon).  $K_k^r = \mathcal{C}_{\Theta_k}(K_*)$  lies in  $\mathcal{K}_k^c$  and has  $\Pi_k(K_k^r) = 0$  and  $\mathcal{A}_{\Theta_k}(K_k^r) - \Pi_k(K_k^r) \geq \mathcal{A}_\omega(K_*)$ .

*Proof.* By Fact 2.9,  $K_k^r$  is a polygon cap with the support values of  $K_*$  on  $\Theta_k^\circ$ , which contains all the sampled normals; so  $\Pi_k(K_k^r) = 0$ . Also  $\mathcal{A}_{\Theta_k}(K_k^r) = \mathcal{A}_{\Theta_k}(K_*) \geq \mathcal{A}_\omega(K_*)$ , by Fact 2.9.  $\square$

**Lemma 4.3** (two samples bound a cap). Let  $\Theta$  be an angle set,  $K \in \mathcal{K}_\Theta^c$ ,  $t \in \Theta$ , and suppose that  $q(h_K(s) - h_*(s))^2 \leq c$  for  $s = t$  and  $s = t + \pi/2$ , with  $q > 0$ ,  $c \geq 0$ . Then  $K \subset [-R, R] \times [0, 1]$  with  $R = H/\sin t + H/\cos t$ ,  $H = |h_*(t)| + |h_*(t + \pi/2)| + c/q + 1$ .

*Proof.* A real number  $a$  with  $a^2 \leq c/q$  has  $|a| \leq c/q + 1$ , as  $2|a| \leq a^2 + 1$ . So  $h_K(t) \leq h_*(t) + c/q + 1 \leq H$ , and likewise  $h_K(t + \pi/2) \leq H$ . A point  $(x, y) \in K$  has  $0 \leq y \leq 1$ , since  $K \subset P_\omega$ . As  $t \in (0, \pi/2)$ ,  $x \cos t + y \sin t \leq H$  gives  $x \leq H/\cos t$ , and  $-x \sin t + y \cos t \leq H$  gives  $-x \leq H/\sin t$ ; both bounds are at most  $R$ , since  $H \geq 0$ .  $\square$

**Lemma 4.4** (the niche lies in a box). Let  $R \geq 0$ . If  $K \in \mathcal{K}_\omega^c$  and  $K \subset [-R, R] \times [0, 1]$ , then  $\mathcal{N}(K) \subset [-R, R] \times [0, 2R]$ ; the same holds for  $\mathcal{N}_\Theta(K) \subset \mathcal{N}(K)$ , for every angle set  $\Theta$ .

*Proof.* Let  $p \in \mathcal{N}(K)$ , so  $p_y \geq 0$  and  $p \in Q_K^-(t)$  for some  $t \in (0, \omega)$ . As  $K \subset [-R, R] \times [0, 1]$ ,  $h_K(t) \leq R \cos t + \sin t \leq R \cos t + 1$  and  $h_K(t + \pi/2) \leq R \sin t + \cos t \leq R \sin t + 1$ . So  $\langle p, u_t \rangle < R \cos t$  and  $\langle p, v_t \rangle < R \sin t$ . Since  $p_y \geq 0$  and  $\langle p, u_t \rangle = p_x \cos t + p_y \sin t$ , we get  $p_x < R$ ; likewise  $\langle p, v_t \rangle = -p_x \sin t + p_y \cos t$  gives  $p_x > -R$ . Finally  $p_y = \sin t \langle p, u_t \rangle + \cos t \langle p, v_t \rangle < 2R \sin t \cos t \leq 2R$ , as  $R \geq 0$ .  $\square$

**Proposition 4.5** (penalized maximizers exist). For every  $k \geq 0$  there is a penalized maximizer at stage  $k$ , and every penalized maximizer  $K$  at stage  $k$  satisfies

$$\mathcal{A}_{\Theta_k}(K) - \Pi_k(K) \geq \mathcal{A}_\omega(K_*).$$

If  $K^{(k)}$  is a penalized maximizer at stage  $k$  for every  $k \geq 0$ , there is  $R_0 \geq 0$  such that  $K^{(k)} \subset [-R_0, R_0] \times [0, 1]$  for every  $k$ .

*Proof.* Let  $\Psi_k = \mathcal{A}_{\Theta_k} - \Pi_k$  on  $\mathcal{K}_k^c$ , and let  $\bar{t} = \omega/2$ , the angle of  $\Theta_0$ , which lies in every  $\Theta_k$ . Let  $c = c_{\omega, \bar{t}} > 0$  be the constant of Fact 2.10; it does not depend on  $k$ .

*The polygon caps with  $\Psi_k > 0$  lie in one box.* Let  $C \in \mathcal{K}_k^c$  with  $\Psi_k(C) > 0$ . As  $\Pi_k(C) \geq 0$ ,  $\mathcal{A}_{\Theta_k}(C) > 0$ , so  $C$  has width at most  $c$  and, as it lies in the strip  $0 \leq y \leq 1$ ,  $|C| \leq c$ . By Fact 2.9,  $\Pi_k(C) < \mathcal{A}_{\Theta_k}(C) = |C| - |\mathcal{N}_{\Theta_k}(C)| \leq c$ . The samples of level 0 lie at  $\bar{t}$  and  $\bar{t} + \pi/2$ , so (4.2) gives  $\nu_0(h_C(s) - h_*(s))^2 \leq \Pi_k(C) \leq c$  for  $s = \bar{t}, \bar{t} + \pi/2$ , and Lemma 4.3 with  $q = \nu_0$  puts  $C$  in  $[-R_0, R_0] \times [0, 1]$ , where

$$R_0 = \frac{H}{\sin \bar{t}} + \frac{H}{\cos \bar{t}} \geq 0, \quad H = |h_*(\bar{t})| + |h_*(\bar{t} + \frac{\pi}{2})| + \frac{c}{\nu_0} + 1,$$

do not depend on  $k$  or  $C$ . Moreover  $\Psi_k(C) \leq \mathcal{A}_{\Theta_k}(C) \leq c$ ; hence  $\Psi_k \leq c$  on  $\mathcal{K}_k^c$ , as  $c > 0$ .

*A maximizing sequence and its limit cap.* By Lemma 4.2,  $M_k = \sup \Psi_k \geq \Psi_k(K_k^I) \geq \mathcal{A}_\omega(K_*) > 0$ , and  $M_k \leq c$ . Take  $C_j \in \mathcal{K}_k^c$  with  $\Psi_k(C_j) > 0$  and  $\Psi_k(C_j) \rightarrow M_k$ . They lie in  $[-R_0, R_0] \times [0, 1]$ , so  $|h_{C_j}| \leq R_0 + 1$ , and as  $\Theta_k^\diamond$  is finite, a subsequence, still written  $(C_j)$ , has  $h_{C_j}(s) \rightarrow h_\infty(s)$  for every  $s \in \Theta_k^\diamond$ . Let  $I = \Theta_k^\diamond \cup \{\omega + \pi, 3\pi/2\}$ , the normal angles of the polygon caps in  $\mathcal{K}_k^c$ , and put  $h_\infty(\omega + \pi) = h_\infty(3\pi/2) = 0$ ; then  $h_{C_j}(s) \rightarrow h_\infty(s)$  for every  $s \in I$ , and  $h_\infty(\omega) = h_\infty(\pi/2) = 1$ . For  $\delta \geq 0$  let  $K_\infty^\delta$  be the set of points  $p$  with  $\langle p, u_s \rangle \leq h_\infty(s) + \delta$  for every  $s \in I$ . It is closed and convex, and bounded, since it lies in the strip  $-\delta \leq y \leq 1 + \delta$  and in  $H_-(t, h_\infty(t) + \delta) \cap H_-(t + \pi/2, h_\infty(t + \pi/2) + \delta)$  for  $t \in \Theta_k \subset (0, \pi/2)$ . Let  $K_\infty = K_\infty^0$ ; it is the set  $\mathcal{C}_{\Theta_k}(h_\infty)$  of Section 2.3. Let  $r \in I$ , and take  $q_j \in C_j$  with  $\langle q_j, u_r \rangle = h_{C_j}(r)$ . As the  $q_j$  lie in  $[-R_0, R_0] \times [0, 1]$ , a subsequence of  $(q_j)$  converges to a point  $q$ ; passing to the limit in  $\langle q_j, u_s \rangle \leq h_{C_j}(s)$ ,  $s \in I$ , gives  $q \in K_\infty$ , and  $\langle q, u_r \rangle = h_\infty(r)$ . So  $K_\infty$  is nonempty and  $h_{K_\infty}(r) = h_\infty(r)$ . Hence  $K_\infty$  is a convex body, with the support values 1, 1, 0, 0 at  $\omega, \pi/2, \omega + \pi, 3\pi/2$ , and an intersection of half-planes with normal angles in  $I$ ; that is,  $K_\infty \in \mathcal{K}_k^c$ . Also  $h_{C_j}(s) \rightarrow h_{K_\infty}(s)$  for every  $s \in I$ .

*$K_\infty$  is a penalized maximizer.* The sampled normals lie in  $\Theta_k^\diamond$ , so  $\Pi_k(C_j) \rightarrow \Pi_k(K_\infty)$ . Let  $\delta > 0$ . For large  $j$ ,  $h_{C_j}(s) < h_\infty(s) + \delta$  at the finitely many  $s \in \Theta_k^\diamond$ , and  $h_{C_j}(s) = h_\infty(s) = 0$  for  $s = \omega + \pi, 3\pi/2$ ; so  $C_j \subset K_\infty^\delta$ . As  $\delta \downarrow 0$  the bounded sets  $K_\infty^\delta$  decrease to  $K_\infty$ , so  $|K_\infty^\delta| \rightarrow |K_\infty|$ , and therefore  $\limsup_j |C_j| \leq |K_\infty|$ . Let  $p \in \mathcal{N}_{\Theta_k}(K_\infty)$ :  $p \in F_\omega$ , and  $\langle p, u_t \rangle < h_\infty(t) - 1$  and  $\langle p, v_t \rangle < h_\infty(t + \pi/2) - 1$  for some  $t \in \Theta_k$ . These strict inequalities hold with  $h_{C_j}$  in place of  $h_\infty$  for large  $j$ , so  $p \in \mathcal{N}_{\Theta_k}(C_j)$  for all large  $j$ . As  $h_{C_j} \leq R_0 + 1$ , all the niches  $\mathcal{N}_{\Theta_k}(C_j)$  lie in the set of points  $p \in F_\omega$  with  $\langle p, u_t \rangle < R_0$  and  $\langle p, v_t \rangle < R_0$  for some  $t \in \Theta_k$ , which is bounded:  $p_y \geq 0$  gives  $p_x < R_0/\cos t$  and  $-p_x < R_0/\sin t$ , and  $p_y = \sin t \langle p, u_t \rangle + \cos t \langle p, v_t \rangle < 2R_0$ . So their areas are finite, and Fatou's lemma gives  $|\mathcal{N}_{\Theta_k}(K_\infty)| \leq \liminf_j |\mathcal{N}_{\Theta_k}(C_j)|$ . Since  $\Psi_k = |\cdot| - |\mathcal{N}_{\Theta_k}(\cdot)| - \Pi_k$  on  $\mathcal{K}_k^c$  (Fact 2.9),

$$M_k = \lim_j \Psi_k(C_j) \leq \limsup_j |C_j| - \liminf_j |\mathcal{N}_{\Theta_k}(C_j)| - \lim_j \Pi_k(C_j) \leq \Psi_k(K_\infty).$$

As  $\Psi_k \leq M_k$  on  $\mathcal{K}_k^c$ ,  $K_\infty$  is a penalized maximizer at stage  $k$ .

*The lower bound and the box.* A penalized maximizer  $K$  at stage  $k$  has  $\Psi_k(K) \geq \Psi_k(K_k^r) \geq \mathcal{A}_\omega(K_*)$  by Lemma 4.2. If  $K^{(k)}$  is a penalized maximizer at stage  $k$  for every  $k$ , then  $\Psi_k(K^{(k)}) \geq \mathcal{A}_\omega(K_*) > 0$ , and the first step puts every  $K^{(k)}$  in  $[-R_0, R_0] \times [0, 1]$ .  $\square$

#### 4.2. Convergence to the given cap.

**Lemma 4.6** (limits of polygon caps with nested angle sets). *Let  $k_1 < k_2 < \dots$  and  $R \geq 0$ , and let  $C_j \in \mathcal{K}_{k_j}^c$  lie in  $[-R, R] \times [0, 1]$  and converge in  $d_H$  to a convex body  $K_\infty$ , whose niche  $\mathcal{N}(K_\infty)$  and sofa area  $\mathcal{A}_\omega(K_\infty)$  are given by the formulas of Section 2.2. Then for every  $\varepsilon > 0$ ,  $\mathcal{A}_{\Theta_{k_j}}(C_j) \leq \mathcal{A}_\omega(K_\infty) + \varepsilon$  for all large  $j$ .*

*Proof.* Let  $\varepsilon > 0$ . By the continuity of the area (Fact 2.1(c)),  $|C_j| \leq |K_\infty| + \varepsilon/2$  for large  $j$ . Let  $p \in \mathcal{N}(K_\infty)$ :  $p \in F_\omega \cap Q_{K_\infty}^-(t)$  for some  $t \in (0, \omega)$ . The two strict inequalities that define  $Q_{K_\infty}^-(t)$  hold at  $p$  with a margin  $\delta > 0$  at some dyadic angle  $t'$  near  $t$ , since  $h_{K_\infty}$  is continuous. For large  $j$ ,  $t' \in \Theta_{k_j}$  and  $\sup|h_{C_j} - h_{K_\infty}| < \delta/2$ , so  $p \in F_\omega \cap Q_{C_j}^-(t') \subset \mathcal{N}_{\Theta_{k_j}}(C_j)$ . The sets  $\mathcal{N}_{\Theta_{k_j}}(C_j)$  lie in  $[-R, R] \times [0, 2R]$  by Lemma 4.4 (so their areas are finite), and Fatou's lemma gives  $|\mathcal{N}(K_\infty)| \leq \liminf_j |\mathcal{N}_{\Theta_{k_j}}(C_j)|$ ; so  $|\mathcal{N}(K_\infty)| \leq |\mathcal{N}_{\Theta_{k_j}}(C_j)| + \varepsilon/2$  for large  $j$ . As  $\mathcal{A}_{\Theta_{k_j}}(C_j) = |C_j| - |\mathcal{N}_{\Theta_{k_j}}(C_j)|$  (Fact 2.9), for large  $j$

$$\mathcal{A}_{\Theta_{k_j}}(C_j) \leq |K_\infty| - |\mathcal{N}(K_\infty)| + \varepsilon = \mathcal{A}_\omega(K_\infty) + \varepsilon.$$

$\square$

**Lemma 4.7** (dyadic supports determine a cap). *If  $K, K' \in \mathcal{K}_\omega^c$  and  $h_K(s) = h_{K'}(s)$  for all  $s \in \bigcup_m (\Theta_m \cup (\Theta_m + \pi/2))$ , then  $K = K'$ .*

*Proof.* By continuity and density,  $h_K = h_{K'}$  on  $J_\omega$ , and  $h_K = h_{K'} = 0$  at  $\omega + \pi$  and  $3\pi/2$ . For a cap  $K$ ,  $F_\omega = H_K(\omega + \pi) \cap H_K(3\pi/2)$ , so by (2.2)  $K$  is the intersection of its supporting half-planes  $H_K(s)$ ,  $s \in J_\omega \cup \{\omega + \pi, 3\pi/2\}$ ; the same holds for  $K'$ , and these half-planes are those of  $K$ .  $\square$

**Proposition 4.8** (selection). *If  $K_*$  maximizes  $\mathcal{A}_\omega$  on  $\mathcal{K}_\omega^c$ , there are  $k_1 < k_2 < \dots$ , penalized maximizers  $K_j$  at stage  $k_j$ , and  $R_0 \geq 0$  such that  $K_j \subset [-R_0, R_0] \times [0, 1]$  for every  $j$  and  $K_j \rightarrow K_*$  in  $d_H$ .*

*Proof.* By Proposition 4.5, for every  $k$  there is a penalized maximizer  $K^{(k)}$  at stage  $k$ , and there is  $R_0 \geq 0$  with  $K^{(k)} \subset [-R_0, R_0] \times [0, 1]$  for every  $k$ . By Blaschke's theorem (Fact 2.1(c)) there are  $k_1 < k_2 < \dots$  with  $K_j = K^{(k_j)} \rightarrow K_\infty$  for a convex body  $K_\infty$ ; the  $K_j$  lie in  $[-R_0, R_0] \times [0, 1]$ .

$K_\infty$  is a cap with rotation angle  $\omega$ . The support values at  $\omega, \pi/2, \omega + \pi, 3\pi/2$  pass to the limit. Let  $(a, b)$  be a gap of  $J_\omega \cup \{\omega + \pi, 3\pi/2\}$  in the circle; it has length at most  $\pi/2$ . A polygon cap  $K_j$  has no normal angle strictly between  $a$  and  $b$ , so the supporting lines  $l_{K_j}(a)$  and  $l_{K_j}(b)$  meet at a vertex  $q$  of  $K_j$  and  $h_{K_j}(r) = \langle q, u_r \rangle$  for  $a \leq r \leq b$ ; with  $\sin(b - a)u_r =$

$\sin(b-r)u_a + \sin(r-a)u_b$  this gives  $\sin(b-a)h_{K_j}(r) = \sin(b-r)h_{K_j}(a) + \sin(r-a)h_{K_j}(b)$ . The identity passes to  $h_{K_\infty}$ , and its coefficients are nonnegative, so  $H_{K_\infty}(a) \cap H_{K_\infty}(b) \subset H_{K_\infty}(r)$ . Hence  $K_\infty = \bigcap_r H_{K_\infty}(r)$  is the intersection of the  $H_{K_\infty}(r)$  with  $r \in J_\omega \cup \{\omega + \pi, 3\pi/2\}$ .

*The penalties tend to 0.* By Proposition 4.5,  $0 \leq \Pi_{k_j}(K_j) \leq \mathcal{A}_{\Theta_{k_j}}(K_j) - \mathcal{A}_\omega(K_*)$ . Let  $\varepsilon > 0$ . By Lemma 4.6 with  $R = R_0$ , and the maximality of  $K_*$ ,  $\mathcal{A}_{\Theta_{k_j}}(K_j) \leq \mathcal{A}_\omega(K_\infty) + \varepsilon \leq \mathcal{A}_\omega(K_*) + \varepsilon$  for all large  $j$ . So  $\Pi_{k_j}(K_j) \leq \varepsilon$  for all large  $j$ , and  $\Pi_{k_j}(K_j) \rightarrow 0$ .

$K_\infty = K_*$ . Let  $t \in \Theta_m$ . For  $k_j \geq m$ , (4.2) gives  $\nu_m(h_{K_j}(s) - h_*(s))^2 \leq \Pi_{k_j}(K_j)$  for  $s = t, t + \pi/2$ . As  $j \rightarrow \infty$ ,  $\Pi_{k_j}(K_j) \rightarrow 0$  and  $h_{K_j}(s) \rightarrow h_{K_\infty}(s)$ , and  $\nu_m > 0$ ; so  $h_{K_\infty}(s) = h_*(s)$ . Hence  $h_{K_\infty} = h_*$  at all dyadic normals, and  $K_\infty = K_*$  by Lemma 4.7.  $\square$

For a selection as in Proposition 4.8, put  $\eta_j = d_H(K_j, K_*) \rightarrow 0$ ; at every sampled normal,  $|h_{K_j}(s) - h_*(s)| \leq \eta_j$ .

## 5. PENALIZED MAXIMIZERS ARE ALMOST BALANCED

A maximum polygon cap is balanced [1, Thm. 3.4.9]:  $\sigma_K(t) = \tau_K(t)$  for every  $t \in \Theta^\circ$ , since otherwise pushing one side would increase  $\mathcal{A}_\Theta$ . A penalized maximizer need not be balanced, but pushing a side changes the penalty by a controlled amount, and the argument gives a one-sided bound on the *defect*  $\zeta_K(t) = \sigma_K(t) - \tau_K(t)$ . In this section we fix  $k \geq 0$ ,  $\Theta = \Theta_k$ , a penalized maximizer  $K \in \mathcal{K}_k^c$  at stage  $k$ , and  $\eta \geq 0$  with  $|h_K(s) - h_*(s)| \leq \eta$  at every sampled normal  $s$ . Let  $\bar{\nu} = \sum_s \nu_k(s)$  be the total weight;  $\bar{\nu} \leq 1$  by (4.1).

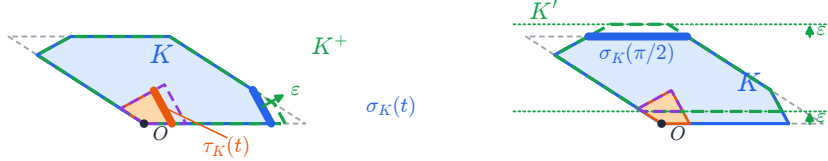


FIGURE 4. Pushing one side of a polygon cap  $K$  with angle set  $\{\omega/2\}$ ,  $\omega = 1$ ; the grey dashed parallelogram is  $P_\omega$ , and the niche of  $K$  is orange. Left: raising the height at the floating normal  $t = \omega/2$  by  $\varepsilon$  gives the cap  $K^+$  (green, dashed) and its niche (purple, dashed); the cap gains a strip of area  $\sigma_K(t)\varepsilon + O(\varepsilon^2)$  along its edge with normal  $t$ , and the niche a strip of area  $\tau_K(t)\varepsilon + O(\varepsilon^2)$  along the side on the inner wall  $b_K(t)$ . Right: raising the pinned height at  $\pi/2$  by  $\varepsilon$  moves both lines of the horizontal strip up by  $\varepsilon$  (green, dotted) and gives  $K'$  (green, dashed), with its niche (purple, dashed); the cap gains a strip along its top edge, of length  $\sigma_K(\pi/2)$ , and loses one along its bottom edge, and  $K'$  is a translate of a polygon cap.

### 5.1. Floating normals.

**Lemma 5.1** (raising a floating height). *Let  $K \in \mathcal{K}_\Theta^c$ ,  $t \in \Theta^\diamond \setminus \{\omega, \pi/2\}$ ,  $\varepsilon \geq 0$ , and  $K^+ = \mathcal{C}_\Theta(h_K^{t,\varepsilon})$ . Then  $K^+ \in \mathcal{K}_\Theta^c$ ,  $K \subset K^+$ ,  $h_{K^+}(s) = h_K(s)$  for  $s \in \Theta^\diamond \setminus \{t\}$ , and  $h_K(t) \leq h_{K^+}(t) \leq h_K(t) + \varepsilon$ .*

*Proof.* Since  $h_K(\omega) = h_K(\pi/2) = 1$ ,  $P_{h_K^{t,\varepsilon}} = P_\omega$ , so

$$K^+ = P_\omega \cap \bigcap_s H_-(s, h_K^{t,\varepsilon}(s)),$$

the intersection over the floating normals  $s$ . As  $h_K \leq h_K^{t,\varepsilon}$ ,  $K = \mathcal{C}_\Theta(h_K) \subset K^+ \subset P_\omega$ . So  $K^+$  is a convex body (it is bounded, since the half-planes with normals  $t'$  and  $t' + \pi/2$ ,  $t' \in \Theta$ , bound it also when  $\omega = \pi/2$ ), it has the support values 1, 1, 0, 0 of  $P_\omega$  at  $\omega, \pi/2, \omega + \pi, 3\pi/2$ , and it is an intersection of half-planes with the normal angles of a polygon cap. So  $K^+ \in \mathcal{K}_\Theta^c$ . Next,  $h_{K^+} \geq h_K$  since  $K \subset K^+$ , and  $h_{K^+}(s) \leq h_K^{t,\varepsilon}(s)$  since  $K^+ \subset H_-(s, h_K^{t,\varepsilon}(s))$ .  $\square$

**Proposition 5.2** (floating defects). *For every floating normal  $t \in \Theta^\diamond \setminus \{\omega, \pi/2\}$ ,*

$$\sigma_K(t) - \tau_K(t) \leq 2\eta \nu_k(t).$$

*Proof.* Let  $K_\varepsilon^+ = \mathcal{C}_\Theta(h_K^{t,\varepsilon})$ . By Lemma 5.1,  $K_\varepsilon^+$  is a polygon cap that has the support values of  $K$  on  $\Theta^\diamond \setminus \{t\}$ , and differs from  $K$  at  $t$  by at most  $\varepsilon$ ; so by (4.4),  $\Pi_k(K_\varepsilon^+) - \Pi_k(K) \leq \nu_k(t)(2\eta\varepsilon + \varepsilon^2)$ . Penalized maximality gives  $\mathcal{A}_\Theta(K_\varepsilon^+) - \mathcal{A}_\Theta(K) \leq \Pi_k(K_\varepsilon^+) - \Pi_k(K)$ . By Fact 2.12(c) (with  $C = K_\varepsilon^+$ ,  $v = 0$ ) and (a), for small  $\varepsilon$

$$\mathcal{A}_\Theta(K_\varepsilon^+) \geq \mathcal{A}_\Theta(h_K^{t,\varepsilon}) \geq \mathcal{A}_\Theta(K) + (\sigma_K(t) - \tau_K(t))\varepsilon - M\varepsilon^2.$$

Hence  $(\sigma_K(t) - \tau_K(t))\varepsilon - M\varepsilon^2 \leq \nu_k(t)(2\eta\varepsilon + \varepsilon^2)$ . Divide by  $\varepsilon$  and let  $\varepsilon \rightarrow 0$ .  $\square$

**5.2. Pinned normals.** In this subsection  $\omega < \pi/2$ , so that the two pinned normals  $\omega$  and  $\pi/2$  are distinct and every cap contains  $O$  and  $o_\omega$  (Lemma 2.5); in particular  $h_K \geq 0$ . Moving a pinned normal moves both lines of a strip, and the result is in general only a translate of a polygon cap.

**Lemma 5.3** (the pinned sandwich). *Let  $K \in \mathcal{K}_\Theta^c$ ,  $t \in \{\omega, \pi/2\}$ ,  $\varepsilon \in [0, 1]$ , and  $K' = \mathcal{C}_\Theta(h_K^{t,\varepsilon})$ . Then*

$$(1 - \varepsilon)K + \varepsilon o_\omega \subset K' \subset (1 + \varepsilon)K.$$

*If  $R \geq 0$  and  $|h_K(s)| \leq R$  for all  $s$ , then  $|h_{K'}(s) - h_K(s)| \leq 2R\varepsilon$  for every  $s$ .*

*Proof.* Let  $t'$  be the other pinned normal. Then  $K' = \{p : \varepsilon \leq \langle p, u_t \rangle \leq 1 + \varepsilon, 0 \leq \langle p, u_{t'} \rangle \leq 1, \langle p, u_s \rangle \leq h_K(s) \text{ for floating } s\}$ . Let  $p \in K$  and  $q = (1 - \varepsilon)p + \varepsilon o_\omega \in K'$ . Then  $q$  satisfies every inequality of  $K'$  except possibly the one for  $t$ , and  $\langle q, u_t \rangle = (1 - \varepsilon)\langle p, u_t \rangle + \varepsilon \in [\varepsilon, 1]$  because  $\langle o_\omega, u_t \rangle = 1$  and  $0 \leq \langle p, u_t \rangle \leq 1$ . So  $q \in K'$ . Conversely, let  $p \in K'$ .

For floating  $s$ ,  $\langle p/(1+\varepsilon), u_s \rangle \leq h_K(s)/(1+\varepsilon) \leq h_K(s)$  as  $h_K(s) \geq 0$ ;  $\langle p, u_t \rangle/(1+\varepsilon) \in [0, 1]$  and  $\langle p, u_{t'} \rangle/(1+\varepsilon) \in [0, 1]$ . So  $p/(1+\varepsilon) \in K$ . Taking support functions,  $(1-\varepsilon)h_K(s) + \varepsilon\langle o_\omega, u_s \rangle \leq h_{K'}(s) \leq (1+\varepsilon)h_K(s)$ . As  $o_\omega \in K$ ,  $-\langle o_\omega, u_s \rangle = \langle o_\omega, u_{s+\pi} \rangle \leq h_K(s+\pi) \leq R$ ; with  $|h_K(s)| \leq R$  this gives  $-2R\varepsilon \leq h_{K'}(s) - h_K(s) \leq R\varepsilon$ .  $\square$

**Lemma 5.4** (moving back to standard position). *Let  $K \in \mathcal{K}_\Theta^c$ ,  $t \in \{\omega, \pi/2\}$ ,  $\varepsilon \geq 0$ , and suppose that  $K' = \mathcal{C}_\Theta(h_K^{t,\varepsilon})$  is a translate  $C + v$  of a polygon cap  $C \in \mathcal{K}_\Theta^c$ ,  $v \in \mathbb{R}^2$ . Then  $|\langle v, u_s \rangle| \leq (2/\cos\omega + 1)\varepsilon$  for all  $s$ . If moreover  $\varepsilon \leq 1$ ,  $R \geq 0$  and  $|h_K(s)| \leq R$  for all  $s$ , then*

$$|h_C(s) - h_K(s)| \leq \gamma(R)\varepsilon \quad \text{for every } s, \quad \gamma(R) = 2R + \frac{2}{\cos\omega} + 1.$$

*Proof.* Let  $t'$  be the other pinned normal. For  $s \in \{\omega, \pi/2\}$ ,  $K'$  lies in the strip  $a_s \leq \langle p, u_s \rangle \leq a_s + 1$ , where  $a_t = \varepsilon$  and  $a_{t'} = 0$ . As  $C$  is a cap, it has width exactly 1 in the direction  $u_s$  (its support values at  $s$  and  $s + \pi$  are 1 and 0), and so has  $C + v$ ; hence  $\langle v, u_s \rangle = a_s$ . With  $s = \pi/2$  and  $s = \omega$  this gives  $v_y \in [0, \varepsilon]$  and  $v_x \cos\omega + v_y \sin\omega \in [0, \varepsilon]$ , so  $|v_x| \leq 2\varepsilon/\cos\omega$  and  $|\langle v, u_s \rangle| \leq |v_x| + |v_y| \leq (2/\cos\omega + 1)\varepsilon$ . Finally  $h_C(s) = h_{K'}(s) - \langle v, u_s \rangle$ , and Lemma 5.3 gives  $|h_{K'}(s) - h_K(s)| \leq 2R\varepsilon$ ; so  $|h_C(s) - h_K(s)| \leq 2R\varepsilon + (2/\cos\omega + 1)\varepsilon = \gamma(R)\varepsilon$ .  $\square$

**Proposition 5.5** (pinned defects). *Let  $\omega < \pi/2$ ,  $R \geq 0$  with  $|h_K(s)| \leq R$  for all  $s$ , and  $t \in \{\omega, \pi/2\}$ . Then*

$$\sigma_K(t) - \tau_K(t) \leq 2\bar{\nu}\eta\gamma(R), \quad \gamma(R) = 2R + \frac{2}{\cos\omega} + 1.$$

*Proof.* If  $\sigma_K(t) = 0$ , the left side is  $-\tau_K(t) \leq 0$ . Let  $\sigma_K(t) > 0$ . By Fact 2.12(b), for small  $\varepsilon \in (0, 1]$  the set  $\mathcal{C}_\Theta(h_K^{t,\varepsilon})$  is a translate  $C_\varepsilon + v_\varepsilon$  of a polygon cap  $C_\varepsilon$ . By Lemma 5.4,  $|h_{C_\varepsilon}(s) - h_K(s)| \leq \gamma(R)\varepsilon$  at every  $s$ , so (4.3) gives  $\Pi_k(C_\varepsilon) - \Pi_k(K) \leq \bar{\nu}(2\eta\gamma(R)\varepsilon + \gamma(R)^2\varepsilon^2)$ . Penalized maximality compares  $K$  with  $C_\varepsilon$ , and Fact 2.12(c), (a) give

$$\mathcal{A}_\Theta(C_\varepsilon) \geq \mathcal{A}_\Theta(h_K^{t,\varepsilon}) \geq \mathcal{A}_\Theta(K) + (\sigma_K(t) - \tau_K(t))\varepsilon - M\varepsilon^2.$$

So  $(\sigma_K(t) - \tau_K(t))\varepsilon - M\varepsilon^2 \leq \bar{\nu}(2\eta\gamma(R)\varepsilon + \gamma(R)^2\varepsilon^2)$ ; divide by  $\varepsilon$  and let  $\varepsilon \rightarrow 0$ .  $\square$

### 5.3. A two-sided bound.

**Corollary 5.6** (two-sided defect bound). *Let  $\omega < \pi/2$ , and  $R \geq 0$  with  $|h_K(s)| \leq R$  for all  $s$ . For every  $t \in \Theta^\circ$ ,*

$$|\sigma_K(t) - \tau_K(t)| \leq \frac{2\bar{\nu}\eta + 4\bar{\nu}\eta\gamma(R)}{\sin t}, \quad \gamma(R) = 2R + \frac{2}{\cos\omega} + 1.$$

*Proof.* Put  $\zeta(s) = \sigma_K(s) - \tau_K(s)$ , and  $e(s) = 2\eta\nu_k(s)$  for floating  $s$ ,  $e(s) = 2\bar{\nu}\eta\gamma(R)$  for the two pinned  $s$ . By Propositions 5.2 and 5.5,  $\zeta \leq e$  on  $\Theta^\circ$ , and  $e \geq 0$  with  $\sum_s e(s) \leq 2\bar{\nu}\eta + 4\bar{\nu}\eta\gamma(R) =: E$ . By Fact 2.11(c), the

sums  $\sum_s \sigma_K(s) \sin s$  and  $\sum_s \tau_K(s) \sin s$  both equal  $A_K^-(0)_x - C_K^+(\omega)_x$ , so  $\sum_s \sin s \zeta(s) = 0$ ; and  $0 < \sin s \leq 1$  on  $\Theta^\diamond \subset (0, \pi)$ . So

$$\begin{aligned} \sin t \zeta(t) &= - \sum_{s \neq t} \sin s \zeta(s) \geq - \sum_{s \neq t} \sin s e(s) \geq -E, \\ \sin t \zeta(t) &\leq \sin t e(t) \leq E. \end{aligned}$$

□

## 6. THE PINNED BOUNDS AND THE RIGHT ANGLE

Baek shows that a balanced maximum sofa  $S_\omega$  with  $|S_\omega| \geq 11/5$  and rotation angle  $\omega \in [\omega_0, \pi/2]$  has a rotated copy that turns by a right angle [1, Thm. 1.5.2]. The proof uses the balance only through two inequalities,  $w_K^\circ \leq \sigma_K(\pi/2)$  and  $z_K^\circ \leq \sigma_K(\omega)$ , which bound the wedge gaps on the two floors  $y = 0$  and  $l(\omega, 0)$  by the lengths of the sides of the cap on the opposite walls  $y = 1$  and  $l(\omega, 1)$  [1, Thm. 4.1.4]. We prove these inequalities for  $K_*$  from the defect bounds of Section 5, and then follow Baek's proof with them as hypotheses. In Sections 6.1 and 6.2 we have  $\omega < \pi/2$ , except in Lemma 6.6(c), and  $K_*$  is as in Section 4; Proposition 6.9 also covers  $\omega = \pi/2$ .

**6.1. The pinned bounds.** For a cap  $K \in \mathcal{K}_\omega^c$  and  $t \in (0, \omega)$  let

$$w_K(t) = h_K(0) - \frac{h_K(t) - 1}{\cos t}, \quad z_K(t) = h_K(\omega + \frac{\pi}{2}) - \frac{h_K(t + \pi/2) - 1}{\cos(\omega - t)},$$

the *wedge gaps*. With  $A = (h_K(0), 0) = A_K^-(0)$  and  $C = h_K(\omega + \pi/2)v_\omega = C_K^+(\omega)$  the ends of the upper boundary of  $K$  (see (2.3)),  $w_K(t) = \langle A - W_K(t), u_0 \rangle$ , where  $W_K(t) = ((h_K(t) - 1)/\cos t, 0)$  is the point where the inner wall  $b_K(t)$  meets the  $x$ -axis, and  $z_K(t) = \langle C - Z_K(t), v_\omega \rangle$ , where  $Z_K(t)$  is the point where  $d_K(t)$  meets  $l(\omega, 0)$  [1, Def. 2.5.5]. Let  $w_K^\circ = \inf_{t \in (0, \omega)} w_K(t)$  and  $z_K^\circ = \inf_{t \in (0, \omega)} z_K(t)$  [1, Def. 4.1.1]. As  $t \rightarrow \omega$ ,  $w_K(t) \rightarrow h_K(0)$ , so  $w_K^\circ \leq h_K(0)$ ; as  $t \rightarrow 0$ ,  $z_K(t) \rightarrow h_K(\omega + \pi/2)$ , so  $z_K^\circ \leq h_K(\omega + \pi/2)$ .

**Lemma 6.1** (wedge gaps and the polyline). *Every polygon cap  $K \in \mathcal{K}_\Theta^c$  satisfies  $w_K^\circ \leq \tau_K(\pi/2)$  and  $z_K^\circ \leq \tau_K(\omega)$ .*

This is the first half of the proof of [1, Thm. 4.1.2].

*Proof.* Choose  $t_0 \in \Theta$  with  $w_K(t_0) = \min_{t \in \Theta} w_K(t)$ . A point  $(x, 0)$  of  $\mathcal{N}_\Theta(K)$  lies in  $F_\omega$ , so  $x \geq 0$ , and in some  $Q_K^-(t)$ ,  $t \in \Theta$ , so  $x \cos t < h_K(t) - 1$ , that is  $x < h_K(0) - w_K(t) \leq h_K(0) - w_K(t_0)$ . Hence  $\mathcal{N}_\Theta(K) \cap l(\pi/2, 0) \subset [0, h_K(0) - w_K(t_0)) \times \{0\}$ , and by Fact 2.11(b)

$$\sigma_K(3\pi/2) - \tau_K(\pi/2) = \mathcal{H}^1(\mathcal{N}_\Theta(K) \cap l(\pi/2, 0)) \leq \max(h_K(0) - w_K(t_0), 0).$$

The bottom edge  $e_K(3\pi/2)$  contains  $O$  and  $A$  (Lemmas 2.5 and 2.6), so  $\sigma_K(3\pi/2) \geq h_K(0)$ . If  $w_K(t_0) \geq h_K(0)$ , then  $\tau_K(\pi/2) \geq \sigma_K(3\pi/2) \geq h_K(0) \geq w_K^\circ$ . Otherwise  $\tau_K(\pi/2) \geq \sigma_K(3\pi/2) - h_K(0) + w_K(t_0) \geq w_K(t_0) \geq w_K^\circ$ , as  $t_0 \in \Theta \subset (0, \omega)$ . The second inequality is the same argument on the line  $l(\omega, 0)$ , with  $C$ ,  $z_K$  and  $h_K(\omega + \pi/2)$  in place of  $A$ ,  $w_K$  and  $h_K(0)$ : a

point  $\mu v_\omega$  of  $\mathcal{N}_\Theta(K)$  has  $\mu \geq 0$ , as it lies in  $F_\omega$ , and  $\mu \cos(\omega - t) = \langle \mu v_\omega, v_t \rangle < h_K(t + \pi/2) - 1$  for some  $t \in \Theta$ , that is  $\mu < h_K(\omega + \pi/2) - z_K(t)$ ; the edge  $e_K(\omega + \pi)$  contains  $O$  and  $C$ , so  $\sigma_K(\omega + \pi) \geq h_K(\omega + \pi/2)$ ; and Fact 2.11(b) is used with  $t = \omega$ .  $\square$

**Lemma 6.2** (continuity). *For caps  $K, K' \in \mathcal{K}_\omega^c$  with  $d_H(K, K') = \varepsilon$ ,  $|w_K^\circ - w_{K'}^\circ| \leq (1 + \sec \omega)\varepsilon$  and  $|z_K^\circ - z_{K'}^\circ| \leq (1 + \sec \omega)\varepsilon$ .*

This is [1, Lemma 4.1.1], with the analogous bound for  $z^\circ$ .

*Proof.* For  $t \in (0, \omega)$ ,  $|w_K(t) - w_{K'}(t)| \leq \varepsilon + \varepsilon \sec t \leq (1 + \sec \omega)\varepsilon$ ; similarly for  $z$ , since  $\cos(\omega - t) \geq \cos \omega$ . The infima differ by at most as much.  $\square$

Baek obtains [1, Thm. 4.1.4] from the weak convergence of surface area measures [1, Thm. 4.1.3] and the Portmanteau theorem. Instead we use the following bound on edge lengths.

**Lemma 6.3** (upper semicontinuity of edge lengths). *Let convex bodies  $K_j$  converge to a convex body  $K$  in  $d_H$ , let  $t \in \mathbb{R}$ , and let real numbers  $a_j \rightarrow a$  satisfy  $a_j \leq \sigma_{K_j}(t)$  for every  $j$ . Then  $a \leq \sigma_K(t)$ .*

*Proof.* For  $0 < \delta < \pi$ ,

$$h_K(t + \delta) \geq \langle v_K^+(t), u_{t+\delta} \rangle = h_K(t) \cos \delta + \langle v_K^+(t), v_t \rangle \sin \delta,$$

$$h_K(t - \delta) \geq \langle v_K^-(t), u_{t-\delta} \rangle = h_K(t) \cos \delta - \langle v_K^-(t), v_t \rangle \sin \delta.$$

Adding, and using  $\sigma_K(t) = \langle v_K^+(t) - v_K^-(t), v_t \rangle$ ,

$$\sigma_K(t) \leq \hat{\sigma}_K(\delta) = \frac{h_K(t + \delta) + h_K(t - \delta) - 2h_K(t) \cos \delta}{\sin \delta}.$$

The same holds for each  $K_j$ . Since  $h_{K_j} \rightarrow h_K$  pointwise,  $\hat{\sigma}_{K_j}(\delta) \rightarrow \hat{\sigma}_K(\delta)$ , and  $a_j \leq \sigma_{K_j}(t) \leq \hat{\sigma}_{K_j}(\delta)$  gives  $a \leq \hat{\sigma}_K(\delta)$  for every  $\delta \in (0, \pi)$ . As  $\delta \downarrow 0$ ,

$$\frac{h_K(t + \delta) - h_K(t) \cos \delta}{\sin \delta} \rightarrow h'_K(t^+), \quad \frac{h_K(t) \cos \delta - h_K(t - \delta)}{\sin \delta} \rightarrow h'_K(t^-),$$

because  $(1 - \cos \delta)/\delta \rightarrow 0$  and  $\delta/\sin \delta \rightarrow 1$ . By Fact 2.1(a),  $\hat{\sigma}_K(\delta) \rightarrow h'_K(t^+) - h'_K(t^-) = \langle v_K^+(t) - v_K^-(t), v_t \rangle = \sigma_K(t)$ , so  $a \leq \sigma_K(t)$ .  $\square$

**Proposition 6.4** (pinned bounds). *Let  $\omega \in (0, \pi/2)$  and let  $K_* \in \mathcal{K}_\omega^c$  be a maximizing cap with  $\mathcal{A}_\omega(K_*) > 0$ . Then  $w_{K_*}^\circ \leq \sigma_{K_*}(\pi/2)$  and  $z_{K_*}^\circ \leq \sigma_{K_*}(\omega)$ .*

*Proof.* Take the selection  $K_j \rightarrow K_*$  of Proposition 4.8, with  $K_j \subset [-R_0, R_0] \times [0, 1]$ , and  $\eta_j = d_H(K_j, K_*) \rightarrow 0$ . Every point  $p$  of this box has  $|\langle p, u_s \rangle| \leq R_0 + 1$ , so  $|h_{K_j}(s)| \leq R_0 + 1 = R$  for all  $s$ , where  $R \geq 0$ ; at every sampled normal  $|h_{K_j} - h_*| \leq \eta_j$ , and the total weight is  $\bar{\nu} \leq 1$  by (4.1). By Corollary 5.6 at  $t = \pi/2$ , with  $e_j = (2 + 4\gamma(R))\eta_j$ ,  $|\sigma_{K_j}(\pi/2) - \tau_{K_j}(\pi/2)| \leq e_j$ ; so by Lemma 6.1

$$w_{K_j}^\circ - e_j \leq \tau_{K_j}(\pi/2) - e_j \leq \sigma_{K_j}(\pi/2).$$

By Lemma 6.2,  $|w_{K_j}^\circ - w_{K_*}^\circ| \leq (1 + \sec \omega)\eta_j$ , so  $w_{K_j}^\circ - e_j \rightarrow w_{K_*}^\circ$ , and Lemma 6.3 with  $t = \pi/2$  and  $a_j = w_{K_j}^\circ - e_j$  gives the first inequality. For the

second use  $t = \omega$ , where Corollary 5.6 divides by  $\sin \omega > 0$ : by Lemma 6.1,  $z_{K_j}^\circ - e_j / \sin \omega \leq \sigma_{K_j}(\omega)$ , and Lemmas 6.2 and 6.3 conclude as before.  $\square$

**6.2. The triangle in the niche.** Except in Lemma 6.6(c), let  $\omega \in [\omega_0, \pi/2)$ , so that  $\omega > \pi/4$ . Put  $c = c_\omega = \sec \omega - \tan \omega > 0$ , so that  $o_\omega - v_0 = cu_0$  and  $o_\omega - u_\omega = cv_\omega$  [1, Prop. 4.2.1], and let  $\Delta_\omega$  be the triangle with vertices  $O$ ,  $cu_0$ ,  $cv_\omega$ , which cuts off the corner  $O$  of  $P_\omega$ . Let  $d_{\min} = 5/4$  if  $\omega < \tan^{-1}(11/5)$  and  $d_{\min} = 11/10$  otherwise, and for  $d \in [0, \tan \omega]$  let  $R_{\omega,d} = P_\omega \cap H_-(0, d+c) \cap H_-(\omega + \pi/2, d+c)$  be  $P_\omega$  without its two acute corners.

**Fact 6.5** (numerical inequalities at the corner). *Let  $\omega \in [\omega_0, \pi/2)$ .*

- (a)  $|R_{\omega,d_{\min}}| < 11/5$ . [1, Lemma 4.2.2]
- (b) *If  $d \in [d_{\min}, \tan \omega]$  and  $r_y = 1 - d \cot \omega \geq 0$ , then the points  $q_0 = o_\omega - v_0 + du_0$  and  $q_1 = o_\omega - gu_0$ ,  $g = \sqrt{1 - r_y^2}$ , satisfy*

$$\langle q_0 - cu_0, u_{\pi/2-\omega} \rangle > 1 \quad \text{and} \quad \langle q_1 - cv_\omega, v_{\pi/2-\omega} \rangle > 1.$$

[1, Lemma 4.2.4]

**Lemma 6.6** (the triangle lies in the niche). (a) *Let  $\omega \in [\omega_0, \pi/2)$ , let  $K$  be a compact set, and let  $q_0, q_1 \in K$  satisfy  $\langle q_0 - cu_0, u_{\pi/2-\omega} \rangle > 1$  and  $\langle q_1 - cv_\omega, v_{\pi/2-\omega} \rangle > 1$ . Then  $O$ ,  $cu_0$  and  $cv_\omega$  lie in  $Q_K^-(\pi/2 - \omega)$ .*

(b) *Let  $\omega \in [\omega_0, \pi/2)$  and let  $K \in \mathcal{K}_\omega^c$  satisfy  $w_K^\circ \leq \sigma_K(\pi/2)$  and  $h_K(0) \geq d_{\min} + c$ . Then  $O$ ,  $cu_0$  and  $cv_\omega$  lie in  $Q_K^-(\pi/2 - \omega)$ .*

(c) *For every  $\omega \in \mathbb{R}$ , every compact set  $K$  and every  $t \in \mathbb{R}$ , the reflection  $M_\omega$  maps  $Q_{M_\omega(K)}^-(t)$  into  $Q_K^-(\omega - t)$ .*

(d) *Let  $\omega \in [\omega_0, \pi/2)$  and let  $K \in \mathcal{K}_\omega^c$  satisfy  $\mathcal{A}_\omega(K) \geq 11/5$ ,  $w_K^\circ \leq \sigma_K(\pi/2)$  and  $z_K^\circ \leq \sigma_K(\omega)$ . Then for some  $t_\sharp \in (0, \omega)$  the three vertices  $O$ ,  $cu_0$ ,  $cv_\omega$  of  $\Delta_\omega$  lie in the closure of  $Q_K^-(t_\sharp)$ .*

Part (d) is [1, Thm. 4.2.5] for a cap that satisfies the pinned bounds, in place of a balanced maximum cap; the proof there uses the balance only through them. Parts (a) and (b) are steps of that proof, and (c) is a consequence of [1, Prop. 2.5.4].

*Proof.* (a) Put  $t = \pi/2 - \omega$ . As  $q_0, q_1 \in K$ ,  $h_K(t) \geq \langle q_0, u_t \rangle$  and  $h_K(t + \pi/2) \geq \langle q_1, v_t \rangle$ . So every point  $p$  with  $\langle p, u_t \rangle \leq \langle cu_0, u_t \rangle$  and  $\langle p, v_t \rangle \leq \langle cv_\omega, v_t \rangle$  lies in  $Q_K^-(t)$ , because the hypotheses give

$$\begin{aligned} \langle p, u_t \rangle &\leq \langle cu_0, u_t \rangle < \langle q_0, u_t \rangle - 1 \leq h_K(t) - 1, \\ \langle p, v_t \rangle &\leq \langle cv_\omega, v_t \rangle < \langle q_1, v_t \rangle - 1 \leq h_K(t + \pi/2) - 1. \end{aligned}$$

Now  $\langle cu_0, u_t \rangle = c \sin \omega > 0$  and  $\langle cv_\omega, v_t \rangle = c \cos(\omega - t) = c \sin 2\omega > 0$ , while  $\langle cu_0, v_t \rangle = -c \sin t = -c \cos \omega < 0$  and  $\langle cv_\omega, u_t \rangle = c \sin(t - \omega) = c \cos 2\omega \leq 0$ , since  $\pi/4 < \omega < \pi/2$ . So  $O$ ,  $cu_0$  and  $cv_\omega$  satisfy both conditions.

(b) Put  $d = h_K(0) - c \geq d_{\min}$  and  $q_0 = (h_K(0), 0)$ , which is  $A_K^-(0) \in K$  by Lemma 2.6; then  $q_0 = (c+d)u_0 = o_\omega - v_0 + du_0$ . As  $q_0 \in P_\omega$ ,  $h_K(0) \cos \omega \leq 1$ ,

so  $d \cos \omega \leq 1 - c \cos \omega = \sin \omega$ . Hence  $d \leq \tan \omega$ , and  $r_y = 1 - d \cot \omega$  lies in  $[0, 1]$ . The point  $r = (h_K(0), r_y)$  lies on  $l_K(0)$  and on  $l(\omega, 1)$ , since  $\langle r, u_\omega \rangle = (c + d) \cos \omega + \sin \omega - d \cos \omega = 1$ . Let  $g = \sqrt{1 - r_y^2}$  and  $s = q_0 - g u_0$ : the triangle  $s, q_0, r$  has a right angle at  $q_0$ , and  $r - s = (g, r_y)$  is a unit vector.

For  $t \in (0, \omega)$ ,  $\sin \omega u_t = \sin(\omega - t) u_0 + \sin t u_\omega$  with nonnegative coefficients, and every  $p \in K$  has  $\langle p, u_0 \rangle \leq h_K(0) = \langle r, u_0 \rangle$  and  $\langle p, u_\omega \rangle \leq 1 = \langle r, u_\omega \rangle$ , as  $K \subset P_\omega$ . So  $h_K(t) \leq \langle r, u_t \rangle$ , and since  $\langle r - s, u_t \rangle \leq \|r - s\| = 1$ ,

$$h_K(t) - 1 \leq \langle r, u_t \rangle - \langle r - s, u_t \rangle = \langle s, u_t \rangle = (h_K(0) - g) \cos t.$$

So  $W_K(t)$  lies on or to the left of  $s$ , that is  $w_K(t) \geq g$ . Hence  $0 \leq g \leq w_K^\circ \leq \sigma_K(\pi/2)$ .

The top edge  $e_K(\pi/2)$  lies on the line  $l(\pi/2, 1)$ , whose points in  $P_\omega$  have abscissa at most  $c$ . Its end  $v_K^-(\pi/2)$  is such a point, and its other end  $v_K^+(\pi/2) = v_K^-(\pi/2) - \sigma_K(\pi/2) u_0$  (Fact 2.1(b), with  $v_{\pi/2} = -u_0$ ) has abscissa at most  $c - \sigma_K(\pi/2) \leq c - g$ . So  $q_1 = o_\omega - g u_0 = (c - g, 1)$  lies on the segment from  $v_K^+(\pi/2)$  to  $o_\omega$ , which lies in  $K$ , since  $o_\omega \in K$  (Lemma 2.5) and  $K$  is convex.

Finally  $d \in [d_{\min}, \tan \omega]$  and  $r_y \geq 0$ , so by Fact 6.5(b) the points  $q_0, q_1 \in K$  satisfy the hypotheses of (a), and (a) gives the claim.

(c) The reflection  $M_\omega$  is a symmetric linear map with  $M_\omega u_s = u_{\omega + \pi/2 - s}$ , so  $\langle M_\omega p, u_s \rangle = \langle p, u_{\omega + \pi/2 - s} \rangle$  for all  $p$  and  $s$ ; in particular  $h_{M_\omega(K)}(s) = h_K(\omega + \pi/2 - s)$ . Let  $p \in Q_{M_\omega(K)}^-(t)$ , that is  $\langle p, u_t \rangle < h_K(\omega + \pi/2 - t) - 1$  and  $\langle p, u_{t + \pi/2} \rangle < h_K(\omega - t) - 1$ . Then

$$\begin{aligned} \langle M_\omega p, u_{\omega - t} \rangle &= \langle p, u_{t + \pi/2} \rangle < h_K(\omega - t) - 1, \\ \langle M_\omega p, u_{\omega - t + \pi/2} \rangle &= \langle p, u_t \rangle < h_K(\omega - t + \pi/2) - 1, \end{aligned}$$

so  $M_\omega p \in Q_K^-(\omega - t)$ .

(d) If  $h_K(0) < d_{\min} + c$  and  $h_K(\omega + \pi/2) < d_{\min} + c$ , then  $K \subset R_{\omega, d_{\min}}$ , as  $K \subset P_\omega$ , and  $|K| \leq |R_{\omega, d_{\min}}| < 11/5$  by Fact 6.5(a), whereas  $|K| \geq \mathcal{A}_\omega(K) \geq 11/5$ . So  $h_K(0) \geq d_{\min} + c$  or  $h_K(\omega + \pi/2) \geq d_{\min} + c$ . In the first case (b) puts  $O$ ,  $cu_0$  and  $cv_\omega$  in  $Q_K^-(\pi/2 - \omega)$ , hence in its closure, and  $t_\sharp = \pi/2 - \omega$  lies in  $(0, \omega)$ , as  $\omega > \pi/4$ .

In the second case we use the mirror image  $K^m = M_\omega(K)$ , a cap with  $h_{K^m}(s) = h_K(\omega + \pi/2 - s)$  and  $\sigma_{K^m}(\pi/2) = \sigma_K(\omega)$  (Fact 2.4(d)). Then  $h_{K^m}(0) = h_K(\omega + \pi/2) \geq d_{\min} + c$ , and the formulas for the wedge gaps give  $w_{K^m}(t) = z_K(\omega - t)$  for  $t \in (0, \omega)$ , so  $w_{K^m}^\circ = z_K^\circ \leq \sigma_K(\omega) = \sigma_{K^m}(\pi/2)$ . By (b),  $O$ ,  $cu_0$  and  $cv_\omega$  lie in  $Q_{K^m}^-(\pi/2 - \omega)$ . As  $M_\omega$  fixes  $O$  and exchanges  $u_0$  and  $v_\omega$ , (c) puts  $O$ ,  $cv_\omega$  and  $cu_0$  in  $Q_K^-(\omega - (\pi/2 - \omega))$ , hence in its closure, and  $t_\sharp = 2\omega - \pi/2$  lies in  $(0, \omega)$ , as  $\pi/4 < \omega < \pi/2$ .  $\square$

### 6.3. A rotated copy turns by the right angle.

**Lemma 6.7** (the width of the sofa). *Let  $\omega \in (0, \pi/2)$ ,  $c = c_\omega$  and  $K \in \mathcal{K}_\omega^c$ , and suppose that  $O$ ,  $cu_0$  and  $cv_\omega$  lie in the closure of  $Q_K^-(t_\sharp)$  for some  $t_\sharp \in (0, \omega)$ . Then*

- (a) every point  $\lambda u_0 + \mu v_\omega$  with  $\lambda, \mu \geq 0$  and  $\lambda + \mu < c$  lies in  $\mathcal{N}(K)$ ;  
(b)  $\langle p - q, u_t \rangle \leq 1$  for all  $p \in K$ ,  $q \in K \setminus \mathcal{N}(K)$  and  $t \in [\omega, \pi/2]$ .

The width bound (b) is used in the proof of [1, Thm. 1.5.2], for the pentagon  $P_\omega \setminus \Delta_\omega$ .

*Proof.* (a) The closure of  $Q_K^-(t_\#)$  lies in the closed quarter-plane of the points  $p$  with  $\langle p, u_{t_\#} \rangle \leq \alpha$  and  $\langle p, v_{t_\#} \rangle \leq \beta$ , where  $\alpha = h_K(t_\#) - 1$  and  $\beta = h_K(t_\# + \pi/2) - 1$ . It contains  $cu_0$  and  $cv_\omega$ , and  $\langle cu_0, u_{t_\#} \rangle = c \cos t_\# > 0$ ,  $\langle cv_\omega, v_{t_\#} \rangle = c \cos(\omega - t_\#) > 0$ , so  $\alpha > 0$  and  $\beta > 0$ . The point  $p = \lambda u_0 + \mu v_\omega$  is  $(\lambda/c)cu_0 + (\mu/c)cv_\omega$ , with  $\lambda/c, \mu/c \geq 0$  and  $(\lambda + \mu)/c < 1$ . So  $\langle p, u_{t_\#} \rangle \leq (\lambda + \mu)\alpha/c < \alpha$  and  $\langle p, v_{t_\#} \rangle \leq (\lambda + \mu)\beta/c < \beta$ , and  $p \in Q_K^-(t_\#)$ . Also  $p \in F_\omega$ , since  $\langle p, u_\omega \rangle = \lambda \cos \omega \geq 0$  and  $p_y = \mu \cos \omega \geq 0$ ; as  $t_\# \in (0, \omega)$ ,  $p \in \mathcal{N}(K)$ .

(b) Write  $q = \lambda u_0 + \mu v_\omega$ . As  $q \in P_\omega$ ,  $\lambda = \langle q, u_\omega \rangle / \cos \omega \geq 0$  and  $\mu = q_y / \cos \omega \geq 0$ , and  $\lambda + \mu \geq c$  by (a). For  $t \in [\omega, \pi/2]$  both  $\cos t$  and  $\sin(t - \omega)$  are nonnegative, so  $\langle q, u_t \rangle = \lambda \cos t + \mu \sin(t - \omega) \geq c \min(\cos t, \sin(t - \omega))$ . Also  $\cos \omega u_t = \cos t u_\omega + \sin(t - \omega) u_{\pi/2}$ , and  $\langle p, u_\omega \rangle \leq 1$ ,  $p_y \leq 1$  as  $p \in P_\omega$ , so  $\cos \omega \langle p, u_t \rangle \leq \cos t + \sin(t - \omega)$ , that is  $\langle p, u_t \rangle \leq c \cos t + \sin t = \langle o_\omega, u_t \rangle$ , as  $c \cos \omega = 1 - \sin \omega$ . If  $\cos t \leq \sin(t - \omega)$ , then  $\langle p - q, u_t \rangle \leq \sin t \leq 1$ . Otherwise, using  $c(1 + \sin \omega) = \cos \omega$  and  $c \cos \omega = 1 - \sin \omega$ ,

$$\langle p - q, u_t \rangle \leq c \cos t + \sin t - c \sin(t - \omega) = \cos(t - \omega) \leq 1. \quad \square$$

**Lemma 6.8** (turning inside the horizontal side). *Let  $S$  be a moving sofa with rotation angle  $\omega < \pi/2$  such that  $\langle p - q, u_t \rangle \leq 1$  for all  $p, q \in S$  and  $t \in [\omega, \pi/2]$ . Then  $R_\psi S$ ,  $\psi = \pi/2 - \omega$ , is a moving sofa with rotation angle  $\pi/2$ .*

*Proof.* Let  $(\theta, \mathbf{c})$  be a movement of  $S$  with rotation angle  $\omega$ , as in Definition 2.2; its translation  $\mathbf{c}$  is set in bold, to distinguish it from  $c_\omega$ . As  $S$  is compact, there is  $R$  with  $\|p\| \leq R$  for all  $p \in S$ . For  $\rho \in [0, \psi]$  let  $b(\rho) = (1 - 2R, h_S(3\pi/2 - \rho))$ , which depends continuously on  $\rho$ , as  $h_S$  is continuous. Then  $R_\rho S + b(\rho) \subset H_L$ : for  $p \in S$  the abscissa of  $R_\rho p + b(\rho)$  is  $p_x \cos \rho - p_y \sin \rho + 1 - 2R \leq |p_x| + |p_y| + 1 - 2R \leq 1$ ; and, as  $(R_\rho p)_y = \langle p, u_{\pi/2 - \rho} \rangle$  and  $u_{3\pi/2 - \rho} = -u_{\pi/2 - \rho}$ , its ordinate is

$$\langle p, u_{\pi/2 - \rho} \rangle + h_S(3\pi/2 - \rho) = \langle p, u_{\pi/2 - \rho} \rangle - \min_{q \in S} \langle q, u_{\pi/2 - \rho} \rangle,$$

which lies in  $[0, 1]$  by the hypothesis with  $t = \pi/2 - \rho \in [\omega, \pi/2]$ . Define  $\theta'$  and  $\mathbf{c}'$  on  $[0, 1]$  by

$$\begin{aligned} \theta'(s) &= -3\psi s, & \mathbf{c}'(s) &= b(\psi - 3\psi s) & \text{on } [0, \tfrac{1}{3}], \\ \theta'(s) &= -\psi, & \mathbf{c}'(s) &= (2 - 3s)b(0) + (3s - 1)\mathbf{c}(0) & \text{on } [\tfrac{1}{3}, \tfrac{2}{3}], \\ \theta'(s) &= \theta(3s - 2) - \psi, & \mathbf{c}'(s) &= \mathbf{c}(3s - 2) & \text{on } [\tfrac{2}{3}, 1]. \end{aligned}$$

The definitions agree at  $s = 1/3$  and at  $s = 2/3$ , so  $\theta'$  and  $\mathbf{c}'$  are continuous;  $\theta'(0) = 0$  and  $\theta'(1) = \theta(1) - \psi = -\pi/2$ . The image of  $R_\psi S$  under  $p \mapsto R_{\theta'(s)} p + \mathbf{c}'(s)$  is  $R_{\theta'(s) + \psi} S + \mathbf{c}'(s)$ . On  $[0, 1/3]$  it is  $R_\rho S + b(\rho) \subset H_L$  with

$\rho = \psi(1 - 3s)$ , so the sofa turns clockwise by  $\psi$  inside  $H_L$ . On  $[1/3, 2/3]$  it is  $S + (2 - 3s)b(0) + (3s - 1)c(0)$ , which lies in  $H_L$ , since  $H_L$  is convex and contains  $S + b(0)$  and the starting position  $S + c(0)$  of the movement of  $S$ . On  $[2/3, 1]$  it is the position of  $S$  at the time  $3s - 2$  of its movement, which lies in  $L$ , and in  $V_L$  at  $s = 1$ . So  $(\theta', c')$  is a movement of  $R_\psi S$  with rotation angle  $\pi/2$ , and  $R_\psi S$  is closed and connected.  $\square$

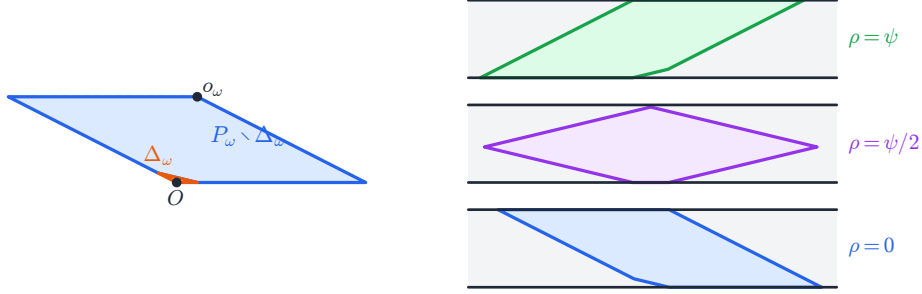


FIGURE 5. Lemmas 6.7 and 6.8, for  $\omega = 1.1$ . Left: the parallelogram  $P_\omega$  (dashed), its corner triangle  $\Delta_\omega$  (orange) and the pentagon  $P_\omega \setminus \Delta_\omega$  (blue); the sofa  $K \setminus \mathcal{N}(K)$  lies in the pentagon, up to the far side of  $\Delta_\omega$ . Right: the pentagon turned by  $\rho = \psi, \psi/2$  and  $0, \psi = \pi/2 - \omega$ , in a horizontal strip of width one: it fits for every  $\rho \in [0, \psi]$ , since its width in the direction  $u_t$  is at most 1 for  $t \in [\omega, \pi/2]$ .

**Proposition 6.9** (a right-angle monotone sofa). *Let  $\omega \in [\omega_0, \pi/2]$  and let  $T$  be a monotone sofa with rotation angle  $\omega$  and  $|T| = |G|$  (such as the sofa  $T$  of Proposition 3.2). Then  $R_\psi T, \psi = \pi/2 - \omega$ , is a moving sofa with rotation angle  $\pi/2$ . Consequently there are  $w_1 \in \mathbb{R}^2$  and a monotone sofa  $U$  with rotation angle  $\pi/2$  such that*

$$R_\psi T + w_1 \subset U, \quad |U| = |G|,$$

*and the cap  $\mathcal{C}(U) \in \mathcal{K}_{\pi/2}^c$  is maximizing, with  $\mathcal{A}(\mathcal{C}(U)) = |G|$ .*

*Proof.* If  $\omega = \pi/2$ , then  $\psi = 0$ , and  $T$  is a moving sofa with rotation angle  $\pi/2$  by Fact 2.4(a). Let  $\omega < \pi/2$  and  $K = \mathcal{C}(T)$ . By Fact 2.4(a), (b) and (c),  $T$  is a moving sofa with rotation angle  $\omega$  in standard position,  $K$  is a cap,  $T = K \setminus \mathcal{N}(K)$  and  $\mathcal{A}_\omega(K) = |T| = |G|$ ; so  $K$  is maximizing by Fact 2.8, and  $\mathcal{A}_\omega(K) > 11/5 > 0$ . Proposition 6.4, applied to  $K_* = K$ , gives the pinned bounds  $w_K^\circ \leq \sigma_K(\pi/2)$  and  $z_K^\circ \leq \sigma_K(\omega)$ . With them and  $\mathcal{A}_\omega(K) \geq 11/5$ , Lemma 6.6(d) gives  $t_\sharp \in (0, \omega)$  such that  $O, cu_0$  and  $cv_\omega$  lie in the closure of  $Q_K^-(t_\sharp)$ . As  $T = K \setminus \mathcal{N}(K)$ , Lemma 6.7(b) gives  $\langle p - q, u_t \rangle \leq 1$  for  $p, q \in T$  and  $t \in [\omega, \pi/2]$ , and Lemma 6.8 shows that  $R_\psi T$  is a moving sofa with rotation angle  $\pi/2$ . Also  $|R_\psi T| = |T| = |G|$ , and the second part is Proposition 3.2 for the moving sofa  $S' = R_\psi T$ .  $\square$

By Propositions 3.2 and 6.9, there are  $w_0, w_1 \in \mathbb{R}^2$  such that, with  $g_1(p) = R_\psi(p + w_0) + w_1$ ,

$$g_1(S) \subset U, \quad |U| = |G|, \quad (6.1)$$

where  $U$  is a monotone sofa with rotation angle  $\pi/2$  whose cap is maximizing.

## 7. CURVATURE BOUNDS AND THE INJECTIVITY CONDITION

Baek proves the injectivity condition for a balanced maximum cap  $K$  with rotation angle  $\pi/2$  from the inequality  $\sigma_K \leq \int_0^{\pi/2} k_0(g_K^+(t)) dt$  on  $[0, \pi/2]$  [1, Thm. 6.4.3]. This inequality is the limit of  $\sigma_{K_n}(t) \leq k_0(g_{K_n}^+(t))\delta + O(\delta^2)$  for maximum polygon caps  $K_n$  with step size  $\delta$  [1, Thm. 6.3.3], whose proof uses the balance  $\sigma = \tau$ . For the polygon caps of Section 4 the defect bound of Proposition 5.2 replaces the balance, and its errors add up to at most  $2\eta$  by (4.1). In this section we use Section 4, Proposition 5.2 and (4.1) with  $\omega = \pi/2$  and  $\mathcal{A} = \mathcal{A}_{\pi/2}$ :  $K_* \in \mathcal{K}_{\pi/2}^c$  is a maximizing cap with  $\mathcal{A}(K_*) > 0$ ,  $\Theta_k = \{j\delta : 0 < j < n\}$  with  $n = 2^{k+1}$  and step size  $\delta = \delta_k = \pi/(2n) \leq \pi/4$ . All the normals of  $\Theta_k$  are floating. A polygon cap  $K \in \mathcal{K}_k^c$  has no edge with normal angle 0 or  $\pi$  and none with normal in  $(-\pi/2, 0)$ ; its edge normals are among  $\delta, 2\delta, \dots, (2n-1)\delta$  and  $3\pi/2$ . For an angle  $u$  let  $H_K^b(u)$  and  $H_K^d(u)$  be as in Section 2.5, so that  $Q_K^-(u)$  is the complement of  $H_K^b(u) \cup H_K^d(u)$ .

### 7.1. One normal of a polygon cap.

**Lemma 7.1** (the inner wall near the corner). *Let  $K \in \mathcal{K}_k^c$ ,  $t \in \Theta_k$ ,  $q = \tan(\delta/2)$ , and  $\mathbf{x} = \mathbf{x}_K(t)$ . Let  $\vec{b} = \vec{b}_K(t) = \{\mathbf{x} - sv_t : s \geq 0\}$ . Then the set of  $s \geq 0$  with  $\mathbf{x} - sv_t \in H_K^d(t - \delta)$  is  $[0, \alpha_-]$  and the set of  $s \geq 0$  with  $\mathbf{x} - sv_t \in H_K^d(t + \delta)$  is  $[0, \alpha_+]$ , where*

$$\alpha_- = \tan \delta (g_K^-(t) - 1 + q), \quad \alpha_+ = \tan \delta (1 - g_K^+(t) + q)$$

(an interval is empty if its right end is negative); and the set of  $s \in \mathbb{R}$  with  $\mathbf{x} - sv_t \in H_K^b(t - \delta) \cap H_K^b(t + \delta)$  is an interval, empty or of length  $2q - \sigma_K(t)$ .

This is [1, Lemma 6.3.2], together with the length  $2 \tan(\delta/2) - \sigma_K(t)$  that Baek asserts in the proof of [1, Thm. 6.3.3]. They are stated there for maximum polygon caps, but the proofs use only that no normal angle of  $K$  lies in  $(t - \delta, t) \cup (t, t + \delta) \cup (t + \frac{\pi}{2} - \delta, t + \frac{\pi}{2}) \cup (t + \frac{\pi}{2}, t + \frac{\pi}{2} + \delta)$ , which holds for every  $K \in \mathcal{K}_k^c$ .

*Proof.* Since no normal angle of  $K$  lies between  $t + \pi/2$  and  $t + \delta + \pi/2$ , the supporting lines  $l_K(t + \pi/2)$  and  $l_K(t + \delta + \pi/2)$  meet at a vertex  $r$  of  $K$ , namely  $r = C_K^+(t) = \mathbf{y}_K(t) - g_K^+(t)u_t$ , and  $h_K(t + \delta + \pi/2) = \langle r, v_{t+\delta} \rangle$ . As  $\mathbf{x} - r = (g_K^+(t) - 1)u_t - v_t$ ,

$$\begin{aligned} \langle \mathbf{x}, v_{t+\delta} \rangle - h_K(t + \delta + \frac{\pi}{2}) + 1 &= \langle \mathbf{x} - r, v_{t+\delta} \rangle + 1 \\ &= -(g_K^+(t) - 1) \sin \delta - \cos \delta + 1. \end{aligned}$$

Now  $\langle \mathbf{x} - sv_t, v_{t+\delta} \rangle = \langle \mathbf{x}, v_{t+\delta} \rangle - s \cos \delta$ , so  $\mathbf{x} - sv_t \in H_K^d(t+\delta)$  iff  $s \leq \alpha_+ = [(1 - g_K^+(t)) \sin \delta + 1 - \cos \delta] / \cos \delta$ , and  $(1 - \cos \delta) / \cos \delta = \tan \delta \tan(\delta/2)$ . For  $t - \delta$  use  $r' = C_K^-(t) = \mathbf{y}_K(t) - g_K^-(t)u_t$  in the same way. Next,  $\langle \mathbf{x} - sv_t, u_{t \mp \delta} \rangle = \langle \mathbf{x}, u_{t \mp \delta} \rangle \pm s \sin \delta$ , so the conditions  $\langle p, u_{t-\delta} \rangle \geq h_K(t-\delta) - 1$  and  $\langle p, u_{t+\delta} \rangle \geq h_K(t+\delta) - 1$  on  $p = \mathbf{x} - sv_t$  read  $s \geq s_-$  and  $s \leq s_+$  with

$$\begin{aligned} s_+ - s_- &= \frac{2 \cos \delta (h_K(t) - 1) - h_K(t+\delta) - h_K(t-\delta) + 2}{\sin \delta} \\ &= \frac{2(1 - \cos \delta) - \sigma_K(t) \sin \delta}{\sin \delta} = 2q - \sigma_K(t), \end{aligned}$$

where we used  $u_{t+\delta} + u_{t-\delta} = 2 \cos \delta u_t$ ,  $\langle \mathbf{x}, u_t \rangle = h_K(t) - 1$  and  $h_K(t+\delta) + h_K(t-\delta) - 2 \cos \delta h_K(t) = \sigma_K(t) \sin \delta$ . The last identity holds because no normal angle of  $K$  lies in  $(t-\delta, t) \cup (t, t+\delta)$ : the vertices  $v_K^-(t) = v_K(t-\delta, t)$  and  $v_K^+(t) = v_K(t, t+\delta)$  lie on  $l_K(t-\delta)$  and on  $l_K(t+\delta)$ , so that  $h_K(t \pm \delta) = h_K(t) \cos \delta \pm \langle v_K^\pm(t), v_t \rangle \sin \delta$ , while  $\sigma_K(t) = \langle v_K^+(t) - v_K^-(t), v_t \rangle$  by Fact 2.1(b).  $\square$

**Lemma 7.2** (the polygon bound at one normal). *For  $K \in \mathcal{K}_k^c$  and  $t \in \Theta_k$ ,*

$$\begin{aligned} \tau_K(t) &\leq \max\left(\tan \delta (g_K^-(t) - 1 + q), \tan \delta (1 - g_K^+(t) + q), 0\right) \\ &\quad + \max(2q - \sigma_K(t), 0), \quad q = \tan \frac{\delta}{2}. \end{aligned}$$

*Proof.* Let  $X = \partial \mathcal{N}_{\Theta_k}(K) \cap \vec{b}_K(t)$ ; by Fact 2.11(a),  $\tau_K(t) = \mathcal{H}^1(X)$ . Let  $I = \{t - \delta, t, t + \delta\}$ . The points of  $X$  that lie in  $H_K^d(u)$  for some  $u \in I$  are of the first kind. On  $\vec{b}_K(t)$  the half-plane  $H_K^d(t)$  contains only the corner  $\mathbf{x}_K(t)$ , so by Lemma 7.1 the points of the first kind form a set of length at most  $\max(0, \alpha_-, \alpha_+)$ . Let  $p \in X$  not be of the first kind. As  $X \subset \text{cl} \mathcal{N}_{\Theta_k}(K) \subset \{y \geq 0\}$ ,  $p_y \geq 0$ , and  $p_y = 0$  for at most one point of  $X$ , since the line  $b_K(t)$  is not horizontal. Let  $p_y > 0$ . For  $u \in \Theta_k$ ,  $p \notin Q_K^-(u)$ : otherwise the open set  $\{y > 0\} \cap Q_K^-(u)$ , which lies in  $\mathcal{N}_{\Theta_k}(K)$ , would be a neighborhood of  $p$  contained in the niche, and  $p$  would not be a boundary point. Also  $Q_K^-(0)$  and  $Q_K^-(\pi/2)$  lie in  $\{y < 0\}$ , since one of the two inequalities that define each of them is  $y < h_K(\pi/2) - 1 = 0$ . As  $t - \delta, t + \delta \in \Theta_k \cup \{0, \pi/2\}$  and  $Q_K^-(u)$  is the complement of  $H_K^b(u) \cup H_K^d(u)$ , we get  $p \in H_K^b(t-\delta) \cap H_K^b(t+\delta)$ . These are the points of the second kind, and by Lemma 7.1 they form a set of length at most  $\max(0, 2q - \sigma_K(t))$ . So  $X$  consists of points of the two kinds and of at most one point on the  $x$ -axis, and  $\mathcal{H}^1(X)$  is at most the sum of the two lengths.  $\square$

**Lemma 7.3** (with a defect). *Let  $K \in \mathcal{K}_k^c$ ,  $t \in \Theta_k$ ,  $e \geq 0$ ,  $D \geq 0$  with  $g_K^+(t) \leq D$  and  $\sigma_K(t) \leq \tau_K(t) + e$ . Then*

$$\sigma_K(t) \leq k_0(g_K^+(t)) \delta + (D + 4)\delta^2 + e.$$

*Proof.* Let  $M = |g_K^+(t) - 1|$ . As  $C_K^+(t) \in K$ ,  $g_K^+(t) = h_K(t) - \langle C_K^+(t), u_t \rangle \geq 0$ ; with  $g_K^+(t) \leq D$  this gives  $M \leq D + 1$ . By Fact 2.1(b) and  $v_{t+\pi/2} = -u_t$ ,

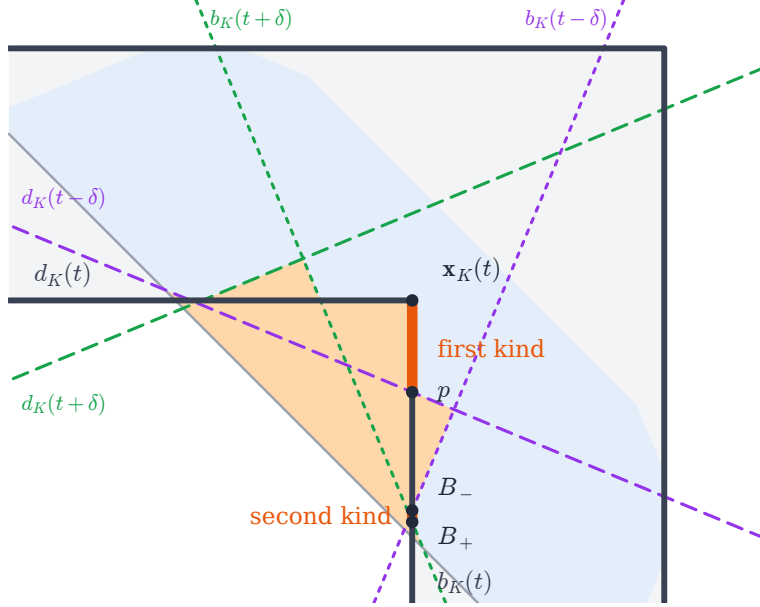


FIGURE 6. Lemmas 7.1 and 7.2, for the polygon cap with the angle set  $\Theta_1$  ( $n = 4$ ,  $\delta = \pi/8$ ) and the support values of Gerver's cap, at  $t = \pi/4$ , drawn in the frame of the supporting hallway  $L_K(t)$  (the polygon niche is shaded). The part of  $\vec{b}_K(t)$  above the dashed line  $d_K(t - \delta)$  is the segment  $[p, \mathbf{x}_K(t)]$  (points of the first kind); further down  $\vec{b}_K(t)$  runs between the dotted lines  $b_K(t - \delta)$  and  $b_K(t + \delta)$ , from  $B_-$  to  $B_+$  (points of the second kind). The side of the niche on  $\vec{b}_K(t)$  lies in the union of the two segments.

$C_K^+(t) = C_K^-(t) - \sigma_K(t + \pi/2)u_t$ , so  $g_K^+(t) - g_K^-(t) = \sigma_K(t + \pi/2) \geq 0$ , and each of  $g_K^-(t) - 1 + q$  and  $1 - g_K^+(t) + q$  is at most  $M + q$ . So the first maximum in Lemma 7.2 is at most  $\tan \delta (M + q) \leq \delta M + (D + 3)\delta^2$ , using  $\tan \delta \leq \delta + \delta^2$ ,  $q \leq \delta$  and  $\delta \leq 1$ . If  $\sigma_K(t) \geq 2q$ , the second maximum is 0 and  $\sigma_K(t) \leq \tau_K(t) + e \leq \delta M + (D + 3)\delta^2 + e$ . If  $\sigma_K(t) < 2q$ , then  $\sigma_K(t) \leq \delta M + (D + 3)\delta^2 + (2q - \sigma_K(t)) + e$  and  $2q \leq \delta + \delta^2$ , so  $\sigma_K(t) \leq \frac{M+1}{2}\delta + \frac{D+4}{2}\delta^2 + \frac{e}{2}$ . Both  $M$  and  $(M + 1)/2$  are at most  $k_0(g_K^+(t))$ .  $\square$

## 7.2. From normals to intervals.

**Lemma 7.4** (arms between two steps). *Let  $D \geq 0$ , let  $K \in \mathcal{K}_k^c$  with  $\|p - q\| \leq D$  for all  $p, q \in K$ , and let  $t \in \{0\} \cup \Theta_k$ . Then for every  $s \in (t, t + \delta)$*

$$g_K^+(s) = g_K^-(s), \quad g_K^+(s) \leq g_K^+(t), \quad g_K^-(t + \delta) \leq g_K^-(s),$$

and  $g_K^+(t) - g_K^-(t + \delta) \leq D\delta$ .

This is [1, Lemma 6.4.1], with  $D$  in place of the bound 5 for maximum polygon caps.

*Proof.* Since there is no normal angle of  $K$  strictly between  $t$  and  $t + \delta$ , nor between  $t + \pi/2$  and  $t + \delta + \pi/2$ , the supporting lines  $l_K(s)$ ,  $s \in [t, t + \delta]$ , pass through the vertex  $A = v_K(t, t + \delta)$  of  $K$ , and the lines  $l_K(s + \pi/2)$  through the vertex  $C = v_K(t + \pi/2, t + \delta + \pi/2)$ ; moreover  $C_K^+(t) = C$ ,  $C_K^-(t + \delta) = C$ , and  $C_K^\pm(s) = C$  for  $s \in (t, t + \delta)$ . As  $\langle \mathbf{y}_K(s), u_s \rangle = h_K(s) = \langle A, u_s \rangle$ , the function  $G(s) = \langle A - C, u_s \rangle$  equals  $g_K^+$  at  $t$ ,  $g_K^-$  at  $t + \delta$ , and both  $g_K^+$  and  $g_K^-$  on  $(t, t + \delta)$ . Its derivative  $G'(s) = \langle A - C, v_s \rangle$  is at most 0 on  $[t, t + \delta]$ , because  $\langle A, v_s \rangle \leq h_K(s + \pi/2) = \langle C, v_s \rangle$ , and at least  $-\|A - C\| \geq -D$ , as  $A, C \in K$ . So  $G$  is nonincreasing on  $[t, t + \delta]$ , which gives the inequalities for  $s \in (t, t + \delta)$ , and  $G(s) + Ds$  is nondecreasing there, which gives  $G(t) - G(t + \delta) \leq D\delta$ .  $\square$

**Lemma 7.5** (summing the steps). *Let  $B, C, D \geq 0$ , let  $K \in \mathcal{K}_k^c$  with  $\|p - q\| \leq D$  for all  $p, q \in K$  and  $k_0(g_K^+) \leq B$  on  $[0, \pi/2]$ , and suppose that for every  $t \in \{0\} \cup \Theta_k$*

$$\sigma_K(t) \leq k_0(g_K^+(t))\delta + C\delta^2 + e(t), \quad e(t) \geq 0.$$

*Then for  $0 \leq a \leq b \leq \pi/2$*

$$\sigma_K([a, b]) \leq \int_a^b k_0(g_K^+(u))du + \left(2B + \frac{\pi}{2}(C + D)\right)\delta + \sum_{j=0}^{n-1} e(j\delta),$$

*and for  $a \geq -\pi/2$  with  $a' = \max(a, 0) \leq b \leq \pi/2$*

$$\sigma_K((a, b)) \leq \int_{a'}^b k_0(g_K^+(u))du + \left(2B + \frac{\pi}{2}(C + D)\right)\delta + \sum_{j=0}^{n-1} e(j\delta).$$

*Proof.* Let  $t \in \{0\} \cup \Theta_k$  and  $s \in (t, t + \delta)$ . By Lemma 7.4,  $g_K^+(s) \leq g_K^+(t)$  and  $g_K^+(t) - g_K^+(s) = g_K^+(t) - g_K^-(s) \leq g_K^+(t) - g_K^-(t + \delta) \leq D\delta$ ; as  $k_0$  is 1-Lipschitz,  $k_0(g_K^+(s)) \geq k_0(g_K^+(t)) - D\delta$ . Integrating over  $(t, t + \delta)$  gives  $k_0(g_K^+(t))\delta \leq \int_t^{t+\delta} k_0(g_K^+(u))du + D\delta^2$ , hence

$$\sigma_K(t) \leq \int_t^{t+\delta} k_0(g_K^+(u))du + (C + D)\delta^2 + e(t).$$

For  $0 \leq j < n$ ,  $K$  has no normal angle in  $(j\delta, j\delta + \delta)$ , so  $v_K^+$  is constant on  $[j\delta, j\delta + \delta]$ , equal to a vertex  $w$ . There  $h_K(s) = \langle w, u_s \rangle$ , and the function  $G_K(s) = \langle w, v_s \rangle + \int_0^s h_K$  of Fact 2.1(b) has the derivative  $-\langle w, u_s \rangle + h_K(s) = 0$ ; so  $\sigma_K((j\delta, j\delta + \delta)) = 0$  and  $\sigma_K([j\delta, j\delta + \delta)) = \sigma_K(j\delta)$ . Adding the bounds at  $t = j\delta$  for  $i \leq j < m$  gives, for integers  $0 \leq i \leq m \leq n$ ,

$$\sigma_K([i\delta, m\delta)) \leq \int_{i\delta}^{m\delta} k_0(g_K^+(u))du + (m - i)(C + D)\delta^2 + \sum_{j=i}^{m-1} e(j\delta).$$

Now let  $0 \leq a \leq b \leq \pi/2$ ,  $i = \lfloor a/\delta \rfloor$  and  $m = \lceil b/\delta \rceil$ . Then  $i\delta \leq a < i\delta + \delta$ ,  $b \leq m\delta < b + \delta$  and  $i \leq m \leq n$ , as  $b \leq \pi/2 = n\delta$ ; so  $[a, b) \subset [i\delta, m\delta)$ . Hence  $\sigma_K([a, b)) \leq \sigma_K([i\delta, m\delta))$ , and the first bound follows: the integrals of  $k_0(g_K^+)$

over  $[i\delta, a]$  and over  $[b, m\delta]$ , intervals of length less than  $\delta$  in  $[0, \pi/2]$ , are at most  $B\delta$  each;  $(m-i)(C+D)\delta^2 \leq n(C+D)\delta^2 = \frac{\pi}{2}(C+D)\delta$ ; and the errors  $e(j\delta)$  with  $i \leq j < m$  are among the nonnegative terms of  $\sum_{j=0}^{n-1} e(j\delta)$ . For the second bound,  $K$  has no normal angle in  $(-\pi/2, 0)$ , so  $\sigma_K((-\pi/2, 0)) = 0$  in the same way. As  $(a, b) \subset (-\pi/2, 0) \cup [a', b)$ ,  $\sigma_K((a, b)) \leq \sigma_K([a', b))$ , and the first bound applies to  $[a', b)$ .  $\square$

**Lemma 7.6** (the bound on an open interval in the limit). *Let  $K_j$  be polygon caps (with any angle sets) that converge in  $d_H$  to  $K \in \mathcal{K}_{\pi/2}^c$ , and let  $\varepsilon_j \rightarrow 0$  be such that, for every  $j$  and all  $-\pi/2 \leq a < b \leq \pi/2$  with  $a' = \max(a, 0) \leq b$ ,*

$$\sigma_{K_j}((a, b)) \leq \int_{a'}^b k_0(g_{K_j}^+) + \varepsilon_j.$$

*Then  $\sigma_K((a, b)) \leq \int_{a'}^b k_0(g_K^+)$  for all such  $a$  and  $b$ .*

With Lemma 7.7, this is the passage to the limit in the proof of [1, Thm. 6.4.3]. Baek obtains the lower semicontinuity of the mass of an open interval from the weak convergence of  $\sigma_{K_n}$  and the Portmanteau theorem; the proof here uses support functions instead.

*Proof.* Let  $-\pi/2 \leq a < b \leq \pi/2$  with  $a' = \max(a, 0) \leq b$ . By Fact 2.15(a),  $\int_0^{\pi/2} |g_{K_j}^+ - g_K^+| \rightarrow 0$ ; as  $k_0$  is 1-Lipschitz and  $[a', b] \subset [0, \pi/2]$ , the right sides  $c_j = \int_{a'}^b k_0(g_{K_j}^+) + \varepsilon_j$  converge to  $c = \int_{a'}^b k_0(g_K^+)$ . For  $0 < \varepsilon < \pi$  let  $\Phi_K(\varepsilon) = \langle v_K(b - \varepsilon, b), v_b \rangle - \langle v_K(a, a + \varepsilon), v_a \rangle + \int_a^b h_K$ , and define  $\Phi_{K_j}(\varepsilon)$  in the same way. A point  $p = h_K(b)u_b + mv_b$  of  $e_K(b)$  lies in  $H_K(b - \varepsilon)$ , that is,  $h_K(b) \cos \varepsilon - m \sin \varepsilon \leq h_K(b - \varepsilon)$ , so  $\langle v_K(b), v_b \rangle \geq \langle v_K(b - \varepsilon, b), v_b \rangle$ ; likewise  $\langle v_K^+(a), v_a \rangle \leq \langle v_K(a, a + \varepsilon), v_a \rangle$ . By (2.1),  $\Phi_K(\varepsilon) \leq \sigma_K((a, b))$ , and in the same way  $\Phi_{K_j}(\varepsilon) \leq \sigma_{K_j}((a, b)) \leq c_j$ . Fix  $\varepsilon$ . The points  $v_{K_j}(b - \varepsilon, b)$  and  $v_{K_j}(a, a + \varepsilon)$  depend continuously on the support values of  $K_j$  at  $a, a + \varepsilon, b - \varepsilon$  and  $b$ , and  $h_{K_j} \rightarrow h_K$  uniformly; so  $\Phi_{K_j}(\varepsilon) \rightarrow \Phi_K(\varepsilon)$ , and therefore  $\Phi_K(\varepsilon) \leq c$ . As  $\varepsilon \downarrow 0$ ,  $\Phi_K(\varepsilon) \rightarrow \sigma_K((a, b))$  by Fact 2.1(a) and (2.1), hence  $\sigma_K((a, b)) \leq c$ .  $\square$

**Lemma 7.7** (passing to the limit). *Let  $K_j$  be polygon caps (with any angle sets) that converge in  $d_H$  to  $K \in \mathcal{K}_{\pi/2}^c$ , and let  $\varepsilon_j \rightarrow 0$  be such that, for all  $-\pi/2 \leq a < b \leq \pi/2$  with  $a' = \max(a, 0) \leq b$ ,*

$$\sigma_{K_j}((a, b)) \leq \int_{a'}^b k_0(g_{K_j}^+) + \varepsilon_j.$$

*Then  $\sigma_K(E) \leq \int_E k_0(g_K^+(t))dt$  for every Borel set  $E \subset [0, \pi/2)$ .*

*Proof.* By Lemma 7.6,  $\sigma_{K_j}((a, b)) \leq \int_{a'}^b k_0(g_{K_j}^+)$  for all  $-\pi/2 \leq a < b \leq \pi/2$  with  $a' = \max(a, 0) \leq b$ . Let  $\mu$  and  $\nu$  be the restrictions of  $\sigma_K$  and of  $k_0(g_K^+(t))dt$  to  $[0, \pi/2)$ ; the claim is  $\mu \leq \nu$ . Both are finite measures on  $\mathbb{R}$ :  $\sigma_K([0, \pi/2))$  is finite, and  $k_0(g_K^+) \leq g_K^+ + 1$  is bounded, since  $g_K^+(t) = h_K(t) - \langle C_K^+(t), u_t \rangle$  lies between 0 and  $h_K(t) + h_K(t + \pi)$ , as  $C_K^+(t) \in K$ .

Let  $a < b$  be real,  $a' = \max(a, 0)$  and  $b' = \min(b, \pi/2)$ . If  $b' \leq a'$ , then  $(a, b) \cap [0, \pi/2)$  is empty. Otherwise  $(a, b) \cap [0, \pi/2)$  contains  $(a', b')$  and lies in  $(\max(a, -\pi/2), b')$ , an interval to which the bound above applies, so

$$\mu((a, b)) \leq \sigma_K((\max(a, -\pi/2), b')) \leq \int_{a'}^{b'} k_0(g_K^+) \leq \nu((a, b)).$$

As  $\mu$  and  $\nu$  are finite and  $(a, b]$  is the intersection of the decreasing intervals  $(a, b+1/i)$ , also  $\mu((a, b]) \leq \nu((a, b])$  for all  $a < b$ . Hence  $F(x) = \nu((-\infty, x]) - \mu((-\infty, x])$  is nondecreasing; it is also right-continuous, and its Lebesgue–Stieltjes measure  $dF$  satisfies  $\nu = \mu + dF$ , since both sides give every interval  $(a, b]$  the mass  $\nu((a, b])$ ; hence  $\mu \leq \mu + dF = \nu$ .  $\square$

### 7.3. The curvature bounds.

**Definition 7.8** (curvature bounds). A cap  $K \in \mathcal{K}_{\pi/2}^c$  satisfies the *curvature bounds* if, as measures,

$$\sigma_K \leq k_0(g_K^+(t))dt \text{ on } [0, \pi/2), \quad \sigma_K \leq k_0(f_K^-(t - \pi/2))dt \text{ on } (\pi/2, \pi].$$

**Proposition 7.9** (maximizing caps satisfy the curvature bounds). *Every maximizing cap  $K_* \in \mathcal{K}_{\pi/2}^c$  with  $\mathcal{A}(K_*) > 0$  satisfies the curvature bounds.*

*Proof.* Take the selection  $K_j \rightarrow K_*$  of Proposition 4.8, with stages  $k_j$ , step sizes  $\delta_j \rightarrow 0$ , in  $[-R_0, R_0] \times [0, 1]$ , and  $\eta_j = d_H(K_j, K_*) \rightarrow 0$ . Let  $D = 2R_0 + 2$ . Two points of the box differ by at most  $2R_0$  in  $x$  and by at most 1 in  $y$ , so  $\|p - q\| \leq D$  for all  $p, q \in K_j$ . Also  $|h_{K_j}| \leq R_0 + 1$ , and as  $C_{K_j}^+(t) \in K_j$ ,

$$0 \leq g_{K_j}^+(t) = h_{K_j}(t) - \langle C_{K_j}^+(t), u_t \rangle \leq h_{K_j}(t) + h_{K_j}(t + \pi) \leq D;$$

since  $k_0(x) \leq |x| + 1$ ,  $k_0(g_{K_j}^+) \leq B = D + 1$ . Let  $e_j(t) = 2\eta_j \nu_{k_j}(t) \geq 0$ . For  $t \in \Theta_{k_j}$ , Proposition 5.2 gives  $\sigma_{K_j}(t) \leq \tau_{K_j}(t) + e_j(t)$ , so by Lemma 7.3

$$\sigma_{K_j}(t) \leq k_0(g_{K_j}^+(t))\delta_j + (D + 4)\delta_j^2 + e_j(t).$$

This holds also at  $t = 0$ , where the right side is nonnegative and  $\sigma_{K_j}(0) = 0$ , since  $K_j$  has no normal angle in  $(-\pi/2, \delta_j)$  and so  $v_{K_j}^+(0) = v_{K_j}^-(0)$ . The grid points  $i\delta_j$ ,  $0 \leq i < 2^{k_j+1}$ , are distinct, and the weights  $\nu_{k_j}(s)$  add up to at most 1 by (4.1), so the errors  $e_j(i\delta_j)$  add up to at most  $2\eta_j$ . By Lemma 7.5 with  $C = D + 4$ , for  $-\pi/2 \leq a < b \leq \pi/2$  with  $a' \leq b$ ,

$$\sigma_{K_j}((a, b)) \leq \int_{a'}^b k_0(g_{K_j}^+) + (2B + \pi(D + 2))\delta_j + 2\eta_j.$$

By Lemma 7.7,  $K_*$  satisfies the first curvature bound. The reflection  $K_*^m$  is a cap with the same sofa area (Fact 2.4(d)), hence maximizing with positive sofa area, and the argument above, applied to  $K_*^m$  in place of  $K_*$ , shows that it satisfies the first curvature bound. Let  $E \subset (\pi/2, \pi]$  be a Borel set,

so that  $\pi - E \subset [0, \pi/2)$ . By Fact 2.4(d),  $\sigma_{K_*}(E) = \sigma_{K_*^m}(\pi - E)$ , and by Fact 2.14(c),  $g_{K_*^m}^+(s) = f_{K_*}^-(\pi/2 - s)$ ; hence

$$\sigma_{K_*}(E) \leq \int_{\pi-E} k_0(f_{K_*}^-(\frac{\pi}{2} - s)) ds = \int_E k_0(f_{K_*}^-(t - \frac{\pi}{2})) dt,$$

which is the second curvature bound for  $K_*$ .  $\square$

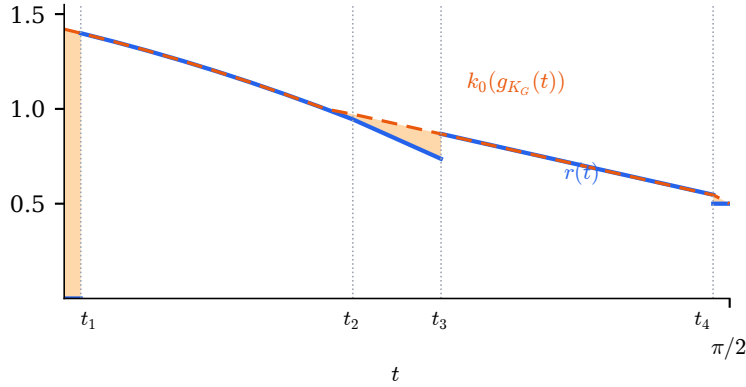


FIGURE 7. The first curvature bound for Gerver's cap, which maximizes  $\mathcal{A}$ , as  $\mathcal{A}(K_G) = |G|$  (Fact 2.17(b), Fact 2.4(c), Fact 2.8). The density  $r(t)$  of  $\sigma_{K_G}$  on  $[0, \pi/2)$  (blue) and  $k_0(g_{K_G}(t))$  (orange, dashed) are computed from Romik's rotation path. In the figure the blue curve lies below the orange one, and the two agree on  $[t_1, t_*]$  and on  $[t_3, t_4]$ , where  $t_* = 0.616\dots$  is the angle at which  $g_{K_G}$  equals 2; the gap between them is shaded.

#### 7.4. The injectivity condition.

**Lemma 7.10** (integral inequalities). *Let  $K \in \mathcal{K}_{\pi/2}^c$  satisfy the curvature bounds. Then  $K$  satisfies condition (1) of Definition 2.13, and*

$$\begin{aligned} f_K(t) - 1 &\geq \int_0^t m_0(g_K(u)) du \quad (0 \leq t < \frac{\pi}{2}), \\ g_K(t) - 1 &\geq \int_t^{\pi/2} m_0(f_K(u)) du \quad (0 < t \leq \frac{\pi}{2}). \end{aligned}$$

*Proof.* Each restriction of  $\sigma_K$  is a finite measure, dominated by a measure with a density, hence absolutely continuous with respect to Lebesgue measure; so it has a density (Radon–Nikodym), and condition (1) holds. By Fact 2.14(d),  $f_K, g_K$  are defined and continuous on  $[0, \pi/2]$ ,  $f_K^+ = f_K$  on  $[0, \pi/2)$ , and  $f_K(0) = g_K(\pi/2) = 1$ . For  $t \in [0, \pi/2)$ ,  $(0, t] \subset [0, \pi/2)$ , so the

first curvature bound and Fact 2.14(b) give

$$\begin{aligned} f_K(t) - 1 &= f_K^+(t) - f_K^+(0) = \int_0^t g_K - \sigma_K((0, t]) \\ &\geq \int_0^t (g_K - k_0(g_K)) = \int_0^t m_0(g_K). \end{aligned}$$

For  $a \leq b$ ,

$$g_K^+(b) - g_K^+(a) = \sigma_K((a + \frac{\pi}{2}, b + \frac{\pi}{2}]) - \int_a^b f_K^+.$$

As  $\langle \mathbf{y}_K(s), u_s \rangle = h_K(s)$ ,  $\langle \mathbf{y}_K(s), v_s \rangle = h_K(s + \pi/2)$ ,  $A_K^+(s) = v_K^+(s)$ ,  $C_K^+(s) = v_K^+(s + \pi/2)$  and  $u_s = -v_{s+\pi/2}$ , we have  $g_K^+(s) = h_K(s) + \langle v_K^+(s + \pi/2), v_{s+\pi/2} \rangle$  and  $f_K^+(s) = h_K(s + \pi/2) - \langle v_K^+(s), v_s \rangle$ . By Fact 2.1(b),  $\langle v_K^+(d), v_d \rangle - \langle v_K^+(c), v_c \rangle = \sigma_K((c, d]) - \int_c^d h_K$  for  $c \leq d$ . The product rule [1, Lemma 5.1.3] for  $h_K(s) = \langle v_K^+(s), u_s \rangle$ , with  $v_K^+(d) - v_K^+(c) = \int_{(c, d]} v_s d\sigma_K(s)$  (Fact 2.1(b)) and  $\frac{d}{ds} u_s = v_s$ , gives  $h_K(b) - h_K(a) = \int_{(a, b]} \langle v_s, u_s \rangle d\sigma_K(s) + \int_a^b \langle v_K^+(s), v_s \rangle ds = \int_a^b \langle v_K^+(s), v_s \rangle ds$ . Hence

$$\begin{aligned} g_K^+(b) - g_K^+(a) &= h_K(b) - h_K(a) + \sigma_K((a + \frac{\pi}{2}, b + \frac{\pi}{2}]) - \int_{a+\pi/2}^{b+\pi/2} h_K, \\ \int_a^b f_K^+ &= \int_{a+\pi/2}^{b+\pi/2} h_K - (h_K(b) - h_K(a)), \end{aligned}$$

and the identity follows. Let  $t \in (0, \pi/2]$ , and take  $a = t$ ,  $b = \pi/2$ . As  $g_K = g_K^+$ ,  $g_K(\pi/2) = 1$ ,  $f_K^+ = f_K$  on  $[t, \pi/2)$  and  $(t + \pi/2, \pi] \subset (\pi/2, \pi]$ , the second curvature bound gives

$$\begin{aligned} g_K(t) - 1 &= \int_t^{\pi/2} f_K - \sigma_K((t + \frac{\pi}{2}, \pi]) \\ &\geq \int_t^{\pi/2} f_K - \int_{t+\pi/2}^{\pi} k_0(f_K^-(u - \frac{\pi}{2})) du = \int_t^{\pi/2} m_0(f_K). \quad \square \end{aligned}$$

**Lemma 7.11** (the lower sequence). *Let  $f, g : [0, \pi/2] \rightarrow [0, \infty)$  be continuous with  $f(t) - 1 \geq \int_0^t m_0(g)$  for  $t \in [0, \pi/2)$  and  $g(t) - 1 \geq \int_t^{\pi/2} m_0(f)$  for  $t \in (0, \pi/2]$ . Then for every  $j \geq 0$ ,  $f_j(t) \leq f(t)$  on  $[0, \pi/2)$  and  $f_j(\pi/2 - t) \leq g(t)$  on  $(0, \pi/2]$ .*

*Proof.* Induction on  $j$ , the case  $j = 0$  being  $f_0 = 0 \leq f, g$ . If it holds for  $j$ , then for  $t \in [0, \pi/2)$ , since  $m_0$  is nondecreasing and  $f_j(\pi/2 - u) \leq g(u)$  for  $u \in (0, t]$ ,  $\mathcal{F}f_j(t) = 1 + \int_0^t m_0(f_j(\pi/2 - u)) du \leq 1 + \int_0^t m_0(g) \leq f(t)$ , and with  $f_j \leq f$  we get  $f_{j+1} = \max(f_j, \mathcal{F}f_j) \leq f$  on  $[0, \pi/2)$ . For  $t \in (0, \pi/2]$ ,  $\mathcal{F}f_j(\pi/2 - t) = 1 + \int_t^{\pi/2} m_0(f_j(s)) ds \leq 1 + \int_t^{\pi/2} m_0(f) \leq g(t)$ , and with  $f_j(\pi/2 - t) \leq g(t)$  this gives  $f_{j+1}(\pi/2 - t) \leq g(t)$ .  $\square$

**Proposition 7.12** (injectivity). *Every  $K \in \mathcal{K}_{\pi/2}^c$  that satisfies the curvature bounds satisfies the injectivity condition, and  $f_K, g_K > 1$  on  $(0, \pi/2)$ .*

*Proof.* By Lemma 7.10,  $K$  satisfies (1) and the integral inequalities; by Fact 2.14(d),  $f_K$  and  $g_K$  are continuous on  $[0, \pi/2]$ , and they are nonnegative, since  $f_K(t) = h_K(t + \pi/2) - \langle A_K(t), v_t \rangle$  and  $g_K(t) = h_K(t) - \langle C_K(t), u_t \rangle$  with  $A_K(t), C_K(t) \in K$ . By Lemma 7.11 with  $j = 11$  and Fact 2.15(b), for  $t \in (0, \pi/2)$ ,  $1 < f_{11}(t) \leq f_K(t)$  and  $1 < f_{11}(\pi/2 - t) \leq g_K(t)$ . By Fact 2.14(d),  $\mathbf{x}_K$  is continuously differentiable on  $[0, \pi/2]$  with  $\mathbf{x}'_K = -(f_K - 1)u_t + (g_K - 1)v_t$ , which gives (2) and (3).  $\square$

**Corollary 7.13** (maximizing right-angle caps are in  $\mathcal{K}^i$ ). *Every  $K \in \mathcal{K}_{\pi/2}^c$  with  $\mathcal{A}(K) = |G|$  lies in  $\mathcal{K}^i$ .*

*Proof.*  $K$  is maximizing (Fact 2.8). Gerver's cap  $K_G \in \mathcal{K}_{\pi/2}^c$  has  $\mathcal{A}(K_G) = |G|$  (Fact 2.17(b), Fact 2.4(c)), so  $\mathcal{A}(K) \geq \mathcal{A}(K_G) = |G| > 0$ . Hence  $K$  satisfies the curvature bounds (Proposition 7.9) and the injectivity condition (Proposition 7.12); and  $|K| = \mathcal{A}(K) + |\mathcal{N}(K)| \geq |G| \geq 11/5$ .  $\square$

## 8. EQUALITY IN BAEK'S UPPER BOUND

By Corollary 7.13, every cap  $K \in \mathcal{K}_{\pi/2}^c$  with  $\mathcal{A}(K) = |G|$  lies in  $\mathcal{K}^i$ . For a cap  $K \in \mathcal{K}^i$ , write  $K_0 = K_G$  for Gerver's cap and  $K_1 = K$ , let  $\xi_0 = \xi_G$  and  $\xi_1 = \xi_K$  be their canonical triples (Section 2.5), and for  $\lambda \in [0, 1]$  let  $K_\lambda$ ,  $B_\lambda$ ,  $D_\lambda$  be the components of  $\xi_\lambda = (1 - \lambda)\xi_0 + \lambda\xi_1$ . Thus  $K_\lambda = (1 - \lambda)K_0 + \lambda K_1$ ,  $B_\lambda = (1 - \lambda)B_{K_0} + \lambda B_{K_1}$  and  $D_\lambda = (1 - \lambda)D_{K_0} + \lambda D_{K_1}$ . Write  $h_i = h_{K_i}$  for  $i = 0, 1$ , and  $\Delta = h_1 - h_0$ .

### 8.1. Equality in the concavity of $\mathcal{Q}$ .

**Lemma 8.1** (equality in the Mamikon terms). *Let  $K \in \mathcal{K}^i$  with  $\mathcal{A}(K) = |G|$ . Then for every  $\lambda \in [0, 1]$*

$$\begin{aligned} \mathcal{S}_{K_\lambda} &= (1 - \lambda)\mathcal{S}_{K_0} + \lambda\mathcal{S}_{K_1}, \\ \mathcal{R}_{B_\lambda} &= (1 - \lambda)\mathcal{R}_{B_0} + \lambda\mathcal{R}_{B_1}, \\ \mathcal{L}_{D_\lambda} &= (1 - \lambda)\mathcal{L}_{D_0} + \lambda\mathcal{L}_{D_1}. \end{aligned}$$

*Proof.* By Fact 2.16(a), (c),  $|G| = \mathcal{A}(K) \leq \mathcal{Q}(\xi_1) \leq \mathcal{Q}(\xi_0) = |G|$ , so  $\mathcal{Q}(\xi_1) = \mathcal{Q}(\xi_0)$ , and  $\mathcal{Q} \leq \mathcal{Q}(\xi_0)$  on  $\mathcal{T}$ . Let  $\lambda \in [0, 1]$ . The set  $\mathcal{T}$  is closed under Minkowski combinations, so  $\xi_\lambda \in \mathcal{T}$ . By Fact 2.16(b),  $\mathcal{Q} = \Lambda - \mathcal{S} - \mathcal{R} - \mathcal{L}$  with  $\Lambda$  convex-linear and  $\mathcal{S}$ ,  $\mathcal{R}$ ,  $\mathcal{L}$  convex, so  $\mathcal{Q}$  is concave on  $\mathcal{T}$ , and

$$\mathcal{Q}(\xi_\lambda) \geq (1 - \lambda)\mathcal{Q}(\xi_0) + \lambda\mathcal{Q}(\xi_1) = \mathcal{Q}(\xi_0) \geq \mathcal{Q}(\xi_\lambda).$$

So  $\mathcal{Q}(\xi_\lambda) = (1 - \lambda)\mathcal{Q}(\xi_0) + \lambda\mathcal{Q}(\xi_1)$ , and since  $\Lambda$  is convex-linear this says

$$\begin{aligned} [(1 - \lambda)\mathcal{S}_{K_0} + \lambda\mathcal{S}_{K_1} - \mathcal{S}_{K_\lambda}] &+ [(1 - \lambda)\mathcal{R}_{B_0} + \lambda\mathcal{R}_{B_1} - \mathcal{R}_{B_\lambda}] \\ &+ [(1 - \lambda)\mathcal{L}_{D_0} + \lambda\mathcal{L}_{D_1} - \mathcal{L}_{D_\lambda}] = 0. \end{aligned}$$

Each bracket is  $\geq 0$  by the convexity of  $\mathcal{S}$ ,  $\mathcal{R}$  and  $\mathcal{L}$ , so each is 0.  $\square$

**8.2. Equal tangent displacements.** In this subsection,  $K_0$  and  $K_1$  are any two convex bodies,  $\lambda \in (0, 1)$ ,  $K_\lambda = (1 - \lambda)K_0 + \lambda K_1$ ,  $h_i = h_{K_i}$  and  $\Delta = h_1 - h_0$ . Let  $a < b < a + \pi$ , and let  $\mathbf{z}_K$  be curves on  $[a, b]$  as in Section 2.5, one for every convex body  $K$ , that are convex-linear in  $K$ :  $\mathbf{z}_{K_\lambda} = (1 - \lambda)\mathbf{z}_{K_0} + \lambda\mathbf{z}_{K_1}$ . By Fact 2.16(b), the curves of the four terms of  $\mathcal{S}_K$  are of this kind. For  $i \in \{0, \lambda, 1\}$  write  $\varrho_i = \varrho_{K_i, \mathbf{z}_{K_i}}$  and  $\mathcal{M}_i = \mathcal{M}_{K_i}(a, b; \mathbf{z}_{K_i})$ . As  $v_{K_\lambda}^+(t) = (1 - \lambda)v_{K_0}^+(t) + \lambda v_{K_1}^+(t)$  for every  $t$  [1, Thm. 7.1.2],  $\varrho_\lambda = (1 - \lambda)\varrho_0 + \lambda\varrho_1$  on  $[a, b]$ . By Mamikon's theorem [1, Thm. 7.4.1],  $\varrho_{K, \mathbf{z}}$  is almost everywhere measurable and bounded on  $[a, b]$  for every convex body  $K$  and every curve  $\mathbf{z}$  as in Section 2.5; so  $\varrho_0^2$ ,  $\varrho_1^2$  and  $\varrho_0\varrho_1$  are integrable on  $[a, b]$ . Since  $(1 - \lambda)x^2 + \lambda y^2 - ((1 - \lambda)x + \lambda y)^2 = \lambda(1 - \lambda)(x - y)^2$ ,

$$(1 - \lambda)\mathcal{M}_0 + \lambda\mathcal{M}_1 - \mathcal{M}_\lambda = \frac{\lambda(1 - \lambda)}{2} \int_a^b (\varrho_0(t) - \varrho_1(t))^2 dt. \quad (8.1)$$

As  $\lambda(1 - \lambda) > 0$ , the equality  $\mathcal{M}_\lambda = (1 - \lambda)\mathcal{M}_0 + \lambda\mathcal{M}_1$  implies that  $\varrho_0 = \varrho_1$  almost everywhere on  $(a, b)$ . If moreover  $\varrho_0$  and  $\varrho_1$  are continuous on  $(a, b)$ , then  $\varrho_0 = \varrho_1$  at every point of  $(a, b)$ , because the set where they differ is open and has measure 0.

Now let  $K_0, K_1 \in \mathcal{K}_{\pi/2}^c$  satisfy condition (1) of Definition 2.13, and let  $[a, b] \subset [0, \pi/2]$  or  $[a, b] \subset [\pi/2, \pi]$ , so that  $(a, b) \subset (0, \pi/2)$  or  $(a, b) \subset (\pi/2, \pi)$ . By Fact 2.14(d),  $v_{K_i}^+(t) = v_{K_i}^-(t)$  for  $t \in [0, \pi/2] \cup (\pi/2, \pi]$ ; so, by the one-sided derivatives of Fact 2.1(a),  $h_i$  is differentiable at every such  $t$ , with  $h_i'(t) = \langle v_{K_i}^+(t), v_t \rangle$ . Also by Fact 2.14(d),  $v_{K_i}^+ = A_{K_i}$  on  $(0, \pi/2)$  and  $v_{K_i}^+(t) = C_{K_i}(t - \pi/2)$  for  $t \in (\pi/2, \pi)$ , where  $A_{K_i}$  and  $C_{K_i}$  are continuous on  $[0, \pi/2]$ . So, for  $i = 0, 1$ ,  $v_{K_i}^+$  is continuous on  $(a, b)$ , and so is  $\varrho_i(t) = \langle \mathbf{z}_{K_i}(t) - v_{K_i}^+(t), v_t \rangle$ .

**Lemma 8.2** (the tangent equations). *Let  $K_0, K_1 \in \mathcal{K}_{\pi/2}^c$  satisfy condition (1) of Definition 2.13, let  $\lambda \in (0, 1)$ , and let  $a < b$  with  $[a, b] \subset [0, \pi/2]$  or  $[a, b] \subset [\pi/2, \pi]$ .*

- (a) *If  $T - \pi < a$ ,  $b \leq T$  and  $\mathcal{M}_{K_\lambda}(a, b; \mathbf{l}_{K_\lambda}^T) = (1 - \lambda)\mathcal{M}_{K_0}(a, b; \mathbf{l}_{K_0}^T) + \lambda\mathcal{M}_{K_1}(a, b; \mathbf{l}_{K_1}^T)$ , then there are  $p, q \in \mathbb{R}$  with  $\Delta(t) = p \cos t + q \sin t$  for  $t \in [a, b]$ , and  $\Delta(T) = p \cos T + q \sin T$ .*
- (b) *If  $\mathcal{M}_{K_\lambda}(a, b; \mathbf{y}_{K_\lambda}) = (1 - \lambda)\mathcal{M}_{K_0}(a, b; \mathbf{y}_{K_0}) + \lambda\mathcal{M}_{K_1}(a, b; \mathbf{y}_{K_1})$ , then  $\Delta(t) = \Delta(b) - \int_t^b \Delta(u + \pi/2) du$  for  $t \in [a, b]$ .*

*Proof.* Both  $\Delta$  and  $t \mapsto \Delta(t + \pi/2)$  are continuous, as support functions are.

(a) Here  $b \leq T < a + \pi$  and  $[a, b] \subset (T - \pi, T]$ . For  $t \in [a, b]$ ,  $\mathbf{l}_K^T(t)$  lies on  $l_K(t)$ : it is the point  $v_K(t, T)$  of  $l_K(t)$  if  $t < T$ , and  $v_K^-(T) \in l_K(T)$  if  $t = T$ . By [1, Thm. 8.3.1],  $\mathbf{l}_K^T$  is continuous and of bounded variation on  $[a, b]$ , and by [1, Thm. 8.3.2],  $\mathbf{l}_{K_\lambda}^T = (1 - \lambda)\mathbf{l}_{K_0}^T + \lambda\mathbf{l}_{K_1}^T$  on  $[a, b]$ . So these curves are of the kind considered above, and  $\varrho_0 = \varrho_1$  on  $(a, b)$ . Let  $t \in (a, b)$ . Then  $0 < T - t < \pi$ ,  $\mathbf{l}_{K_i}^T(t) = v_{K_i}(t, T)$  and  $\langle v_{K_i}(t, T), v_t \rangle = (h_i(T) - h_i(t) \cos(T - t)) / \sin(T - t)$ ,

so

$$\varrho_i(t) = \frac{h_i(T) - h_i(t) \cos(T-t)}{\sin(T-t)} - h'_i(t).$$

Multiplied by  $\sin(T-t)$ , the equality  $\varrho_0(t) = \varrho_1(t)$  is

$$\sin(T-t)\Delta'(t) + \cos(T-t)\Delta(t) = \Delta(T).$$

Then  $\chi(t) = (\Delta(t) - \Delta(T) \cos(T-t)) / \sin(T-t)$  has  $\chi' = [\Delta' \sin(T-t) + \Delta \cos(T-t) - \Delta(T)] / \sin^2(T-t) = 0$  on  $(a, b)$ , so  $\chi$  is a constant  $\kappa$  there, and  $\Delta(t) = \Delta(T) \cos(T-t) + \kappa \sin(T-t)$  for  $t \in (a, b)$ . Both sides are continuous, so this holds on  $[a, b]$ . It reads  $\Delta(t) = p \cos t + q \sin t$  with  $p = \Delta(T) \cos T + \kappa \sin T$  and  $q = \Delta(T) \sin T - \kappa \cos T$ , and then  $p \cos T + q \sin T = \Delta(T)$ .

(b) Here  $b-a \leq \pi/2 < \pi$ . By the formula  $\mathbf{y}_K(t) = h_K(t)u_t + h_K(t+\pi/2)v_t$  for the outer corner (Section 2.2),  $\langle \mathbf{y}_K(t), u_t \rangle = h_K(t)$ , that is,  $\mathbf{y}_K(t) \in l_K(t)$ ; and  $\mathbf{y}_K$  is convex-linear in  $K$ , as  $h_K$  is. As  $K$  is bounded,  $h_K$  is Lipschitz, and so is  $\mathbf{y}_K$  on  $[a, b]$ , which makes  $\mathbf{y}_K$  continuous and of bounded variation there. These curves are of the kind considered above, and  $\varrho_0 = \varrho_1$  on  $(a, b)$ . As  $\langle \mathbf{y}_{K_i}(t), v_t \rangle = h_i(t+\pi/2)$ , we have  $\varrho_i(t) = h_i(t+\pi/2) - h'_i(t)$ , and  $\varrho_0 = \varrho_1$  says  $\Delta'(t) = \Delta(t+\pi/2)$  for  $t \in (a, b)$ . For  $t \in [a, b]$ ,  $\Delta$  is continuous on  $[t, b]$  and has the continuous derivative  $u \mapsto \Delta(u+\pi/2)$  on  $(t, b)$ , so  $\Delta(b) - \Delta(t) = \int_t^b \Delta(u+\pi/2) du$ .  $\square$

### 8.3. The cap is a horizontal translate.

**Proposition 8.3** (the difference of support functions). *Let  $K \in \mathcal{K}^i$  with  $A(K) = |G|$ . Then there is  $s \in \mathbb{R}$  with  $h_K(t) - h_{K_G}(t) = s \cos t$  for  $t \in [0, \pi]$ .*

*Proof.* We use the notation of the beginning of this section, with  $K_0 = K_G$ ,  $K_1 = K$  and  $\Delta = h_K - h_{K_G}$ . By Lemma 8.1 with  $\lambda = 1/2$ ,  $\mathcal{S}_{K_{1/2}} = \frac{1}{2}\mathcal{S}_{K_0} + \frac{1}{2}\mathcal{S}_{K_1}$ . Each of the four terms of  $\mathcal{S}$  is convex (Fact 2.16(b)), so its value at  $K_{1/2}$  is at most the mean of its values at  $K_0$  and  $K_1$ ; as the sums are equal, each term takes at  $K_{1/2}$  the mean of its values at  $K_0$  and  $K_1$ . The cap  $K_0 = K_G$  lies in  $\mathcal{K}^i$  by Fact 2.17(b), and so does  $K_1 = K$ ; so both satisfy condition (1). Also  $0 < \varphi < \pi/4$  (Appendix B). So Lemma 8.2 applies with  $\lambda = 1/2$ : part (a) to the first, third and fourth terms, with  $(a, b, T)$  equal to  $(0, \varphi, \pi/2)$ ,  $(\pi/2 - \varphi, \pi/2, \pi - \varphi)$  and  $(\pi/2, \pi, \pi)$ , and part (b) to the second, with  $(a, b) = (\varphi, \pi/2 - \varphi)$ . It gives  $p_j, q_j \in \mathbb{R}$  such that

- (i)  $\Delta(t) = p_1 \cos t + q_1 \sin t$  for  $t \in [0, \varphi]$ , and  $\Delta(\pi/2) = q_1$ ;
- (ii)  $\Delta(t) = \Delta(\pi/2 - \varphi) - \int_t^{\pi/2 - \varphi} \Delta(u + \pi/2) du$  for  $t \in [\varphi, \pi/2 - \varphi]$ ;
- (iii)  $\Delta(t) = p_3 \cos t + q_3 \sin t$  for  $t \in [\pi/2 - \varphi, \pi/2]$ , and  $\Delta(\pi - \varphi) = p_3 \cos(\pi - \varphi) + q_3 \sin(\pi - \varphi)$ ;
- (iv)  $\Delta(t) = p_4 \cos t + q_4 \sin t$  for  $t \in [\pi/2, \pi]$ , and  $\Delta(\pi) = -p_4$ .

Moreover  $\Delta(\pi/2) = 1 - 1 = 0$ , since every cap with rotation angle  $\pi/2$  has the support value 1 at  $\pi/2$ . Let  $s = -\Delta(\pi)$ .

On  $[\pi/2, \pi]$ : by (iv) at  $t = \pi/2$ ,  $q_4 = \Delta(\pi/2) = 0$ ; so  $\Delta(t) = p_4 \cos t$ , and  $p_4 = -\Delta(\pi) = s$ .

On  $[\pi/2 - \varphi, \pi/2]$ : by (iii) at  $t = \pi/2$ ,  $q_3 = \Delta(\pi/2) = 0$ , so  $\Delta(\pi - \varphi) = p_3 \cos(\pi - \varphi)$ . As  $\pi - \varphi \in [\pi/2, \pi]$ , the previous step gives  $\Delta(\pi - \varphi) = s \cos(\pi - \varphi)$ , and  $\cos(\pi - \varphi) = -\cos \varphi \neq 0$ ; so  $p_3 = s$ .

On  $[\varphi, \pi/2 - \varphi]$ : let  $b = \pi/2 - \varphi$ . For  $u \in [\varphi, b]$  we have  $u + \pi/2 \in [\pi/2, \pi]$ , so  $\Delta(u + \pi/2) = s \cos(u + \pi/2) = -s \sin u$  by the first step, and  $\Delta(b) = s \cos b$  by the second. By (ii),  $\Delta(t) = s \cos b + s \int_t^b \sin u du = s \cos t$ .

On  $[0, \varphi]$ : by (i),  $q_1 = \Delta(\pi/2) = 0$ , so  $\Delta(\varphi) = p_1 \cos \varphi$ , while  $\Delta(\varphi) = s \cos \varphi$  by the third step. As  $\cos \varphi \neq 0$ ,  $p_1 = s$ . So  $\Delta(t) = s \cos t$  on each of the four intervals, which cover  $[0, \pi]$ .  $\square$

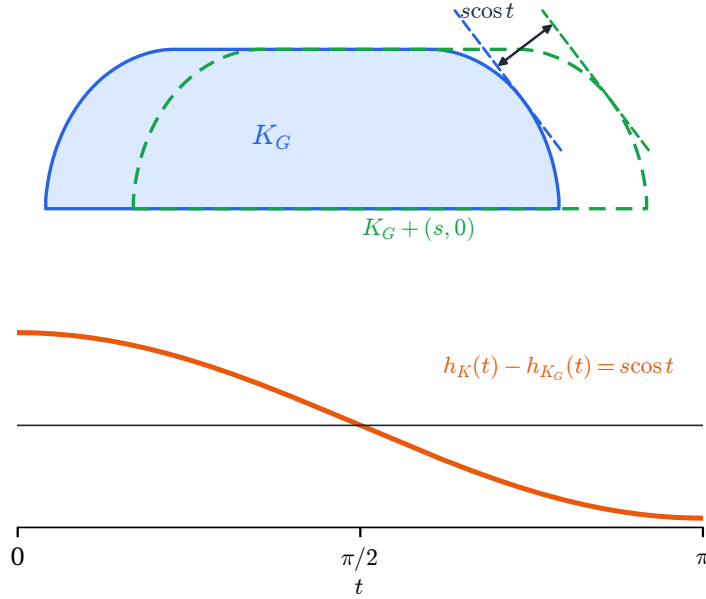


FIGURE 8. Proposition 8.3 and Lemma 8.4. Top: Gerver's cap  $K_G$  and its horizontal translate by  $(s, 0)$ ; their supporting lines at the normal  $t$  lie  $s \cos t$  apart. Bottom: the difference  $h_K - h_{K_G} = s \cos t$  of the support functions on  $[0, \pi]$ .

**Lemma 8.4** (caps with translated supports). *Let  $C \in \mathcal{K}_{\pi/2}^c$  and  $s \in \mathbb{R}$ .*

- (a) *Let  $K$  be a cap in  $\mathcal{K}_{\pi/2}^c$  with  $h_K(t) - h_C(t) = s \cos t$  for  $t \in [0, \pi]$ . Then  $K = C + (s, 0)$ ,  $\mathcal{N}(K) = \mathcal{N}(C) + (s, 0)$  and  $K \setminus \mathcal{N}(K) = (C \setminus \mathcal{N}(C)) + (s, 0)$ .*
- (b) *The translate  $C + (s, 0)$  lies in  $\mathcal{K}_{\pi/2}^c$ ,  $h_{C+(s,0)}(t) - h_C(t) = s \cos t$  for every  $t$ , and  $\mathcal{A}(C + (s, 0)) = \mathcal{A}(C)$ .*

*Proof.* (a) A point  $p$  lies in  $K$  if and only if  $p_y \geq 0$  and  $\langle p, u_t \rangle \leq h_K(t)$  for  $t \in [0, \pi]$ , and likewise for  $C$ . Indeed,  $K$  lies in  $P_{\pi/2} = \mathbb{R} \times [0, 1]$ , and  $\langle p, u_t \rangle \leq h_K(t)$  for  $p \in K$ . Conversely,  $K$  is an intersection of closed half-planes with normal angles in  $J_{\pi/2} \cup \{3\pi/2\} = [0, \pi] \cup \{3\pi/2\}$ ; each of them contains  $K$ , hence contains the supporting half-plane  $H_K(t)$  at its normal angle  $t$ , and  $H_K(3\pi/2) = \{p : p_y \geq 0\}$  because  $h_K(3\pi/2) = 0$ . For  $t \in [0, \pi]$ ,  $\langle p - (s, 0), u_t \rangle = \langle p, u_t \rangle - s \cos t$  and  $h_C(t) = h_K(t) - s \cos t$ , and  $p - (s, 0)$  has the ordinate of  $p$ ; so  $p \in K$  if and only if  $p - (s, 0) \in C$ , that is,  $K = C + (s, 0)$ .

By the definition of the niche,  $p \in \mathcal{N}(K)$  if and only if  $p_y \geq 0$  and there is  $t \in (0, \pi/2)$  with  $\langle p, u_t \rangle < h_K(t) - 1$  and  $\langle p, v_t \rangle < h_K(t + \pi/2) - 1$ , and likewise for  $C$ . For  $t \in (0, \pi/2)$  we have  $h_K(t) = h_C(t) + s \cos t$  and  $h_K(t + \pi/2) = h_C(t + \pi/2) - s \sin t$ , while  $\langle p - (s, 0), u_t \rangle = \langle p, u_t \rangle - s \cos t$  and  $\langle p - (s, 0), v_t \rangle = \langle p, v_t \rangle + s \sin t$ . So  $p \in \mathcal{N}(K)$  if and only if  $p - (s, 0) \in \mathcal{N}(C)$ , that is,  $\mathcal{N}(K) = \mathcal{N}(C) + (s, 0)$ . Together with  $K = C + (s, 0)$ , this gives  $K \setminus \mathcal{N}(K) = (C \setminus \mathcal{N}(C)) + (s, 0)$ .

(b) As  $h_{C+w}(t) = h_C(t) + \langle w, u_t \rangle$ ,  $h_{C+(s,0)}(t) - h_C(t) = s \cos t$ . This vanishes at  $t = \pi/2$  and  $t = 3\pi/2$ , so  $h_{C+(s,0)}(\pi/2) = 1$  and  $h_{C+(s,0)}(3\pi/2) = 0$ . The convex body  $C + (s, 0)$  is the intersection of the translates by  $(s, 0)$  of the closed half-planes whose intersection is  $C$ , and these have the same normal angles; so  $C + (s, 0) \in \mathcal{K}_{\pi/2}^c$ . By (a) with  $K = C + (s, 0)$ ,  $\mathcal{N}(C + (s, 0)) = \mathcal{N}(C) + (s, 0)$ , and as translations preserve area,  $\mathcal{A}(C + (s, 0)) = |C| - |\mathcal{N}(C)| = \mathcal{A}(C)$ .  $\square$

**Theorem 8.5** (maximizing right-angle caps). *Let  $K$  be a cap in  $\mathcal{K}_{\pi/2}^c$ . Then  $\mathcal{A}(K) = |G|$  if and only if  $K = K_G + (s, 0)$  for some  $s \in \mathbb{R}$ , and in this case  $K \setminus \mathcal{N}(K) = G + (s, 0)$ . The maximizing caps in  $\mathcal{K}_{\pi/2}^c$  (Definition 3.1) are the horizontal translates  $K_G + (s, 0)$  of Gerver's cap, and every monotone sofa  $U$  with rotation angle  $\pi/2$  and  $|U| = |G|$  is a horizontal translate  $G + (s, 0)$  of  $G$ .*

*Proof.* If  $\mathcal{A}(K) = |G|$ , then  $K \in \mathcal{K}^i$  by Corollary 7.13, Proposition 8.3 gives  $s \in \mathbb{R}$  with  $h_K(t) - h_{K_G}(t) = s \cos t$  for  $t \in [0, \pi]$ , and  $K = K_G + (s, 0)$  by Lemma 8.4(a) with  $C = K_G$ . Conversely, if  $K = K_G + (s, 0)$ , then  $\mathcal{A}(K) = \mathcal{A}(K_G)$  by Lemma 8.4(b), and  $\mathcal{A}(K_G) = |G|$  by Fact 2.4(c), as  $G$  is a monotone sofa with cap  $K_G$  (Fact 2.17(b)). In this case Lemma 8.4(b) and then (a), applied to  $K_G + (s, 0)$  and  $K_G$ , give  $K \setminus \mathcal{N}(K) = (K_G \setminus \mathcal{N}(K_G)) + (s, 0) = G + (s, 0)$ , since  $G = K_G \setminus \mathcal{N}(K_G)$  (Fact 2.17(b)).

By Fact 2.8,  $\mathcal{A} \leq |G|$  on  $\mathcal{K}_{\pi/2}^c$ , and  $K_G \in \mathcal{K}_{\pi/2}^c$  has  $\mathcal{A}(K_G) = |G|$ . So a cap  $K \in \mathcal{K}_{\pi/2}^c$  is maximizing if and only if  $\mathcal{A}(K) = |G|$ , that is, if and only if  $K = K_G + (s, 0)$  for some  $s$ ; and every such translate lies in  $\mathcal{K}_{\pi/2}^c$  by Lemma 8.4(b). Finally, let  $U$  be a monotone sofa with rotation angle  $\pi/2$  and  $|U| = |G|$ . Its cap  $K = \mathcal{C}(U)$  lies in  $\mathcal{K}_{\pi/2}^c$ , with  $U = K \setminus \mathcal{N}(K)$  and  $\mathcal{A}(K) = |U| = |G|$  (Fact 2.4(b), (c)). So  $K \in \mathcal{K}^i$  by Corollary 7.13, and Proposition 8.3 and Lemma 8.4(a) with  $C = K_G$  give  $s \in \mathbb{R}$  with  $U = K \setminus \mathcal{N}(K) = (K_G \setminus \mathcal{N}(K_G)) + (s, 0) = G + (s, 0)$ .  $\square$

**8.4. A second proof of optimality.** The pinned bounds (Proposition 6.4, for  $\omega < \pi/2$ ) and the curvature bounds (Proposition 7.9, for  $\omega = \pi/2$ ) hold for every maximizing cap with positive sofa area, and their proofs do not use the value of the largest sofa area. The proof of Theorem 1.1 uses Baek's theorem (Fact 2.8) to show that a cap with sofa area  $|G|$  is maximizing. Applied to the caps of Fact 2.7, the two bounds give the properties of step (3) of Section 1.3 without Fact 2.8, and with Fact 2.16 they determine the largest sofa area. This gives a second proof of Fact 2.8. The proofs in this subsection use the Facts of Section 2 other than Fact 2.8, and the results of Sections 4 to 8 other than Proposition 6.9, Corollary 7.13, and Theorem 8.5, which are the only ones there whose proofs use Fact 2.8.

**Lemma 8.6** (a maximizing right-angle cap). *Let  $K \in \mathcal{K}_{\pi/2}^c$  be a maximizing cap. Then  $K \in \mathcal{K}^i$  and  $\mathcal{A}(K) = |G|$ , and there is  $s \in \mathbb{R}$  with  $K = K_G + (s, 0)$  and  $K \setminus \mathcal{N}(K) = G + (s, 0)$ .*

*Proof.* Gerver's cap  $K_G$  lies in  $\mathcal{K}_{\pi/2}^c$  and has  $\mathcal{A}(K_G) = |G|$  (Fact 2.17(b), Fact 2.4(c)). As  $K$  is maximizing,  $\mathcal{A}(K) \geq \mathcal{A}(K_G) = |G|$ , and  $|G| \geq 2.2192$  by Fact 2.17(a). So  $\mathcal{A}(K) > 0$ , and  $K$  satisfies the curvature bounds (Proposition 7.9) and the injectivity condition (Proposition 7.12). Also  $|K| = \mathcal{A}(K) + |\mathcal{N}(K)| \geq |G| \geq 11/5$ , so  $K \in \mathcal{K}^i$ . Hence

$$|G| = \mathcal{A}(K_G) \leq \mathcal{A}(K) \leq \mathcal{Q}(\xi_K) \leq \mathcal{Q}(\xi_G) = |G|,$$

where the first inequality holds as  $K$  is maximizing, the second by Fact 2.16(a), and the third and the last equality by Fact 2.16(c). So  $\mathcal{A}(K) = |G|$ . Now Proposition 8.3 gives  $s \in \mathbb{R}$  with  $h_K(t) - h_{K_G}(t) = s \cos t$  for  $t \in [0, \pi]$ , and Lemma 8.4(a), with  $C = K_G$ , gives  $K = K_G + (s, 0)$  and  $K \setminus \mathcal{N}(K) = (K_G \setminus \mathcal{N}(K_G)) + (s, 0) = G + (s, 0)$ , as  $G = K_G \setminus \mathcal{N}(K_G)$  (Fact 2.17(b)).  $\square$

**Lemma 8.7** (a right-angle motion). *Let  $\omega \in [\omega_0, \pi/2]$ , and let  $T$  be a monotone sofa with rotation angle  $\omega$  and  $|T| \geq 11/5$  whose cap  $\mathcal{C}(T)$  is maximizing. Then  $R_\psi T$ ,  $\psi = \pi/2 - \omega$ , is a moving sofa with rotation angle  $\pi/2$ .*

*Proof.* If  $\omega = \pi/2$ , then  $\psi = 0$ , and  $T$  is a moving sofa with rotation angle  $\pi/2$  by Fact 2.4(a). Let  $\omega < \pi/2$  and  $K = \mathcal{C}(T)$ . By Fact 2.4(a), (b) and (c),  $T$  is a moving sofa with rotation angle  $\omega$  in standard position,  $K$  is a cap,  $T = K \setminus \mathcal{N}(K)$  and  $\mathcal{A}_\omega(K) = |T| \geq 11/5 > 0$ . As  $K$  is maximizing, Proposition 6.4 gives the pinned bounds  $w_K^\circ \leq \sigma_K(\pi/2)$  and  $z_K^\circ \leq \sigma_K(\omega)$ . The proof of Proposition 6.9 uses  $|T| = |G|$  only to obtain these bounds and  $\mathcal{A}_\omega(K) \geq 11/5$ , and it concludes as before: Lemma 6.6(d) gives  $t_\sharp \in (0, \omega)$  such that  $O$ ,  $cu_0$  and  $cv_\omega$  lie in the closure of  $Q_K^-(t_\sharp)$ , Lemma 6.7(b) gives  $\langle p - q, u_t \rangle \leq 1$  for  $p, q \in T$  and  $t \in [\omega, \pi/2]$ , and Lemma 6.8 shows that  $R_\psi T$  is a moving sofa with rotation angle  $\pi/2$ .  $\square$

**Theorem 8.8** (optimality from the maximizing caps). *(a) Every cap  $C \in \mathcal{K}_{\pi/2}^c$  has  $\mathcal{A}(C) \leq |G|$ . For a cap  $K \in \mathcal{K}_{\pi/2}^c$ , the following are equivalent:  $K$  is maximizing;  $\mathcal{A}(K) = |G|$ ;  $K = K_G + (s, 0)$  for some  $s \in \mathbb{R}$ .*

- (b)  $G$  is a moving sofa, and every moving sofa  $S$  has  $|S| \leq |G|$ .  
(c) For every  $\omega \in (0, \pi/2]$  and every cap  $K \in \mathcal{K}_\omega^c$ ,  $\mathcal{A}_\omega(K) \leq |G|$ .

*Proof.* (a) Let  $K_{\pi/2}$  be the cap of Fact 2.7 for  $\omega = \pi/2$ . It is maximizing, so  $\mathcal{A}(K_{\pi/2}) = |G|$  by Lemma 8.6, and every  $C \in \mathcal{K}_{\pi/2}^c$  has  $\mathcal{A}(C) \leq \mathcal{A}(K_{\pi/2}) = |G|$ . Hence a cap  $K \in \mathcal{K}_{\pi/2}^c$  with  $\mathcal{A}(K) = |G|$  is maximizing, and a maximizing cap is a translate  $K_G + (s, 0)$  by Lemma 8.6. Conversely,  $K_G + (s, 0)$  lies in  $\mathcal{K}_{\pi/2}^c$  and has  $\mathcal{A}(K_G + (s, 0)) = \mathcal{A}(K_G) = |G|$  by Lemma 8.4(b), Fact 2.17(b) and Fact 2.4(c).

(b) By Fact 2.17(b),  $G$  is a monotone sofa with rotation angle  $\pi/2$ , hence a moving sofa (Fact 2.4(a)). For the bound, first let  $S$  be a moving sofa with rotation angle  $\pi/2$ , and let  $K_{\pi/2}$  and  $S_{\pi/2} = K_{\pi/2} \setminus \mathcal{N}(K_{\pi/2})$  be as in Fact 2.7. Then

$$|S| \leq |S_{\pi/2}| = \mathcal{A}(K_{\pi/2}) = |G|,$$

where the inequality holds by Fact 2.7, the first equality by Fact 2.4(c), as  $\mathcal{C}(S_{\pi/2}) = K_{\pi/2}$ , and the second by Lemma 8.6, as  $K_{\pi/2}$  is maximizing. Now let  $S$  be any moving sofa. If  $|S| < 11/5$ , then  $|S| < |G|$ , as  $|G| \geq 2.2192$  (Fact 2.17(a)). Otherwise  $S$  is a moving sofa with some rotation angle  $\omega \in [\omega_0, \pi/2]$  (Fact 2.3(a)). Let  $K_\omega$  and  $S_\omega = K_\omega \setminus \mathcal{N}(K_\omega)$  be as in Fact 2.7. Then  $|S_\omega| \geq |S| \geq 11/5$ , and the cap  $\mathcal{C}(S_\omega) = K_\omega$  of the monotone sofa  $S_\omega$  is maximizing. By Lemma 8.7,  $R_\psi S_\omega$ ,  $\psi = \pi/2 - \omega$ , is a moving sofa with rotation angle  $\pi/2$ , and so

$$|S| \leq |S_\omega| = |R_\psi S_\omega| \leq |G|,$$

where the first inequality holds by Fact 2.7, the equality as rotations preserve area, and the last inequality by the first case.

- (c) Let  $K_\omega$  and  $S_\omega$  be as in Fact 2.7. Every  $K \in \mathcal{K}_\omega^c$  has

$$\mathcal{A}_\omega(K) \leq \mathcal{A}_\omega(K_\omega) = |S_\omega| \leq |G|,$$

where the first inequality holds by Fact 2.7, the equality by Fact 2.4(c), as  $\mathcal{C}(S_\omega) = K_\omega$ , and the last inequality by (b), as  $S_\omega$  is a moving sofa (Fact 2.4(a)).  $\square$

With  $|G| > 11/5$  (Fact 2.17(a)), Theorem 8.8 contains Fact 2.8. The proofs of Theorem 1.1 and Corollary 9.4 use Fact 2.8 only in Propositions 3.2 and 6.9, Corollary 7.13, and Theorem 8.5 and in the proof of Corollary 9.4. They go through with Theorem 8.8 in place of Fact 2.8, with Lemma 8.7 for Proposition 6.9, and with Lemma 8.6 for the cap of the monotone sofa  $U$ ; this is how the formalization proves them a second time (Section 12). So Theorem 1.1 and Corollary 9.4 hold without [1, Thm. 1.1.1].

The second proof uses Baek's caps  $K_\omega$  and  $K_{\pi/2}$  of step (2) only through Fact 2.7, whose statement does not involve their balance. Baek derives the two properties of step (3) for these caps from their balance [1, Thm. 1.5.2, Thm. 8.1.1 (2)]; here they follow from maximality, by Lemmas 8.6 and 8.7. For the other steps of the chain, the second proof uses Baek's results: Fact 2.4 for step (1), Fact 2.7 for the caps of step (2a) and Fact 2.16 for step (4),

with Fact 2.3 for the bound on the rotation angle. The proof also uses  $|G| = \mathcal{A}(K_G) \leq \mathcal{A}(K_{\pi/2})$ , which is not in (1.1) and holds as  $K_{\pi/2}$  is maximizing; with step (4) it gives the value of the maximum, and with Proposition 8.3 the maximizing caps.

Section 11 reaches the conclusions of Lemma 8.6 and Theorem 8.5 without Lemma 8.1, Lemma 8.2 and Proposition 8.3, from an estimate in the proof of stability (Theorem 11.1).

## 9. RECOVERING THE SOFA; PROOF OF THEOREM 1.1

By (6.1) and Theorem 8.5, a rigid image of  $S$  lies in a horizontal translate of  $G$  and has the same area. A closed subset of a set of finite area, with the same area, is the whole set if that set is the closure of its interior (Lemma 9.1). So it remains to show that  $G$  is the closure of its interior.

**Lemma 9.1** (recovering a set from its area). *Let  $X \subset Y \subset \mathbb{R}^2$  with  $X$  closed,  $Y = \text{cl}(\text{int } Y)$ ,  $|Y| < \infty$  and  $|X| = |Y|$ . Then  $X = Y$ .*

*Proof.* As  $X$  is closed, hence measurable, and  $|Y| < \infty$ ,  $|Y \setminus X| = |Y| - |X| = 0$ . The set  $\text{int } Y \setminus X$  is open and lies in  $Y \setminus X$ , so it has area 0; as a nonempty open set has positive area, it is empty. Hence  $\text{int } Y \subset X$ , and since  $X$  is closed,  $Y = \text{cl}(\text{int } Y) \subset X \subset Y$ .  $\square$

**Lemma 9.2** (removing the region under an envelope). *Let  $K \subset \mathbb{R}^2$  with  $K = \text{cl}(\text{int } K)$  and  $[a, b] \times [0, 1] \subset K$ . Let  $\Gamma$  be a compact set of points with abscissa in  $[a, b]$  and height in  $[0, 1]$ , and let  $\Gamma_{<}$  be the set of points  $q$  with  $q_y \geq 0$  that lie strictly below a point  $\gamma$  of  $\Gamma$  with  $\gamma_x = q_x$ . If  $X = K \setminus \Gamma_{<}$  is closed, then  $X = \text{cl}(\text{int } X)$ .*

*Proof.* Let  $\bar{\Gamma}$  be the set of points  $q$  with  $q_y \geq 0$  that lie on or below a point of  $\Gamma$  with the same abscissa. As the heights of  $\Gamma$  are nonnegative,  $\bar{\Gamma}$  is the image of the compact set  $\Gamma \times [0, 1]$  under  $(\gamma, \lambda) \mapsto (\gamma_x, \lambda\gamma_y)$ , so it is compact; and  $\Gamma_{<} \subset \bar{\Gamma}$ .

First, every point  $p \in K \setminus \bar{\Gamma}$  lies in  $\text{cl}(\text{int } X)$ . Indeed,  $p$  has a neighborhood outside the closed set  $\bar{\Gamma}$ , and since  $K = \text{cl}(\text{int } K)$ ,  $p$  is a limit of interior points of  $K$  outside  $\bar{\Gamma}$ . These points form an open subset of  $K \setminus \bar{\Gamma} \subset K \setminus \Gamma_{<} = X$ , so they are interior points of  $X$ .

Now let  $p \in X$ . If  $p \notin \bar{\Gamma}$ , then  $p \in \text{cl}(\text{int } X)$  by the first step. If  $p \in \bar{\Gamma}$ , there is  $\gamma \in \Gamma$  with  $\gamma_x = p_x$  and  $p_y \leq \gamma_y$ , so  $p_x \in [a, b]$  and  $p_y < 1$ . For  $0 < s < 1 - p_y$ , the point  $(p_x, p_y + s)$  lies in the rectangle  $[a, b] \times [0, 1] \subset K$ , and not in  $\bar{\Gamma}$ : a point  $\delta \in \Gamma$  with  $\delta_x = p_x$  and  $\delta_y \geq p_y + s > p_y$  would put  $p$  in  $\Gamma_{<}$ . By the first step these points lie in  $\text{cl}(\text{int } X)$ , and so does their limit  $p$  as  $s \rightarrow 0$ .

So  $X \subset \text{cl}(\text{int } X)$ , and  $\text{cl}(\text{int } X) \subset X$  since  $X$  is closed.  $\square$

**Proposition 9.3** (Gerver's sofa is regular closed).  $G = \text{cl}(\text{int } G)$ .

*Proof.* Let  $\Gamma = \mathbf{D}([t_0, t_2]) \cup \mathbf{x}([t_1, t_4]) \cup \mathbf{B}([t_3, t_5])$ ,  $a = \mathbf{D}(0)_x$  and  $b = \mathbf{B}(\pi/2)_x$  (Figure 9). The set  $\Gamma$  is compact, as the union of the images of three closed

intervals under the continuous curves  $\mathbf{D}$ ,  $\mathbf{x}$  and  $\mathbf{B}$ . By Fact 2.17(c), the abscissa increases along  $\mathbf{D}$  and  $\mathbf{B}$  and, as  $t$  grows, decreases along  $\mathbf{x}$ , and the endpoints are joined, so

$$a = \mathbf{D}(0)_x < \mathbf{D}(t_2)_x = \mathbf{x}(t_4)_x < \mathbf{x}(t_1)_x = \mathbf{B}(t_3)_x < \mathbf{B}(t_5)_x = b,$$

and the abscissa of every point of  $\Gamma$  lies in  $[a, b]$ . The ordinate of  $\mathbf{D}$  is nondecreasing, since  $\mathbf{D}'$  is a positive multiple of  $u_t$  and  $\sin t \geq 0$ ; it starts at 0 and ends at  $\mathbf{x}(t_4)_y < 1$ . The ordinate of  $\mathbf{B}$  is nonincreasing, since  $\mathbf{B}'$  is a negative multiple of  $v_t$  and  $\cos t \geq 0$ ; it ends at 0 and starts at  $\mathbf{x}(t_1)_y < 1$ . The ordinate of  $\mathbf{x}$  on  $[t_1, t_4]$  lies in  $(0, 1)$  by Fact 2.18. So the heights of  $\Gamma$  lie in  $[0, 1)$ .

The points  $(a, 1) = \mathbf{D}(0) + v_0 = \mathbf{C}(0)$  and  $(b, 1) = \mathbf{B}(\pi/2) + u_{\pi/2} = \mathbf{A}(\pi/2)$  lie in  $K_G$ , since  $\mathbf{A} = A_{K_G}$  and  $\mathbf{C} = C_{K_G}$  on  $[0, \pi/2]$  (Fact 2.17(b)). As  $K_G$  is convex, it contains the segment from  $(a, 1)$  to  $(b, 1)$ . A cap  $K$  with rotation angle  $\pi/2$  is closed downward above the floor: if  $p \in K$  and  $0 \leq y \leq p_y$ , then  $(p_x, y) \in K$ . Indeed,  $K$  is an intersection of closed half-planes with normal angles in  $[0, \pi] \cup \{3\pi/2\}$ . Lowering  $p$  does not increase  $\langle p, u_s \rangle$  for  $s \in [0, \pi]$ , as  $\sin s \geq 0$ , and a closed half-plane with normal angle  $3\pi/2$  that contains  $K$  contains  $\{y \geq 0\}$ , as  $h_K(3\pi/2) = 0$ . Hence  $K_G \supset [a, b] \times [0, 1]$ . As  $a < b$ ,  $K_G$  is a closed convex set with nonempty interior, so  $K_G = \text{cl}(\text{int } K_G)$ .

By Fact 2.17(b) and (c),  $G = K_G \setminus \mathcal{N}(K_G) = K_G \setminus \Gamma_{<}$ , and  $G$  is closed, being a moving sofa (Fact 2.17(b), Fact 2.4(a)). So Lemma 9.2 applies with  $K = K_G$ .  $\square$

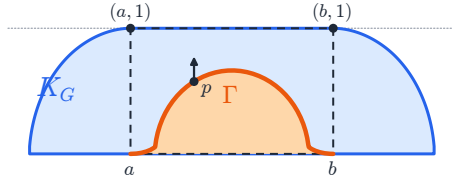


FIGURE 9. Proposition 9.3. Gerver's cap  $K_G$  with its niche (orange) below the curve  $\Gamma$  formed by  $\mathbf{D}$ ,  $\mathbf{x}$  and  $\mathbf{B}$ ; the dashed rectangle  $[a, b] \times [0, 1]$  lies in  $K_G$ , and  $\Gamma$  stays below height one (Fact 2.18). Every point  $p$  of  $\Gamma$  is a limit of points of  $G$  above it.

*Proof of Theorem 1.1.* Let  $S$  be a moving sofa with  $|S| = |G|$ . By Proposition 3.2 and Proposition 6.9, there are  $\omega \in [\omega_0, \pi/2]$  and  $w_0, w_1 \in \mathbb{R}^2$  such that, with  $\psi = \pi/2 - \omega$  and  $g_1(p) = R_\psi(p + w_0) + w_1$ , (6.1) holds:  $g_1(S) \subset U$  for a monotone sofa  $U$  with rotation angle  $\pi/2$  and  $|U| = |G|$ . By Theorem 8.5, there is  $s \in \mathbb{R}$  with

$$g_1(S) \subset U = G + (s, 0).$$

Let  $g(p) = g_1(p) - (s, 0) = R_\psi p + v$  with  $v = R_\psi w_0 + w_1 - (s, 0)$ . Then  $g(S) \subset G$ , and  $\psi \in [0, \pi/2 - \omega_0]$ . The set  $g(S)$  is closed, since  $S$  is closed and  $g$  is a homeomorphism, and  $|g(S)| = |S| = |G|$ , since rotations and translations preserve area; and  $|G| < \infty$  by Fact 2.17(a). By Proposition 9.3,  $G = \text{cl}(\text{int } G)$ , so  $g(S) = G$  by Lemma 9.1, that is,  $R_\psi S + v = G$ . This is the assertion with  $\theta = \psi$ .  $\square$

**Corollary 9.4** (the maximal sofas). *A moving sofa  $S$  has maximal area, that is,  $|S'| \leq |S|$  for every moving sofa  $S'$ , if and only if  $R_\theta S + v = G$  for some  $\theta \in \mathbb{R}$  and  $v \in \mathbb{R}^2$ .*

*Proof.* By Fact 2.8,  $|S'| \leq |G|$  for every moving sofa  $S'$ , and  $G$  is a moving sofa (Fact 2.17(b), Fact 2.4(a)). If  $S$  has maximal area, then  $|G| \leq |S| \leq |G|$ , and Theorem 1.1 gives  $\theta$  and  $v$ . Conversely, if  $R_\theta S + v = G$ , then  $|S| = |G| \geq |S'|$  for every moving sofa  $S'$ , since rotations and translations preserve area.  $\square$

**Lemma 9.5** (the widths of Gerver's sofa). (a) *There are  $a, b, x_- \in \mathbb{R}$  with  $a < 0$  and  $b - x_- > 1$  such that the points  $(a, 1)$ ,  $(b, 1)$ ,  $(x_-, 0)$  and  $(1, 0)$  lie in  $G$ .*

(b) *For every  $r \in [0, \pi]$  with  $r \neq \pi/2$  there are  $p, q \in G$  with  $\langle p - q, u_r \rangle > 1$ .*

(c) *If  $w \in \mathbb{R}^2$  is a unit vector and  $\langle p - q, w \rangle \leq 1$  for all  $p, q \in G$ , then  $w_x = 0$ .*

*Proof.* (a) Let  $a = \mathbf{D}(0)_x$ ,  $b = \mathbf{B}(\pi/2)_x$  and  $x_- = \mathbf{x}(\pi/2)_x - 1$ , and let  $\Gamma$  be as in the proof of Proposition 9.3. A point  $q \in \Gamma_<$  lies strictly below a point  $\gamma \in \Gamma$  with  $\gamma_x = q_x$ , so  $\gamma_y > q_y \geq 0$ . Write  $\gamma = \mathbf{D}(t)$  with  $t \in [t_0, t_2]$ ,  $\gamma = \mathbf{x}(t)$  with  $t \in [t_1, t_4]$ , or  $\gamma = \mathbf{B}(t)$  with  $t \in [t_3, t_5]$ . As  $\mathbf{D}(t_0)$  and  $\mathbf{B}(t_5)$  lie on the  $x$ -axis,  $t \neq t_0$  in the first case and  $t \neq t_5$  in the third. By the strict monotonicity of the abscissa along  $\mathbf{D}$  on  $[t_0, t_2]$ , along  $\mathbf{x}$  on  $[t_1, t_4]$  and along  $\mathbf{B}$  on  $[t_3, t_5]$  (Fact 2.17(c)), and the chain  $a = \mathbf{D}(0)_x < \mathbf{D}(t_2)_x = \mathbf{x}(t_4)_x < \mathbf{x}(t_1)_x = \mathbf{B}(t_3)_x < b$  of the proof of Proposition 9.3,  $q_x = \gamma_x \in (a, b)$ . As  $G = K_G \setminus \mathcal{N}(K_G) = K_G \setminus \Gamma_<$  (Fact 2.17(b), (c)), every point of  $K_G$  with abscissa outside  $(a, b)$  lies in  $G$ .

The points  $(a, 1) = \mathbf{C}(0)$  and  $(b, 1) = \mathbf{A}(\pi/2)$  lie in  $K_G$ , as in the proof of Proposition 9.3. So do  $\mathbf{A}(0) = (1, 0)$  (Appendix B, condition (2)) and  $\mathbf{C}(\pi/2) = (x_-, 0)$  (Fact 2.17(d)), as  $\mathbf{A} = A_{K_G}$  and  $\mathbf{C} = C_{K_G}$  on  $[0, \pi/2]$  (Fact 2.17(b)). By the formula for the inner corner (Section 2.2), applied to  $\mathbf{x} = \mathbf{x}_{K_G}$  (Fact 2.17(b)) at  $t = 0$  and  $t = \pi/2$ ,  $h_{K_G}(0) = \langle \mathbf{x}(0), u_0 \rangle + 1 = 1$ , as  $\mathbf{x}(0) = O$  (Appendix B, condition (2)), and  $h_{K_G}(\pi) = \langle \mathbf{x}(\pi/2), v_{\pi/2} \rangle + 1 = -x_-$ . So every point  $p$  of  $K_G$  has  $p_x = \langle p, u_0 \rangle \leq h_{K_G}(0) = 1$  and  $-p_x = \langle p, u_\pi \rangle \leq h_{K_G}(\pi) = -x_-$ ; in particular  $x_- \leq a$  and  $b \leq 1$ . Hence none of  $a, b, x_-$  and  $1$  lies in  $(a, b)$ , and the four points lie in  $G$ .

By condition (3) of the injectivity condition, which  $K_G$  satisfies with  $\mathbf{x}_{K_G} = \mathbf{x}$  (Fact 2.17(b)),  $\alpha < 0 < \beta$  on  $(0, \pi/2)$ . So the abscissa of  $\mathbf{x}$ , whose derivative is  $\alpha \cos t - \beta \sin t$ , is decreasing on  $[0, \pi/2]$ . With the chain above,  $a < \mathbf{x}(t_4)_x < \mathbf{x}(t_1)_x \leq \mathbf{x}(0)_x = 0$  and  $b > \mathbf{x}(t_1)_x > \mathbf{x}(t_4)_x \geq \mathbf{x}(\pi/2)_x = x_- + 1$ .

(b) Let  $a, b$  and  $x_-$  be as in (a). Since  $0 \leq \sin r \leq 1$ ,  $\sin r \geq \sin^2 r$ , and likewise  $|\cos r| \geq \cos^2 r$ ; so  $|\cos r| + \sin r \geq 1$ . For  $r \in [0, \pi/2)$ ,  $\cos r > 0$  and  $\langle (b, 1) - (x_-, 0), u_r \rangle = (b - x_-) \cos r + \sin r > \cos r + \sin r \geq 1$ . For  $r \in (\pi/2, \pi]$ ,  $\cos r < 0$  and  $\langle (a, 1) - (1, 0), u_r \rangle = (1 - a)|\cos r| + \sin r > |\cos r| + \sin r \geq 1$ . These four points lie in  $G$  by (a).

(c) Suppose first that  $w_y \geq 0$ . Then  $w = u_r$  with  $r = \arccos w_x \in [0, \pi]$ , and if  $w_x \neq 0$ , then  $r \neq \pi/2$ , and (b) contradicts the hypothesis. If  $w_y < 0$ , the same applies to  $-w$ , since  $\langle p - q, -w \rangle = \langle q - p, w \rangle \leq 1$  for  $p, q \in G$ .  $\square$

**Corollary 9.6** (no rotation is needed). (a) *If  $S$  is a moving sofa with  $|S| = |G|$ , then  $S + v = G$  for some  $v \in \mathbb{R}^2$ , so that  $S$  is a translate of  $G$ ; in particular,  $S$  is a moving sofa with rotation angle  $\pi/2$ .*

(b) *If  $\theta \in \mathbb{R}$  and  $v \in \mathbb{R}^2$  satisfy  $R_\theta G + v \subset H_L$ , then  $\sin \theta = 0$ . So  $R_\theta G$  is a moving sofa only if  $\theta$  is a multiple of  $\pi$ ; in particular,  $R_{\pi/2} G$  is not a moving sofa.*

*Proof.* (a) By the proof of Theorem 1.1, there are  $\psi \in [0, \pi/2 - \omega_0]$  and  $v \in \mathbb{R}^2$  such that  $g(p) = R_\psi p + v$  maps  $S$  onto  $G$ . A moving sofa lies in a horizontal strip of height one: the start position  $\Phi_0(S)$  of a movement of  $S$  is a translate of  $S$  and lies in  $H_L$  (Definition 2.2), so  $p_y - q_y \leq 1$  for  $p, q \in S$ . For  $p, q \in S$ ,

$$\langle g(p) - g(q), v_\psi \rangle = \langle R_\psi(p - q), R_\psi v_0 \rangle = \langle p - q, v_0 \rangle = p_y - q_y \leq 1,$$

so  $\langle x - y, u_{\psi + \pi/2} \rangle \leq 1$  for all  $x, y \in G = g(S)$ . If  $\psi > 0$ , then  $\psi + \pi/2 \in (\pi/2, \pi)$ , as  $\psi \leq \pi/2 - \omega_0 < \pi/2$ , and Lemma 9.5(b) contradicts this. So  $\psi = 0$ , and  $S + v = g(S) = G$ . As  $G$  is a moving sofa with rotation angle  $\pi/2$  (Fact 2.17(b), Fact 2.4(a)), so is its translate  $S = G - v$ .

(b) For  $p, q \in G$ , the points  $R_\theta p + v$  and  $R_\theta q + v$  of  $H_L$  have ordinates in  $[0, 1]$ , so the ordinate of  $R_\theta(p - q)$ , which is  $\langle p - q, w \rangle$  with the unit vector  $w = (\sin \theta, \cos \theta)$ , is at most 1. By Lemma 9.5(c),  $\sin \theta = w_x = 0$ . If  $R_\theta G$  is a moving sofa, the start position of a movement of it is a translate of  $R_\theta G$  that lies in  $H_L$  (Definition 2.2), so  $\sin \theta = 0$  by the first statement. As  $\sin(\pi/2) = 1$ ,  $R_{\pi/2} G$  is not a moving sofa.  $\square$

## 10. STABILITY

This section proves that a moving sofa whose area is close to  $|G|$  is, after a translation, close to  $G$ , at the rate of the square root of the missing area, and that this rate is optimal. As in Section 8,  $\varphi$  is Gerver's angle,  $K_G$  is Gerver's cap and  $\mathcal{A} = \mathcal{A}_{\pi/2}$ . The *deficit* of a set  $S$  of finite area is  $\varepsilon(S) = |G| - |S|$ , which is nonnegative for moving sofas by Baek's theorem (Fact 2.8); the proofs below take this inequality from Theorem 11.4(b), which proves it again in Section 11. Two sets  $X, Y \subset \mathbb{R}^2$  are *r-close* if every point of each lies within Euclidean distance  $r$  of a point of the other; for nonempty compact sets, this means that their Hausdorff distance is at most  $r$ . For a nonempty compact set  $S$ , let  $h_S(t) = \max_{p \in S} \langle p, u_t \rangle$ . The *normalized* translate  $S^\natural = S + (h_S(\pi) - h_G(\pi), 1 - h_S(\pi/2))$  has  $h_{S^\natural}(\pi/2) = 1$  and

$h_{S^\natural}(\pi) = h_G(\pi)$ , and  $S$  is *normalized* if  $S^\natural = S$ . Let  $X \triangle Y = (X \setminus Y) \cup (Y \setminus X)$  be the symmetric difference.

**Theorem 10.1** (stability of Gerver's sofa). *There are  $C, C', \varepsilon_0 > 0$  such that for every moving sofa  $S$  with  $\varepsilon = \varepsilon(S) < \varepsilon_0$ , the sets  $S^\natural$  and  $G$  are  $C\sqrt{\varepsilon}$ -close, and  $|S^\natural \triangle G| \leq C'\sqrt{\varepsilon}$ .*

**Theorem 10.2** (the rotation angle). *There are  $C'', \varepsilon_0 > 0$  such that every moving sofa  $S$  with rotation angle  $\omega \in [\omega_0, \pi/2]$  and  $\varepsilon(S) < \varepsilon_0$  satisfies  $0 \leq \pi/2 - \omega \leq C''\varepsilon(S)$ .*

A *rigid motion* is a map  $g(p) = R_\theta p + v$ . For nonempty compact sets  $X, Y$ ,  $d_{\text{rig}}(X, Y)$  is the infimum of the  $d \geq 0$  for which  $X$  and  $g(Y)$  are  $d$ -close for some rigid motion  $g$ . Let  $\mathbb{B}(p, r)$  and  $\mathbb{B}^\circ(p, r)$  be the closed and the open disk of radius  $r$  about  $p$ .

**Theorem 10.3** (the exponent  $1/2$  is optimal). (a) *There are a point  $p$  and  $r_0 > 0$  such that for every  $r \in (0, r_0)$  the set  $G_r = G \setminus \mathbb{B}^\circ(p, r)$  is a moving sofa with  $\varepsilon(G_r) = \pi r^2$  and  $d_{\text{rig}}(G_r, G) = r$ . Moreover,  $G_r$  and  $G$  are  $r$ -close, and  $r \leq d$  whenever  $G_r$  and  $g(G)$  are  $d$ -close for a rigid motion  $g$ .*

(b) *Let  $a > 1/2$ ,  $C \in \mathbb{R}$  and  $\varepsilon_0 > 0$ . There is a moving sofa  $S$  with  $0 < \varepsilon(S) < \varepsilon_0$  such that  $S$  and  $g(G)$  are  $C\varepsilon(S)^a$ -close for no rigid motion  $g$ .*

(c) *For the same  $a, C$  and  $\varepsilon_0$ , there is a moving sofa  $S'$  with  $0 < \varepsilon(S') < \varepsilon_0$  and  $C\varepsilon(S')^a < d_{\text{rig}}(S', G)$ .*

Baek's functional  $\mathcal{Q}$  is given by a formula [1, Def. 8.2.2]: for convex bodies  $K, B$  and  $D$ ,

$$\begin{aligned} \mathcal{Q}(K, B, D) = & |K| + \frac{1}{2} \int_{(3\pi/2, 3\pi/2 + \varphi^L)} h_D d\sigma_D + \mathcal{J}([Y_D, \mathbf{x}_K(\varphi^L)]) \\ & - \mathcal{J}(\mathbf{x}_K|_{[\varphi^R, \varphi^L]}) + \mathcal{J}([\mathbf{x}_K(\varphi^R), X_B]) + \frac{1}{2} \int_{(\pi + \varphi^R, 3\pi/2)} h_B d\sigma_B, \end{aligned} \quad (10.1)$$

where  $X_B = v_B^+(\pi + \varphi^R)$ ,  $Y_D = v_D^-(3\pi/2 + \varphi^L)$  and  $\mathcal{J}([p, q]) = \frac{1}{2} p \times q$ ; the integrals are the curve areas of two arcs of the boundaries of  $D$  and  $B$  [1, Thm. 7.3.2]. Let  $\bar{\mathcal{T}}$  be defined as  $\mathcal{T}$  (Section 2.5), with  $K \in \mathcal{K}_{\pi/2}^c$  in place of  $K \in \mathcal{K}^i$ ; by Lemma 10.6 below,  $\mathcal{Q} \leq |G|$  on  $\bar{\mathcal{T}}$ . For  $K \in \mathcal{K}_{\pi/2}^c$  let  $s_K = h_{K_G}(\pi) - h_K(\pi)$ , so that  $K$  and  $K_G + (s_K, 0)$  have the same leftmost abscissa.

**Theorem 10.4** (the cap estimate). (a) *For every  $\xi = (K, B, D) \in \bar{\mathcal{T}}$ , the caps  $K$  and  $K_G + (s_K, 0)$  are  $2 \sec \varphi \sqrt{|G| - \mathcal{Q}(\xi)}$ -close.*

(b) *For every  $K \in \mathcal{K}^i$ , the caps  $K$  and  $K_G + (s_K, 0)$  are  $2 \sec \varphi \sqrt{|G| - \mathcal{A}(K)}$ -close.*

(c)  $2 \sec \varphi \leq 2500/1249 < 1001/500$ , so (a) and (b) hold with  $1001/500$  in place of  $2 \sec \varphi$ .

Sections 10.1 and 10.2 bound, for every  $\xi = (K, B, D) \in \bar{\mathcal{T}}$ , the distance from  $K$  to  $K_G + (s_K, 0)$  by a multiple of  $\sqrt{|G| - \mathcal{Q}(\xi)}$ : with the constant 80 in Proposition 10.9 and Lemma 10.10, and with  $2 \sec \varphi$  in Theorem 10.4. The proofs of Theorems 10.1 and 10.2 use the bound with  $2 \sec \varphi$ , through Theorem 11.1. Near  $K_G$ , Baek's bound  $\mathcal{A}(K) \leq \mathcal{Q}(\xi_K)$  holds without the injectivity condition (Section 10.3), and a rotation angle  $\omega < \pi/2$  costs area proportional to  $\pi/2 - \omega$  (Section 10.4). Section 10.5 passes from the cap back to the sofa, and a compactness argument, which uses the optimality and uniqueness theorems of Section 11, brings the normalized translate of every moving sofa of small deficit near  $G$  (Section 10.6); this proves Theorems 10.1 and 10.2. Section 11.5 shows that this is not circular. Section 10.7 removes small disks from  $G$  to prove Theorem 10.3.

**10.1. The deficit of  $\mathcal{Q}$  on  $\bar{\mathcal{T}}$ .** Let  $\mathcal{D}$  be a set closed under the Minkowski combinations  $c_\lambda$  (Section 2.5). A functional  $\Psi$  on  $\mathcal{D}$  is *quadratic* if  $\Psi(x) = \beta(x, x)$  with  $\beta$  convex-linear in each argument; its *directional derivative*  $D\Psi(x; y)$  is the right derivative of  $\lambda \mapsto \Psi(c_\lambda(x, y))$  at 0 [1, Def. 7.1.4, Def. 7.1.5]. Baek shows that  $D\Psi(x; y) = \beta(x, y) + \beta(y, x) - 2\beta(x, x)$ , and that a quadratic concave  $\Psi$  is largest at  $x$  if and only if  $D\Psi(x; y) \leq 0$  for all  $y$  [1, Lemma 7.1.4, Thm. 7.1.5]. Let  $E_\Psi(x, y) = 4\Psi(c_{1/2}(x, y)) - 2\Psi(x) - 2\Psi(y)$ . As  $\beta$  is convex-linear in each argument,  $E_\Psi(x, y) = \beta(x, y) + \beta(y, x) - \beta(x, x) - \beta(y, y)$ , and Baek's formula for  $D\Psi$  gives

$$\Psi(x) - \Psi(y) = -D\Psi(x; y) + E_\Psi(x, y). \quad (10.2)$$

For a convex body  $K$  and a curve  $\mathbf{z}$  as in Section 2.5, we call  $\varrho_{K, \mathbf{z}}$  the *displacement* of  $\mathbf{z}$ . For convex bodies  $K_0, K_1$ , the *energy*  $E_S(K_0, K_1)$  is the sum, over the four terms  $\mathcal{M}_K(a, b; \mathbf{z}_K)$  of  $\mathcal{S}_K$ , of  $\frac{1}{2} \int_a^b (\varrho_{K_0, \mathbf{z}_{K_0}} - \varrho_{K_1, \mathbf{z}_{K_1}})^2 dt$ , and the energies  $E_{\mathcal{R}}(B_0, B_1)$  and  $E_{\mathcal{L}}(D_0, D_1)$  are defined likewise. As the curves of these terms are convex-linear in the body (Fact 2.16(b)), (8.1) gives, for all convex bodies and  $\lambda \in [0, 1]$ ,

$$(1 - \lambda)\mathcal{S}_{K_0} + \lambda\mathcal{S}_{K_1} - \mathcal{S}_{(1-\lambda)K_0 + \lambda K_1} = \lambda(1 - \lambda)E_S(K_0, K_1), \quad (10.3)$$

and likewise for  $\mathcal{R}$  and  $\mathcal{L}$ , which are therefore convex. Let  $I = [\varphi, \pi/2 - \varphi]$ .

**Lemma 10.5** ( $\mathcal{Q}$  on  $\bar{\mathcal{T}}$ ).  *$\bar{\mathcal{T}}$  is closed under Minkowski combinations. On  $\bar{\mathcal{T}}$ ,  $\mathcal{Q}(K, B, D) = \Lambda(K) - \mathcal{S}_K - \mathcal{R}_B - \mathcal{L}_D$ , where  $\Lambda$ , Baek's  $\mathcal{P}_K + \mathcal{S}_K$  (Fact 2.16(b)), is convex-linear on  $\mathcal{K}_{\pi/2}^c$ , and  $\mathcal{Q}$  is quadratic and concave.*

*Proof.* Support functions of Minkowski combinations are the same combinations of the support functions, so  $\bar{\mathcal{T}}$  is closed under them. Baek's proof of the decomposition in Fact 2.16(b) [1, Lemma 8.3.4] uses only the equalities at the ends of the intervals, so it holds on  $\bar{\mathcal{T}}$ . On  $\mathcal{K}_{\pi/2}^c$ ,  $|K|$  is the sum of the curve areas of the four arcs of  $\mathcal{S}_K$  and of the areas  $\frac{1}{2}\sigma_K(t)h_K(t)$  of the triangles spanned by  $O$  and the edges  $e_K(t)$ ,  $t = 0, \varphi, \pi/2 - \varphi, \pi/2, \pi$ . Each edge lies on the supporting line that carries the adjacent end segments  $[v_K^+(a), \mathbf{z}(a)]$  and  $[\mathbf{z}(b), v_K^-(b)]$  of the Mamikon terms. In  $\mathcal{S}_K + \mathcal{P}_K$  [1, Def. 8.3.4], the

curve areas of the arcs cancel against  $|K|$ , and what remains are  $\mathcal{J}(\mathbf{y}_K|_I)$ ,  $-\mathcal{J}(\mathbf{x}_K|_I)$ , the triangles of the edges and curve areas of segments on the five lines  $l_K(t)$ ,  $t = 0, \varphi, \pi/2 - \varphi, \pi/2, \pi$ . Joining the collinear pieces on each of these lines gives, with  $W_K$  and  $Z_K$  as in Section 6.1,

$$\begin{aligned} \Lambda(K) = h_K(0) + h_K(\pi) + \frac{2 \sin \varphi - h_K(\varphi) - h_K(\pi - \varphi)}{2 \cos \varphi} \\ + \mathcal{J}(\mathbf{y}_K|_I) - \mathcal{J}(\mathbf{x}_K|_I) + \varsigma_K, \end{aligned}$$

where

$$\begin{aligned} \varsigma_K = \mathcal{J}([I_K^{\pi/2}(\varphi), \mathbf{y}_K(\varphi)]) - \mathcal{J}([W_K(\varphi), \mathbf{x}_K(\varphi)]) \\ - \mathcal{J}([I_K^{\pi-\varphi}(\pi/2), \mathbf{y}_K(\pi/2 - \varphi)]) + \mathcal{J}([Z_K(\pi/2 - \varphi), \mathbf{x}_K(\pi/2 - \varphi)]). \end{aligned}$$

The differences  $\mathbf{y}_K - \mathbf{x}_K = u_t + v_t$ ,  $I_K^{\pi/2}(\varphi) - W_K(\varphi) = ((1 - \sin \varphi)/\cos \varphi, 1)$  and  $I_K^{\pi-\varphi}(\pi/2) - Z_K(\pi/2 - \varphi) = (-(1 - \sin \varphi)/\cos \varphi, 1)$  do not depend on  $K$ , so  $\Lambda$  is convex-linear [1, Lemma 7.1.6]. The terms of (10.1) are quadratic [1, Thm. 7.1.3, Thm. 7.3.2, Thm. 7.1.2, Prop. 7.2.2], so  $\mathcal{Q}$  is quadratic; as  $\Lambda$  is convex-linear and  $\mathcal{S}, \mathcal{R}, \mathcal{L}$  are convex by (10.3),  $\mathcal{Q} = \Lambda - \mathcal{S} - \mathcal{R} - \mathcal{L}$  is concave.  $\square$

**Lemma 10.6** (the directional derivative at Gerver's triple). *For every  $\xi \in \bar{\mathcal{T}}$ ,  $D\mathcal{Q}(\xi_G; \xi) \leq 0$ . Hence  $\mathcal{Q} \leq \mathcal{Q}(\xi_G) = |G|$  on  $\bar{\mathcal{T}}$ .*

*Proof.* Let  $\xi_G = (K_0, B_0, D_0)$  and  $\xi = (K, B, D)$ , let  $\check{\sigma}_X$  be the image of  $\sigma_X$  under  $t \mapsto t - \pi$ , and let  $\Delta = h_K - h_{K_0}$ ,  $\Delta_B(t) = h_B(t + \pi) - h_{B_0}(t + \pi)$  and  $\Delta_D(t) = h_D(t + \pi) - h_{D_0}(t + \pi)$ . Baek computes  $D\mathcal{Q}(\xi_G; \xi)$  for  $\xi \in \mathcal{T}$  [1, Thm. 8.5.6, Thm. 8.5.7], using  $X_{B_0} = \mathbf{x}_{K_0}(\varphi)$  and  $Y_{D_0} = \mathbf{x}_{K_0}(\varphi^L)$  [1, Thm. 8.4.3]. The computation differentiates only the curves of  $\xi_G$ , so it holds on  $\bar{\mathcal{T}}$ , and it gives

$$\begin{aligned} D\mathcal{Q}(\xi_G; \xi) = \int_{[0, \pi]} \Delta d\sigma_{K_0} - \int_{I \cup (I + \pi/2)} \Delta i_{K_0} dt \\ + \int_{(\varphi, \pi/2)} \Delta_B d\check{\sigma}_{B_0} + \int_{(\pi/2, \pi - \varphi)} \Delta_D d\check{\sigma}_{D_0}. \end{aligned}$$

The four terms are the directional derivatives of  $|K|$ , of  $-\mathcal{J}(\mathbf{x}_K|_I)$ , which is an integral of  $\Delta$  against a density  $i_{K_0}$ , of the terms of (10.1) that involve  $B$  and of those that involve  $D$  [1, Thm. 8.5.2, Thm. 8.5.3]. The first holds for all caps, as  $\Delta$  vanishes at  $3\pi/2$ , the only angle in  $(\pi, 2\pi)$  that carries  $\sigma_{K_0}$ .

Baek shows that, on  $[0, \pi]$ ,  $\sigma_{K_0}$  is the sum of the measure with density  $i_{K_0}$  on  $I \cup (I + \pi/2)$ , of  $\check{\sigma}_{B_0}$  on  $[\pi/2 - \theta, \pi/2)$ , of  $\check{\sigma}_{D_0}$  on  $(\pi/2, \pi/2 + \theta]$  and of the atom at  $\pi/2$  [1, Thm. 8.4.5], where  $\theta$  is Gerver's second angle (Section 2.6). Moreover,  $\check{\sigma}_{B_0}$  and  $\check{\sigma}_{D_0}$  vanish on  $(\varphi, \pi/2 - \theta)$  and  $(\pi/2 + \theta, \pi - \varphi)$ , and  $h_{K_0}(t) + h_{B_0}(t + \pi) = 1$  on  $[\pi/2 - \theta, \pi/2]$  and  $h_{K_0}(t) + h_{D_0}(t + \pi) = 1$  on  $[\pi/2, \pi/2 + \theta]$  [1, Thm. 8.4.3]. So the part of the first term against the density  $i_{K_0}$  cancels the second term, the atom contributes  $\Delta(\pi/2)\sigma_{K_0}(\{\pi/2\}) = 0$  as

$h_K(\pi/2) = h_{K_0}(\pi/2)$ , and the rest of the first term combines with the last two terms into

$$\begin{aligned} D\mathcal{Q}(\xi_G; \xi) &= \int_{[\frac{\pi}{2}-\theta, \frac{\pi}{2})} (h_K(t) + h_B(t + \pi) - 1) d\check{\sigma}_{B_0}(t) \\ &\quad + \int_{(\frac{\pi}{2}, \frac{\pi}{2}+\theta]} (h_K(t) + h_D(t + \pi) - 1) d\check{\sigma}_{D_0}(t), \end{aligned}$$

which is nonpositive by the inequalities defining  $\bar{\mathcal{T}}$ . By Lemma 10.5 and [1, Thm. 7.1.5],  $\mathcal{Q} \leq \mathcal{Q}(\xi_G)$  on  $\bar{\mathcal{T}}$ , and  $\mathcal{Q}(\xi_G) = |G|$  (Fact 2.16(c)).  $\square$

**Proposition 10.7** (the deficit certificate). *For every  $\xi = (K, B, D) \in \bar{\mathcal{T}}$ ,*

$$|G| - \mathcal{Q}(\xi) = -D\mathcal{Q}(\xi_G; \xi) + E_{\mathcal{S}}(K_G, K) + E_{\mathcal{R}}(B_{K_G}, B) + E_{\mathcal{L}}(D_{K_G}, D),$$

*and the four terms on the right are nonnegative; so  $E_{\mathcal{S}}(K_G, K) \leq |G| - \mathcal{Q}(\xi)$ .*

*Proof.* Apply (10.2) to  $\mathcal{Q}$ ,  $x = \xi_G$  and  $y = \xi$ . By Lemma 10.5, the convex-linear  $\Lambda$  drops out of  $E_{\mathcal{Q}}(\xi_G, \xi)$ , and (10.3) at  $\lambda = 1/2$  gives  $E_{\mathcal{Q}}(\xi_G, \xi) = E_{\mathcal{S}}(K_G, K) + E_{\mathcal{R}}(B_{K_G}, B) + E_{\mathcal{L}}(D_{K_G}, D)$ . The first term is nonnegative by Lemma 10.6, and the energies are integrals of squares.  $\square$

**10.2. From the energies to the cap; proof of Theorem 10.4.** For any function  $f$  with right derivatives and any  $T \in \mathbb{R}$ , the *tangent residual*  $r_T$  and the *corner residual*  $r_c$  of  $f$  are

$$r_T(t) = \frac{f(T) - f(t) \cos(T - t)}{\sin(T - t)} - f'(t^+), \quad r_c(t) = f(t + \pi/2) - f'(t^+).$$

The *four residuals* of  $f$  are  $r_1 = r_{\pi/2}$  on  $(0, \varphi)$ ,  $r_2 = r_c$  on  $(\varphi, \pi/2 - \varphi)$ ,  $r_3 = r_{\pi - \varphi}$  on  $(\pi/2 - \varphi, \pi/2)$  and  $r_4 = r_{\pi}$  on  $(\pi/2, \pi)$ , and  $E(f) = \frac{1}{2} \sum_j \int r_j^2$ , each integral over the interval of  $r_j$ . By the quotient rule,  $q_T(t) = (f(t) - f(T) \cos(T - t)) / \sin(T - t)$  has the right derivative  $-r_T(t) / \sin(T - t)$ . So, if  $f$  is continuous with integrable right derivatives on  $[s_1, s_2]$ , the fundamental theorem of calculus for right derivatives gives

$$\begin{aligned} q_T(s_1) - q_T(s_2) &= \int_{s_1}^{s_2} \frac{r_T(u)}{\sin(T - u)} du \quad \text{if } \sin(T - u) \neq 0 \text{ on } [s_1, s_2], \quad (10.4) \\ f(s_1) &= f(s_2) - \int_{s_1}^{s_2} f(u + \pi/2) du + \int_{s_1}^{s_2} r_c(u) du. \end{aligned}$$

For  $K_0, K_1 \in \mathcal{K}_{\pi/2}^c$  and  $\Delta = h_{K_1} - h_{K_0}$ , the function *formed with*  $K_0$  and  $K_1$  is  $f(t) = \Delta(t) + \Delta(\pi) \cos t$ . Then  $f = h_{K_1} - h_{K_0+(s,0)}$  with  $s = h_{K_0}(\pi) - h_{K_1}(\pi)$ ,  $f(\pi/2) = f(\pi) = 0$ , and  $f$  has at every  $t$  the right derivative  $f'(t^+) = \langle v_{K_1}^+(t) - v_{K_0}^+(t), v_t \rangle - \Delta(\pi) \sin t$  (Fact 2.1(a)).

**Lemma 10.8** (residuals and displacements). *Let  $K_0, K_1 \in \mathcal{K}_{\pi/2}^c$ , and let  $f$  be formed with them. On the open interval of each term  $\mathcal{M}_K(a, b; \mathbf{z}_K)$  of  $\mathcal{S}_K$ , the corresponding residual of  $f$  is  $\varrho_{K_1, \mathbf{z}_{K_1}} - \varrho_{K_0, \mathbf{z}_{K_0}}$ . The four residuals and their squares are integrable, and  $E(f) = E_{\mathcal{S}}(K_0, K_1)$ .*

*Proof.* For  $T - \pi < t < T$ , the point  $\mathbf{l}_K^T(t) = v_K(t, T)$  lies on  $l_K(t)$  and  $l_K(T)$ , and  $u_T = \cos(T - t)u_t + \sin(T - t)v_t$ , so  $\langle v_K(t, T), v_t \rangle = (h_K(T) - h_K(t) \cos(T - t)) / \sin(T - t)$ ; and  $\langle \mathbf{y}_K(t), v_t \rangle = h_K(t + \pi/2)$  (Section 2.2). Hence  $\varrho_{K, \mathbf{l}_K^T}(t) = (h_K(T) - h_K(t) \cos(T - t)) / \sin(T - t) - \langle v_K^+(t), v_t \rangle$  and  $\varrho_{K, \mathbf{y}_K}(t) = h_K(t + \pi/2) - \langle v_K^+(t), v_t \rangle$ . So  $\varrho_{K, \mathbf{l}_K^T}$  and  $\varrho_{K, \mathbf{y}_K}$  are the residuals  $r_T$  and  $r_c$  of  $h_K$ , whose right derivative is  $\langle v_K^+(t), v_t \rangle$  (Fact 2.1(a)). Residuals are linear in the function, and multiples of  $\cos t$ , whose right derivatives are the same multiples of  $-\sin t$ , have zero residuals; as  $f = h_{K_1} - h_{K_0} + \Delta(\pi) \cos t$ , the residuals of  $f$  are the differences of the displacements. By Mamikon's theorem [1, Thm. 7.4.1], the displacements are measurable and bounded, so the residuals and their squares are integrable.  $\square$

**Proposition 10.9** (the support bound). *Let  $0 < \varphi < \pi/4$ , not necessarily Gerver's angle, and form the four residuals with this  $\varphi$ . Let  $f$  be continuous, with right derivatives on  $(0, \pi)$ ,  $f(\pi/2) = f(\pi) = 0$ , and four residuals that are integrable and square integrable. For  $t \in [0, \pi]$ :*

- (a)  $|f(t)| \leq 20 \sum_{j=1}^4 \int |r_j| \leq 80 \sqrt{E(f)}$ ;
- (b)  $f(t)^2 \leq \mathcal{G}_\varphi(t) \sum_j \int r_j^2 \leq 2 \sec^2 \varphi \sum_j \int r_j^2$ , so  $|f(t)| \leq 2 \sec \varphi \sqrt{E(f)}$ , where  $\mathcal{G}_\varphi(t)$  is  $\cos^2 t (2 \sec^2 \varphi - \tan t)$  for  $t \leq \varphi$ ,  $\cos t (2 \sec \varphi - \sin t)$  for  $\varphi < t \leq \pi/2 - \varphi$ ,  $\sin t \cos t + 2 \tan \varphi \cos^2 t$  for  $\pi/2 - \varphi < t \leq \pi/2$ , and  $-\sin t \cos t$  for  $t > \pi/2$ .

For Gerver's angle  $\varphi$ , if  $\xi = (K, B, D) \in \bar{\mathcal{T}}$  and  $f$  is formed with  $K_G$  and  $K$ , then  $|f| \leq 80 \sqrt{|G| - \mathcal{Q}(\xi)}$  on  $[0, \pi]$ .

*Proof.* Let  $b' = \pi/2 - \varphi$  and  $T = \pi - \varphi$ . As  $f(\pi/2) = f(\pi) = 0$ , (10.4) gives

$$\begin{aligned} f(t) &= -\sin t \int_{\pi/2}^t \frac{r_4(u)}{\sin u} du, & t \in [\pi/2, \pi), \\ f(t) &= -\frac{\cos t}{\cos \varphi} f(T) + \sin(T - t) \int_t^{\pi/2} \frac{r_3(u) du}{\sin(T - u)}, & t \in [b', \pi/2], \\ f(t) &= f(b') - \int_t^{b'} f(u + \frac{\pi}{2}) du + \int_t^{b'} r_2, & t \in [\varphi, b'], \\ f(t) &= \cos t \left( \frac{f(\varphi)}{\cos \varphi} + \int_t^\varphi \frac{r_1(u)}{\cos u} du \right), & t \in [0, \varphi]. \end{aligned}$$

(a) Let  $m_j = \int |r_j|$  and  $M = \sum_j m_j$ . On  $[\pi/2, \pi]$ , the first formula gives  $|f| \leq m_4$ , as  $0 < \sin t \leq \sin u$  for  $\pi/2 \leq u \leq t$ . On  $[b', \pi/2]$ , the second gives  $|f(t)| \leq 3|f(T)| + 2m_3 \leq 5M$ , as  $|\cos t| \leq \cos \varphi$ ,  $\sin(T - u) \geq 1/2$  and  $|f(T)| \leq m_4$ . On  $[\varphi, b']$ , the third gives  $|f| \leq 5M + 2m_4 + m_2 \leq 8M$ , as  $|f(u + \pi/2)| \leq m_4$  and  $b' - \varphi \leq 2$ . On  $[0, \varphi]$ , the fourth gives  $|f| \leq 2|f(\varphi)| + 2m_1 \leq 18M \leq 20M$ , as  $\cos u \geq \cos \varphi \geq 1/2$ . The intervals have length at most 2, so  $m_j^2 \leq 2 \int r_j^2$  by the Cauchy-Schwarz inequality, and  $M^2 \leq 4 \sum_j m_j^2 \leq 8 \sum_j \int r_j^2 = 16E(f)$ .

(b) Let  $k(u) = (\sec \varphi + \cos u) / \sin u$ . On  $(\pi/2, \pi)$ ,  $r_4 = f \cot u - f'$ , so  $(kf)' = -f - k r_4$ ; integrating this, and inserting the first two formulas, turns

the third into

$$f(t) = \int_t^{b'} r_2 + \int_{b'}^{\pi/2} \frac{r_3(u)du}{\sin(T-u)} + (\sec \varphi - \sin t) \int_{\pi/2}^{\pi/2+t} \frac{r_4(u)du}{\sin u} + \int_{\pi/2+t}^T k r_4,$$

the terms with  $f(T)$  cancelling as  $k(T) = \tan \varphi$ . With  $f(\varphi)$  and  $f(T)$  inserted, each formula writes  $f(t)$  as a sum of integrals of kernels against the residuals over disjoint intervals. By the Cauchy–Schwarz inequality,  $f(t)^2$  is at most the sum of the squared  $L^2$  norms of the kernels times  $\sum_j \int r_j^2$ , and  $\int \csc^2 = -\cot$ ,  $\int \sec^2 = \tan$  and  $\int k^2 = -(\sec^2 \varphi + 1) \cot u - 2 \sec \varphi \csc u - u$  give this sum as  $\mathcal{G}_\varphi(t)$ ; on  $[\pi/2, \pi)$ , for instance, it is  $\sin^2 t (\cot(\pi/2) - \cot t) = -\sin t \cos t$ . Finally  $\mathcal{G}_\varphi \leq 2 \sec^2 \varphi$ , as  $\tan t \geq 0$  on  $[0, \varphi]$ ,  $\cos t (2 \sec \varphi - \sin t) \leq 2 \sec \varphi \leq 2 \sec^2 \varphi$ ,  $|\sin t \cos t| \leq 1/2$  and  $1/2 + 2 \tan \varphi \leq 2 + 2 \tan^2 \varphi$ .

For the last claim, Lemma 10.8 gives the hypotheses and  $E(f) = E_S(K_G, K)$ , which is at most  $|G| - \mathcal{Q}(\xi)$  (Proposition 10.7).  $\square$

**Lemma 10.10** (downward support values, and distances). *(a) For  $K \in \mathcal{K}_{\pi/2}^c$ ,  $h_K(t)$  is  $h_K(0) \cos t$  if  $\sin t \leq 0 \leq \cos t$ , and  $-h_K(\pi) \cos t$  if  $\sin t, \cos t \leq 0$ . So  $d_H(K, K') = \max_{[0, \pi]} |h_K - h_{K'}|$  for  $K, K' \in \mathcal{K}_{\pi/2}^c$ , and if  $f$  is formed with caps of  $\mathcal{K}_{\pi/2}^c$  and  $|f| \leq \rho$  on  $[0, \pi]$ , then  $|f| \leq \rho$  on  $\mathbb{R}$ .*

*(b) Convex bodies  $K, L$  with  $d_H(K, L) \leq r$  are  $r$ -close; and if nonempty compact sets  $X, Y$  are  $r$ -close, then  $|h_X - h_Y| \leq r$ .*

*Proof.* (a) Every  $p \in K$  has  $p_y \geq 0$  and  $-h_K(\pi) \leq p_x \leq h_K(0)$ , with equality at  $A_K^-(0) = (h_K(0), 0)$  and  $C_K^+(\pi/2) = (-h_K(\pi), 0)$  (Lemma 2.6). This gives  $h_K(t)$  for  $\sin t \leq 0$ , where  $f(t)$  is then  $f(0) \cos t$  or 0. (b) For  $p \in K$ ,  $\langle p, u_t \rangle \leq h_L(t) + r = h_{L+r\mathbb{B}(0,1)}(t)$  for all  $t$ , so  $p \in L + r\mathbb{B}(0, 1)$ , and symmetrically. If  $p \in X$  has  $\langle p, u_t \rangle = h_X(t)$  and  $q \in Y$  has  $\|p - q\| \leq r$ , then  $h_Y(t) \geq \langle q, u_t \rangle \geq h_X(t) - r$ , and symmetrically.  $\square$

*Proof of Theorem 10.4.* (a) Let  $f$  be formed with  $K_G$  and  $K$ , so that  $f = h_K - h_{K_G+(s_K, 0)}$ . By Lemma 10.8, Proposition 10.9(b) applies and gives  $|f| \leq 2 \sec \varphi \sqrt{E_S(K_G, K)}$  on  $[0, \pi]$ , and  $E_S(K_G, K) \leq |G| - \mathcal{Q}(\xi)$  by Proposition 10.7. By Lemma 10.10(a) the bound holds on  $\mathbb{R}$ , so  $d_H(K, K_G + (s_K, 0)) \leq 2 \sec \varphi \sqrt{|G| - \mathcal{Q}(\xi)}$ , and Lemma 10.10(b) gives the closeness. (b) For  $K \in \mathcal{K}^i$ ,  $\xi_K \in \mathcal{T} \subset \bar{\mathcal{T}}$  and  $\mathcal{A}(K) \leq \mathcal{Q}(\xi_K)$  (Fact 2.16(a)); so (a) applies to  $\xi_K$ , and  $|G| - \mathcal{Q}(\xi_K) \leq |G| - \mathcal{A}(K)$ . (c) As  $\varphi \leq 0.04$  (Fact 2.17(a)),  $\cos \varphi \geq 1 - \varphi^2/2 \geq 1249/1250$ .  $\square$

**10.3. The upper bound near Gerver's cap.** Baek proves  $\mathcal{A}(K) \leq \mathcal{Q}(\xi_K)$  for  $K \in \mathcal{K}^i$  [1, Thm. 8.2.4]; this subsection proves it for the caps of  $\mathcal{K}_{\pi/2}^c$  near  $K_G$ . If  $d_H(K, K') \leq \delta$ , then  $\|\mathbf{x}_K(t) - \mathbf{x}_{K'}(t)\| \leq 2\delta$ , by the formula for the inner corner (Section 2.2). Let  $\Gamma$  be as in Fact 2.17(c), and let  $a_\Gamma = \mathbf{D}(0)_x$  and  $b_\Gamma = \mathbf{B}(\pi/2)_x$ , the numbers  $a < b$  of the proof of Proposition 9.3. The top edge  $e_{K_G}(\pi/2)$  of  $K_G$  is the segment from  $C_{K_G}^+(0) = \mathbf{C}(0) = (a_\Gamma, 1)$  to  $A_{K_G}^-(\pi/2) = \mathbf{A}(\pi/2) = (b_\Gamma, 1)$  (Fact 2.17(b) and the proof of Proposition 9.3).

Recall that the width of a cap  $K$  is  $h_K(0) + h_K(\pi)$ ; its *cut half-planes* are  $\check{H}_K^R = H_K^b(\varphi)$  and  $\check{H}_K^L = H_K^d(\pi/2 - \varphi)$  [1, Def. 8.1.5].

**Proposition 10.11** (the upper bound near  $K_G$ ). *There is  $\delta \in (0, 1]$  such that every  $K \in \mathcal{K}_{\pi/2}^c$  with  $d_H(K, K_G) \leq \delta$  satisfies  $\xi_K = (K, B_K, D_K) \in \bar{\mathcal{T}}$ ,  $\mathcal{N}(K) \subset K$  and  $\mathcal{A}(K) \leq \mathcal{Q}(\xi_K) \leq |G|$ , and  $K$  and  $K_G + (s_K, 0)$  are  $2 \sec \varphi \sqrt{|G| - \mathcal{A}(K)}$ -close.*

**Lemma 10.12** (Gerver's roof). *There are  $\bar{\gamma} < 1$ ,  $L \geq 0$  and an  $L$ -Lipschitz  $\gamma : [a_\Gamma, b_\Gamma] \rightarrow [0, \bar{\gamma}]$  with  $\gamma(a_\Gamma) = \gamma(b_\Gamma) = 0$  and graph  $\Gamma$ , such that  $\mathcal{N}(K_G) = \{(x, y) : x \in [a_\Gamma, b_\Gamma], 0 \leq y < \gamma(x)\}$ . Moreover  $-h_{K_G}(\pi) < a_\Gamma < b_\Gamma < h_{K_G}(0)$  and  $[a_\Gamma, b_\Gamma] \times [0, 1] \subset K_G$ .*

*Proof.* The chords of  $\Gamma$  have bounded slopes: on  $[t_0, t_2]$ ,  $\mathbf{D}'$  is a positive multiple of  $u_t$  (Fact 2.17(c)) and  $\cos t \geq 1/2$ ; likewise for  $\mathbf{B}$  on  $[t_3, t_5]$ ; and on  $[t_1, t_4]$  the abscissa of  $\mathbf{x}$  has the derivative  $\alpha \cos t - \beta \sin t$ , which is negative (Lemma 9.5), hence bounded away from 0. As the three pieces occupy consecutive intervals of abscissas and meet at their ends,  $\Gamma$  is the graph of an  $L$ -Lipschitz  $\gamma$  on  $[a_\Gamma, b_\Gamma]$ , with  $\gamma(a_\Gamma) = \mathbf{D}(0)_y = 0$  and  $\gamma(b_\Gamma) = \mathbf{B}(\pi/2)_y = 0$ . The proof of Proposition 9.3 gives the heights in  $[0, 1)$ , with a largest,  $\bar{\gamma}$ , and  $[a_\Gamma, b_\Gamma] \times [0, 1] \subset K_G$ ; the niche is that of Fact 2.17(c). As  $\sigma_{K_G}$  has no atom at  $\pi$  (Fact 2.14(d)), the edge  $e_{K_G}(\pi)$  is the point  $(-h_{K_G}(\pi), 0)$  (Lemma 2.6). The point  $(a_\Gamma, 1)$  of  $K_G$  is not this point, so  $\langle (a_\Gamma, 1), u_\pi \rangle < h_{K_G}(\pi)$ , that is,  $-h_{K_G}(\pi) < a_\Gamma$ . Likewise  $b_\Gamma < h_{K_G}(0)$ .  $\square$

**Lemma 10.13** (persistence on edges). *Let  $K_0 \in \mathcal{K}_{\pi/2}^c$  and  $\mathbb{B} = \mathbb{B}(0, 1)$ .*

- (a) *If  $I' \subset [0, \pi]$  is compact and  $F$  is continuous on  $(K_0 + \mathbb{B}) \times I'$ , with  $F(p, t) > 0$  for  $t \in I'$  and  $p \in e_{K_0}(t)$ , then there is  $\delta \in (0, 1]$  such that  $F(p, t) > 0$  for all  $K \in \mathcal{K}_{\pi/2}^c$  with  $d_H(K, K_0) \leq \delta$ ,  $t \in I'$  and  $p \in e_K(t)$ .*
- (b) *If  $F$  is continuous on  $\mathbb{R}^2$  and  $F > 0$  on  $e_{K_0}(t_0)$ , then there are  $\delta, \rho > 0$  such that  $F(p) > 0$  for all such  $K$ ,  $t \in [0, \pi]$  with  $|t - t_0| \leq \rho$ , and  $p \in e_K(t)$ .*
- (c) *If  $K_0 \in \mathcal{K}^i$  and  $0 < s_1 \leq s_2 < \pi/2$ , there are  $c, \delta > 0$  such that every  $K \in \mathcal{K}_{\pi/2}^c$  with  $d_H(K, K_0) \leq \delta$  has, for  $t \in [s_1, s_2]$ ,  $h_K(t + \pi/2) - \langle p, v_t \rangle \geq 1 + c$  for  $p \in e_K(t)$  and  $h_K(t) - \langle p, u_t \rangle \geq 1 + c$  for  $p \in e_K(t + \pi/2)$ . So  $f_K^\pm, g_K^\pm \geq 1 + c$  on  $[s_1, s_2]$ , and there the abscissa of  $\mathbf{x}_K$  has right derivatives at most  $-c$ .*

*Proof.* (a) The function  $\Theta(p, t) = \text{dist}(p, K_0) + |\langle p, u_t \rangle - h_{K_0}(t)|$  vanishes exactly when  $p \in e_{K_0}(t)$ , so it has a positive minimum  $m$  on the compact set where  $F \leq 0$ ; let  $\delta = \min(1, m/4)$ . A cap  $K$  with  $d_H(K, K_0) \leq \delta$  lies in  $K_0 + \mathbb{B}$  and is  $\delta$ -close to  $K_0$  (Lemma 10.10), so  $\Theta(p, t) \leq 2\delta < m$  for  $p \in e_K(t)$ . (b) The proof of (a) applies to  $\Theta(p) = \text{dist}(p, K_0) + |\langle p, u_{t_0} \rangle - h_{K_0}(t_0)|$ , which is at most  $2\delta + 4R|t - t_0|$  for  $p \in e_K(t)$ , where  $R$  bounds the norms on  $K_0 + \mathbb{B}$ . (c) By condition (3) of the injectivity condition and Fact 2.14(d),  $f_{K_0}$  and  $g_{K_0}$  are continuous and exceed 1 on  $(0, \pi/2)$ , hence exceed  $1 + m$  on  $[s_1, s_2]$  for

some  $m > 0$ , and  $e_{K_0}(t)$ ,  $e_{K_0}(t + \pi/2)$  are the points  $A_{K_0}(t)$ ,  $C_{K_0}(t)$ . Part (a) for  $F(p, t) = h_{K_0}(t + \pi/2) - \langle p, v_t \rangle - 1 - m/2$  and its analogue, with  $\delta \leq m/4$ , gives  $c = m/4$ . The right derivative of  $\mathbf{x}_K$  is  $-(f_K^+ - 1)u_t + (g_K^+ - 1)v_t$  (Fact 2.14(a)), with abscissa at most  $-c(\cos t + \sin t) \leq -c$ .  $\square$

**Lemma 10.14** (the niche of a nearby cap). *(a) For every  $\eta > 0$  there is  $\delta \in (0, 1]$  such that every  $K \in \mathcal{K}_{\pi/2}^c$  with  $d_H(K, K_G) \leq \delta$  has  $a_\Gamma - \eta \leq p_x \leq b_\Gamma + \eta$  for all  $p \in \mathcal{N}(K)$ .  
(b) There is  $\delta \in (0, 1]$  such that  $\mathcal{N}(K) \subset K$  for every  $K \in \mathcal{K}_{\pi/2}^c$  with  $d_H(K, K_G) \leq \delta$ .  
(c) Let  $K \in \mathcal{K}_{\pi/2}^c$ . If  $d_H(K, K_G) \leq 1/20$ , then  $K$  has width at least  $21/10$ . If  $K$  has width at least  $21/10$ , then for every  $\varphi \in [0.039, 0.04]$ , Gerver's angle among them,  $W_K(\varphi)$  and  $Z_K(\pi/2 - \varphi)$  lie on the bottom edge of  $K$ , strictly between its ends, and  $K \cap H_K^b(\varphi) \cap H_K^d(\pi/2 - \varphi) = \emptyset$ .*

*Proof.* (a) A point  $p \in \mathcal{N}(K)$  has  $p_y \geq 0$  and lies in  $Q_K^-(t)$  for some  $t \in (0, \pi/2)$ , so  $Z_K(t)_x < p_x < W_K(t)_x$ , where  $Z_K(t)_x = (1 - h_K(t + \pi/2))/\sin t$  and  $W_K(t)_x = (h_K(t) - 1)/\cos t$ . For  $K_G$ ,  $\mathbf{x}(t)_y = \sin t \cos t (W_{K_G}(t)_x - Z_{K_G}(t)_x)$  is positive (by Fact 2.18 on  $[t_1, t_4]$ , and elsewhere as  $\mathbf{x}(t)$  lies above  $\mathbf{D}(t)$  or  $\mathbf{B}(t)$ ), so the open segment of the  $x$ -axis between  $Z_{K_G}(t)$  and  $W_{K_G}(t)$  lies in  $\mathcal{N}(K_G)$ , and  $a_\Gamma \leq Z_{K_G}(t)_x$  and  $W_{K_G}(t)_x \leq b_\Gamma$  (Lemma 10.12). As the top edge of  $K_G$  is  $[(a_\Gamma, 1), (b_\Gamma, 1)]$ , Lemma 10.13(b) at  $t_0 = \pi/2$  gives  $\rho \in (0, \pi/4]$  and  $\delta_0$  such that, if  $\delta \leq \delta_0$  and  $t \leq \rho$ , each  $q \in e_K(t + \pi/2)$  has  $q_x > a_\Gamma - \eta$ , and  $Z_K(t)_x = q_x + (1 - q_y \cos t)/\sin t \geq q_x$ . For  $t > \rho$ ,  $|Z_K(t)_x - Z_{K_G}(t)_x| \leq \delta/\sin \rho \leq \eta$  if  $\delta \leq \eta \sin \rho$ . So  $p_x > a_\Gamma - \eta$ , and symmetrically  $p_x < b_\Gamma + \eta$ .

(b) Let  $p \in \mathcal{N}(K)$ , so that  $p \in Q_K^-(t)$  for some  $t \in (0, \pi/2)$ . Let  $M' = \max \mathbf{x}_y < 1$  (Fact 2.18) and  $\delta \leq (1 - M')/4$ . As  $(0, 1) = u_t \sin t + v_t \cos t$ ,  $p_y < \mathbf{x}_K(t)_y \leq \mathbf{x}(t)_y + 2\delta \leq y' = (1 + M')/2$ . By compactness, there is  $m > 0$  with  $h_{K_G}(t) - \langle q, u_t \rangle \geq m$  for all  $q \in [a_\Gamma, b_\Gamma] \times [0, y']$  and  $t \in [0, \pi]$ , as  $(q_x, 1) \in K_G$  and  $-h_{K_G}(\pi) < a_\Gamma < b_\Gamma < h_{K_G}(0)$ . With  $\delta \leq m/4$  and (a) for  $\eta = m/4$ ,  $p$  lies within  $m/4$  of such a  $q$ , so  $\langle p, u_t \rangle \leq h_{K_G}(t) - 3m/4 \leq h_K(t)$  for  $t \in [0, \pi]$ , and  $p \in K$  by (2.2).

(c) By Lemma 10.10, a cap  $K' \in \mathcal{K}_{\pi/2}^c$  lies in  $[-h_{K'}(\pi), h_{K'}(0)] \times [0, 1]$ , so its area is at most its width. Hence  $K_G$  has width at least  $|K_G| \geq |G| > 11/5$  (Fact 2.17(a)), and  $K$ , whose support values differ from those of  $K_G$  by at most  $1/20$ , has width at least  $11/5 - 2/20 = 21/10$ . As  $h_K(\varphi) \geq h_K(0) \cos \varphi$  and  $(21/10) \cos \varphi > 1$ ,  $W_K(\varphi)_x > -h_K(\pi)$ ; and  $W_K(\varphi)_x < h_K(0)$ , as  $h_K(\varphi) \leq h_K(0) \cos \varphi + h_K(\pi/2) \sin \varphi < h_K(0) \cos \varphi + 1$  by the sublinearity of  $h_K$ . Likewise for  $Z_K(\pi/2 - \varphi)$ . A point  $p$  of  $K$  in  $H_K^b(\varphi) \cap H_K^d(\pi/2 - \varphi)$  would satisfy

$$2 \sin \varphi \geq \langle p, u_\varphi + u_{\pi-\varphi} \rangle \geq h_K(\varphi) + h_K(\pi - \varphi) - 2 \geq (21/10) \cos \varphi - 2,$$

where the first inequality holds as  $u_\varphi + u_{\pi-\varphi} = (0, 2 \sin \varphi)$  and  $p_y \leq 1$ , the second by the definition of the two half-planes, and the third as  $h_K(\varphi) \geq h_K(0) \cos \varphi$  and  $h_K(\pi - \varphi) \geq h_K(\pi) \cos \varphi$ ; as  $\varphi \leq 0.04$ , this is false.  $\square$

**Lemma 10.15** (the canonical triple and the cuts). *There is  $\delta \in (0, 1]$  such that every  $K \in \mathcal{K}_{\pi/2}^c$  with  $d_H(K, K_G) \leq \delta$  satisfies:*

- (a)  $\xi_K \in \bar{\mathcal{T}}$ ;
- (b)  $\mathbf{x}_K(t) \notin \check{H}_K^R$  for  $t \in (\varphi, \pi/2]$ , and  $\mathbf{x}_K(t) \notin \check{H}_K^L$  for  $t \in [0, \pi/2 - \varphi]$ ;
- (c)  $\mathbf{x}_K(t)_y > 0$  for  $t \in [\varphi, \pi/2 - \varphi]$ .

*Proof.* Take  $c$  and  $\delta \leq 1/20$  from Lemma 10.13(c) for  $K_0 = K_G$  and  $[\varphi, \pi/2 - \varphi]$ ; so  $g_K^+(\varphi) > 1$ ,  $f_K^-(\pi/2 - \varphi) > 1$ , and  $K$  has width at least  $21/10$  (Lemma 10.14(c)). (a) The inequalities defining  $\bar{\mathcal{T}}$  hold for  $B_K$  and  $D_K$  by definition. For the equalities at the ends  $\pi/2$  and  $0$ ,  $B_K$  contains  $(h_K(0), 0)$  and  $D_K$  contains  $(-h_K(\pi), 0)$ : for  $t \in [\varphi, \pi/2]$ ,  $h_K(t) \leq h_K(0) \cos t + h_K(\pi/2) \sin t \leq h_K(0) \cos t + 1$  by the sublinearity of  $h_K$ , and likewise for  $D_K$ . For the equality at  $\varphi$  we follow Baek's proof of [1, Lemma 8.1.7]:  $b_K(\varphi)$  contains  $W_K(\varphi) \in K$  (Lemma 10.14), and the point  $p$  of  $K \cap b_K(\varphi)$  farthest in the direction  $v_\varphi$  lies in some  $e_K(\vartheta)$ ,  $\vartheta \in [\varphi, \varphi + \pi]$ . If  $\vartheta > \varphi + \pi/2$ , then  $\langle p, u_\varphi \rangle \leq \langle v_K^+(\varphi + \pi/2), u_\varphi \rangle$ , that is,  $g_K^+(\varphi) \leq 1$ . So  $\vartheta \leq \varphi + \pi/2$ , and the sublinearity of  $h_K$  between  $\varphi$ ,  $\vartheta$  and  $\pi/2$  (where  $h_K = 1 \geq p_y$ ) gives  $\langle p, u_t \rangle \geq h_K(t) - 1$  for  $t \in [\varphi, \pi/2]$ , so  $p \in B_K$ . The mirror argument with  $f_K^-(\pi/2 - \varphi) > 1$  treats  $D_K$ . (b) Let  $t_* = (\varphi + \pi/2)/2$ . For  $t \in (\varphi, t_*]$ , the right derivative of  $\langle \mathbf{x}_K, u_\varphi \rangle$  is  $-(f_K^+ - 1) \cos(t - \varphi) - (g_K^+ - 1) \sin(t - \varphi) \leq -c$ , so  $\langle \mathbf{x}_K(t), u_\varphi \rangle < \langle \mathbf{x}_K(\varphi), u_\varphi \rangle = h_K(\varphi) - 1$ . On  $[t_*, \pi/2]$ ,  $K_G$  satisfies the strict inequality  $\langle \mathbf{x}(t), u_\varphi \rangle < h_{K_G}(\varphi) - 1$  [1, Lemma 8.1.6] with a positive minimal gap, and this persists for small  $\delta$ , as  $\langle \mathbf{x}_K(t), u_\varphi \rangle$  and  $h_K(\varphi)$  differ from their values for  $K_G$  by at most  $2\delta$  and  $\delta$ . The left cut is symmetric. (c)  $\mathbf{x}(t)_y$  has a positive minimum  $m'$  on  $[\varphi, \pi/2 - \varphi]$  (Fact 2.18), and  $\mathbf{x}_K(t)_y \geq \mathbf{x}(t)_y - 2\delta > 0$  if  $\delta < m'/2$ .  $\square$

**Lemma 10.16** (the three regions of the niche). *Let  $\varphi$  be any number in  $[0.039, 0.04]$ , and form  $\bar{\mathcal{T}}$ ,  $\xi_K$ ,  $I$ ,  $\mathcal{Q}$  and the cut half-planes with it. Let  $K \in \mathcal{K}_{\pi/2}^c$  have width at least  $21/10$ ,  $\xi_K \in \bar{\mathcal{T}}$ ,  $\mathcal{N}(K) \subset K$ , the properties (b) and (c) of Lemma 10.15, and those of Lemma 10.13(c) on  $[\varphi, \pi/2 - \varphi]$  for some  $c > 0$ . Then  $\mathcal{A}(K) \leq \mathcal{Q}(\xi_K)$ .*

*Proof.* We follow Baek's proof, which bounds the areas of three regions of the niche [1, Lemma 8.2.2, Lemma 8.2.3, Thm. 8.2.4]. Write  $B = B_K$ ,  $D = D_K$ ,  $x^R = \mathbf{x}_K(\varphi)$ ,  $x^L = \mathbf{x}_K(\pi/2 - \varphi)$ ,  $W = W_K(\varphi)$  and  $Z = Z_K(\pi/2 - \varphi)$ . As  $\mathcal{N}(K) \subset K$ , Lemma 10.14(c) splits  $\mathcal{N}(K)$  into its parts in  $\check{H}_K^R$ , in  $\check{H}_K^L$  and outside both. For  $t \in (\varphi, \pi/2]$ ,  $\check{H}_K^R \cap Q_K^-(t) = \check{H}_K^R \setminus H_K^b(t)$ , as  $\mathbf{x}_K(t) \notin \check{H}_K^R$  and  $u_\varphi = \cos(t - \varphi)u_t - \sin(t - \varphi)v_t$ . So the points of  $K$  outside  $B$  in the supporting half-planes of  $B$  with normals  $\pi + \varphi$  and  $3\pi/2$  lie in  $\mathcal{N}(K) \cap \check{H}_K^R$ ; among them are the points of the region between the arc of  $B$  with normals in  $(\pi + \varphi, 3\pi/2)$  and its end tangents, which meet at  $W \in K$ . This region has

area  $\mathcal{J}([X_B, W]) - \frac{1}{2} \int_{(\pi+\varphi, 3\pi/2)} h_B d\sigma_B$  [1, Lemma 7.3.5], so  $|\mathcal{N}(K) \cap \check{H}_K^R|$  is at least this area. Likewise  $|\mathcal{N}(K) \cap \check{H}_K^L| \geq \mathcal{J}([Z, Y_D]) - \frac{1}{2} \int_{(3\pi/2, 3\pi/2+\varphi^L)} h_D d\sigma_D$ . By Lemma 10.13(c), the abscissa of  $\mathbf{x}_K$  decreases on  $I$  with right derivatives at most  $-c$ , so  $\mathbf{x}_K|_I$  is the graph of a Lipschitz function of the abscissa. The region  $\Omega$  between this graph and the  $x$ -axis and the triangles  $T^R$  with vertices  $(x_x^R, 0)$ ,  $W$ ,  $x^R$  and  $T^L$  with vertices  $Z$ ,  $(x_x^L, 0)$ ,  $x^L$  are disjoint, and lie in  $\mathcal{N}(K)$  outside the cut half-planes: points below  $\mathbf{x}_K(t)$  lie in  $Q_K^-(t)$ , and Lemma 10.15(b) applies. By changes of variables and integration by parts with right derivatives,

$$\begin{aligned} |\Omega \cup T^R \cup T^L| &= - \int_I (\mathbf{x}_K)_x' (\mathbf{x}_K)_y dt + \frac{1}{2} \tan \varphi ((x_y^R)^2 + (x_y^L)^2) \\ &= \mathcal{J}([W, x^R]) + \mathcal{J}(\mathbf{x}_K|_I) + \mathcal{J}([x^L, Z]). \end{aligned}$$

Adding the three bounds and joining the collinear points  $x^R, X_B, W$  on  $b_K(\varphi)$  and  $Y_D, x^L, Z$  on  $d_K(\pi/2 - \varphi)$  gives  $|\mathcal{N}(K)| \geq |K| - \mathcal{Q}(\xi_K)$ .  $\square$

*Proof of Proposition 10.11.* Let  $\delta$  be the least of  $1/20$  and of those of Lemma 10.15, Lemma 10.14(b) and Lemma 10.13(c) for  $K_0 = K_G$  and  $[\varphi, \pi/2 - \varphi]$ ; recall  $K_G \in \mathcal{K}^i$  (Fact 2.17(b)) and  $\varphi \in [0.039, 0.04]$  (Fact 2.17(a)). By Lemma 10.14(c),  $K$  has width at least  $21/10$ . Then Lemmas 10.14 to 10.16 give  $\xi_K \in \bar{\mathcal{T}}$ ,  $\mathcal{N}(K) \subset K$  and  $\mathcal{A}(K) \leq \mathcal{Q}(\xi_K)$ . Lemma 10.6 and Theorem 10.4(a), stated together as Theorem 11.1, give  $\mathcal{Q}(\xi_K) \leq |G|$  and show that  $K$  and  $K_G + (s_K, 0)$  are  $2 \sec \varphi \sqrt{|G| - \mathcal{Q}(\xi_K)}$ -close, hence  $2 \sec \varphi \sqrt{|G| - \mathcal{A}(K)}$ -close, as  $\mathcal{A}(K) \leq \mathcal{Q}(\xi_K)$ .  $\square$

**10.4. The cost of a missing final angle.** For  $K \in \mathcal{K}_{\pi/2}^c$  and  $t \in \mathbb{R}$ , the coordinates of  $p - \mathbf{x}_K(t)$  in the frame  $(u_t, v_t)$  are  $\ell_K^b(t, p) = \langle p, u_t \rangle - h_K(t) + 1$  and  $\ell_K^d(t, p) = \langle p, v_t \rangle - h_K(t + \pi/2) + 1$ ; we write  $\ell_K(t, p)$  for the pair  $(\ell_K^b(t, p), \ell_K^d(t, p))$ . Then  $Q_K^-(t)$  is the set where both coordinates are negative. For  $\omega \in [0, \pi/2]$ , let  $\mathcal{N}_\omega(K)$  be the set of  $p$  with  $p_y \geq 0$  and  $p \in Q_K^-(t)$  for some  $t \in (0, \omega)$ , so that  $\mathcal{N}_{\pi/2}(K) = \mathcal{N}(K)$ , and let  $O_\omega(K) = (K \setminus \mathcal{N}_\omega(K)) \setminus (K \setminus \mathcal{N}(K))$  be the *omitted wedges*. A set  $S$  satisfies the constraints of  $(K, \omega)$  if  $S \subset K$ ,  $\max(\ell_K^b(t, p), \ell_K^d(t, p)) \geq 0$  for  $p \in S$  and  $t \in [0, \omega]$ , and  $\ell_K^b(\omega, p) \geq 0$  for  $p \in S$ .

**Proposition 10.17** (the terminal comparison). *There are  $c > 0$ ,  $\delta \in (0, 1]$ ,  $\psi_0 > 0$  and  $R \geq 1$  such that for every  $K \in \mathcal{K}_{\pi/2}^c$  with  $d_H(K, K_G) \leq \delta$ , every  $\omega \in [0, \pi/2]$  with  $\psi = \pi/2 - \omega \leq \psi_0$  and every measurable set  $S$  that satisfies the constraints of  $(K, \omega)$ , with  $\varepsilon = |G| - |S|$  and  $U = K \setminus \mathcal{N}(K)$ :*

- (a)  $|S| \leq \mathcal{A}(K) - c\psi$ ,  $\psi \leq \varepsilon/c$  and  $|G| - \mathcal{A}(K) \leq \varepsilon$ ;
- (b)  $|S \setminus U| \leq \varepsilon$  and  $|U \setminus S| \leq 2\varepsilon$ ;
- (c)  $\max(\ell_K^b(t, p), \ell_K^d(t, p)) \geq -4R\psi$  for  $p \in S$  and  $t \in (0, \pi/2)$ .

**Lemma 10.18** (a rectangle missed by the sofa). *There are  $c_0, \delta, \psi_0 > 0$  such that for every  $K \in \mathcal{K}_{\pi/2}^c$  with  $d_H(K, K_G) \leq \delta$ , every  $\psi \in (0, \psi_0]$  and every set*

$S$  with  $\ell_K^b(\pi/2 - \psi, p) \geq 0$  for  $p \in S$ , there is a measurable  $F \subset K \setminus \mathcal{N}(K)$  with  $F \cap S = \emptyset$  and  $|F| = c_0\psi$ .

*Proof.* Let  $x_0 = -h_{K_G}(\pi)$ ,  $d = (a_\Gamma - x_0)/2 > 0$  (Lemma 10.12),  $l = x_0 + d/2$  and  $r = a_\Gamma - d$ . As  $(x_0, 0), (a_\Gamma, 1) \in K_G$ , the rectangle  $[l, r] \times [0, 1/4]$  lies in  $K_G$ , so by compactness there is  $m > 0$  such that the points of  $[l, r] \times [0, 1/8]$  have distance at least  $m$  from the supporting lines of  $K_G$  with normals in  $[0, \pi]$ . For  $\delta \leq m/2$  this rectangle lies in  $K$  by (2.2), and by Lemma 10.14(a) with  $\eta = d/4$  it misses  $\mathcal{N}(K)$ . By Lemma 10.13(b) for the top edge of  $K_G$ ,  $q = v_K^+(\pi/2)$  has  $q_y = 1$  and  $q_x > a_\Gamma - d/4$ , for small  $\delta$ . Let  $c_0 = (r - l)d/8$ ,  $\psi_0 = \min(1, d/4, 1/d)$  and  $F = [l, r] \times [0, \psi d/8]$ . For  $p \in F$ , as  $h_K(\pi/2 - \psi) \geq q_x \sin \psi + \cos \psi$ ,  $q_x - p_x \geq 3d/4$ ,  $\sin \psi \geq \psi/2$  and  $1 - \cos \psi \leq \psi^2/2 \leq \psi d/8$ ,

$$\ell_K^b(\pi/2 - \psi, p) \leq -(q_x - p_x) \sin \psi + p_y \cos \psi + 1 - \cos \psi \leq -\frac{3\psi d}{8} + \frac{\psi d}{8} + \frac{\psi d}{8} < 0,$$

so  $p \notin S$ .  $\square$

**Lemma 10.19** (the omitted wedges). *For all  $\eta > 0$  and  $R \geq 1$  there are  $\delta \in (0, 1]$  and  $\psi_0 > 0$  such that  $|O_\omega(K)| \leq 4\eta(3R + 1)(\pi/2 - \omega)$  for every  $K \in \mathcal{K}_{\pi/2}^c$  with  $d_H(K, K_G) \leq \delta$  and  $\|p\| \leq R$  on  $K$ , and every  $\omega \in [0, \pi/2]$  with  $\pi/2 - \omega \leq \psi_0$ .*

*Proof.* A point  $p \in O_\omega(K)$  has  $p_y \geq 0$  and both coordinates of  $\ell_K(t, p)$  negative at some  $t \in [\omega, \pi/2)$ , so  $p_y < \mathbf{x}_K(t)_y$  (proof of Lemma 10.14). As  $h_K$  is  $2R$ -Lipschitz,  $h_K(\pi/2) = 1$ ,  $|h_K| \leq R$  and  $\cos t \leq \pi/2 - t$ ,  $\mathbf{x}_K(t)_y = (h_K(t) - 1) \sin t + (h_K(t + \pi/2) - 1) \cos t \leq (3R + 1)(\pi/2 - \omega)$ . By Lemma 10.14(a),  $a_\Gamma - \eta \leq p_x \leq b_\Gamma + \eta$ . Every  $x \in (a_\Gamma, b_\Gamma)$  is the abscissa of a point of  $\Gamma$  other than  $\mathbf{D}(t_0)$  and  $\mathbf{B}(t_5)$ , and these points have positive height:  $\mathbf{D}_y$  increases strictly from 0 on  $[t_0, t_2]$ ,  $\mathbf{B}_y$  decreases strictly to 0 on  $[t_3, t_5]$  (Fact 2.17(c)), and  $\mathbf{x}_y > 0$  on  $[t_1, t_4]$  (Fact 2.18). So the segment  $(a_\Gamma, b_\Gamma) \times \{0\}$  lies in  $\mathcal{N}(K_G)$ , and by compactness there is  $\nu > 0$  such that each  $x \in [a_\Gamma + \eta, b_\Gamma - \eta]$  has a  $t \in (\nu, \pi/2 - \nu)$  at which both coordinates of  $\ell_{K_G}(t, (x, 0))$  are less than  $-\nu$ . If  $\delta \leq \nu/4$ , the points of  $[a_\Gamma + \eta, b_\Gamma - \eta] \times [0, \delta]$  then lie in  $\mathcal{N}_\omega(K)$  when  $\omega > \pi/2 - \nu$ . With  $\psi_0 = \min(\nu/2, \delta/(3R + 1))$ ,  $O_\omega(K)$  lies in  $([a_\Gamma - \eta, a_\Gamma + \eta] \cup [b_\Gamma - \eta, b_\Gamma + \eta]) \times [0, (3R + 1)(\pi/2 - \omega)]$ .  $\square$

*Proof of Proposition 10.17.* Caps with  $d_H(K, K_G) \leq 1$  lie in  $K_G + \mathbb{B}(0, 1) \subset \mathbb{B}(0, R)$  for some  $R \geq 1$ . Let  $c_0$  be as in Lemma 10.18,  $c = c_0/2$  and  $\eta = c_0/(16(3R + 1))$ , and let  $\delta, \psi_0$  be the least of those of Lemmas 10.18 and 10.19 and Proposition 10.11. By Proposition 10.11,  $|U| = \mathcal{A}(K) \leq |G|$ . The set  $V = K \setminus \mathcal{N}_\omega(K)$  contains  $U$  and, by the constraints,  $S$ , and  $|V \setminus U| = |O_\omega(K)| \leq c\psi$ . Lemma 10.18, or  $F = \emptyset$  if  $\psi = 0$ , gives  $F \subset U$ , disjoint from  $S$ , with  $|F| = 2c\psi$ , so

$$|S| + 2c\psi \leq |V| = |U| + |V \setminus U| \leq \mathcal{A}(K) + c\psi,$$

where the first inequality holds as  $S$  and  $F$  are disjoint subsets of  $V$ , and the last as  $|U| = \mathcal{A}(K)$  and  $|V \setminus U| \leq c\psi$ . With  $\mathcal{A}(K) \leq |G|$  this gives (a),

then  $|S \setminus U| \leq |V \setminus U| \leq c\psi \leq \varepsilon$  and  $|U \setminus S| = |U| - |S| + |S \setminus U| \leq 2\varepsilon$ . For (c),  $\ell_K^b(\cdot, p)$  and  $\ell_K^d(\cdot, p)$  are  $4R$ -Lipschitz for  $p \in K$ , and their maximum is nonnegative at  $\omega$ .  $\square$

### 10.5. From the cap back to the sofa.

**Lemma 10.20** (a right-angle cap of a sofa). *Let  $S$  be a moving sofa with rotation angle  $\omega$  and  $h_S(\pi/2) = 1$ , and let  $K = \{p : p_y \geq 0, \langle p, u_t \rangle \leq h_S(t) \text{ for } t \in [0, \pi]\}$ .*

- (a)  $S \subset \mathbb{R} \times [0, 1]$ ,  $h_S(\omega) + h_S(\omega + \pi) \leq 1$ , and  $S \subset L_S(t)$  for  $t \in [0, \omega]$ .
- (b)  $K$  is a cap in  $\mathcal{K}_{\pi/2}^c$  with  $S \subset K$  and  $h_K = h_S$  on  $[0, \pi]$ , and, if  $\omega \in [0, \pi/2]$ ,  $S$  satisfies the constraints of  $(K, \omega)$ . If  $S$  and  $G$  are  $\delta$ -close, then  $d_H(K, K_G) \leq \delta$ .

*Proof.* (a) As in the proof of Corollary 9.6,  $p_y - q_y \leq 1$  for  $p, q \in S$ , so  $S \subset \mathbb{R} \times [0, 1]$ , and the end position  $\Phi_1(S) \subset V_L$  gives  $\langle q - p, u_\omega \rangle \leq 1$ . For  $t \in [0, \omega]$ , some  $s$  has  $\theta(s) = -t$ , and  $\Phi_s(S) \subset L$  puts  $S$  in a translate of  $R_t L$ , hence in  $L_S(t)$  [1, Prop. 2.2.3]. (b)  $K$  is closed, convex, contained in  $[-h_S(\pi), h_S(0)] \times [0, 1]$ , contains  $S$  and, with each point  $p$ , the point  $(p_x, 0)$ ; as an intersection of half-planes with normals in  $[0, \pi] \cup \{3\pi/2\}$ , it is a cap, with  $h_K = h_S$  on  $[0, \pi]$ . If  $\omega \leq \pi/2$  and  $t \in [0, \omega]$ , then  $t$  and  $t + \pi/2$  lie in  $[0, \pi]$ , so  $\mathbf{x}_K(t) = \mathbf{x}_S(t)$ ; as  $S \subset L_S(t)$ , at each point of  $S$  one of  $\ell_K^b(t, \cdot)$  and  $\ell_K^d(t, \cdot)$  is nonnegative. Also  $\ell_K^b(\omega, \cdot) \geq 0$  on  $S$  by (a). Finally  $h_{K_G} = h_G$  on  $[0, \pi]$  (Fact 2.4(b), Fact 2.17(b)), and Lemma 10.10 gives  $d_H(K, K_G) = \max_{[0, \pi]} |h_S - h_G| \leq \delta$ .  $\square$

**Lemma 10.21** (margins of Gerver's sofa). (a) *There is  $d_0 > 0$  such that  $h_{K_G}(t) - \langle p, u_t \rangle \geq d_0$  for  $p \in \mathcal{N}(K_G)$  and  $t \in [0, \pi]$ .* (b) *There are  $c_1, \tau > 0$  such that every  $p \in \mathcal{N}(K_G)$  has a  $t \in (0, \pi/2)$  at which both coordinates of  $\ell_{K_G}(t, p)$  are at most  $-\min(c_1(\gamma(p_x) - p_y), \tau)$ .* (c) *There are  $\kappa, r_0 > 0$  such that for every  $p \in G$  and  $\rho \in (0, r_0]$  some disk  $\mathbb{B}(z, \kappa\rho)$  lies in  $G \cap \mathbb{B}(p, \rho)$ .*

*Proof.* (a) By Lemma 10.12,  $\mathcal{N}(K_G) \subset [a_\Gamma, b_\Gamma] \times [0, \bar{\gamma}]$ , and  $h_{K_G}(t) - \langle p, u_t \rangle > 0$  on this rectangle for  $t \in [0, \pi]$ : at 0 and  $\pi$  as  $-h_{K_G}(\pi) < a_\Gamma < b_\Gamma < h_{K_G}(0)$ , and otherwise as  $(p_x, 1) \in K_G$  makes it at least  $(1 - p_y) \sin t$ ; compactness gives  $d_0$ . (b) Let  $q = (p_x, \gamma(p_x)) \in \Gamma$  and  $d = q_y - p_y$ . Moving down by  $d$  lowers  $\ell_{K_G}^b(t, \cdot)$  by  $d \sin t$  and  $\ell_{K_G}^d(t, \cdot)$  by  $d \cos t$ . If  $q = \mathbf{x}(t)$ ,  $t \in [t_1, t_4]$ , both coordinates of  $\ell_{K_G}(t, q)$  vanish, and  $\min(\sin t, \cos t)$  is bounded below. For  $q = \mathbf{B}(t)$ ,  $t \in [t_3, t_5]$ ,  $\ell_{K_G}^b(t, q) = 0$ ,  $\sin t \geq \sin t_3$ , and  $\ell_{K_G}^d(t, q) = \alpha(t)$  is at most a negative constant, as  $\alpha < 0$  on  $(0, \pi/2)$  and  $\mathbf{B}(\pi/2)_x = \mathbf{x}(\pi/2)_x - \alpha(\pi/2) > \mathbf{x}(t_1)_x > \mathbf{x}(\pi/2)_x$ . The case  $q = \mathbf{D}(t)$  is symmetric, with  $\beta$ . (c) By Lemma 10.12,  $G$  is the union of the convex bodies  $K_G \cap \{x \leq a_\Gamma\}$ ,  $K_G \cap \{x \geq b_\Gamma\}$  and  $[a_\Gamma, b_\Gamma] \times [\bar{\gamma}, 1]$ , which have interior points, and of  $\{(x, y) : x \in [a_\Gamma, b_\Gamma], \gamma(x) \leq y \leq (1 + \bar{\gamma})/2\}$ . A convex body  $C$  with  $\mathbb{B}(z_0, s) \subset C \subset \mathbb{B}(z_0, D)$  has the property with  $\kappa = s/(D + s)$  and  $r_0 = D + s$ :

the homothety of ratio  $\rho/(D+s)$  about  $p$  maps  $\mathbb{B}(z_0, s)$  into  $C \cap \mathbb{B}(p, \rho)$ . For a point  $p$  of the last set,  $\rho \leq \min(b_\Gamma - a_\Gamma, (1 - \bar{\gamma})/2)$  and  $w = \rho/(4(L+2))$ , the disk  $\mathbb{B}(z, w)$ ,  $z = (p_x \pm 2w, p_y + (3L+2)w)$  with the sign towards the farther end of  $[a_\Gamma, b_\Gamma]$ , lies in  $G \cap \mathbb{B}(p, \rho)$ , as  $\gamma$  is  $L$ -Lipschitz.  $\square$

**Lemma 10.22** (the two directed distances). *Let  $K_0, K \in \mathcal{K}_{\pi/2}^c$  and  $\delta \geq 0$  with  $d_H(K, K_0) \leq \delta$ , and let  $U_0 = K_0 \setminus \mathcal{N}(K_0)$  and  $U = K \setminus \mathcal{N}(K)$ .*

- (a) *Suppose that  $K_0$  has, in place of  $K_G$ , the properties of Lemma 10.12 (with some  $a_\Gamma, b_\Gamma, \bar{\gamma}, L, \gamma$ ) and of Lemma 10.21(a), (b) (with some  $d_0, c_1, \tau$ ). Let  $\zeta \geq 0$ ,  $\delta < d_0$ ,  $\delta + \zeta < \tau$ , and let  $S \subset K$  have  $\max(\ell_K^b(t, p), \ell_K^d(t, p)) \geq -\zeta$  for  $p \in S$  and  $t \in (0, \pi/2)$ . Then every point of  $S$  lies within  $\max(1, 1/c_1)(\delta + \zeta)$  of a point of  $U_0$ .*
- (b) *Suppose that  $U_0$  has the property of Lemma 10.21(c) (with some  $\kappa, r_0$ ). Let  $A \geq 0$ ,  $\varepsilon > 0$ ,  $\rho = (4(A+1)/\kappa)\sqrt{\varepsilon} \leq r_0$  and  $2\delta \leq A\sqrt{\varepsilon}$ , and let  $S$  be a set with  $|U \setminus S| \leq 2\varepsilon$ . Then every point of  $U_0$  lies within  $\rho$  of a point of  $S$ .*

*Proof.* (a) Let  $p \in S \setminus U_0$ ; the coordinates of  $\ell_K$  and  $\ell_{K_0}$  differ by at most  $\delta$ . If  $p \in \mathcal{N}(K_0)$ , let  $d = \gamma(p_x) - p_y$  and take  $t$  from Lemma 10.21(b); were  $c_1 d > \delta + \zeta$ , both coordinates of  $\ell_K(t, p)$  would be less than  $-\zeta$ , so  $d \leq (\delta + \zeta)/c_1$ , and  $(p_x, \gamma(p_x)) \in U_0$  is at distance  $d$ . Otherwise  $p \notin K_0$ , and some  $q \in K_0$  has  $\|p - q\| \leq \delta$  (Lemma 10.10); were  $q \in \mathcal{N}(K_0)$ , then  $\langle p, u_t \rangle \leq \langle q, u_t \rangle + \delta \leq h_{K_0}(t) - d_0 + \delta < h_{K_0}(t)$  for  $t \in [0, \pi]$ , and  $p \in K_0$  by (2.2). So  $q \in U_0$ . (b) First,  $U$  contains every  $p'$  with  $\mathbb{B}(p', 2\delta) \subset U_0$ :  $p' + \delta u_t \in K_0$  gives  $\langle p', u_t \rangle \leq h_K(t)$  for  $t \in [0, \pi]$ , so  $p' \in K$ ; and if both coordinates of  $\ell_K(t, p')$  were negative, then  $p' - \delta u_t - \delta v_t$ , at distance  $\sqrt{2}\delta$ , would have both coordinates of  $\ell_{K_0}$  negative and lie in  $\mathcal{N}(K_0)$ . Now let  $p \in U_0$  have no point of  $S$  in  $\mathbb{B}(p, \rho)$ , take  $\mathbb{B}(z, \kappa\rho) \subset U_0 \cap \mathbb{B}(p, \rho)$ , and let  $\mu = \kappa\rho/2 = 2(A+1)\sqrt{\varepsilon} \geq 2\delta$ . The square of side  $\mu$  centered at  $z$  lies in  $\mathbb{B}(z, \mu)$ , so the disks of radius  $2\delta$  about its points lie in  $\mathbb{B}(z, \kappa\rho)$ ; hence it lies in  $U \setminus S$ , and  $4(A+1)^2\varepsilon = \mu^2 \leq |U \setminus S| \leq 2\varepsilon$ , which is false.  $\square$

**Proposition 10.23** (local recovery of the sofa). *There are  $C, c, \delta, \psi_0, \varepsilon_0 > 0$ , with  $\delta \leq 1$  and  $\varepsilon_0 \leq 1$ , such that for every  $K \in \mathcal{K}_{\pi/2}^c$  with  $h_K(\pi) = h_{K_G}(\pi)$  and  $d_H(K, K_G) \leq \delta$ , every  $\omega \in [0, \pi/2]$  with  $\pi/2 - \omega \leq \psi_0$  and every measurable set  $S$  that satisfies the constraints of  $(K, \omega)$  and has  $\varepsilon = |G| - |S| \in (0, \varepsilon_0)$ , the sets  $S$  and  $G$  are  $C\sqrt{\varepsilon}$ -close and  $\pi/2 - \omega \leq \varepsilon/c$ .*

*Proof.* Take  $c, \delta_1, \psi_0, R$  from Proposition 10.17,  $\delta_2$  from Proposition 10.11,  $\delta = \min(\delta_1, \delta_2)$ , and  $d_0, c_1, \tau, \kappa, r_0$  from Lemma 10.21. Let  $a = 2 \sec \varphi$ ,  $B' = 4R/c$ ,  $C_1 = \max(1, 1/c_1)(a + B')$ ,  $C_2 = 4(2a + 1)/\kappa$  and  $C = \max(1, C_1, C_2)$ , and choose  $\varepsilon_0 \leq 1$  such that  $a\sqrt{\varepsilon} < d_0$ ,  $(a + B')\sqrt{\varepsilon} < \tau$  and  $C_2\sqrt{\varepsilon} < r_0$  for  $\varepsilon < \varepsilon_0$ . By Proposition 10.17,  $\pi/2 - \omega \leq \varepsilon/c$ ,  $|G| - \mathcal{A}(K) \leq \varepsilon$ ,  $|(K \setminus \mathcal{N}(K)) \setminus S| \leq 2\varepsilon$ , and (c) holds with  $4R(\pi/2 - \omega) \leq B'\sqrt{\varepsilon}$ , as  $\varepsilon \leq 1$ . As  $s_K = 0$ , Proposition 10.11 and Lemma 10.10 give  $d_H(K, K_G) \leq a\sqrt{\varepsilon}$ . By Lemmas 10.12 and 10.21,  $K_G$  has the properties of Lemma 10.22, with

$K_G \setminus \mathcal{N}(K_G) = G$ . Part (a), with  $a\sqrt{\varepsilon}$  for  $\delta$  and  $\zeta = B'\sqrt{\varepsilon}$ , puts  $S$  within  $C_1\sqrt{\varepsilon}$  of  $G$ ; part (b), with  $A = 2a$ , puts  $G$  within  $C_2\sqrt{\varepsilon}$  of  $S$ .  $\square$

**Lemma 10.24** (the symmetric difference). *For every  $C > 0$  there are  $C' > 0$  and  $\varepsilon_0 \in (0, 1]$  such that every compact set  $S$  that is  $C\sqrt{\varepsilon}$ -close to  $G$ , where  $\varepsilon = |G| - |S| \in (0, \varepsilon_0]$ , satisfies  $|S \Delta G| \leq C'\sqrt{\varepsilon}$ .*

*Proof.* Fix a disk  $\mathbb{B}(z, r) \subset K_G$ . Then  $h_{K_G} + d \leq (1 + d/r)h_{K_G} - (d/r)\langle z, u_t \rangle$ , so  $K_G + d\mathbb{B}(0, 1)$  lies in the image of  $K_G$  under the homothety of ratio  $1 + d/r$  about  $z$ . So the points within  $d \leq 1$  of  $K_G$  outside it have area at most  $((1 + d/r)^2 - 1)|K_G| \leq A_0d$  for a constant  $A_0$ . A point  $p \in \mathcal{N}(K_G)$  within  $d$  of a point  $q \in G$  has  $\gamma(p_x) - p_y \leq (L + 1)d$ : if  $q_x \in [a_\Gamma, b_\Gamma]$ , then  $q_y \geq \gamma(q_x) \geq \gamma(p_x) - Ld$ , and otherwise  $\gamma(p_x) \leq Ld$ , while  $q_y \geq 0$ . With  $C\sqrt{\varepsilon_0} \leq 1$ ,  $d = C\sqrt{\varepsilon}$  and  $d' = (L + 1)d$ , the points of  $S \setminus G$  in  $\mathcal{N}(K_G)$  lie in  $\{(x, y) : x \in [a_\Gamma, b_\Gamma], \gamma(x) - 2d' < y < \gamma(x) + d'\}$ , of area  $3d'(b_\Gamma - a_\Gamma)$ . So  $|S \setminus G| \leq Ad$  with  $A = A_0 + 3(L + 1)(b_\Gamma - a_\Gamma)$ , and, as  $\varepsilon \leq 1$ ,  $|S \Delta G| = |G| - |S| + 2|S \setminus G| \leq (1 + 2AC)\sqrt{\varepsilon}$ .  $\square$

#### 10.6. Compactness and uniqueness; proof of Theorems 10.1 and 10.2.

Let  $d_\infty$  be the Hausdorff distance between nonempty compact sets for the norm  $\|(x, y)\|_\infty = \max(|x|, |y|)$ . As  $\|p\| \leq 2\|p\|_\infty$ , sets with  $d_\infty \leq r$  are  $2r$ -close; and the nonempty compact subsets of a compact set form a compact space for  $d_\infty$ . For a compact set  $S$ ,  $\ell_S^b$ ,  $\ell_S^d$  and  $\ell_S$  are defined as for caps, with  $h_S$  in place of  $h_K$ .

**Lemma 10.25** (limits of normalized sofas). (a) *Let  $S$  be a compact connected set and  $\omega \geq 0$  with  $S \subset \mathbb{R} \times [0, 1]$ ,  $h_S(\pi/2) = 1$ ,  $h_S(\omega) + h_S(\omega + \pi) \leq 1$  and  $\max(\ell_S^b(t, p), \ell_S^d(t, p)) \geq 0$  for  $p \in S$  and  $t \in [0, \omega]$ . Then  $S$  is a moving sofa with rotation angle  $\omega$ .*

(b) *A moving sofa  $S$  with rotation angle  $\omega \geq \pi/4$ ,  $h_S(\pi/2) = 1$  and  $h_S(\pi) = h_G(\pi)$  lies in  $[-h_G(\pi), -h_G(\pi) + 6] \times [0, 1]$ .*

(c) *Let  $S_n$  be moving sofas with rotation angles  $\omega_n \in [\omega_0, \pi/2]$ ,  $h_{S_n}(\pi/2) = 1$  and  $h_{S_n}(\pi) = h_G(\pi)$ , and let  $S$  be a nonempty compact set with  $d_\infty(S_n, S) \rightarrow 0$  and  $\omega_n \rightarrow \omega$ . Then  $S$  is a moving sofa with rotation angle  $\omega \in [\omega_0, \pi/2]$ ,  $h_S(\pi/2) = 1$  and  $h_S(\pi) = h_G(\pi)$ ; and if  $|S_n| \rightarrow M$ , then  $M \leq |S|$ .*

*Proof.* (a) Let  $\theta(s) = -s\omega$  and  $c(s) = -R_{-s\omega}\mathbf{x}_S(s\omega)$ . Then  $\Phi_s(p) = R_{-s\omega}(p - \mathbf{x}_S(s\omega))$  has the coordinates  $\ell_S(s\omega, p)$ , both at most 1 and one nonnegative, so  $\Phi_s(S) \subset L$ . At  $s = 0$  they are  $(p_x - h_S(0) + 1, p_y)$ , so  $\Phi_0(S) \subset H_L$ ; at  $s = 1$ ,  $\ell_S^b(\omega, p) \in [0, 1]$ , as  $\langle p, u_\omega \rangle \geq -h_S(\omega + \pi) \geq h_S(\omega) - 1$ , so  $\Phi_1(S) \subset V_L$ . (b) By Lemma 10.20,  $S \subset \mathbb{R} \times [0, 1]$  and  $S \subset L_S(\pi/4)$ . As  $S$  is connected and  $\ell_S^b(\pi/4, \cdot) - \ell_S^d(\pi/4, \cdot)$  takes both signs on  $S$  (where  $\langle p, u_{\pi/4} \rangle$  and where  $\langle p, v_{\pi/4} \rangle$  is largest), some  $z \in S$  has  $\ell_S^b = \ell_S^d \geq 0$  at  $\pi/4$ , so  $(h_S(\pi/4) + h_S(3\pi/4) - 2)/\sqrt{2} = \mathbf{x}_S(\pi/4)_y \leq z_y \leq 1$ . For  $p, q \in S$ , as  $p_y, q_y \geq 0$ ,  $p_x - q_x \leq \sqrt{2}(h_S(\pi/4) + h_S(3\pi/4)) \leq 6$ ; and  $\min_S p_x = -h_G(\pi)$ . (c) Support values converge along converging directions,

as  $|h_{S_n}(t_n) - h_S(t)| \leq 2d_\infty(S_n, S) + |h_S(t_n) - h_S(t)|$  (Lemma 10.10). So  $h_S(\pi/2) = 1$ ,  $h_S(\pi) = h_G(\pi)$ , and  $h_S(\omega) + h_S(\omega + \pi) \leq 1$  by Lemma 10.20; and  $S \subset \mathbb{R} \times [0, 1]$ .  $S$  is connected, as disjoint closed sets that cover  $S$  and meet it have disjoint open neighborhoods that would eventually cover  $S_n$  and meet it. For  $p \in S$  and  $t \in [0, \omega]$ , take  $p_n \in S_n$  with  $p_n \rightarrow p$  and  $t_n = (t/\omega)\omega_n$ ; the inequality  $\max(\ell_{S_n}^b, \ell_{S_n}^d)(t_n, p_n) \geq 0$  (Lemma 10.20) passes to the limit, and (a) applies. Finally, for  $r > |S|$ , an open set  $\mathcal{O} \supset S$  with  $|\mathcal{O}| < r$  contains  $S_n$  for large  $n$ , so  $M \leq r$ .  $\square$

**Proposition 10.26** (sofas of small deficit are near  $G$ ). *For all  $\rho, \psi_0 > 0$  there is  $\varepsilon_0 > 0$  such that for every moving sofa  $S$  with rotation angle  $\omega \in [\omega_0, \pi/2]$  and  $\varepsilon(S) < \varepsilon_0$ , the sets  $S^\natural$  and  $G$  are  $\rho$ -close, and  $\pi/2 - \omega < \psi_0$ .*

*Proof.* Otherwise there are moving sofas  $S_n$  with rotation angles  $\omega_n \in [\omega_0, \pi/2]$  and  $\varepsilon(S_n) < 1/(n+1)$  that violate the conclusion. The translates  $S_n^\natural$  are moving sofas with the same rotation angles, lie in the box of Lemma 10.25(b), as  $\omega_0 \geq \pi/4$ , and have  $|S_n^\natural| \rightarrow |G|$ , as  $0 \leq \varepsilon(S_n) < 1/(n+1)$  (Theorem 11.4(b)). By compactness, a subsequence has  $d_\infty(S_n^\natural, S) \rightarrow 0$  and  $\omega_n \rightarrow \omega$  for a nonempty compact set  $S$ ; by Lemma 10.25,  $S$  is a moving sofa with rotation angle  $\omega$ ,  $h_S(\pi/2) = 1$ ,  $h_S(\pi) = h_G(\pi)$  and  $|S| \geq |G|$ , so  $|S| = |G|$  by Theorem 11.4(b). Theorem 11.7(a) gives  $S + w = G$  for some  $w$ , and the support values at  $\pi/2$  and  $\pi$  give  $w = 0$ . As  $G$  is a moving sofa with rotation angle  $\omega$ ,  $\langle q - p, u_\omega \rangle \leq 1$  on  $G$  (Lemma 10.20), so  $\omega = \pi/2$  by Lemma 9.5(b). For large  $n$ ,  $d_\infty(S_n^\natural, G) < \rho/2$  and  $\pi/2 - \omega_n < \psi_0$ , a contradiction.  $\square$

*Proof of Theorems 10.1 and 10.2.* Take  $C, c, \delta, \psi_0, \varepsilon_1$  from Proposition 10.23,  $C', \varepsilon_2$  from Lemma 10.24 for this  $C$ ,  $\varepsilon_3$  from Proposition 10.26 for  $\rho = \delta$  and this  $\psi_0$ , and  $\varepsilon_0 = \min(\varepsilon_1, \varepsilon_2, \varepsilon_3)$ . Let  $S$  be a moving sofa with rotation angle  $\omega \in [\omega_0, \pi/2]$  and  $\varepsilon = \varepsilon(S) < \varepsilon_0$ . If  $\varepsilon = 0$ , then  $S^\natural = G$  and  $\omega = \pi/2$ , as in the proof of Proposition 10.26. Otherwise  $\varepsilon > 0$ , by Theorem 11.4(b). Then  $N = S^\natural$  is a moving sofa with rotation angle  $\omega$ , and by Proposition 10.26  $N$  and  $G$  are  $\delta$ -close and  $\pi/2 - \omega < \psi_0$ . The set  $K$  of Lemma 10.20 for  $N$ , a cap in  $\mathcal{K}_{\pi/2}^c$ , has  $h_K(\pi) = h_{K_G}(\pi)$  (as  $h_{K_G} = h_G$  on  $[0, \pi]$ ) and  $d_H(K, K_G) \leq \delta$ , and  $N$  satisfies the constraints of  $(K, \omega)$ . So Proposition 10.23 shows that  $N$  and  $G$  are  $C\sqrt{\varepsilon}$ -close and  $\pi/2 - \omega \leq \varepsilon/c$ , and Lemma 10.24 gives  $|N \triangle G| \leq C'\sqrt{\varepsilon}$ . This proves Theorem 10.2, with  $C'' = 1/c$ . For Theorem 10.1, let  $\varepsilon'_0 = \min(\varepsilon_0, |G| - 11/5) > 0$  (Fact 2.17(a)): a moving sofa with deficit less than  $\varepsilon'_0$  has area more than  $11/5$ , hence a rotation angle in  $[\omega_0, \pi/2]$  (Fact 2.3).  $\square$

### 10.7. Punctured sofas; proof of Theorem 10.3.

**Lemma 10.27** (removing a disk). *Let  $X$  be a closed connected set,  $p \in \mathbb{R}^2$  and  $r > 0$  with  $\mathbb{B}(p, r) \subset X$ , and let  $Y = X \setminus \mathbb{B}^\circ(p, r)$ . Then  $Y$  is closed and connected, and  $r$ -close to  $X$ ;  $|X| - |Y| = \pi r^2$  if  $X$  is compact; and if  $X$  is a moving sofa with rotation angle  $\omega$ , so is  $Y$ .*

*Proof.* The boundary circle of  $\mathbb{B}^\circ(p, r)$  lies in  $Y$  and is connected. If closed sets  $A, B$  cover  $Y$  and  $Y \cap A \cap B = \emptyset$ , the circle lies in one of them, say  $A$ ; then  $(Y \cap A) \cup \mathbb{B}(p, r)$  and  $Y \cap B$  are disjoint closed sets that cover  $X$ , so  $Y \cap B = \emptyset$ . A point  $x \in \mathbb{B}(p, r)$  lies within  $r$  of the point of the circle on the ray from  $p$  through  $x$ . The disk  $\mathbb{B}^\circ(p, r)$  has area  $\pi r^2$ , and a movement of  $X$  moves the closed connected subset  $Y$ .  $\square$

**Lemma 10.28** (rigid copies keep interior points). *Let  $X$  be a compact set and  $p$  an interior point of  $X$ . There is  $\eta > 0$  such that  $p \in g(X)$  for every rigid motion  $g$  for which  $X$  and  $g(X)$  are  $d$ -close for some  $d < \eta$ .*

*Proof.* Otherwise there are  $g_n(x) = R_{\vartheta_n}x + v_n$  with  $X$  and  $g_n(X)$   $d_n$ -close,  $d_n \rightarrow 0$  and  $p \notin g_n(X)$ . The  $v_n$  are bounded, so a subsequence has  $(\cos \vartheta_n, \sin \vartheta_n, v_n) \rightarrow (\cos \vartheta, \sin \vartheta, v)$ ; let  $g(x) = R_\vartheta x + v$ . For  $x \in X$ ,  $g_n^{-1}(x)$  lies within  $d_n$  of  $X$ , so  $g^{-1}(x) \in X$ . Hence  $g^{-1}$  maps the interior of  $X$  onto an open subset of  $X$  that contains  $g^{-1}(p)$ , and, as  $g_n^{-1}(p) \rightarrow g^{-1}(p)$ ,  $p \in g_n(X)$  for large  $n$ , a contradiction.  $\square$

*Proof of Theorem 10.3.* (a) As  $G$  is the closure of its interior (Proposition 9.3), it has an interior point  $p$ ; let  $\mathbb{B}(p, R) \subset G$ , let  $\eta$  be as in Lemma 10.28 for  $X = G$ , and let  $r_0 = \min(R/2, \eta/3)$  and  $r \in (0, r_0)$ . As  $G$  is a moving sofa (Fact 2.17(b), Fact 2.4(a)), Lemma 10.27 shows that  $G_r$  is a moving sofa,  $r$ -close to  $G$ , with  $\varepsilon(G_r) = \pi r^2$ . If  $G_r$  and  $g(G)$  are  $d$ -close with  $d < r$ , then  $G$  and  $g(G)$  are  $(r + d)$ -close and  $r + d < \eta$ , so  $p \in g(G)$  by Lemma 10.28; then  $p$  lies within  $d < r$  of a point of  $G_r$ , which is false. So  $r \leq d$ , and  $d_{\text{rig}}(G_r, G) = r$ , attained by the identity. (b), (c) As  $2a - 1 > 0$ , some  $r \in (0, r_0)$  has  $\pi r^2 < \varepsilon_0$  and  $C\pi^a r^{2a-1} < 1$ , that is,  $C(\pi r^2)^a < r$ . Then  $S' = G_r$  has  $0 < \varepsilon(S') < \varepsilon_0$  and  $C\varepsilon(S')^a < r = d_{\text{rig}}(S', G)$ , which is (c). Item (c) implies (b): if  $S'$  and  $g(G)$  were  $C\varepsilon(S')^a$ -close for a rigid motion  $g$ , then  $d_{\text{rig}}(S', G) \leq C\varepsilon(S')^a$ . So  $S = S'$  serves for (b).  $\square$

## 11. ONE CERTIFICATE FOR OPTIMALITY, UNIQUENESS AND STABILITY

The estimates of Sections 10.1 and 10.2 combine into one estimate for Baek's functional  $\mathcal{Q}$  on the domain  $\bar{\mathcal{T}}$  of Section 10, which we call the coercive certificate (Theorem 11.1). For  $\xi \in \bar{\mathcal{T}}$ , call  $|G| - \mathcal{Q}(\xi)$  the deficit of  $\mathcal{Q}$  at  $\xi$ . The first part of the certificate says that the deficit is nonnegative. The second part bounds the distance from the cap  $K$  of  $\xi$  to a horizontal translate of Gerver's cap by  $2 \sec \varphi$  times the square root of the deficit; so a small deficit forces a small distance, which is what *coercive* refers to.

Optimality, uniqueness and stability rest on these two parts. Let  $K \in \mathcal{K}_{\pi/2}^c$  be a maximizing cap. Then  $|G| \leq \mathcal{A}(K) \leq \mathcal{Q}(\xi_K) \leq |G|$ , where the first inequality holds as  $K$  is maximizing, the second by Baek's bound, which applies as  $K \in \mathcal{K}^i$  (Proposition 11.3), and the third by the first part. So the largest sofa area of a cap in  $\mathcal{K}_{\pi/2}^c$  is  $|G|$ , and the argument of Section 8.4 turns this into optimality (Section 11.3). The deficit at  $\xi_K$  is 0, so by the second part  $K$  is a horizontal translate of  $K_G$ ; the argument of Sections 3 and 9

turns this into uniqueness (Sections 11.2 and 11.4). For a cap  $K$  near  $K_G$ , Section 10.3 proves  $\mathcal{A}(K) \leq \mathcal{Q}(\xi_K)$  without the injectivity condition, and the second part then bounds the distance from  $K$  to a horizontal translate of  $K_G$  by a multiple of  $\sqrt{|G| - \mathcal{A}(K)}$  (Proposition 10.11). Section 10 derives stability from this bound and from the optimality and uniqueness proved here (Section 11.5). Theorem 11.8 collects the three results.

The proof of uniqueness in Sections 3 to 9 and the second proof of optimality in Section 8.4 find the maximizing caps in  $\mathcal{K}_{\pi/2}^c$  through the case of equality in the concavity of  $\mathcal{Q}$  (Lemma 8.1, Lemma 8.2 and Proposition 8.3). Below, Proposition 11.3 finds them from the certificate. It takes the place of Lemma 8.6 in the proof of optimality and, through Lemma 11.5(c), of Theorem 8.5 in the proof of uniqueness, with the other changes listed after Theorem 8.8; the rest of the two proofs is unchanged. So the proofs of this section use neither Baek's theorem (Fact 2.8), nor Lemma 8.1, Lemma 8.2 and Proposition 8.3, nor the results by which Baek derives step (3) from the balance of his caps (Section 8.4).

**11.1. The certificate.** Recall from Section 10 that  $\bar{\mathcal{T}}$  is defined as  $\mathcal{T}$  (Section 2.5), with  $K \in \mathcal{K}_{\pi/2}^c$  in place of  $K \in \mathcal{K}^i$ ; that  $s_K = h_{K_G}(\pi) - h_K(\pi)$ , so that  $K$  and  $K_G + (s_K, 0)$  have the same leftmost abscissa; and that two sets are  $r$ -close if every point of each lies within Euclidean distance  $r$  of a point of the other.

**Theorem 11.1** (the coercive certificate). *Let  $\xi = (K, B, D) \in \bar{\mathcal{T}}$ . Then  $\mathcal{Q}(\xi) \leq |G|$ , and the caps  $K$  and  $K_G + (s_K, 0)$  are  $2 \sec \varphi \sqrt{|G| - \mathcal{Q}(\xi)}$ -close.*

*Proof.* The first assertion is Lemma 10.6, and the second is Theorem 10.4(a).  $\square$

The first part comes from the concavity of  $\mathcal{Q}$  on  $\bar{\mathcal{T}}$  and the sign of its directional derivatives at Gerver's triple  $\xi_G$  (Lemmas 10.5 and 10.6). The second comes from three results. The deficit certificate (Proposition 10.7) bounds the energy  $E_{\mathcal{S}}(K_G, K)$  by  $|G| - \mathcal{Q}(\xi)$ . Lemma 10.8 and Proposition 10.9(b) bound  $h_K - h_{K_G + (s_K, 0)}$  on  $[0, \pi]$  by  $2 \sec \varphi$  times the square root of this energy, and Lemma 10.10 turns this bound into closeness. By (10.3), the energy measures how far  $\mathcal{S}$  is from being convex-linear on the segment from  $K_G$  to  $K$ . Section 8 shows that  $\mathcal{S}$  is convex-linear there when  $K \in \mathcal{K}^i$  and  $\mathcal{A}(K) = |G|$  (Lemma 8.1), and solves the resulting equations for  $h_K - h_{K_G}$  (Lemma 8.2 and Proposition 8.3); the certificate replaces these steps by a bound.

## 11.2. Maximizing right-angle caps.

**Lemma 11.2** (zero deficit of  $\mathcal{Q}$ ). *Let  $\xi = (K, B, D) \in \bar{\mathcal{T}}$  with  $\mathcal{Q}(\xi) = |G|$ . Then  $K = K_G + (s_K, 0)$ .*

*Proof.* By Theorem 11.1,  $K$  and  $K_G + (s_K, 0)$  are 0-close. So every point of each set lies at distance 0 from a point of the other, that is, belongs to the other, and each set contains the other.  $\square$

**Proposition 11.3** (maximizing right-angle caps from the certificate). *Let  $K \in \mathcal{K}_{\pi/2}^c$  be a maximizing cap.*

- (a)  $K \in \mathcal{K}^i$ , and  $\mathcal{A}(K) = \mathcal{Q}(\xi_K) = |G|$ .
- (b) There is  $s \in \mathbb{R}$  with  $K = K_G + (s, 0)$  and  $K \setminus \mathcal{N}(K) = G + (s, 0)$ .

*Proof.* (a) Gerver's cap  $K_G$  lies in  $\mathcal{K}_{\pi/2}^c$  and has  $\mathcal{A}(K_G) = |G|$  (Fact 2.17(b), Fact 2.4(c)). As  $K$  is maximizing,  $\mathcal{A}(K) \geq \mathcal{A}(K_G) = |G|$ , and  $|G| \geq 2.2192$  by Fact 2.17(a). So  $\mathcal{A}(K) > 0$ , and  $K$  satisfies the curvature bounds (Proposition 7.9) and the injectivity condition (Proposition 7.12). Also  $|K| = \mathcal{A}(K) + |\mathcal{N}(K)| \geq |G| \geq 11/5$ , so  $K \in \mathcal{K}^i$ . By Fact 2.16(a), the canonical triple  $\xi_K$  lies in  $\mathcal{T}$ , hence in  $\bar{\mathcal{T}}$  as  $\mathcal{K}^i \subset \mathcal{K}_{\pi/2}^c$ , and Baek's bound  $\mathcal{A}(K) \leq \mathcal{Q}(\xi_K)$  holds. The first part of Theorem 11.1 gives  $\mathcal{Q}(\xi_K) \leq |G|$ . Hence

$$|G| \leq \mathcal{A}(K) \leq \mathcal{Q}(\xi_K) \leq |G|,$$

and the three numbers are equal.

(b) By (a),  $K \in \mathcal{K}^i$ , so  $\xi_K \in \mathcal{T} \subset \bar{\mathcal{T}}$ , and  $\mathcal{Q}(\xi_K) = |G|$ . So  $K = K_G + (s, 0)$  with  $s = s_K$  by Lemma 11.2. Lemma 8.4(b) and then (a), applied to  $K_G + (s, 0)$  and  $C = K_G$ , give  $K \setminus \mathcal{N}(K) = (K_G \setminus \mathcal{N}(K_G)) + (s, 0) = G + (s, 0)$ , since  $G = K_G \setminus \mathcal{N}(K_G)$  (Fact 2.17(b)).  $\square$

Proposition 11.3 contains the conclusion of Lemma 8.6. Its proof of  $K \in \mathcal{K}^i$  is that of Lemma 8.6, but the bound  $\mathcal{Q}(\xi_K) \leq |G|$  comes from the certificate instead of Fact 2.16(c), and the shape of  $K$  from Lemma 11.2 instead of Proposition 8.3.

### 11.3. Optimality.

**Theorem 11.4** (optimality from the certificate). (a) *Every cap  $C \in \mathcal{K}_{\pi/2}^c$  has  $\mathcal{A}(C) \leq |G|$ . For a cap  $K \in \mathcal{K}_{\pi/2}^c$ , the following are equivalent:  $K$  is maximizing;  $\mathcal{A}(K) = |G|$ ;  $K = K_G + (s, 0)$  for some  $s \in \mathbb{R}$ .*

(b)  *$G$  is a moving sofa, and every moving sofa  $S$  has  $|S| \leq |G|$ .*

(c) *For every  $\omega \in (0, \pi/2]$  and every cap  $K \in \mathcal{K}_{\omega}^c$ ,  $\mathcal{A}_{\omega}(K) \leq |G|$ .*

*Proof.* (a) Let  $K_{\pi/2}$  be the cap of Fact 2.7 for  $\omega = \pi/2$ . It is maximizing, so  $\mathcal{A}(K_{\pi/2}) = |G|$  by Proposition 11.3(a), and every  $C \in \mathcal{K}_{\pi/2}^c$  has  $\mathcal{A}(C) \leq \mathcal{A}(K_{\pi/2}) = |G|$ . A maximizing cap  $K$  is a translate  $K_G + (s, 0)$  by Proposition 11.3(b). A translate  $K = K_G + (s, 0) \in \mathcal{K}_{\pi/2}^c$  has  $\mathcal{A}(K) = \mathcal{A}(K_G) = |G|$  by Lemma 8.4(b), Fact 2.17(b) and Fact 2.4(c). Finally, a cap  $K$  with  $\mathcal{A}(K) = |G|$  is maximizing, since  $\mathcal{A}(C) \leq |G|$  for every  $C \in \mathcal{K}_{\pi/2}^c$ .

(b) By Fact 2.17(b),  $G$  is a monotone sofa with rotation angle  $\pi/2$ , hence a moving sofa (Fact 2.4(a)). For the bound, first let  $S$  be a moving sofa with rotation angle  $\pi/2$ , and let  $K_{\pi/2}$  and  $S_{\pi/2} = K_{\pi/2} \setminus \mathcal{N}(K_{\pi/2})$  be as in Fact 2.7. Then

$$|S| \leq |S_{\pi/2}| = \mathcal{A}(K_{\pi/2}) = |G|,$$

where the inequality holds by Fact 2.7, the first equality by Fact 2.4(c), as  $\mathcal{C}(S_{\pi/2}) = K_{\pi/2}$ , and the second by Proposition 11.3(a), as  $K_{\pi/2}$  is maximizing. Now let  $S$  be any moving sofa. If  $|S| < 11/5$ , then  $|S| < |G|$ , as  $|G| \geq 2.2192$  (Fact 2.17(a)). Otherwise  $S$  is a moving sofa with some rotation angle  $\omega \in [\omega_0, \pi/2]$  (Fact 2.3(a)). Let  $K_\omega$  and  $S_\omega = K_\omega \setminus \mathcal{N}(K_\omega)$  be as in Fact 2.7. Then  $|S_\omega| \geq |S| \geq 11/5$ , and the cap  $\mathcal{C}(S_\omega) = K_\omega$  of the monotone sofa  $S_\omega$  is maximizing. By Lemma 8.7,  $R_\psi S_\omega$ ,  $\psi = \pi/2 - \omega$ , is a moving sofa with rotation angle  $\pi/2$ , and so

$$|S| \leq |S_\omega| = |R_\psi S_\omega| \leq |G|,$$

where the first inequality holds by Fact 2.7, the equality as rotations preserve area, and the last inequality by the first case.

(c) Let  $K_\omega$  and  $S_\omega$  be as in Fact 2.7. Every  $K \in \mathcal{K}_\omega^c$  has

$$\mathcal{A}_\omega(K) \leq \mathcal{A}_\omega(K_\omega) = |S_\omega| \leq |G|,$$

where the first inequality holds by Fact 2.7, the equality by Fact 2.4(c), as  $\mathcal{C}(S_\omega) = K_\omega$ , and the last inequality by (b), as  $S_\omega$  is a moving sofa (Fact 2.4(a)).  $\square$

These are the assertions and the proofs of Theorem 8.8, with Proposition 11.3 in place of Lemma 8.6.

#### 11.4. Uniqueness.

**Lemma 11.5** (monotone sofas of area  $|G|$ ). *Let  $\omega \in (0, \pi/2]$ .*

- (a) *If  $T$  is a monotone sofa with rotation angle  $\omega$  and  $|T| = |G|$ , then its cap  $\mathcal{C}(T)$  is maximizing.*
- (b) *If  $S'$  is a moving sofa with rotation angle  $\omega$  and  $|S'| = |G|$ , then there are  $w \in \mathbb{R}^2$  and a monotone sofa  $T$  with rotation angle  $\omega$  such that  $S' + w \subset T$  and  $|T| = |G|$ .*
- (c) *If  $U$  is a monotone sofa with rotation angle  $\pi/2$  and  $|U| = |G|$ , then  $U = G + (s, 0)$  for some  $s \in \mathbb{R}$ .*

*Proof.* (a) The monotone sofa  $T$  is a moving sofa with rotation angle  $\omega$  in standard position (Fact 2.4(a)), so  $\mathcal{C}(T)$  is a cap (Fact 2.4(b)), and  $\mathcal{A}_\omega(\mathcal{C}(T)) = |T| = |G|$  (Fact 2.4(c)). Every  $C \in \mathcal{K}_\omega^c$  has  $\mathcal{A}_\omega(C) \leq |G| = \mathcal{A}_\omega(\mathcal{C}(T))$  by Theorem 11.4(c), so  $\mathcal{C}(T)$  is maximizing.

(b) By Fact 2.3(b),  $S' + w$  is in standard position for some  $w$ , and it is a moving sofa with rotation angle  $\omega$ . By Fact 2.4(a),  $T = \mathcal{I}(S' + w)$  is a moving sofa with rotation angle  $\omega$  that contains  $S' + w$ , and it is a monotone sofa. Translations preserve area and  $S' + w \subset T$ , so  $|G| = |S' + w| \leq |T|$ ; and  $|T| \leq |G|$  by Theorem 11.4(b), as  $T$  is a moving sofa.

(c) By (a), the cap  $K = \mathcal{C}(U) \in \mathcal{K}_{\pi/2}^c$  is maximizing, so Proposition 11.3(b) gives  $s \in \mathbb{R}$  with  $K \setminus \mathcal{N}(K) = G + (s, 0)$ . As  $U = K \setminus \mathcal{N}(K)$  (Fact 2.4(c)),  $U = G + (s, 0)$ .  $\square$

Parts (a) and (b), with Fact 2.3(a), Fact 2.4(c) and Fact 2.17(a), give Proposition 3.2, and part (c) is the last assertion of Theorem 8.5.

**Proposition 11.6** (a rigid image in Gerver's sofa). *Let  $S$  be a moving sofa with  $|S| = |G|$ . Then there are  $\omega \in [\omega_0, \pi/2]$  and  $v \in \mathbb{R}^2$  such that  $R_\psi S + v \subset G$ , where  $\psi = \pi/2 - \omega \in [0, \pi/2 - \omega_0]$ .*

*Proof.* As  $|S| = |G| \geq 2.2192$  (Fact 2.17(a)),  $S$  is a moving sofa with some rotation angle  $\omega \in [\omega_0, \pi/2]$  (Fact 2.3(a)). By Lemma 11.5(b), there are  $w_0 \in \mathbb{R}^2$  and a monotone sofa  $T$  with rotation angle  $\omega$  such that  $S + w_0 \subset T$  and  $|T| = |G|$ . By Lemma 11.5(a), the cap  $\mathcal{C}(T)$  is maximizing, and  $|T| = |G| \geq 11/5$ ; so  $R_\psi T$  is a moving sofa with rotation angle  $\pi/2$  (Lemma 8.7). It has area  $|G|$ , as rotations preserve area. So Lemma 11.5(b), for  $S' = R_\psi T$  and the rotation angle  $\pi/2$ , gives  $w_1 \in \mathbb{R}^2$  and a monotone sofa  $U$  with rotation angle  $\pi/2$  such that  $R_\psi T + w_1 \subset U$  and  $|U| = |G|$ , and  $U = G + (s, 0)$  for some  $s$  by Lemma 11.5(c). With  $v = R_\psi w_0 + w_1 - (s, 0)$ ,

$$R_\psi S + v = R_\psi(S + w_0) + w_1 - (s, 0) \subset R_\psi T + w_1 - (s, 0) \subset U - (s, 0) = G. \quad \square$$

**Theorem 11.7** (uniqueness from the certificate). *Let  $S$  be a moving sofa.*

- (a) *If  $|S| = |G|$ , then  $S + v = G$  for some  $v \in \mathbb{R}^2$ .*
- (b) *The following are equivalent: (i)  $|S| = |G|$ ; (ii)  $|S'| \leq |S|$  for every moving sofa  $S'$ ; (iii)  $R_\theta S + v = G$  for some  $\theta \in \mathbb{R}$  and  $v \in \mathbb{R}^2$ ; (iv)  $S + v = G$  for some  $v \in \mathbb{R}^2$ .*

*Proof.* (a) Let  $\omega$ ,  $\psi$  and  $v$  be as in Proposition 11.6, and  $g(p) = R_\psi p + v$ . The set  $g(S)$  is closed, since  $S$  is closed and  $g$  is a homeomorphism; it lies in  $G$ ;  $|g(S)| = |S| = |G|$ , as rotations and translations preserve area; and  $|G| < \infty$  (Fact 2.17(a)). As  $G = \text{cl}(\text{int } G)$  (Proposition 9.3),  $g(S) = G$  by Lemma 9.1. As in the proof of Corollary 9.6(a),  $p_y - q_y \leq 1$  for  $p, q \in S$ , since  $S$  lies in a horizontal strip of height one, and  $\langle g(p) - g(q), v_\psi \rangle = \langle p - q, v_0 \rangle = p_y - q_y$ ; so  $\langle x - y, u_{\psi+\pi/2} \rangle \leq 1$  for all  $x, y \in G = g(S)$ . If  $\psi > 0$ , then  $\psi + \pi/2 \in (\pi/2, \pi)$ , as  $\psi \leq \pi/2 - \omega_0 < \pi/2$ , and Lemma 9.5(b) gives  $p, q \in G$  with  $\langle p - q, u_{\psi+\pi/2} \rangle > 1$ , a contradiction. So  $\psi = 0$ , and  $S + v = g(S) = G$ .

(b) The first part of the proof of (a), which shows  $g(S) = G$ , gives (i)  $\Rightarrow$  (iii), and (a) is (i)  $\Rightarrow$  (iv); (iii) and (iv) imply (i), since rotations and translations preserve area. By Theorem 11.4(b),  $G$  is a moving sofa and  $|S'| \leq |G|$  for every moving sofa  $S'$ . So (ii) gives  $|G| \leq |S| \leq |G|$ , which is (i), and (i) gives  $|S'| \leq |G| = |S|$  for every moving sofa  $S'$ , which is (ii).  $\square$

These are the proofs of Theorem 1.1, of Corollary 9.4 and of the first assertion of Corollary 9.6(a), with Theorem 11.4 in place of Fact 2.8, Lemma 8.7 in place of Proposition 6.9, and Lemma 11.5(c) in place of Theorem 8.5; Theorem 11.7 contains these three results.

**11.5. Stability.** The proofs of Theorems 10.1 and 10.2 in Section 10 use optimality, uniqueness and the certificate at three places, and at each of them they use the results of this section.

- (A) The deficit  $\varepsilon(S) = |G| - |S|$  of a moving sofa  $S$  is nonnegative, by Theorem 11.4(b). The proof of Proposition 10.26 uses this, and so does the distinction between the cases  $\varepsilon = 0$  and  $\varepsilon > 0$  in the proof of the two theorems.
- (B) In the proof of Proposition 10.26, the limit  $S$  of the normalized sofas  $S_n^\natural$  is a moving sofa with  $|S| \geq |G|$ . Then  $|S| = |G|$  by Theorem 11.4(b), and Theorem 11.7(a) gives  $S + w = G$  for some  $w$ . In the case  $\varepsilon = 0$  of the proof of the two theorems, Theorem 11.7(a), applied in the same way to the normalized translate  $S^\natural$  of the given sofa  $S$ , gives  $S^\natural = G$ .
- (C) The proof of Proposition 10.11 applies Theorem 11.1 to the canonical triple of a cap  $K \in \mathcal{K}_{\pi/2}^c$  near  $K_G$ .

These are the only places where Sections 10.3 to 10.6 use optimality, uniqueness, Lemma 10.6 or Theorem 10.4; their other steps use these results only through Propositions 10.11 and 10.26. In turn, the proofs of Theorems 11.4 and 11.7 use from Section 10 only Lemma 10.6 and Theorem 10.4(a), through the certificate, and Sections 10.1 and 10.2 prove these two results without optimality or uniqueness. So (A) and (B) rest on the certificate through Theorems 11.4 and 11.7, and (C) on the certificate itself; there is no circularity.

**Theorem 11.8** (optimality, uniqueness and stability). *(a)  $G$  is a moving sofa, and every moving sofa  $S$  has  $|S| \leq |G|$ .  
 (b) A moving sofa  $S$  has  $|S| = |G|$  if and only if  $S + v = G$  for some  $v \in \mathbb{R}^2$ .  
 (c) The assertions of Theorems 10.1 and 10.2 hold.*

*Proof.* (a) is Theorem 11.4(b). (b) If  $|S| = |G|$ , then  $S + v = G$  for some  $v$  by Theorem 11.7(a); conversely, translations preserve area. (c) Section 10 proves Theorems 10.1 and 10.2 with (A)–(C) above, which come from Theorems 11.1, 11.4 and 11.7.  $\square$

## 12. FORMALIZATION, THE USE OF AI, AND OPEN QUESTIONS

**12.1. The formalization.** The repository [14] (Apache-2.0) contains five Lean 4 libraries on top of Mathlib (Lean and Mathlib v4.35.0-rc3):

- **MovingSofaOptimality** (51 files, about 40,000 lines): Baek’s paper [1], organized by its chapters, with the results that it cites (Schneider’s area formula for planar convex bodies; the existence and uniqueness of the solution of Romik’s system in the box of Fact 2.17) and the structure of Gerver’s sofa (Fact 2.17(b),(c)), which [1] states without proof (Thm. 8.4.1) or proves invalidly (Thm. 6.1.2);
- **MovingSofaUniqueness** (11 files, about 5,900 lines): Sections 4 to 9, one module per step of the argument, and three modules for the second proof of Section 8.4;
- **MovingSofaBridge** (4 files, about 2,600 lines): the connection with the definitions of formal-conjectures [4], which are different (a motion there is a path of isometries that starts at the identity, and Gerver’s sofa is

defined from Gerver’s four constants); it shows that a set is a moving sofa for formal-conjectures if and only if it lies in the horizontal side of the hallway and is a moving sofa in the sense of Appendix D, and that the optimal areas and the Gerver’s sofas of the two formulations agree, so that formal-conjectures’ statements follow from Baek’s theorem and from Theorem 1.1, with the existence and uniqueness of Gerver’s four constants, which the bridge proves; the uniqueness statement, which it lists as open, follows from Theorem 1.1;

- **MovingSofaStability** (87 files, about 12,500 lines): Section 10, and the certificate of Section 11 (Theorem 11.1);
- **MovingSofaExtremal** (7 files, about 660 lines): Section 11; the optimality and uniqueness theorems that **MovingSofaStability** uses come from it.

The file **Challenge.lean** states fifteen theorems, among them Baek’s theorem (**Baek.gerver\_sofa\_optimal**), Theorem 1.1 (**Baek.gerver\_sofa\_unique**), Theorem 10.1, Theorem 10.2 and Theorem 10.3(b) (**Baek.gerver\_sofa\_stable**, **Baek.gerver\_sofa\_angle\_stable**, **Baek.gerver\_sofa\_stability\_exponent**). The eight theorems of the namespace **Baek** use only Mathlib’s vocabulary and the definitions of Appendices B and D; the other seven use the definitions of formal-conjectures, restated verbatim in a namespace of their own (an anonymous topology instance is given a name), and the three theorems of the namespace **Bridge** also use Baek’s definitions. **Solution.lean**, with the modules that it imports, proves them: the **Challenge** is a file of statements, so its fifteen theorems end in **sorry** by design, and **Comparator** checks that **Solution.lean** proves exactly the statements of **Challenge.lean**. **SolutionCoercive.lean** proves the fifteen statements again, in a namespace of its own, taking optimality, uniqueness and the stability of Theorems 10.1 and 10.2 from **MovingSofaExtremal**; it imports neither **Solution.lean** nor the module **Main** of **MovingSofaUniqueness**. **lake build** succeeds, and an audit checks that every declaration of the five libraries and of **SolutionCoercive.lean** depends only on the axioms **propext**, **Classical.choice** and **Quot.sound**. The project is registered in the Palomar registry as entry PALOMAR-2026-10-02-000008. Its version 4 registers an earlier commit, whose **Challenge** states the twelve theorems other than the three of stability. The continuous integration passed on the commit cited in [14]. To rebuild, run **lake exe cache get && lake build**, then the commands listed in the repository’s **README.md**. The **Challenge** states Theorem 1.1, Theorem 10.1, Theorem 10.2 and Theorem 10.3(b); the other results of this paper are proved in the libraries, and Appendix D lists the declarations.

The machine check shows that Theorem 1.1, the theorems of Section 10 and those of Section 11, as the Lean statements formulate them, follow from Lean’s axioms and Mathlib. The formal proof contains Baek’s argument and the argument of this paper. The machine check does not check that the definitions say what they should. Appendix D reproduces the Lean definitions

(moving sofa, hallway, the rotation path, Gerver’s sofa, and the definitions used by the stability theorems) in mathematical notation, so that they can be read and compared with the intended meaning. Every proof of a result of [1] in the library follows Baek’s argument, except at sixteen steps, each forced by an error or gap of the paper, by mathematics that Mathlib lacks (the Jordan curve theorem, Green’s theorem, the Brunn–Minkowski inequality, mixed volumes), or by the representations chosen in the formalization (the surface area measure as a Lebesgue–Stieltjes measure, the area functional of a convex arc as  $\frac{1}{2} \int h d\sigma$ , no orientations of curves). A route check in continuous integration compares the results used by each Lean proof with those cited by Baek’s proof; the other details were compared with Baek’s TeX source by AI agents, not by a person.

The second proof of optimality (Section 8.4) is formalized in three modules of its own. They prove Theorem 8.8 and, from it, Theorem 1.1 and Corollary 9.4 again, and they do not import the module `Main` of `MovingSofaUniqueness`, which proves Theorem 1.1 from Baek’s theorem. A separate audit checks that no declaration of these modules depends, directly or through other declarations, on `Main`, on [1, Thm. 1.1.1], or on the results from which Baek derives step (3) for the caps of step (2): [1, Thm. 1.5.2, Thm. 8.1.1 (2)] and, among the results they rest on, [1, Thm. 4.1.2, 4.1.4, 4.2.5, 6.1.1, 6.3.3, 6.4.3, 6.5.6] and [1, Cor. 6.4.4].

The certificate of Section 11 is the theorem `coercive_certificate` of `MovingSofaStability`, and the theorem `gerver_sofa_optimal_unique_stable` of `MovingSofaExtremal` states Theorem 11.8. The stability library uses the certificate and the theorems of `MovingSofaExtremal` at the three places of Section 11.5. So `Solution.lean`, and not only `SolutionCoercive.lean`, proves Theorems 10.1 and 10.2 through the theorems of `MovingSofaExtremal`; its proof of Theorem 10.3(b) uses neither optimality nor uniqueness. A third audit, `scripts/AuditCoerciveRoute.lean`, which the continuous integration runs, follows the proofs through the bodies of all the declarations of the repository, private and generated ones included. It checks that each of the 958 declarations of `MovingSofaExtremal`, `MovingSofaStability` and `SolutionCoercive.lean` uses only the three axioms above. It also checks that none of them reaches Baek’s theorem [1, Thm. 1.1.1], the results listed in the previous paragraph, `Solution.lean`, or a declaration of the module `Main` of `MovingSofaUniqueness`, of the module `Rigidity` (which proves Lemmas 8.1 and 8.2 and, with `Main`, Proposition 8.3) or of the three modules of the second proof. It checks that none of the 40 declarations of the five modules of `MovingSofaExtremal` that prove optimality and uniqueness reaches the module that states the stability theorems or a module that imports it (55 modules in all), so that the argument is not circular. Twenty further checks confirm that the proofs use what they should; for instance, the classification of the maximizing caps reaches the certificate and Proposition 10.7, and the stability theorems reach the certificate and the uniqueness theorem of `MovingSofaExtremal`. Eight

checks confirm that the search finds, in the older proofs, results that these are known to use, which shows that the search sees proof bodies. Finally, the audit checks that the fifteen theorems of `SolutionCoercive.lean` have the types of those of `Solution.lean`.

**12.2. The use of AI.** AI systems wrote the first version of the proof, the formalization and this text, at the author’s request; none of them is an author of this paper. Appendix E names them, says what each did and what has been checked by whom. The argument has not yet been refereed.

**12.3. Open questions.** *Stability constants.* The constants of Theorems 10.1 and 10.2 are not explicit; the threshold  $\varepsilon_0$ , for instance, comes from a compactness argument (Proposition 10.26). The best constants are not known. Theorem 10.3 shows that the exponent  $1/2$  is optimal for the Hausdorff distance; whether it is optimal for the area of the symmetric difference is not known.

*Other corners.* For a hallway whose corner has an angle  $\alpha \neq \pi/2$  one can ask for the largest area as a function of  $\alpha$  and for the extremal shapes. Each step of Baek’s method has an analogue to prove, and its numerical inequalities, such as those of Fact 6.5, are specific to the right angle.

*The ambidextrous sofa.* Romik [10] found a sofa of area  $1.6449\dots$  that can turn both left and right around a right-angled corner. As far as we know, it is not known whether it is optimal. Baek’s reduction to monotone sofas uses one hallway; for a sofa that must turn both ways, the cap, the niche and the bound  $\mathcal{Q}$  all need two-sided analogues.

**Acknowledgments.** This paper builds on the results of [1]. The audit of [1] owes twelve findings to the notes of another formalization of that paper [9], and was also compared with the blueprint of a third [12].

## APPENDIX A. PICTURES OF BAEK’S ARGUMENT

The figures of this appendix illustrate the notions and the steps of Baek’s proof described in Section 1.3.

## APPENDIX B. GERVER’S SOFA

We recall Romik’s description of Gerver’s sofa [10, Section 4], in the form used by [1] and by the formalization. Let  $t_0 = 0 < t_1 = \varphi < t_2 = \theta < t_3 = \pi/2 - \theta < t_4 = \pi/2 - \varphi < t_5 = \pi/2$ . A *parameter tuple* consists of the angles  $\varphi, \theta$ , numbers  $a_1, a_2, b_1, b_2, c_1, c_2, d_1, d_2, e_1, e_2$  and points  $\kappa_1, \dots, \kappa_5 \in \mathbb{R}^2$  (22 real parameters). The *rotation path*  $\mathbf{x} : [0, \pi/2] \rightarrow \mathbb{R}^2$  is  $\mathbf{x} = \mathbf{x}_i$  on the  $i$ th

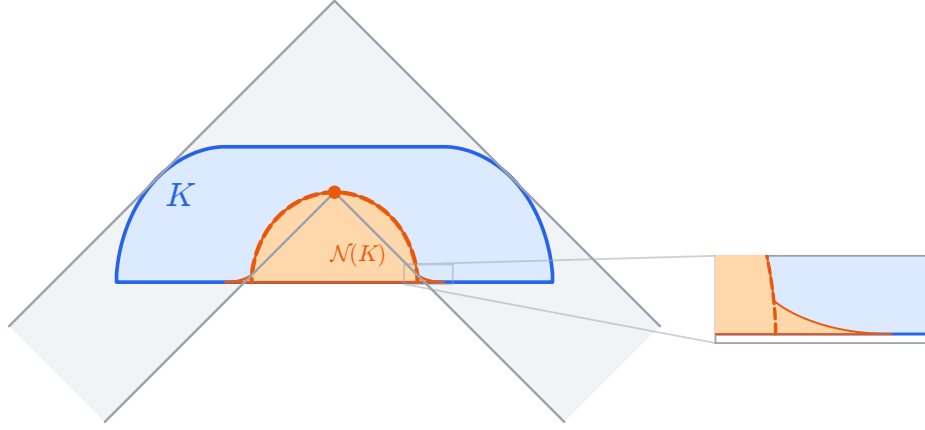


FIGURE 10. Gerver's cap  $K$ , the union of the blue and the orange regions, and its niche  $\mathcal{N}(K)$  (orange); the sofa  $G = K \setminus \mathcal{N}(K)$  is the blue region. Grey: the copy of the hallway at the angle  $t = \pi/4$ . Its outer walls touch  $K$ , and its inner corner (orange dot) lies on the dashed curve, the path of the inner corner of the copies as  $t$  runs from 0 to  $\pi/2$ . Near its two ends the niche extends beyond this path, down to the floor, where the inner walls, not the corner, touch the sofa; the inset magnifies the right end, and the left end is its mirror image.

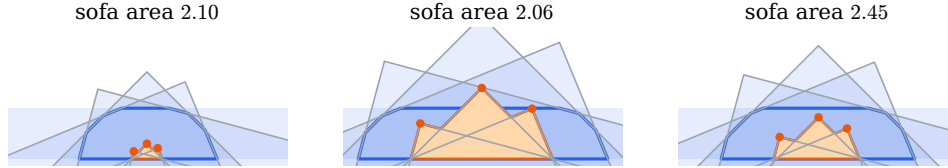


FIGURE 11. Three placements of copies of the hallway at the same three angles  $t = \pi/8, \pi/4, 5\pi/12$  (orange dots: their inner corners). Each copy shades in light blue the region inside its two outer walls, and so do the start and end positions, which keep the sofa in the strip  $0 \leq y \leq 1$ ; the polygonal cap, outlined in blue, is the darkest region, where all of them overlap, and the polygonal niche (orange) is the part of the cap beyond both inner walls of at least one copy. Left: copies close in give a small cap. Middle: copies far out give a large cap, but also a large niche. Right: a maximum polygon cap for these angles.

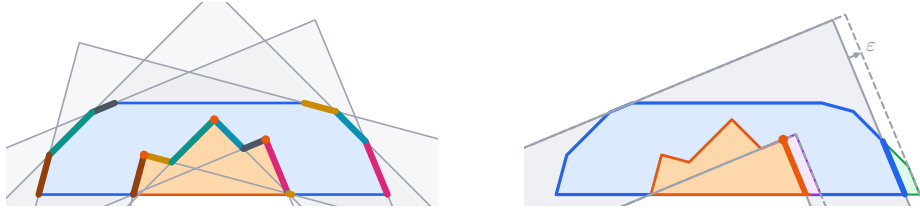


FIGURE 12. Balance of the maximum polygon cap of Figure 11. Left: the three copies of the hallway (light grey; orange dots: their inner corners); each slanted side of the cap lies on an outer wall of one of the copies, and is drawn thick in the same colour as the side of the niche on the inner wall of the same copy parallel to it; the two have the same length. A side of the niche may consist of several pieces, as for the amber pair. Right: Gerver's balancing argument. The copy at the angle  $\pi/8$  (light grey) is moved outward by  $\varepsilon$  (dashed), which moves one of its outer walls and the inner wall parallel to it and leaves the other two walls in place. This adds to the cap a strip along its side on that wall (green), and to the niche a strip along its side on the inner wall (purple); the two sides are drawn thick. The sofa area changes by about  $\varepsilon$  times the difference of their lengths, so if they differed, moving the walls outward or inward would increase it; here the two sides have the same length, 0.63, as they do at each of the six slanted sides of this cap.

phase, where the phases are  $[0, t_1)$ ,  $[t_1, t_2)$ ,  $[t_2, t_3]$ ,  $(t_3, t_4]$  and  $(t_4, t_5]$ , and

$$\begin{aligned} \mathbf{x}_1(t) &= R_t(a_1 \cos t + a_2 \sin t - 1, -a_2 \cos t + a_1 \sin t - \tfrac{1}{2}) + \kappa_1, \\ \mathbf{x}_2(t) &= R_t(-\tfrac{t^2}{4} + b_1 t + b_2, \tfrac{t}{2} - b_1 - 1) + \kappa_2, \\ \mathbf{x}_3(t) &= R_t(c_1 - t, c_2 + t) + \kappa_3, \\ \mathbf{x}_4(t) &= R_t(-\tfrac{t}{2} + d_1 - 1, -\tfrac{t^2}{4} + d_1 t + d_2) + \kappa_4, \\ \mathbf{x}_5(t) &= R_t(e_1 \cos t + e_2 \sin t - \tfrac{1}{2}, -e_2 \cos t + e_1 \sin t - 1) + \kappa_5. \end{aligned}$$

The pieces of a solution agree at the ends of the phases, by condition (3) below, so the conventions at the ends matter only for other tuples. For a differentiable curve  $\mathbf{x}$  let  $\alpha = \langle \mathbf{x}', u_t \rangle$ ,  $\beta = \langle \mathbf{x}', v_t \rangle$ , and let  $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$  be as in Section 2.6. The tuple *solves Romik's equations* if  $0 < \varphi < \theta < \pi/4$  and:

- (1) (left-right symmetry)  $e_1 = a_1$ ,  $e_2 = -a_2$ ,  $d_1 = \frac{\pi}{4} - b_1$ ,  $d_2 = b_2 + \frac{\pi}{4}(2b_1 - \frac{\pi}{4})$ ,  $c_2 = c_1 - \frac{\pi}{2}$ ;
- (2) (the start)  $\kappa_1 = (1 - a_1, \frac{1}{4})$  and  $a_2 = -\frac{1}{4}$ , so that  $\mathbf{x}(0) = O$  and  $\mathbf{A}(0) = (1, 0)$ ;



FIGURE 13. Step (3a) for an actual sofa: a polygonal cap with rotation angle  $\omega = 1.1$  (about  $63^\circ$ ) minus its niche. A copy of the sofa turned by  $\pi/2 - \omega$  (green) first turns back inside the horizontal side, before the corner, through the position halfway (teal), to the sofa's own starting position (purple). This is possible because the small triangle at the inner corner (orange) lies in the hollow of the sofa, which keeps its height at most 1 throughout this turn. The copy then follows the sofa's own movement and turns by  $\omega$  around the corner (blue halfway, grey at the end, and again slid down the vertical side). In total, it turns by the full right angle.

- (3) (continuous differentiability)  $\mathbf{x}_i(t_i) = \mathbf{x}_{i+1}(t_i)$  and  $\mathbf{x}'_i(t_i) = \mathbf{x}'_{i+1}(t_i)$  for  $i = 1, 2, 3, 4$ ;
  - (4) (the contact conditions)  $\mathbf{x}_1(t_1) = \mathbf{B}_4(t_3)$  and  $\mathbf{x}_5(t_4) = \mathbf{D}_2(t_2)$ , where  $\mathbf{B}_4 = \mathbf{x}_4 + \alpha_4 v_t$  and  $\mathbf{D}_2 = \mathbf{x}_2 - \beta_2 u_t$  are the contact curves of  $\mathbf{x}_4$  and  $\mathbf{x}_2$ .
- These are Romik's equations (27)–(44). *Gerver's sofa* is the shape of the rotation path of the solution with  $\varphi \in [0.039, 0.04]$  and  $\theta \in [0.68, 0.69]$ , which is unique in this box (Fact 2.17(a)). Its numerical values, to ten decimal

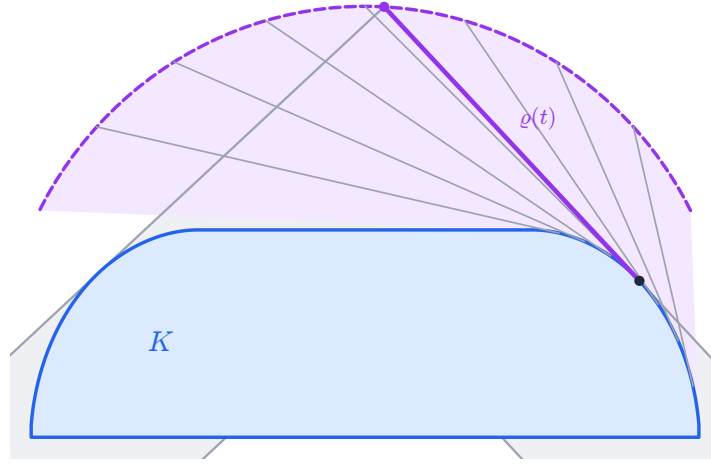


FIGURE 14. One of the terms of  $\mathcal{Q}$  in step (4), for Gerver's cap  $K$ . For an angle  $t$ , the segment of length  $\varrho(t)$  runs along an outer wall of the copy of the hallway at the angle  $t$ , from the point where  $K$  touches that wall (black dot) to the outer corner of the copy (purple dot); the copy at the angle  $t$  is drawn in light grey, with its two outer walls touching  $K$ . The thin grey segments are the same segment for other angles. As  $t$  runs from  $\varphi$  to  $\pi/2 - \varphi$ , where  $\varphi = 0.039\dots$  is Gerver's angle (Appendix B) and this is the range of angles in which the inner corner of the hallway touches Gerver's sofa, the segment sweeps the shaded region, whose area is  $\frac{1}{2} \int \varrho(t)^2 dt$  by Mamikon's theorem; the dashed curve is the path of the outer corner.

places, are

$$\begin{aligned} \varphi &= 0.0391773648, & \theta &= 0.6813015094, \\ a_1 &= e_1 = 1.2103224221, & a_2 &= -e_2 = -0.25, \\ b_1 &= -0.5276245980, & b_2 &= 0.9202583852, \\ c_1 &= 0.6260455228, & c_2 &= -0.9447508039, \\ d_1 &= 1.3130227614, & d_2 &= -0.5253826704, \end{aligned}$$

$$\begin{aligned} \kappa_1 &= (-0.2103224221, 0.25), & \kappa_5 &= (-1.0172040368, 0.25), \\ \kappa_2 &= (-0.9191792928, 0.4724066198), \\ \kappa_3 &= (-0.6137632294, 0.8896264790), \\ \kappa_4 &= (-0.3083471661, 0.4724066198), \end{aligned}$$

and  $\mathbf{x}(\pi/2) = (-1.2275264589, 0)$ ,  $\max_t \mathbf{x}(t)_y = \mathbf{x}(\pi/4)_y = 0.6642678135$ ,  $\mathbf{x}(t_1)_y = \mathbf{x}(t_4)_y = 0.0551867605$ . The area is  $|G| = 2.21953166887\dots$  (Gerver, Romik), and  $2.2192 \leq |G| \leq 2.2199$  is proved in [14]. In Gerver's own parametrization the constants  $A = 0.0944265608$  and  $B = 1.3992037273$

together with  $\varphi$  and  $\theta$  solve a system of four equations [3], from which the 22 parameters above are computed.

The contact curves  $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$  are the points where the sides of the hallway touch the sofa. The sofa touches the walls  $a(t)$  and  $c(t)$  at  $\mathbf{A}(t)$  and  $\mathbf{C}(t)$  for all  $t$ ; the wall  $d(t)$  at  $\mathbf{D}(t)$  for  $t \in [t_0, t_2]$ ; the wall  $b(t)$  at  $\mathbf{B}(t)$  for  $t \in [t_3, t_5]$ ; and the inner corner touches it at  $\mathbf{x}(t)$  for  $t \in [t_1, t_4]$ . As in [3, 10], the boundary of  $G$  consists of 18 pieces:  $\mathbf{A}$  on four phases, the top edge,  $\mathbf{C}$  on four phases, an edge on the  $x$ -axis,  $\mathbf{D}$  on two phases,  $\mathbf{x}$  on three phases,  $\mathbf{B}$  on two phases, and a second edge on the  $x$ -axis.

#### APPENDIX C. CORRECTIONS TO BAEK’S PAPER THAT ARE USED HERE

The audit that accompanies the formalization compared every result of [1] with its TeX source and with the formal statement [14]. Every result of the paper holds with the intended statement. Table 1 lists the corrections that concern the results used in this paper, with the labels  $En$  of the audit. The complete list of the audit has 27 items, E22 only pointing to E21, and two missing hypotheses: [1, Thm. 2.1.1] (Schneider’s Theorem 4.2.3:  $\sigma_K(X)$  is the length of  $\bigcup_{t \in X} e_K(t)$ ) needs  $K$  to have interior points, and [1, Thm. 3.1.2] needs a bounded Nef polygon; every use satisfies them (the text itself uses  $\sigma_K$  only as defined in Fact 2.1(b), which does not depend on Thm. 2.1.1). Twelve of the findings in the audit come from the notes of another formalization of Baek’s proof [9].

Three Facts of Section 2 are known to us only through the formalization [14], because the printed proof is missing or invalid, or because the statement is not in [1]: Fact 2.16(a) (E21), Fact 2.17 (E12, E24) and Fact 2.18. Baek states [1, Thm. 8.4.1] without proof (his Remark 8.4.1: the properties are “easy to verify numerically”) and proves [1, Thm. 6.1.2] by a misreading of a theorem of Gerver; the statements are true, and the formalization proves them from Romik’s equations. Romik asserts the uniqueness of the solution of his equations for  $0 < \varphi < \theta < \pi/4$  without proof; the uniqueness in the box  $\varphi \in [0.039, 0.04]$ ,  $\theta \in [0.68, 0.69]$  is proved in [14]. Baek’s [1, Thm. 8.4.1 (2)] describes the niche of Gerver’s cap as the region enclosed by  $\mathbf{B}$  reversed,  $\mathbf{x}|_{[t_1, t_4]}$ ,  $\mathbf{D}$  reversed and the segment of the  $x$ -axis from  $\mathbf{D}(t_0)$  to  $\mathbf{B}(t_5)$ ; that open region omits the open segment of the  $x$ -axis between  $\mathbf{D}(t_0)$  and  $\mathbf{B}(t_5)$ , which lies in the niche, and the exact description is the identity in Fact 2.17(c). Fact 2.17(d) and Fact 2.18 are not in [1]; the former is proved from Romik’s equations, the second identity as  $\mathbf{C}(\pi/2) = \mathbf{x}(\pi/2) - \beta(\pi/2)u_{\pi/2} + v_{\pi/2}$  with  $\beta(\pi/2) = 0$ .

Table 1: Corrections to [1] that concern the results used here.

| Item | Result of [1]                                           | Correction                                                                                                                                                                                                                                                                                                                     | Where here                         |
|------|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| E1   | Def. 2.3.9                                              | the left and right side of a line are defined for lines not parallel to the $x$ -axis (printed: $y$ -axis)                                                                                                                                                                                                                     | Fact 2.4(a)                        |
| E2   | Thm. 2.4.1, 2.4.2                                       | the union over $t \in [0, \omega]$ and over $(0, \omega)$ agree, since $F_\omega \cap Q_K^-(0) = F_\omega \cap Q_K^-(\omega) = \emptyset$ ; $\mathcal{C}(S)$ is bounded also for $\omega = \pi/2$                                                                                                                              | Fact 2.4(b)                        |
| E3   | Prop. 2.5.4                                             | the reflection exchanges the walls ( $a \leftrightarrow c$ , $b \leftrightarrow d$ ) and the ends $W \leftrightarrow Z$ ; $K$ and not $K^m$ stands on the right of the vertex identity                                                                                                                                         | Fact 2.4(d), Lemma 2.6             |
| E4   | Lemma 2.5.6, Thm. 2.5.8 (and Lemma 3.4.8, Remark 3.4.1) | used without proof, true for every cap: $A_K^-(0) = (h_K(0), 0)$ , $C_K^+(\omega) = h_K(\omega + \pi/2)v_\omega$ , and $O \in K$ if $\omega < \pi/2$ ; the same holds for $o_\omega$ , which Lemma 3.4.8 and Remark 3.4.1 use (not in the audit)                                                                               | Lemma 2.5, Lemma 2.6, Fact 2.14(d) |
| E5   | Thm. 3.1.2, Lemma 3.4.7                                 | the theorem needs its simple Nef polygon, a set $\{p : \mathcal{E}(p \in H_1, \dots, p \in H_n)\}$ for closed or open half-planes $H_1, \dots, H_n$ with distinct boundary lines and a monotone Boolean function $\mathcal{E}$ , to be bounded; the two applications of the theorem for a pinned normal change disjoint strips | Fact 2.12(a)                       |
| E6   | Def. 3.2.5                                              | $\mathcal{N}_\Theta(K) = F_\omega \cap \bigcup_{t \in \Theta} Q_K^-(t)$ (printed: $P_\omega$ in place of $F_\omega$ ); with $P_\omega$ , Prop. 3.3.5 and Lemma 3.4.2 are false                                                                                                                                                 | Section 2.3, Fact 2.10             |
| E7   | Prop. 3.3.1 (2)                                         | normal angles in $\Theta^\circ \cup \{\omega + \pi, 3\pi/2\}$ (printed: $\Theta^\circ$ )                                                                                                                                                                                                                                       | Section 2.3, Fact 2.12             |

| Item | Result of [1]                        | Correction (continued)                                                                                                                                                                                                                                                                                      | Where here                                                             |
|------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| E8   | Thm. 3.4.3,<br>3.5.2, 3.5.4          | minor gaps: a Hausdorff limit of polygon caps is a polygon cap (Thm. 3.4.3), resp. a cap (Thm. 3.5.2), and each point of the niche of the limit lies in the niches of the polygon caps from some index on (Thm. 3.5.4)                                                                                      | Fact 2.7                                                               |
| E9   | Lemma 4.2.4                          | the range is $\omega \in [\sec^{-1}(2.2), \pi/2)$ (printed: $[\tan^{-1}(2.2), \pi/2)$ )                                                                                                                                                                                                                     | Fact 6.5                                                               |
| E10  | Thm. 4.2.5,<br>1.5.2                 | the width bound for the pentagon $P_\omega \setminus \Delta_\omega$ is asserted without proof, and $\Delta_\omega \subset \mathcal{N}(K)$ is taken from Thm. 4.2.5, which puts the vertices of $\Delta_\omega$ in the closure of a quarter-plane $Q_K^-(t)$ ; its proof puts them in the open quarter-plane | Lemma 6.6,<br>Lemma 6.7                                                |
| E12  | Thm. 6.1.2                           | the proof (Gerver's Thm. 2 makes $G$ a balanced maximum sofa) is invalid; the statement is true and follows from Romik's equations                                                                                                                                                                          | Fact 2.17(b)                                                           |
| E13  | Prop. 6.2.2                          | $f_{K^m}^\pm(t) = g_K^\mp(\pi/2 - t)$ , and likewise for $g$ (printed: without the change of angle)                                                                                                                                                                                                         | Fact 2.14(c)                                                           |
| E14  | Def. 6.2.2,<br>Thm. 6.2.3            | $\partial^+$ is the right derivative (Def. 6.2.2 calls it the left one), and the second display of Thm. 6.2.3 is about $\partial^- \mathbf{x}_K$                                                                                                                                                            | Fact 2.14(a)                                                           |
| E15  | Lemma 6.3.1                          | cites Thm. 3.5.4 (balanced maximum caps) for $\mathcal{N}(K) \subset K$ of a maximum polygon cap; the applicable result is Thm. 3.4.10, which gives what the lemma needs, and the lemma holds                                                                                                               | not used in the proof (the box of Proposition 4.5 bounds the diameter) |
| E16  | Def. 6.4.1                           | $f_K$ and $g_K$ map $[0, \pi/2]$ to $\mathbb{R}$ (printed: to $\mathbb{R}^2$ )                                                                                                                                                                                                                              | Fact 2.14(d)                                                           |
| E17  | Lemma 6.5.2–<br>6.5.5,<br>Thm. 6.5.6 | $f_K(0) = 1$ is used without being stated, and so is $g_K(\pi/2) = 1$ (not in the audit); Lemma 6.5.5 is used on $(0, \pi/2]$ (printed: $(0, 1]$ )                                                                                                                                                          | Fact 2.14(d),<br>Fact 2.15(b)                                          |

| Item     | Result of [1]                   | Correction (continued)                                                                                                                                                                                                                                                                                                    | Where here                                    |
|----------|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| E19      | Lemma 8.1.5                     | the proof deduces that the bottom edge $e_K(3\pi/2)$ has length $\sigma_K(3\pi/2) \geq 2.2$ from $ K  \geq 2.2$ without a reason; this holds, since the horizontal width of a cap with rotation angle $\pi/2$ does not increase with the height, so that $ K  \leq \sigma_K(3\pi/2)$                                      | Fact 2.16(a)                                  |
| E20      | Lemma 8.3.5                     | the arc $\mathbf{u}_K^{0,\pi}$ is outside the range $b < a + \pi$ of [1, Thm. 7.3.2]; the area is split at the atom $\pi/2$ instead                                                                                                                                                                                       | Fact 2.16(b)                                  |
| E21      | Lemma 8.1.7, Thm. 8.2.4         | $\mathcal{N}(K) \subset K$ is used but not available on $\mathcal{K}^i$ ; both statements hold and have proofs that avoid it                                                                                                                                                                                              | Fact 2.16(a)                                  |
| E23      | Lemma 8.3.6 (3)                 | the tangent line parametrization is evaluated at $\pi/2$ (printed: at $\varphi^L$ , where the statement is false)                                                                                                                                                                                                         | Fact 2.16(b)                                  |
| E24      | Thm. 8.4.1                      | stated without proof; true, and proved from Romik's equations; part (4) holds with one-sided derivatives at the junctions and the ends of the phases. Not in the audit: part (2), read as a Jordan region, omits the open floor segment of $\mathcal{N}(K_G)$ , and the exact description is the identity in Fact 2.17(c) | Fact 2.17                                     |
| E25, E26 | Thm. 8.4.3 (2), Prop. 8.4.4 (4) | $\mathbf{B}(t_3)$ (printed: $\mathbf{D}(t_3)$ ); the measure $\check{\sigma}_D$ lives on $(\pi/2 + t_0, \pi/2 + t_2]$ with a shifted density                                                                                                                                                                              | Fact 2.16(c)                                  |
| E27      | several                         | misprints, among them: the second half-plane of $Q_S^-(t)$ lacks its “−1” (Prop. 2.2.2); the sign of the change of $ \mathcal{N}_\Theta $ in the proof of Lemma 3.4.7; $\mathcal{R}_B$ starts at $\pi + \varphi^R$ , not $\pi/2 + \varphi^R$ (Def. 8.3.3); the statement of Thm. 2.1.3                                    | Section 2.2, Fact 2.1, Fact 2.12, Section 2.5 |

## APPENDIX D. THE LEAN FORMALIZATION: THE STATEMENT AND A DICTIONARY

**D.1. The statement of Theorem 1.1.** The statement of record is `Baek.gerver_sofa_unique` in `Challenge.lean`. It uses Mathlib’s vocabulary and the definitions below, which `Challenge.lean` copies verbatim from `ChallengeDefs.lean`; they are those of this paper. In Lean the plane is  $\mathbb{R} \times \mathbb{R}$ , the area of a set is its Lebesgue measure `volume`, and `rot t` is the counterclockwise rotation by  $t$ .

- `horizSide` =  $\{p : p_1 \leq 1, 0 \leq p_2 \leq 1\} = H_L$ , `vertSide` =  $\{p : 0 \leq p_1 \leq 1, p_2 \leq 1\} = V_L$ , `hallway` =  $H_L \cup V_L = L$ .
- `IsMovingSofa S` holds if  $S$  is closed and connected (hence nonempty) and there are  $\theta : \mathbb{R} \rightarrow \mathbb{R}$  and  $c : \mathbb{R} \rightarrow \mathbb{R}^2$ , continuous on  $[0, 1]$ , with  $\theta(0) = 0$ ,  $\text{rot}_{\theta(0)}p + c(0) \in H_L$  for  $p \in S$ ,  $\text{rot}_{\theta(s)}p + c(s) \in L$  for  $p \in S$  and  $s \in [0, 1]$ , and  $\text{rot}_{\theta(1)}p + c(1) \in V_L$  for  $p \in S$  (Definition 2.2, the rotation angle  $-\theta(1)$  being left free).
- `shapeOfPath x` =  $H_L \cap \bigcap_{t \in [0, \pi/2]} (\mathbf{x}(t) + R_t L) \cap (\mathbf{x}(\pi/2) + R_{\pi/2} V_L)$ , written in Lean with images of the hallway sets under  $p \mapsto \mathbf{x}(t) + \text{rot}_t p$ .
- `GerverParams` holds the 22 parameters of Appendix B; `P.path` is the rotation path glued from the five curves of Appendix B; `P.IsSolution` is the conjunction of  $0 < \varphi < \theta < \pi/4$  and the four groups (1)–(4) of Appendix B (Romik’s equations (27)–(44)); `P.InBox` says  $\varphi \in [0.039, 0.04]$ ,  $\theta \in [0.68, 0.69]$ ; and `gerverSofa P` = `shapeOfPath P.path`.

The theorem states that for every `P : GerverParams` with `P.IsSolution` and `P.InBox`, and every set  $S \subset \mathbb{R} \times \mathbb{R}$  with `IsMovingSofa S` and `volume S = volume (gerverSofa P)`, there are  $\theta \in \mathbb{R}$  and  $v \in \mathbb{R} \times \mathbb{R}$  such that the image  $\{\text{rot}_{\theta}p + v : p \in S\}$  equals `gerverSofa P`. The theorems `Baek.gerver_params_exists` and `Baek.gerver_params_unique` say that the box contains exactly one solution, so that `gerverSofa P` is Gerver’s sofa  $G$ , and `Baek.gerver_sofa_area` says that its area lies between 2.2192 and 2.2199.

**D.2. The statements of stability.** In `Challenge.lean`, the statements of record of Theorem 10.1, Theorem 10.2 and Theorem 10.3(b) are the theorems `Baek.gerver_sofa_stable`, `Baek.gerver_sofa_angle_stable` and `Baek.gerver_sofa_stability_exponent`. Besides the definitions above, they use five more, which `Challenge.lean` also copies from `ChallengeDefs.lean`. Here  $|X|$  is the Lebesgue measure of  $X$  as a real number.

- `IsMovingSofaWithAngle S ω` holds if  $S$  satisfies the conditions of `IsMovingSofa S` with a movement for which also  $\theta(1) = -\omega$ , that is, if  $S$  is a moving sofa with rotation angle  $\omega$  (Definition 2.2).
- `sofaDeficit P S` =  $|\text{gerverSofa P}| - |S|$ , the deficit  $\varepsilon(S)$  of Section 10.
- `normalizedSofa P S` =  $S + (a_G - a_S, 1 - b_S)$ , where  $a_X$  is the infimum of the abscissas of the points of  $X$  and  $b_S$  the supremum of the heights of the points of  $S$ . For a moving sofa  $S$ , which is compact, this is the translate

$S^\natural$  of Section 10: its highest point has height 1, and its leftmost point has the abscissa of the leftmost point of  $G$ .

- **euclideanDist**  $p\ q = \sqrt{(p_1 - q_1)^2 + (p_2 - q_2)^2}$ , and **EuclideanClose**  $r\ S\ T$  says that  $S$  and  $T$  are  $r$ -close (Section 10) for this distance: every point of  $S$  lies at distance at most  $r$  from a point of  $T$ , and every point of  $T$  at distance at most  $r$  from a point of  $S$ .

Let  $P$  satisfy **P.IsSolution** and **P.InBox**, and write  $\varepsilon$  for **sofaDeficit**  $P\ S$ . The three theorems state the following.

- **Baek.gerver\_sofa\_stable**: there are  $C, C', \varepsilon_0 > 0$  such that every  $S$  with **IsMovingSofa**  $S$  and  $\varepsilon < \varepsilon_0$  satisfies **EuclideanClose**  $(C\sqrt{\varepsilon})\ N$  (**gerverSofa**  $P$ ) and  $|N \triangle G| \leq C'\sqrt{\varepsilon}$ , where  $N = \text{normalizedSofa } P\ S$  and  $\triangle$  is the symmetric difference.
- **Baek.gerver\_sofa\_angle\_stable**: there are  $C, \varepsilon_0 > 0$  such that every  $S$  and  $\omega$  with **IsMovingSofaWithAngle**  $S\ \omega$ ,  $\omega \in [\arccos(5/11), \pi/2]$  and  $\varepsilon < \varepsilon_0$  satisfy  $\pi/2 - \omega \leq C\varepsilon$ .
- **Baek.gerver\_sofa\_stability\_exponent**: for all  $a > 1/2$ ,  $C \in \mathbb{R}$  and  $\varepsilon_0 > 0$  there is a set  $S$  with **IsMovingSofa**  $S$  and  $0 < \varepsilon < \varepsilon_0$  such that **EuclideanClose**  $(C\varepsilon^a)\ S\ \{ \text{rot}_\theta p + v : p \in G \}$  fails for every  $\theta \in \mathbb{R}$  and  $v \in \mathbb{R} \times \mathbb{R}$ .

**D.3. Dictionary.** Table 2 lists the declarations that prove the results of this paper; they are in the repository [14]. Each name is followed, directly or after other names of the same file, by its file in parentheses: a file name without a directory, such as **Main**, is a file of **MovingSofaUniqueness/**, a path with a directory, such as **Basic/SurfaceArea**, is a file of **MovingSofaOptimality/**, and a path ending in **.lean** starts at the root of the repository. The theorems of **Challenge.lean** are stated there with **sorry** and proved in **Solution.lean**. Names are written without the namespace of their library: **image\_eq\_gerver\_of\_volume\_eq** is **MovingSofaUniqueness.image\_eq\_gerver\_of\_volume\_eq**. The declarations of the second proof of Section 8.4 are in the namespace **MovingSofaUniqueness.MaximizerRoute**, and they are written with the prefix **MaximizerRoute**, which also tells them apart from the declarations of the same name in **Main**. In the rows of Section 10, a file name without a directory, such as **GlobalStability**, is a file of **MovingSofaStability/**, whose declarations are written without the namespace **MovingSofaStability**: **unrestricted\_stability** (**GlobalStability**) is **MovingSofaStability.unrestricted\_stability** in **MovingSofaStability/GlobalStability.lean**. For the declarations of Section 11 in the library **MovingSofaExtremal**, its namespace **MovingSofaExtremal** is omitted, and the name is followed by the path of the file from the root, as in **area\_le\_gerver** (**MovingSofaExtremal/Optimality.lean**); the path tells them apart from the declarations of the same name in **Main** and **Rigidity**. In the rows of Section 11, the files of **MovingSofaStability/** are also given by their paths from the

root. A result numbered  $n_1.n_2.n_3$  in [1] is formalized under the name `theorem $n_1$ _ $n_2$ _ $n_3$` , or `lemma`, `proposition`, `corollary` or `definition` in place of `theorem`, followed by a suffix, as in `proposition2_5_4_sigma` or `definition8_1_2_exists`, when its parts are separate declarations. Some of the listed declarations, such as `IsCap.zero_mem`, are private to their files.

Table 2: The results of this paper and the Lean declarations that prove them.

| Result                      | Lean declarations                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Theorem 1.1                 | <code>Baek.gerver_sofa_unique</code> (stated in <code>Challenge.lean</code> , proved in <code>Solution.lean</code> ); <code>image_eq_gerver_of_volume_eq</code> , <code>maximizer_contained_in_gerver</code> (Main)                                                                                                                                                                     |
| Lemma 2.5                   | <code>IsCap.zero_mem</code> , <code>IsCap.oPt_mem</code> (Monotone/CapContainsNiche)                                                                                                                                                                                                                                                                                                    |
| Lemma 2.6, (2.3)            | <code>IsCap.corner_mem</code> , <code>IsCap.cPlus_eq_corner</code> , <code>IsCap.aMinus_eq_corner</code> (Monotone/CapContainsNiche)                                                                                                                                                                                                                                                    |
| Definition 3.1              | <code>IsMaxCap</code> (Main)                                                                                                                                                                                                                                                                                                                                                            |
| Proposition 3.2             | <code>maximal_envelope</code> , <code>own_cap_isMax</code> , <code>isMaxCap_of_area_eq</code> (Main), with <code>theorem2_4_3</code> (Monotone/CapNiche) and <code>theorem2_5_10</code> (Monotone/CapContainsNiche); the last statement: <code>theorem1_5_1</code> (Intro/RotationAngleBound)                                                                                           |
| (4.1), (4.2), (4.3), (4.4), | <code>dyadic_totalWeight</code> , <code>dyadic_totalWeight_le_one</code> ; <code>persistent_sample_first</code> , <code>persistent_sample_second</code> ;                                                                                                                                                                                                                               |
| Definition 4.1              | <code>SupportSamples.penalty_change_uniform</code> , with <code>abs_square_increment_le</code> ; <code>SupportSamples.penalty_change_one</code> ; <code>IsPenalizedMax</code> , with $\Pi_k$ given by <code>dyadicPenalty</code> , <code>dyadicSamples</code> , <code>dyadicLevelWeight</code> (Selection) and $\Theta_k$ by <code>dyadicAngleSet</code> (Balanced/BalancedMaximumSofa) |
| Lemma 4.2                   | <code>SupportSamples.penalty_recovery_zero</code> , <code>SupportSamples.recovery_objective_ge</code> , <code>dyadic_recovery_ge</code> (Selection)                                                                                                                                                                                                                                     |
| Lemma 4.3                   | <code>polygon_subset_sampleBox</code> , <code>sampleBoxRadius</code> (Selection)                                                                                                                                                                                                                                                                                                        |
| Lemma 4.4                   | <code>niche_subset_box</code> , <code>polyNiche_subset_box</code> (Selection)                                                                                                                                                                                                                                                                                                           |
| Proposition 4.5             | existence: <code>exists_dyadic_penalizedMax</code> , <code>exists_penalizedMax</code> , <code>penalty_and_area_le_of_positive</code> (Selection), <code>mpc_exists_limit_polycap</code> , <code>mpc_le_limit_objective</code> (Balanced/MaxPolygonCapExists); the lower bound: <code>selected_objective_ge</code> ; the box: <code>selected_sequence_bounded</code> (Selection)         |
| Lemma 4.6                   | <code>dyadic_objective_limsup</code> (Selection)                                                                                                                                                                                                                                                                                                                                        |
| Lemma 4.7                   | <code>caps_eq_of_dyadic_supports</code> , <code>eqOn_of_dyadic_eq</code> , <code>caps_eq_of_upper_supports</code> (Selection), with <code>nef_eq_setOf_supp</code> (Balanced/PolygonCap)                                                                                                                                                                                                |
| Proposition 4.8             | <code>exists_selectedCapSequence</code> , <code>SelectedCapSequence</code> , <code>penalty_tendsto_zero_of_objective</code> (Selection); the limit is a cap: <code>mpc_limit_isCap</code> (Balanced/BalancedMaximumSofa)                                                                                                                                                                |
| Lemma 5.1                   | <code>floatingCap</code> , <code>floatingCap_polygon</code> , <code>subset_floatingCap</code> , <code>floatingCap_support_bounds</code> , <code>floatingCap_support_other</code> , <code>floatingCap_support_self</code> (Variation)                                                                                                                                                    |

| Result                    | Lean declarations (continued)                                                                                                                                                                                                                       |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proposition 5.2           | floating_defect_le, floating_penalty_growth,<br>assigned_comparison_of_actual,<br>polygon_defect_le_penalty_growth (Variation)                                                                                                                      |
| Lemma 5.3                 | pinned_contract_mem, pinned_div_mem,<br>pinned_raw_support_bound (Variation)                                                                                                                                                                        |
| Lemma 5.4                 | translated_strip_support, pinned_translation_bound,<br>pinned_normalized_support_bound (Variation)                                                                                                                                                  |
| Proposition 5.5           | pinned_defect_le, polygon_defect_le_of_zero_facet<br>(Variation)                                                                                                                                                                                    |
| Corollary 5.6             | abs_selected_defect_le, polygon_weighted_defect_zero,<br>selectorDefectBound, selectorDefectBound_sum_le,<br>abs_weighted_defect_le (Variation)                                                                                                     |
| Lemma 6.1                 | ang_wedgeGapWInf_le_tau, ang_wedgeGapZInf_le_tau<br>(Angle/HorizontalSide)                                                                                                                                                                          |
| Lemma 6.2                 | lemma4_1_1, ang_abs_wedgeGapZInf_sub_le<br>(Angle/HorizontalSide)                                                                                                                                                                                   |
| Lemma 6.3                 | ang_le_sigmaAt_of_tendsto (Angle/HorizontalSide)                                                                                                                                                                                                    |
| Proposition 6.4           | pinned_bounds_of_maximal_positive (Variation), with<br>abs_supp_le_box (Basic/ConvexBody)                                                                                                                                                           |
| Lemma 6.6                 | (a) ang_three_points_mem, (b) ang_consumed_of_supp_zero,<br>(c) ang_mirror_mem_qMinus (Angle/RightAngle); (d)<br>consumed_of_pinned (AngleExtension), with<br>ang_wedgeGapWInf_mirror, ang_sigmaAt_mirror<br>(Angle/HorizontalSide)                 |
| Lemma 6.7                 | (a) ang_mem_niche_of_small, (b) ang_width_le_one<br>(Angle/RightAngle)                                                                                                                                                                              |
| Lemma 6.8                 | right_angle_motion_of_width (Angle/RightAngle)                                                                                                                                                                                                      |
| Proposition 6.9,<br>(6.1) | maximal_monotone_has_right_angle (Main),<br>right_angle_motion_of_pinned_bounds (AngleExtension);<br>the monotone sofa $U$ : maximal_envelope, own_cap_isMax<br>(Main); (6.1): maximizer_contained_in_gerver (Main)                                 |
| Lemma 7.1                 | inj_param_minus, inj_param_plus,<br>inj_param_minus_length, inj_param_plus_length,<br>inj_polygon_sigmaAt_eq (Injectivity/DiscreteIneq)                                                                                                             |
| Lemma 7.2                 | polygon_tau_le_geom (Curvature), with<br>inj_not_mem_qMinus (Injectivity/DiscreteIneq)                                                                                                                                                              |
| Lemma 7.3                 | polygon_curvature_with_defect (Curvature)                                                                                                                                                                                                           |
| Lemma 7.4                 | polygon_arm_cell (Curvature)                                                                                                                                                                                                                        |
| Lemma 7.5                 | polygon_step_with_error, polygon_Ico_with_errors,<br>polygon_Ioo_with_errors, polygonGridError (Curvature)                                                                                                                                          |
| Lemma 7.6                 | curvature_Ioo_limit, k0_integral_tendsto (Curvature),<br>with lemma6_4_2, inj_k0_lipschitz,<br>inj_dot_vplus_le_vint, inj_vint_le_dot_vminus,<br>inj_tendsto_vint, inj_tendsto_vint_right_dot,<br>inj_tendsto_vint_left_dot (Injectivity/LimitIneq) |
| Lemma 7.7                 | firstCurvature_of_polygon_errors,<br>firstCurvature_of_Ioo (Curvature),<br>inj_measure_le_of_Ioo, inj_intervalIntegrable_k0_gPlus<br>(Injectivity/LimitIneq)                                                                                        |
| Definition 7.8            | FirstCurvatureBound, SecondCurvatureBound (Curvature)                                                                                                                                                                                               |

| Result           | Lean declarations (continued)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Proposition 7.9  | curvature_of_maximal_positive,<br>firstCurvature_of_maximal_positive, diameter_le_box,<br>sampled_grid_error_le (Curvature), abs_supp_le_box<br>(Basic/ConvexBody), inj_gPlus_le_width, inj_k0_le<br>(Injectivity/LimitIneq); the second bound:<br>secondCurvature_of_maximal_positive,<br>secondCurvature_of_mirror_first, sigma_eq_map_mirror,<br>gPlus_mirror_eq_fMinus (Curvature)                                                                                                                                                                                                                                                                              |
| Lemma 7.10       | injCond1_of_curvature, first_arm_integral_lower,<br>second_arm_integral_lower, gPlus_sub_gPlus,<br>integral_fPlus (Curvature); the product rule: lemma5_1_3<br>(Basic/LebesgueStieltjes), inj_supp_sub_supp,<br>inj_product_coord (Injectivity/ArmLengths)                                                                                                                                                                                                                                                                                                                                                                                                          |
| Lemma 7.11       | lowerSeq_le_of_integral_bounds (Curvature)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Proposition 7.12 | injectivity_of_curvature, arms_strict_of_curvature<br>(Curvature)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Corollary 7.13   | isKi_of_maximal_area, sofaArea_pos_of_isMaxCap,<br>isMaxCap_of_area_eq (Main)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Lemma 8.1        | ki_maximizer_equality_conditions,<br>MamikonSegmentEquality,<br>mamikonSegmentEquality_of_isMax,<br>mamikonSegmentEquality_iff,<br>ki_upperQL_eq_gerver_of_sofaArea_eq,<br>ConvexDomain.eq_on_segment_of_isMax (Rigidity)                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| (8.1), Lemma 8.2 | halfSquareIntegral_combo_gap, displacement_combo,<br>displacement_ae_eq_of_mamikon_eq,<br>displacement_eqOn_of_mamikon_eq,<br>displacement_continuousOn_arc (Rigidity), with<br>theorem7_1_2_vertices (Convex/ConvexDomain) and<br>theorem7_4_1 (Convex/Mamikon); (a)<br>tangentKernel_of_mamikon_eq,<br>tangent_displacement_formula,<br>tangentKernel_of_equation (Rigidity), with theorem8_3_1,<br>theorem8_3_2 (Optimality/Concavity); (b)<br>middleKernel_of_mamikon_eq,<br>outer_displacement_formula, integrated_middle_equation<br>(Rigidity), with opt_outerCorner_cbv<br>(Optimality/UpperBound) and inj_dot_outerCorner_uvec<br>(Injectivity/ArmLengths) |
| Proposition 8.3  | ki_supp_sub_gerver_eq (Main);<br>capKernel_of_triple_midpoint,<br>capKernel_of_mamikonS_eq, CapKernel,<br>CapKernel.eq_horizontal_translation (Rigidity)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

| Result                                                                | Lean declarations (continued)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lemma 8.4                                                             | <p>(a) <code>cap_eq_translate_of_upper_support</code>,<br/> <code>mem_right_cap_iff</code>, <code>mem_cap_sub_horizontal_iff</code>,<br/> <code>mem_right_niche_iff</code>, <code>mem_niche_sub_horizontal_iff</code>,<br/> <code>sofa_eq_translate_of_upper_support</code> (Rigidity); (b)<br/> <code>isCap_translate_horizontal</code>, <code>supp_translate_horizontal</code>,<br/> <code>niche_translate_horizontal</code>,<br/> <code>sofaArea_translate_horizontal</code> (Rigidity); for Section 11,<br/> proved again without Rigidity:<br/> <code>isCap_translate_horizontal</code>, <code>supp_translate_horizontal</code>,<br/> <code>mem_right_niche_iff</code>, <code>mem_niche_sub_horizontal_iff</code>,<br/> <code>niche_translate_horizontal</code>,<br/> <code>sofaArea_translate_horizontal</code>, with the new lemma<br/> <code>cap_minus_niche_translate</code> for the last assertion of (a) when<br/> <math>K = C + (s, 0)</math><br/> (MovingSofaExtremal/HorizontalTranslation.lean)</p> |
| Theorem 8.5                                                           | <p><code>sofaArea_eq_gerver_iff</code>, <code>cap_eq_gerver_translate</code>,<br/> <code>translate_gerver_cap_sdiff_niche</code> (Main); the maximizing<br/> caps: <code>isMaxCap_iff_translate_gerver_cap</code> (Main); the<br/> monotone sofas: <code>right_angle_monotone_eq_gerver</code>,<br/> <code>ki_sofa_eq_gerver_translate</code> (Main)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Lemma 8.6                                                             | <p><code>MaximizerRoute.gerver_le_of_maximizes</code>,<br/> <code>MaximizerRoute.isKi_of_maximizes</code>,<br/> <code>MaximizerRoute.right_angle_maximizer_value</code>,<br/> <code>MaximizerRoute.right_angle_maximizer_eq_gerver</code><br/> (Maximizers)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Lemma 8.7                                                             | <p><code>MaximizerRoute.maximizing_monotone_has_right_angle</code><br/> (Maximizers), with <code>pinned_bounds_of_maximal_positive</code><br/> (Variation) and <code>right_angle_motion_of_pinned_bounds</code><br/> (AngleExtension)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Theorem 8.8                                                           | <p>(a) <code>MaximizerRoute.right_angle_cap_area_le_gerver</code>,<br/> <code>MaximizerRoute.right_angle_maximizes_iff_translate_gerver</code>,<br/> <code>MaximizerRoute.right_angle_sofaArea_eq_gerver_iff</code>,<br/> <code>MaximizerRoute.right_angle_optimality_and_rigidity</code>;<br/> (b) <code>MaximizerRoute.gerver_sofa_optimal</code>,<br/> <code>MaximizerRoute.right_angle_area_le_gerver</code>,<br/> <code>MaximizerRoute.area_le_gerver_of_large</code>,<br/> <code>MaximizerRoute.area_le_gerver</code>,<br/> <code>MaximizerRoute.volume_le_gerver</code>; (c)<br/> <code>MaximizerRoute.cap_area_le_gerver</code> (Optimality)</p>                                                                                                                                                                                                                                                                                                                                                           |
| Theorem 1.1<br>and Corollary 9.4<br>from Theorem 8.8<br>(Section 8.4) | <p><code>MaximizerRoute.image_eq_gerver_of_volume_eq</code>,<br/> <code>MaximizerRoute.isMaximal_iff_image_eq_gerver</code>,<br/> <code>MaximizerRoute.volume_eq_gerver_iff</code>,<br/> <code>MaximizerRoute.gerver_sofa_optimal_and_unique</code>; the<br/> steps: <code>MaximizerRoute.equal_area_envelope</code>,<br/> <code>MaximizerRoute.own_cap_maximizes</code>,<br/> <code>MaximizerRoute.right_angle_monotone_eq_gerver</code>,<br/> <code>MaximizerRoute.maximizer_contained_in_gerver</code><br/> (Alternative); the audit:<br/> scripts/AuditMaximizerRoute.lean</p>                                                                                                                                                                                                                                                                                                                                                                                                                                 |

| Result          | Lean declarations (continued)                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lemma 9.1       | eq_of_subset_of_measure_eq,<br>eq_of_subset_of_null_sdiff,<br>interior_subset_of_null_sdiff (Rigid); its use in the proof<br>of Theorem 1.1: Rigid.recover, Rigid.isClosed_image,<br>Rigid.volume_image (Rigid)                                                                                                                                                                                                                                         |
| Lemma 9.2       | regularClosed_cap_sdiff_envelope, isCompact_envUnder,<br>outside_closed_envelope_subset,<br>mem_closed_of_positive_path (RegularClosed)                                                                                                                                                                                                                                                                                                                 |
| Proposition 9.3 | gerver_regularClosed, envelope_isCompact,<br>envelope_endpoint_order,<br>envelope_bounds_of_path_height, gerver_contactC_zero,<br>gerver_contactA_pi_div_two,<br>gerver_niche_eq_envUnderStrict (RegularClosed); a cap<br>with rotation angle $\pi/2$ is closed downward above the floor:<br>opt_cap_down (Optimality/Domain)                                                                                                                           |
| Corollary 9.4   | isMaximal_iff_image_eq_gerver (Main)                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Lemma 9.5       | (a) gerver_corner_points, envUnderStrict_fst_mem_100<br>(RegularClosed), GerverParams.gs_H_zero,<br>GerverParams.gs_H_pi, GerverParams.gs_K_bounds,<br>GerverParams.gs_InjCond3,<br>GerverParams.gs_antitoneOn_path_fst<br>(Gerver/StructureCap), GerverParams.gs_α_neg,<br>GerverParams.gs_β_pos (Gerver/Frame); (b)<br>gerver_width_gt_one, (c)<br>fst_eq_zero_of_gerver_width_le_one (RegularClosed)                                                 |
| Corollary 9.6   | (a) translate_eq_gerver_of_volume_eq,<br>isMovingSofaWithAngle_pi_div_two_of_volume_eq (Main),<br>snd_sub_le_one_of_isMovingSofa,<br>dot_sub_vvec_le_one_of_mem_image,<br>Rigid.eq_translate_of_angle_eq_zero (Rigid); (b)<br>sin_eq_zero_of_rot_gerver_mem_horizSide,<br>sin_eq_zero_of_isMovingSofa_rot_gerver,<br>not_isMovingSofa_rot_pi_div_two_gerver (Main)                                                                                      |
| Theorem 10.1    | Baek.gerver_sofa_stable (stated in Challenge.lean, proved<br>in Solution.lean); unrestricted_stability,<br>reduced_sofa_stability,<br>normalizedSofa_eq_gerver_of_zero_deficit<br>(GlobalStability), UnrestrictedStability, sofaDeficit,<br>normalizingShift, normalizedSofa,<br>symmetricDifferenceArea (Statement); the sign of the deficit:<br>sofaDeficit_nonneg (Statement); $r$ -close sets:<br>EuclideanClose, DirectedClose (EuclideanGeometry) |
| Theorem 10.2    | Baek.gerver_sofa_angle_stable (stated in Challenge.lean,<br>proved in Solution.lean); terminal_angle_stability,<br>gerver_reduced_angle_eq (GlobalStability),<br>TerminalAngleStability (Statement)                                                                                                                                                                                                                                                     |

| Result            | Lean declarations (continued)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Theorem 10.3      | (b): <code>Baek.gerver_sofa_stability_exponent</code> (stated in <code>Challenge.lean</code> , proved in <code>Solution.lean</code> ); (a) <code>punctured_gerver_family</code> , <code>small_puncture_rigid_minimum</code> ( <code>PuncturedSofa</code> ), <code>puncture</code> ( <code>EuclideanDisks</code> ), <code>rigidHausdorffDistance</code> , <code>rigidHausdorffDistance_eq_of_attained</code> ( <code>PunctureMetric</code> ); (b) <code>no_hausdorff_exponent_gt_half</code> , (c) <code>rigid_distance_not_higher_order</code> , both with <code>exists_puncture_scale</code> , <code>puncture_power_identity</code> ( <code>SharpExponent</code> ) |
| (10.1), $\bar{T}$ | <code>wideUpperQ</code> , <code>InWideL</code> , <code>WideTriple</code> , <code>wideDomain</code> ( <code>WideDomain</code> ); $Q$ on $\mathcal{T}$ : <code>upperQ</code> ( <code>Optimality/UpperBound</code> )                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Theorem 10.4      | (a) <code>sharp_wide_cap_distance_bound</code> , <code>sharp_wide_cap_support_bound</code> ; (b) <code>sharp_ki_cap_distance_bound</code> , <code>sharp_ki_cap_support_bound</code> ; (c) <code>wide_cap_distance_bound_2002</code> , <code>ki_cap_distance_bound_2002</code> ( <code>SharpCapDistance</code> ), <code>cap_constant_lt_2002</code> ( <code>GreenNorm</code> ); $s_K$ and $K_G + (s_K, 0)$ : <code>capReferenceShift</code> , <code>shiftedReferenceCap</code> ; the proof: <code>cap_euclideanClose_of_upper_support</code> ( <code>CapDistance</code> )                                                                                            |
| (10.2)            | <code>segmentEnergy</code> , <code>deficit_eq_neg_dirDeriv_add_energy</code> ( <code>QuadraticDeficit</code> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| (10.3)            | <code>displacementEnergy</code> , <code>mamikon_combo_energy</code> ( <code>MamikonEnergy</code> ), <code>capResidualEnergy</code> , <code>capResidualEnergy_gap</code> , <code>tangent_energy_gap</code> , <code>outer_energy_gap</code> ( <code>WideResidualEnergy</code> ); (8.1), on which it rests, proved again without <code>MovingSofaUniqueness/Rigidity.lean</code> : <code>halfSquareIntegral_combo_gap</code> , <code>displacement_combo</code> , <code>mamikon_eq_halfSquareIntegral</code> ( <code>MamikonFoundation</code> )                                                                                                                         |
| Lemma 10.5        | <code>inWideL_comb</code> , <code>wide_upperQ_decomposition</code> ( <code>WideDomain</code> ), <code>wideUpperQ_quadratic</code> , <code>wideUpperQ_concave</code> ( <code>WideConcavity</code> ), <code>cap_mamikon_upperP_affine</code> , <code>mamikonS_convex_all</code> ( <code>NonsmoothAffinity</code> ), <code>mamikonS_add_upperP_nonsmooth</code> , <code>capAffineBoundary</code> ( <code>NonsmoothBookkeeping</code> ), <code>cap_area_four_arcs</code> , <code>edgeArea_eq_atom</code> ( <code>ArcAtoms</code> ), <code>outer_inner_area_affine</code> ( <code>CBVAlgebra</code> )                                                                    |
| Lemma 10.6        | <code>gerver_wide_firstVariation_nonpos</code> , <code>wideUpperQ_le_gerver</code> , <code>wideGerver_value</code> ( <code>WideGerverCertificate</code> ), <code>wide_reference_firstVariation</code> ( <code>WideFirstVariation</code> ), <code>reference_core_firstVariation</code> ( <code>ReferenceCoreVariation</code> ), <code>area_firstVariation_caps</code> , <code>mixedArea_symm</code> ( <code>MixedArea</code> ), <code>wide_maximum_iff_firstVariation</code> ( <code>WideConcavity</code> )                                                                                                                                                          |
| Proposition 10.7  | <code>wide_deficit_identity</code> , <code>wideDualSlack_nonneg</code> ( <code>WideGerverCertificate</code> ), <code>wideEnergy_eq_integrals</code> , <code>wide_deficit_eq_slack_add_integrals</code> , <code>wide_capResidualEnergy_le_deficit</code> ( <code>WideResidualEnergy</code> )                                                                                                                                                                                                                                                                                                                                                                         |

| Result            | Lean declarations (continued)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| (10.4)            | <code>tangentResidual, cornerResidual (Residuals),<br/>hasDerivWithinAt_tangentQuotient,<br/>tangent_quotient_integral, corner_reconstruct<br/>(ODEReconstruction)</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Lemma 10.8        | <code>the displacements: tangent_displacement_formula,<br/>outer_displacement_formula (MamikonFoundation);<br/>tangent_displacement_sub, outer_displacement_sub,<br/>tangentResidual_pinned, cornerResidual_pinned,<br/>tangentResidual_translation (Residuals), capDifference,<br/>capDifferenceDeriv, tangent_capDifference_integrable,<br/>corner_capDifference_integrable<br/>(ResidualIntegrability), capDifference_data,<br/>fourResidualEnergy_eq_capEnergy (CapCoercivity)</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Proposition 10.9  | <code>the formulas for <math>f</math>: tangent_reconstruct_left,<br/>tangent_reconstruct_right (ODEReconstruction); (a)<br/>FourResidualData, four_arc_mass_bound,<br/>fourResidualMass_le, four_arc_coercivity<br/>(FourArcCoercivity), last_arc_mass_bound,<br/>tangent_regular_arc_bound (ResidualPropagation),<br/>arcMass_sq_le, four_masses_le_four_sqrt (ResidualMass);<br/>(b) sharp_four_arc_coercivity, sharp_green_control<br/>(SharpEvaluation), sharp_last_formula,<br/>sharp_third_formula, sharp_middle_formula,<br/>sharp_first_formula, tail_product_integral<br/>(SharpReconstruction), last_kernel_norm,<br/>tailKernel_sq_integral (TrigKernelIntegrals),<br/>third_evaluation_norm, middle_evaluation_norm,<br/>first_evaluation_norm (SharpKernelNorms),<br/>greenNormSquared, greenNormSquared_le (GreenNorm),<br/>SquareControl (SharpIntegralControl); the last claim:<br/>capDifference_le_energy, wide_cap_support_bound<br/>(CapCoercivity); with Lemma 10.10, the distance with the<br/>constant 80: wide_cap_distance_bound (CapDistance)</code> |
| Lemma 10.10       | <code>(a) cap_lower_right_support, cap_lower_left_support,<br/>capDifference_bound_all (CapDistance),<br/>upperSupportClose_all (RoofMargins); (b)<br/>euclideanClose_of_support_bound,<br/>supp_add_euclideanDisk (SupportDistance),<br/>EuclideanClose.abs_supp_sub_le (EuclideanGeometry)</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Proposition 10.11 | <code>nearby_cap_certificate, nearby_cap_distance<br/>(LocalUpperBound), with coercive_certificate<br/>(CoerciveCertificate)</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Lemma 10.12       | <code>gerver_roof_data, gerverRoofLeft, gerverRoofRight,<br/>cap_top_strict_between_floor_endpoints (GerverRoof),<br/>CapRoofData (RoofGeometry), envelope_slope_bound<br/>(EnvelopeSlope), exists_roof_function,<br/>VerticalSlopeBound (CurveRoof), roof_value_of_envelope<br/>(GerverMargins); <math>\Gamma</math> as a graph: env_exists_curve_fst<br/>(Gerver/Envelope)</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

| Result            | Lean declarations (continued)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| Lemma 10.13       | (a) <code>exposed_face_property_stable</code> ,<br><code>cap_subset_unit_parallel</code> ( <code>ExposedFaceStability</code> ); (b)<br><code>face_property_stable_in_angle</code> ( <code>AngularFaceStability</code> );<br>(c) <code>CoreArmMargin</code> , <code>core_arm_margin_near_reference</code> ,<br><code>CoreArmMargin.oneSided</code> ( <code>LocalArmMargins</code> ),<br><code>CoreArmMargin.velocity_fst_le</code> ( <code>CoreMonotonicity</code> )                                                                                                                                                     |
| Lemma 10.14       | (a) <code>nearby_niche_horizontal_localization</code> ,<br><code>wedge_point_between_feet</code> , <code>gerver_wedge_feet_bounds</code><br>( <code>NicheFeet</code> ), <code>gerver_near_top_contacts</code><br>( <code>AngularFaceStability</code> ); (b) <code>nearby_niche_subset_cap</code> ,<br><code>nearby_niche_height</code> , <code>cap_rectangle_upper_margin</code><br>( <code>NicheContainment</code> ); (c) <code>nearby_bottomWidth</code> ,<br><code>area_le_bottomWidth</code> , <code>cut_feet_mem_of_width</code> ,<br><code>cut_regions_disjoint_of_width</code> ( <code>CapWidthGeometry</code> ) |
| Lemma 10.15       | (a) <code>nearby_canonical_inWideL</code> ,<br><code>canonical_inWideL_of_cut_arms</code> ( <code>CanonicalTriple</code> ),<br><code>canonical_right_contact</code> , <code>canonical_left_contact</code><br>( <code>CanonicalContacts</code> ); (b) <code>CutSeparated</code> ,<br><code>nearby_cutSeparated</code> ( <code>CutSeparation</code> ); (c)<br><code>nearby_core_height_pos</code> ( <code>CoreRegionGeometry</code> )                                                                                                                                                                                     |
| Lemma 10.16       | <code>separated_upperQ_bound</code> ( <code>LocalUpperBound</code> ),<br><code>separated_tail_areas</code> , <code>separated_right_wedge</code><br>( <code>SeparatedWedges</code> ), <code>positive_core_area_le</code><br>( <code>CoreAreaBound</code> ), <code>separated_core_below</code> ,<br><code>core_curveArea_rightVelocity</code> ( <code>CoreRegionGeometry</code> ),<br><code>exists_core_graph</code> ( <code>CoreGraph</code> ), <code>volume_under_core_graph</code><br>( <code>CoreIntegral</code> ), <code>corner_curveArea_integral</code><br>( <code>CornerAnalysis</code> )                         |
| Proposition 10.17 | <code>partialNiche</code> , <code>partialShape</code> , <code>omittedWedges</code><br>( <code>PartialHallways</code> ), <code>PartialSofaConstraints</code> ,<br><code>nearby_terminal_comparison</code> ,<br><code>area_capShape_of_niche_subset</code> ( <code>TerminalComparison</code> ),<br><code>terminal_region_comparison</code> ( <code>TerminalBookkeeping</code> ),<br><code>exists_uniform_cap_radius</code> ,<br><code>approximate_full_angle_slack</code> ( <code>UniformGeometryBounds</code> )                                                                                                          |
| Lemma 10.18       | <code>terminal_floor_loss</code> , <code>nearby_left_floor_rectangle</code> ,<br><code>terminal_excludes_low_left</code> ( <code>TerminalFloor</code> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Lemma 10.19       | <code>nearby_omittedWedges_area</code> ( <code>OmittedWedgeArea</code> ),<br><code>late_corner_height</code> , <code>omittedWedges_height</code> ,<br><code>area_two_windows_le</code> ( <code>PartialHallways</code> ),<br><code>gerver_floor_slab_covered</code> , <code>gerver_floor_mem_niche</code><br>( <code>FloorCoverage</code> )                                                                                                                                                                                                                                                                              |
| Lemma 10.20       | (a) <code>moving_strip_of_top</code> , <code>moving_terminal_width</code> ,<br><code>moving_terminal_lower</code> , <code>moving_supporting_hallways</code> ,<br><code>moving_hallway_slacks</code> , <code>moving_terminal_projection</code><br>( <code>SofaCoordinates</code> ); (b) <code>sofaCap</code> , <code>subset_sofaCap</code> ,<br><code>sofaCap_isCap</code> , <code>sofaCap_upper_support</code> ,<br><code>sofaCap_partial_constraints</code> , <code>sofaCap_close_to_gerver</code><br>( <code>SofaCap</code> ), <code>gerver_upper_support</code> ( <code>SofaCoordinates</code> )                     |

| Result            | Lean declarations (continued)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
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| Lemma 10.21       | (a) <code>CapRoofData.outer_margin</code> ( <code>RoofMargins</code> ); (b) <code>gerver_roof_slack_margin</code> ( <code>GerverMargins</code> ), <code>envelope_downward_slack</code> , <code>envelope_endpoint_speeds</code> ( <code>EnvelopeSlack</code> ); (c) <code>gerver_interiorBalls</code> ( <code>GerverRoof</code> ), <code>CapRoofData.interiorBalls</code> ( <code>RoofGeometry</code> ), <code>convex_interiorBalls_of_ball</code> , <code>convexBody_hasInteriorBalls</code> , <code>exists_interiorBalls_union</code> ( <code>InteriorBalls</code> ), <code>roofStrip_low_ball</code> , <code>roofStrip_hasInteriorBalls</code> ( <code>EpigraphBalls</code> ); the constants together: <code>gerver_recovery_constants</code> ( <code>GerverMargins</code> ) |
| Lemma 10.22       | (a) <code>ApproxHallways</code> , <code>RoofSlackMargin</code> , <code>directed_to_reference_of_margins</code> ( <code>RoofMargins</code> ); (b) <code>HasInteriorBalls</code> , <code>directedClose_sqrt_of_missing_area</code> , <code>reference_to_sofa_recovery</code> ( <code>MissingAreaRecovery</code> ), <code>reference_erosion_subset</code> ( <code>CapShape</code> )                                                                                                                                                                                                                                                                                                                                                                                               |
| Proposition 10.23 | <code>local_positive_sofa_hausdorff</code> ( <code>LocalSofaRecovery</code> ), with <code>nearby_terminal_comparison</code> ( <code>TerminalComparison</code> ) and <code>nearby_cap_distance</code> ( <code>LocalUpperBound</code> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Lemma 10.24       | <code>gerver_symmetricDifference_from_distance</code> , <code>CapRoofData.outer_area_bound</code> , <code>roof_gap_of_close_point</code> , <code>symmetricDifferenceArea_eq</code> ( <code>SymmetricDifference</code> ), <code>exists_parallel_layer_constant</code> , <code>parallel_subset_homothetic</code> ( <code>ConvexParallelArea</code> )                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Lemma 10.25       | (a) <code>moving_of_supporting_constraints</code> , <code>canonical_placement_coordinates</code> ( <code>SofaLimitMotion</code> ); (b) <code>normalizedBox</code> , <code>pinned_sofa_subset_box</code> , <code>moving_horizontal_span_le_six</code> , <code>connected_corner_height</code> ( <code>SofaBounds</code> ); (c) <code>normalized_moving_limit</code> ( <code>SofaLimitMotion</code> ), <code>compactShape_support_tendsto</code> , <code>compactShape_approximate_point</code> , <code>compactShape_subset_limit</code> , <code>compactShape_connected_limit</code> , <code>compactShape_area_limsup</code> , <code>compactShape_euclideanClose</code> ( <code>CompactSetLimits</code> )                                                                          |
| Proposition 10.26 | <code>near_maximizers_enter_neighborhood</code> , <code>maximizing_subsequence</code> ( <code>QualitativeEntry</code> ), <code>pinned_maximizer_eq_gerver</code> ( <code>SofaCoordinates</code> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Lemma 10.27       | <code>puncture_connected</code> , <code>puncture_area_loss</code> ( <code>EuclideanDisks</code> ), <code>preconnected_sdiff_of_connected_frontier</code> , <code>connected_sdiff_ball</code> ( <code>PunctureTopology</code> ), <code>puncture_euclideanClose</code> , <code>puncture_moving</code> , <code>puncture_movingWithAngle</code> , <code>moving_subset_of_closed_connected</code> ( <code>PunctureMetric</code> )                                                                                                                                                                                                                                                                                                                                                   |
| Lemma 10.28       | <code>rigid_copies_retain_interior</code> , <code>rigid_shift_bound</code> ( <code>RigidInterior</code> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

| Result           | Lean declarations (continued)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| Theorem 11.1     | <code>coercive_certificate</code><br><code>(MovingSofaStability/CoerciveCertificate.lean)</code> , the<br>conjunction of <code>wideUpperQ_le_gerver</code> , <code>wideGerver_value</code><br><code>(MovingSofaStability/WideGerverCertificate.lean;</code><br>Lemma 10.6) and <code>sharp_wide_cap_distance_bound</code><br><code>(MovingSofaStability/SharpCapDistance.lean;</code><br>Theorem 10.4(a)); $K_G + (s_K, 0)$ : <code>shiftedReferenceCap</code> ,<br><code>capReferenceShift</code><br><code>(MovingSofaStability/CapDistance.lean)</code>                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Lemma 11.2       | <code>wide_zero_deficit_cap</code><br><code>(MovingSofaExtremal/CoerciveRigidity.lean)</code> , with<br><code>EuclideanClose.eq_of_zero</code><br><code>(MovingSofaStability/EuclideanGeometry.lean)</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Proposition 11.3 | (a) <code>right_angle_maximizer_certificate</code> ,<br><code>right_angle_maximizer_value</code><br><code>(MovingSofaExtremal/CoerciveRigidity.lean)</code> ,<br><code>gerver_le_of_maximizes</code> , <code>isKi_of_maximizes</code><br><code>(MovingSofaExtremal/Geometry.lean)</code> , with theorem8_2_4<br><code>(Optimality/UpperBound)</code> ; $\xi_K$ as a triple of $\bar{T}$ :<br><code>kiExtensionTriple</code><br><code>(MovingSofaStability/MamikonFoundation.lean)</code> ,<br><code>toWideTriple</code> <code>(MovingSofaStability/WideDomain.lean)</code> ; (b)<br><code>right_angle_maximizer_eq_gerver</code><br><code>(MovingSofaExtremal/CoerciveRigidity.lean)</code> , with<br><code>cap_minus_niche_translate</code><br><code>(MovingSofaExtremal/HorizontalTranslation.lean)</code> ;<br>maximizing caps: <code>IsCap</code> <code>(Monotone/CapDefs)</code> together with<br><code>MaximizesCap</code> <code>(MovingSofaExtremal/Geometry.lean)</code> |
| Theorem 11.4     | (a) <code>right_angle_extremal</code><br><code>(MovingSofaExtremal/CoerciveRigidity.lean)</code> ,<br><code>right_angle_cap_area_le_gerver</code> ,<br><code>right_angle_maximizes_iff_translate_gerver</code> ,<br><code>right_angle_sofaArea_eq_gerver_iff</code> ; (b)<br><code>gerver_sofa_optimal</code> , <code>right_angle_area_le_gerver</code> ,<br><code>area_le_gerver_of_large</code> , <code>area_le_gerver</code> ,<br><code>volume_le_gerver</code> ; (c) <code>cap_area_le_gerver</code><br><code>(MovingSofaExtremal/Optimality.lean)</code> ; Fact 2.7 and<br>Lemma 8.7 in this proof: <code>exists_maximizing_cap</code> ,<br><code>maximizing_monotone_has_right_angle</code><br><code>(MovingSofaExtremal/Geometry.lean)</code>                                                                                                                                                                                                                             |
| Lemma 11.5       | (a) <code>own_cap_maximizes</code> , with theorem2_4_1<br><code>(Monotone/CapNiche)</code> for the cap property, (b)<br><code>equal_area_envelope</code> , (c) <code>right_angle_monotone_eq_gerver</code><br><code>(MovingSofaExtremal/Uniqueness.lean)</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Proposition 11.6 | <code>maximizer_contained_in_gerver</code><br><code>(MovingSofaExtremal/Uniqueness.lean)</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| Result                        | Lean declarations (continued)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| Theorem 11.7                  | <p>(a) <code>image_eq_gserver_of_volume_eq</code>, <code>translate_eq_gserver_of_volume_eq</code> (<code>MovingSofaExtremal/Uniqueness.lean</code>), with <code>Rigid.recover</code>, <code>snd_sub_le_one_of_isMovingSofa</code>, <code>dot_sub_vvec_le_one_of_mem_image</code>, <code>Rigid.eq_translate_of_angle_eq_zero</code> (<code>Rigid</code>) and <code>gserver_regularClosed</code>, <code>gserver_width_gt_one</code> (<code>RegularClosed</code>); (b) <code>volume_eq_gserver_iff</code>, <code>isMaximal_iff_image_eq_gserver</code>, <code>gserver_sofa_optimal_and_unique</code> (<code>MovingSofaExtremal/Uniqueness.lean</code>), and (i)<math>\Leftrightarrow</math>(iv) in <code>gserver_sofa_optimal_unique_stable</code> (<code>MovingSofaExtremal/Unified.lean</code>)</p>                                                                                                                                                               |
| Section 11.5,<br>Theorem 11.8 | <p><code>gserver_sofa_optimal_unique_stable</code> (<code>MovingSofaExtremal/Unified.lean</code>), with <code>unrestricted_stability</code>, <code>terminal_angle_stability</code> (<code>MovingSofaStability/GlobalStability.lean</code>); (A) <code>sofaDeficit_nonneg</code> (<code>MovingSofaStability/Statement.lean</code>); (B) <code>maximizing_subsequence</code> (<code>MovingSofaStability/QualitativeEntry.lean</code>), <code>pinned_maximizer_eq_gserver</code> (<code>MovingSofaStability/SofaCoordinates.lean</code>), <code>normalizedSofa_eq_gserver_of_zero_deficit</code> (<code>MovingSofaStability/GlobalStability.lean</code>); (C) <code>nearby_cap_certificate</code>, <code>nearby_cap_distance</code> (<code>MovingSofaStability/LocalUpperBound.lean</code>); the theorems of <code>Challenge.lean</code> through <code>MovingSofaExtremal:SolutionCoercive.lean</code>; the audit: <code>scripts/AuditCoerciveRoute.lean</code></p> |
| Fact 2.1, (2.1)               | <p>(a) <code>hasDerivWithinAt_supp_right</code>, <code>hasDerivWithinAt_supp_left</code>, <code>tendsto_vplus_right</code>, <code>tendsto_vminus_right</code>, <code>tendsto_vplus_left</code>, <code>tendsto_vminus_left</code>, <code>tendsto_vint_right</code>, <code>tendsto_vint_left</code> (<code>Basic/ConvexBody</code>); (b) <code>sigma_Ioc</code>, <code>sigma_periodic</code>, <code>proposition2_1_2</code>, <code>vplus_sub_vplus</code>, <code>theorem5_2_2</code> (<code>Basic/SurfaceArea</code>), <code>theorem7_1_3</code> (<code>Convex/ConvexDomain</code>); (c) <code>mpc_blaschke</code>, <code>mpc_tendsto_area</code> (<code>Balanced/MaxPolygonCapExists</code>); (2.1): <code>leftLim_sigmaFun</code> (<code>Basic/SurfaceArea</code>), <code>inj_sigma_Ioo_toReal</code> (<code>Injectivity/LimitIneq</code>); the support function of a Minkowski combination: <code>supp_comb</code> (<code>Convex/ConvexDomain</code>)</p>       |

| Result          | Lean declarations (continued)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| Definition 2.2  | <code>IsMovement, IsMovingSofaWithAngle, IsMovingSofa</code><br><code>(Sofa/Defs); the definition of the statement of record:</code><br><code>Baek.IsMovingSofa (Challenge.lean), and its equivalence</code><br><code>with the definition of the library: Baek.isMovingSofa_iff_lib</code><br><code>(Solution.lean); a moving sofa is bounded and compact:</code><br><code>isBounded_of_isMovingSofa, isCompact_of_isMovingSofa</code><br><code>(Monotone/SupportingHallway); its translates:</code><br><code>mpc_isMovingSofaWithAngle_translate</code><br><code>(Balanced/BalancedMaximumSofa)</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Fact 2.3        | <code>(a) theorem1_5_1 (Intro/RotationAngleBound); (b)</code><br><code>proposition2_3_1_exists, proposition2_3_1_subset</code><br><code>(Monotone/MonotoneSofa)</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| (2.2), Fact 2.4 | <code>the supporting hallways, their corners, walls and</code><br><code>quarter-planes: proposition2_2_2_hallway,</code><br><code>proposition2_2_2_innerCorner,</code><br><code>proposition2_2_2_outerCorner, proposition2_2_2_wallA,</code><br><code>proposition2_2_2_wallB, proposition2_2_2_wallC,</code><br><code>proposition2_2_2_wallD, proposition2_2_2_qPlus,</code><br><code>proposition2_2_2_qMinus (Monotone/SupportingHallway);</code><br><code>(2.2): IsCap.mem_iff (Monotone/CapContainsNiche), and</code><br><code><math>K \subset P_\omega</math>: IsCap.subset_para</code><br><code>(Monotone/SupportingHallway); (a) theorem2_3_2,</code><br><code>proposition2_3_3, IsMonotoneSofa.isMovingSofaWithAngle</code><br><code>(Monotone/MonotoneSofa); (b) lemma2_3_5_supp</code><br><code>(Monotone/MonotoneSofa), theorem2_4_1, theorem2_4_2</code><br><code>(Monotone/CapNiche); (c) theorem2_4_3</code><br><code>(Monotone/CapNiche), theorem2_5_9, theorem2_5_10</code><br><code>(Monotone/CapContainsNiche); (d) proposition2_5_4_isCap,</code><br><code>proposition2_5_4_supp, proposition2_5_4_vertices,</code><br><code>proposition2_5_4_sets, proposition2_5_4_sigma,</code><br><code>proposition2_5_4_hallway, cn_fan_mirror</code><br><code>(Monotone/CapContainsNiche), sofaArea_mirror</code><br><code>(Curvature)</code> |
| Fact 2.7        | <code>theorem3_5_5, theorem3_5_6</code><br><code>(Balanced/BalancedMaximumSofa); as one statement:</code><br><code>MaximizerRoute.exists_maximizing_cap (Maximizers)</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Fact 2.8        | <code>theorem1_1_1 (MovingSofaOptimality/Main.lean),</code><br><code>area_le_gerver (Main), Baek.gerver_sofa_optimal (stated</code><br><code>in Challenge.lean, proved in Solution.lean); <math> G  &gt; 11/5</math>:</code><br><code>gerverSofa_area_mem (MovingSofaOptimality/Main.lean);</code><br><code>the bound for caps: cap_area_le_gerver (Main), with</code><br><code>theorem3_5_5, theorem3_5_6</code><br><code>(Balanced/BalancedMaximumSofa)</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Fact 2.9        | <code>proposition3_2_1, proposition3_2_1_fix,</code><br><code>nef_supp_polyCap, proposition3_2_2, theorem3_2_3_1e,</code><br><code>theorem3_2_3 (Balanced/PolygonCap); a polygon cap is the</code><br><code>intersection of its supporting half-planes: nef_eq_setOf_supp</code><br><code>(Balanced/PolygonCap)</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Fact 2.10       | <code>lemma3_4_2 (Balanced/MaxPolygonCapExists)</code>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

| Result          | Lean declarations (continued)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fact 2.11       | the polyline $\mathbf{p}_K$ : theorem3_4_4 (Balanced/Polyline); (a) lemma3_4_5_one (Balanced/Polyline); (b) lemma3_4_5_two (Balanced/Polyline); (c) mpc_walk, mpc_walk_total, mpc_sum_sigma_sin (Balanced/CapGeometry), mpc_tau_eq, mpc_sum_tau_sin (Balanced/Polyline)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Fact 2.12       | the translates of polygon caps: proposition3_3_1 (Balanced/PolygonCap); $\mathcal{C}_\Theta(h) = K$ , $\mathcal{N}_\Theta(h) = \mathcal{N}_\Theta(K)$ and $\mathcal{A}_\Theta(h) = \mathcal{A}_\Theta(K)$ for $h = h_K _{\Theta^\circ}$ : proposition3_3_4, proposition3_3_5 (Balanced/PolygonCap); (a) lemma3_4_7 (Balanced/MaximumPolygonCap), theorem3_1_2, IsSimpleNefPolygon (Balanced/NefPolygon); (b) lemma3_4_8 (Balanced/MaximumPolygonCap); (c) proposition3_3_7, theorem3_3_6 (Balanced/PolygonCap)                                                                                                                                                                                                                                                                                      |
| Definition 2.13 | InjCond1, InjCond2, InjCond3, SatisfiesInjectivity (Injectivity/ArmLengths); $\mathcal{K}^i$ : IsKi (Optimality/Domain)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Fact 2.14       | the arm lengths (before Definition 2.13): proposition6_2_1 (Injectivity/ArmLengths), inj_arm_nonneg (Injectivity/DiscreteIneq), inj_gPlus_sub_gMinus (Injectivity/LimitIneq); (a) theorem6_2_3_right, theorem6_2_3_left (Injectivity/ArmLengths); (b) theorem6_2_5, fPlus_sub_fPlus (Injectivity/ArmLengths); (c) proposition6_2_2 (Injectivity/ArmLengths); (d) proposition6_4_5, proposition6_4_6_continuous, proposition6_4_6_deriv (Injectivity/LimitIneq); its end values $f_K(0) = g_K(\pi/2) = 1$ : inj_fK_zero (Injectivity/BoundingArms), inj_cap_consecutive, inj_consecutive (Injectivity/DiscreteIneq), inj_fMinus_eq (Injectivity/ArmLengths), gK_end (Curvature)                                                                                                                      |
| Fact 2.15       | $k_0$ , $m_0$ and $f_j$ : inj_k0_lipschitz (Injectivity/LimitIneq), inj_m0_mono, inj_m0_of_mem, inj_continuous_lowerSeq (Injectivity/BoundingArms); (a) lemma6_4_2 (Injectivity/LimitIneq); (b) lemma6_5_5 (Injectivity/BoundingArms)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Fact 2.16       | $\mathcal{J}$ and $\mathcal{M}$ : curveArea, segArea (Convex/CurveArea), convexCurveArea (Convex/ConvexCurve), mamikon, theorem7_4_1 (Convex/Mamikon); $\mathcal{K}^i$ and $\mathcal{T}$ are closed under Minkowski combinations: theorem8_1_1_convex, proposition8_1_2 (Optimality/Domain); (a) theorem8_1_8, with lemma8_1_5 (Optimality/Domain), theorem8_2_4 (Optimality/UpperBound); (b) lemma8_3_3, lemma8_3_4, lemma8_3_7, with lemma8_3_5, lemma8_3_6, and theorem8_3_1, theorem8_3_2, theorem8_3_8 (Optimality/Concavity), theorem7_1_2_supp, theorem7_1_2_vertices, theorem7_1_2_sigma (Convex/ConvexDomain), theorem7_4_1, theorem7_4_2 (Convex/Mamikon); (c) corollary8_5_8 (MovingSofaOptimality/Main.lean), theorem8_4_6, with theorem8_4_3_two, proposition8_4_4 (Gerver/Properties) |

| Result    | Lean declarations (continued)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fact 2.17 | (a) <code>definition8_1_2_exists</code> , <code>definition8_1_2_unique</code> , <code>gerverSofa_area_mem</code> ( <code>MovingSofaOptimality/Main.lean</code> ), <code>Baek.gerver_params_exists</code> , <code>Baek.gerver_params_unique</code> , <code>Baek.gerver_sofa_area</code> (stated in <code>Challenge.lean</code> , proved in <code>Solution.lean</code> ); (b) <code>theorem8_4_1_monotone</code> , <code>theorem6_1_2</code> ( <code>Gerver/Properties</code> ), <code>theorem8_1_1_gerver</code> ( <code>MovingSofaOptimality/Main.lean</code> ), <code>GerverParams.gm_sofaArea_cap</code> , <code>GerverParams.gm_isKi</code> ( <code>Gerver/Properties</code> ); (c) <code>theorem8_4_1_niche</code> , <code>theorem8_4_1_tangents</code> ( <code>Gerver/Properties</code> ), <code>env_D1_strictMono</code> , <code>env_x1_strictAnti</code> , <code>env_B1_strictMono</code> , <code>env_niche_eq</code> , <code>env_niche_subset_strict</code> , <code>env_subset_niche</code> , <code>env_exists_curve fst</code> ( <code>Gerver/Envelope</code> ), <code>GerverParams.gn_envHyp</code> , <code>GerverParams.gn_niche_eq</code> ( <code>Gerver/Niche</code> ), <code>gerver_niche_eq_envUnderStrict</code> ( <code>RegularClosed</code> ); (d) <code>GerverParams.gs_path_pi_div_two_snd</code> , <code>GerverParams.gs_beta_pi_div_two</code> ( <code>Gerver/Frame</code> ), <code>GerverParams.gs_C_pi_div_two</code> ( <code>Gerver/StructureCap</code> ); the contacts with the walls $b$ and $d$ described in Appendix B: <code>theorem8_4_1_walls</code> ( <code>Gerver/Properties</code> ) |
| Fact 2.18 | <code>GerverParams.path_snd_lt_one</code> ( <code>RegularClosed</code> ), <code>GerverParams.gb_x_pos</code> ( <code>Gerver/NicheBounds</code> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Fact 6.5  | (a) <code>lemma4_2_2</code> , (b) <code>lemma4_2_4</code> ( <code>Angle/RightAngle</code> ); $o_\omega - v_0 = cu_0$ and $o_\omega - u_\omega = cv_\omega$ : <code>proposition4_2_1</code> ( <code>Angle/RightAngle</code> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

## APPENDIX E. THE USE OF AI

ChatGPT Pro 6 (OpenAI) wrote the first version of the proof of Theorem 1.1 on 2 October 2026, at the author's request, together with drafts of its formalization and of the connection with formal-conjectures; neither Lean draft had ever been compiled. Claude Opus 5.5 (Anthropic), in Claude Code with sub-agents, formalized [1] and completed the two drafts, following the author's **formalize-math-paper** procedure, in rounds between 1 and 3 October 2026; 54 proofs of the first draft and 78 proofs of the second did not compile and were replaced and proved again. Before any proof was written, a separate run of Claude Opus 5.5 reviewed every Lean statement of a result of [1] against its TeX source, and a later separate run compared the Lean statements of uniqueness with the informal proof and found no weakening, vacuous definition or hidden assumption. The repository [14] records who did what, with times and sizes. Claude Sonnet 5.5 (Anthropic), in Claude Code, wrote the first version of this manuscript on 4 October 2026 from the Lean library, the illustrated text of the proofs in [14] and the text of [1]. All figures but one were computed from the definitions by the script `docs/paper/figures/make_figures.py` of the repository [14]; the remaining one is drawn in TikZ. The author asked for each of these steps and

decided its scope. None of the three systems is an author of this paper, since none of them can take responsibility for it. The author has proofread and edited the abstract and Section 1, and has not yet read the other sections; they have been checked only by the runs described below.

The text of this paper is not machine-checked; the dictionary of Appendix D gives its correspondence with the Lean statements. On 4 October 2026, six runs of Claude Opus 5.5, each assigned some sections, compared the statements and proofs of the first version of this paper with the Lean statements and with the TeX source of [1]; a run of Claude Sonnet 5.5 checked the claims about the formalization, the dictionary and the references; and a run of Claude Opus 5.5 read the whole paper without access to either. The findings were corrected, and a further run of Claude Opus 5.5 checked the passages rewritten in response. Two more runs of Claude Opus 5.5 checked a rewrite of the prose for the patterns of machine-written text and for changes of claim. The same day, the Lean library was extended with the results of this paper that it did not yet prove, and Sections 2 and 4 to 9 were rewritten, by sub-agents running Claude Opus 5.5, to state what the Lean states and to follow its proofs. Five runs of Claude Opus 5.5, each assigned some sections and appendices, then compared the result with the Lean, and a second round of the same runs compared the corrected text again. The changes that followed the second round were not checked again.

On 5 October 2026, ChatGPT Pro 6 wrote the second proof of optimality (Section 8.4) in Lean, with a plan for presenting it in this paper. The Lean code had never been compiled. It came as a pull request to the repository, which Claude Opus 5.5, in Claude Code, merged at the author’s request. The code compiled without change and passed the audit that came with it, and also an extended audit, written by Claude Opus 5.5, that covers the results from which Baek derives step (3) of Section 1.3. From the Lean, Claude Opus 5.5 then wrote Section 8.4, Fact 2.7 and the passages that refer to them in Sections 1 to 3 and 12 and Appendices C and D. Two more runs of Claude Opus 5.5 checked these passages, one against the Lean and one for their prose, and their findings were applied. At the author’s request, Claude Opus 5.5 also wrote the sentence of the abstract on the second proof.

The same day, ChatGPT Pro 6 wrote the proofs of Section 10 as notes and as Lean code in 85 modules, which had never been compiled; they came as another pull request. Claude Opus 5.5, in Claude Code with sixteen sub-agents, made the code compile and merged it, at the author’s request. Of its proofs, 123 did not compile and were proved again along the draft’s arguments. Twenty lemmas had lost hypotheses that the draft declared as section variables, since Lean omits a section variable from a statement that does not mention it; these hypotheses were added back. A separate run of Claude Opus 5.5 compared every repaired proof with the draft’s and found no change of argument. On 6 October 2026, Claude Opus 5.5 stated the three theorems of stability in `Challenge.lean`, and a sub-agent running Claude Opus 5.5 wrote Section 10 and its rows of Table 2 from the Lean.

The same session also wrote the sentence of the abstract and the passages of Sections 1 and 12 and Appendix D on stability, and corrected a sign in the description of  $\mathcal{M}_K$  in Section 2. Three runs of Claude Opus 5.5 compared the statements and proofs of Section 10 with the Lean, part by part; a further run compared the new text with the Lean and the repository, and another read it for its prose. Their findings were applied.

On the evening of 5 October 2026, ChatGPT Pro 6 also wrote a third pull request, without compiling or running any of it. It contained the library `MovingSofaExtremal`, which derives optimality and uniqueness from the estimates of Sections 10.1 and 10.2; the module `MamikonFoundation` of the stability library, which proves again results of the module `Rigidity` of the first proof on square integrals, Mamikon displacements and the canonical triple; a second solution of twelve statements of the Challenge (`SolutionCoercive.lean`); and the audit `scripts/AuditCoerciveRoute.lean`. It also switched the module `MamikonEnergy` of the stability library from importing `Rigidity` to importing `MamikonFoundation`. On 6 October, at the author's request, Claude Opus 5.5, in Claude Code, merged the manuscript's branch, which held the compiled stability library, into the pull request's and compiled the result. One proof did not compile: a `change` step in the module `HorizontalTranslation` whose two sides were not definitionally equal, which was replaced by a rewrite; one linter warning was also cleared. The same session then did what the pull request had left for later. It stated the certificate as one theorem (Theorem 11.1), added to `MovingSofaExtremal` the theorem that no rotation is needed (Theorem 11.7(a)), and moved two lemmas of that proof, which Corollary 9.6 shares, from the module `Main` to the module `Rigid`. The stability library had used the module `Main` at four places: Baek's theorem twice, for the sign of the deficit and in the compactness step; the first proof of uniqueness once, to identify the limit of the compactness step and the normalized sofa of zero deficit; and the lemma that a moving sofa lies in a horizontal strip of height one. The session switched these to `MovingSofaExtremal` and to the moved lemmas, and made the local cap estimate of the stability library use the certificate. It then stated Theorem 11.8, extended `SolutionCoercive.lean` to the fifteen statements of the Challenge, and rewrote and ran the audit. The pull request was merged the same day. A sub-agent running Claude Opus 5.5 wrote Section 11, and the passages that refer to it in the other sections and in Appendices D and E, from the Lean; the session wrote the sentence of the abstract on the single estimate. Three runs of Claude Opus 5.5, launched by the sub-agent, compared the new text with the Lean and the repository, part by part, and a fourth checked the passages rewritten after their reports. Two more runs of Claude Opus 5.5, neither able to edit, then read the new text, one against the Lean and the repository and one for its prose. The findings of all these runs were applied; the edits that followed the last two runs were not checked again. No person has read Section 11. At the time of writing, the argument of this paper has not been refereed.

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